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CONIC SECTION

JEE EXERCISES

Exercise J-I

STANDARD & SHIFTED PARABOLA

1. The equation of the circle drawn with the focus of the parabola $(x-1)^2 - 8y = 0$ as its centre and touching the parabola at its vertex is :

परवलय $(x-1)^2 - 8y = 0$ की नाभि को केन्द्र मान कर बनाये गये वृत्त जोकि परवलय को उसके शीर्ष पर स्पर्श करता है, का समीकरण gS

(A) $x^2 + y^2 - 4y = 0$ (B) $x^2 + y^2 - 4y + 1 = 0$

(C) $x^2 + y^2 - 2x - 4y = 0$ (D) $x^2 + y^2 - 2x - 4y + 1 = 0$

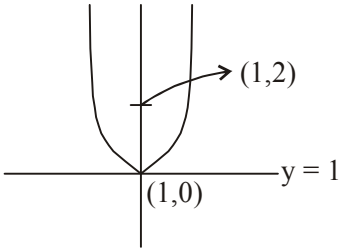
Ans. D

Sol. $(x-1)^2 = 8y \Rightarrow 4a = 8 \Rightarrow a = 2$

Vertex = (1,0) focus = (1,2)

distance between focus a& vertex is r.

r = 2



Equation of circle

$(x-1)^2 + (y-2)^2 = 4$

$x^2 + y^2 - 2x - 4y + 1 = 0$

2. If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$ passes through the points of intersection of the parabola $y^2 = 4ax$ and $x^2 = 4ay$, then :

यदि $a \neq 0$ तथा रेखा $2bx + 3cy + 4d = 0$ परवलय के प्रतिच्छेदन बिन्दु से जाती हो तो $y^2 = 4ax$ and $x^2 = 4ay$, तब :

(A) $d^2 + (3b - 2c)^2 = 0$ (B) $d^2 + (3b + 2c)^2 = 0$

(C) $d^2 + (2b - 3c)^2 = 0$ (D) $d^2 + (2b + 3c)^2 = 0$

Ans. D

Sol. Given parabolas are

$y^2 = 4ax$ ..... (i)

$x^2 = 4ay$ ..... (ii)

Putting the value of y from (ii) in (i), we get

$\frac{x^2}{16a^2} = 4ax \Rightarrow x(x^3 - 64a^3) = 0 \Rightarrow x = 0, 4a$

from (ii), y = 0, 4a. Let A ≡ (0, 0); B ≡ (4a, 4a)

Since, given line $2bx + 3cy + 4d = 0$ passes through A and B,

$\therefore$ d = 0 and $8ab + 12ac = 0$

$\Rightarrow$ $2b + 3c = 0$. ($\because$ $a \neq 0$)

Obviously, $d^2 + (2b + 3c)^2 = 0$

PARAMETRIC COORDINATES OF PARABOLA

3. PN is an ordinate of the parabola $y^2 = 4ax$ (P on $y^2 = 4ax$ and N on x-axis). A straight line is drawn parallel to the axis to bisect NP and meets the curve in Q. NQ meets the tangent at the vertex A in apoint T such that AT = kNP, then the value of k is : (where A is the vertex)

परवलय $y^2=4ax$ की कोटि PN है जहां P परवलय पर स्थित है। अक्ष के समान्तर एक सरल रेखा NP को समद्विभाजित करती है तथा वक्र को Q पर मिलती है। NQ शीर्ष A पर खींची गयी स्पर्श रेखा को बिन्दु T पर इस प्रकार मिलती है कि AT = kNP हो, तो k का मान है : (जहां A शीर्ष है)

(A) 3/2 (B) 2/3 (C) 1 (D) 1/3

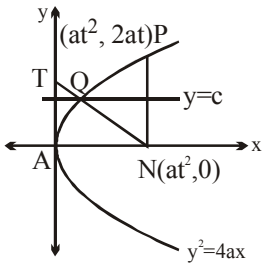
Ans. B

Sol. Equation of PN : $x = at^2$

$y = c$ bisects PN

$\therefore \quad c = at$

which cuts the parabola at Q



$\Rightarrow \quad c^2 = 4ax \qquad \therefore \quad x = \dfrac{c^2}{4a}$

$\therefore \quad Q\left(\dfrac{c^2}{4a}, c\right) = Q\left(\dfrac{at^2}{4}, at\right)$

$\therefore \quad$ Equation of NQ: $y - 0 = \dfrac{at - 0}{\dfrac{at^2}{4} - at^2}(x - at^2)$

$$y = \frac{-4}{3t}(x - at^2)$$

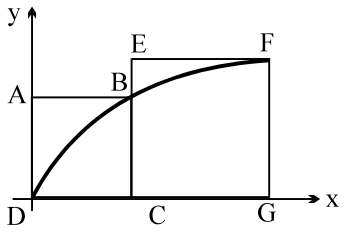
which cuts x = 0 at $\left(0, \dfrac{4at}{3}\right)$

$\therefore \quad T = \dfrac{4at}{3}$ and $NP = 2at$

$\therefore \quad \dfrac{AT}{NP} = \dfrac{4at/3}{2at} = \dfrac{2}{3}$ Ans.

4. ABCD and EFGC are squares and the curve $y = k\sqrt{x}$ passes through the origin D and the points B and F. The ratio $\dfrac{FG}{BC}$ is :

चित्र में ABCD और EFGC वर्ग है और वक्र $y = k\sqrt{x}$ मूल बिन्दु D और बिन्दुओं B और F से गुजरता है तब $\frac{FG}{BC}$ है :



(A) $\frac{\sqrt{5}+1}{2}$ (B) $\frac{\sqrt{3}+1}{2}$ (C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{3}+1}{4}$

Ans. A

Sol. $y^2 = k^2x \quad \Rightarrow \quad y^2 = 4ax$ where $k^2 = 4a$

$B = \left(at_1^2, 2at_1\right);\ F = \left(at_2^2, 2at_2\right) \qquad \{t_1 > 0, t_2 > 0\}$

to find $\frac{FG}{BC} = \frac{2at_2}{2at_1} = \frac{t_2}{t_1}$

now $DC = BC \quad \Rightarrow \quad at_1^2 = 2at_1 \quad \Rightarrow \quad t_1 = 2$

also $at_2^2 - at_1^2 = 2at_2$

$t_2^2 - 4 = 2t_2$

$t_2^2 - 2t_2 - 4 = 0$

$t_2 = \frac{2 \pm \sqrt{4+16}}{2}$ but $t_2 > 0 \quad \therefore \quad t_2 \neq \frac{2-\sqrt{20}}{2}$

$t_2 = \left(\sqrt{5}+1\right)$

$\therefore \quad \frac{t_2}{t_1} = \frac{\sqrt{5}+1}{2}$ Ans.

5. Let P be the point (1, 0) and Q a point on the locus $y^2 = 8x$. The locus of mid point of PQ is :
माना परवलय $y^2 = 8x$ पर बिन्दु Q है तथा कोई बिन्दु P(1, 0) तब PQ के मध्य बिन्दु का बिन्दुपथ होगा :
(A) $y^2 - 4x + 2 = 0$ (B) $y^2 + 4x + 2 = 0$ (C) $x^2 + 4y + 2 = 0$ (D) $x^2 - 4y + 2 = 0$

Ans. A

Sol. $2h = \frac{\alpha^2}{8} + 1$

$8(2h-1)=\alpha^2$ $\qquad$ $2k=\alpha$

$8(2h-1)=4k^2$

$4h-2=k^2$

$y^2-4x+2=0$

6. The vertex A of the parabola $y^2=4ax$ is joined to any point P on it and PQ is drawn at right angles to AP to meet the axis in Q. Projection of PQ on the axis is equal to :

(A) twice the length of latus rectum (B) the length of latus rectum

(C) half the length of latus rectum (D) one fourth of the length of latus rectum

परवलय $y^2=4ax$ के शीर्ष A को परवलय पर स्थित बिन्दु P से मिलने पर यह PQ पर समकोण बनाता है जबकि Q अक्ष पर है तब PQ का अक्ष पर प्रक्षेप होगा :

(A) नाभिलम्ब का दुगुना (B) नाभिलम्ब की लम्बाई के बराबर

(C) नाभिलम्ब की लम्बाई का आधा (D) नाभिलम्ब की लम्बाई का एक चौथाई

Ans. B

Sol. hyperbola $\frac{x^2}{16}-\frac{y^2}{48}=1$

7. The graph of the equation $x+y=x^3+y^3$ is the union of :

(A) line and an ellipse (B) line and a parabola

(C) line and hyperbola (D) line and a point

समीकरण $x+y=x^3+y^3$ निम्न वक्रों का संयुक्त समीकरण है :

(A) रेखा तथा दीर्घवृत्त (B) रेखा तथा परवलय

(C) रेखा तथा अतिपरवलय (D) रेखा तथा बिन्दु

Ans. A

Sol. Given $(x+y)=(x+y)(x^2-xy+y^2) \Rightarrow (x+y)(x^2-xy+y^2-1)=0$

$\therefore$ Either $x+y=0$ or $x^2-xy+y^2-1=0$

Now $x+y=0$ represent a line.

Also $\Delta=\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}=\begin{vmatrix} 1 & \frac{-1}{2} & 0 \\ \frac{-1}{2} & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix}=\frac{-3}{4}\neq 0$ and $h=\frac{-1}{2}$, $a=1=b$, so $h^2<ab$

$\therefore$ $x^2-xy+y^2-1=0$ represent an ellipse .

Hence locus of P(x, y) satisfying above equation is the union of line and an ellipse.

STANDARD & SHIFTED ELLIPSE

8. A focus of an ellipse is at the origin. The directrix is the line x = 4 and the eccentricity is $\frac{1}{2}$. Then the length of the semi-major axis is :

एक दीर्घवृत्त की एक नाभि मूल बिन्दु पर है। रेखा x = 4 उसकी नियता है तथा उसकी उत्केन्द्रता $\frac{1}{2}$ है। तो उसके अर्ध–दीर्घ अक्ष की y Ec lekbZ gS

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{5}{3}$ (D) $\frac{8}{3}$

Ans. D

Sol. $e = \frac{5}{2} \quad \Rightarrow \quad \frac{a}{e} - ae = 4 \quad \Rightarrow \quad a\left[2 - \frac{1}{2}\right] = 4 \quad \Rightarrow \quad a \cdot \frac{3}{2} = 4$

$a = \frac{8}{3} \quad \Rightarrow \quad a\left[\frac{1}{2} - 2\right] = 4$

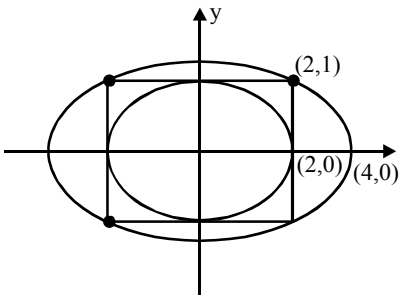
9. The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the coordinate axes, which in turn is inscribed in another ellipse that passes through the point (4, 0). Then the equation of the ellipse is : **[AIEEE-2009]**

दीर्घवृत $x^2 + 4y^2 = 4$ निर्देशांक अक्षों से सरेखित एक आयत के अन्तर्गत है जो स्वयं बिन्दु (4, 0) से जाने वाले दूसरे दीर्घवृत के अन्तर्गत gSA rc bl nh?kZoÙ kZ dk lehdj.k gS **[AIEEE-2009]**

(A) $4x^2 + 48y^2 = 48$ (B) $4x^2 + 64y^2 = 48$ (C) $x^2 + 16y^2 = 16$ (D) $x^2 + 12y^2 = 16$

Ans. D

Sol.



Ellipse $x^2 + 4y^2 = 4$ (Given)

Eq^n. of the ellipse (required)

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$$

Ellipse passes through (2, 1)

therefore $\frac{4}{16}+\frac{1}{b^2}=1 \Rightarrow b^2=\frac{4}{3}$

$$\frac{x^2}{16}+\frac{y^2}{4/3}=1 \Rightarrow \frac{x^2}{16}+\frac{3y^2}{4}=1$$

$\Rightarrow x^2 + 12y^2 = 16$

10. An ellipse is drawn by taking a diameter of the circle $(x-1)^2 + y^2 = 1$ as its semi-minor axis and a diameter of the circle $x^2 + (y-2)^2 = 4$ as its semi-major axis. If the centre of the ellipse is at the origin and its axes are the coordinate axes, then the equation of the ellipse is : **[AIEEE-2012]**

वृत $(x-1)^2 + y^2 = 1$ के एक व्यास को अर्ध लघु अक्ष लेकर तथा वृत $x^2 + (y-2)^2 = 4$ के एक व्यास को अर्ध दीर्घ अक्ष लेकर एक nh?kZo`Ùk [kapk gS tks fn nh?kZo`Ùk dk dsUnz ewy fcUnq gS rFkk bl dk v{kh; v{kksa ds vuqfn'k gSA nh?kZo`Ùk dk lehdj.k gS **[AIEEE-2012]**

(A) $x^2 + 4y^2 = 16$ (B) $4x^2 + y^2 = 4$ (C) $x^2 + 4y^2 = 8$ (D) $4x^2 + y^2 = 8$

Ans. A

Sol. Let the equation of ellipse be

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

from the given conditions

$a = 4$ and $b = 2$

$\therefore$ Eq of ellipse is

$$\frac{x^2}{16}+\frac{y^2}{4}=1$$

or $x^2 + 4y^2 = 16$

STANDARD & SHIFTED HYPERBOLA

11. For the hyperbola $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$, which of the following remains constant α varies?

(A) Abscissae of vertices (B) Abscissae of foci

(C) Eccentricity (D) Directrix

अतिपरवलय $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$, के लिये α का मान परिवर्तित करने पर निम्न में से कौन अचर रहेगा?

(A) शीर्षों के भुज (B) नाभियों के भुज

(C) उत्केन्द्रता (D) नियता

Ans. B

Sol. The given equation of hyperbola is

$$\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$$

$\Rightarrow$ $a = \cos\alpha$, $b = \sin\alpha$

$\Rightarrow$ $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \tan^2\alpha} = \sec\alpha$

$\Rightarrow$ $ae = 1$

$\therefore$ foci $(\pm 1, 0)$

$\therefore$ foci remain constant with respect to α.

12. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then the value of b^2 is :

दीर्घवृत्त $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ तथा अतिपरवलय $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ की नाभियां सम्पाती हों तो b^2 का मान होगा :

(A) 9 (B) 1 (C) 5 (D) 7

Ans. D

Sol. For ellipse $a^2 = 16$ $\Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{\sqrt{16 - b^2}}{4}$ $\Rightarrow$ focii $\equiv (\pm ae, 0) = \left(\pm\sqrt{16 - b^2}, 0\right)$

For hyperbola, $e = \sqrt{1 + \frac{b^2}{a^2}} = \frac{5}{4}$ $\therefore$ focii $\equiv (\pm ae, 0) = (\pm 3, 0)$

$\therefore$ $\sqrt{16 - b^2} = 3$ $\Rightarrow$ $b^2 = 7$

13. In eccentricity of conjugate hyperbola of the given hyperbola $\left|\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2}\right| = 3$ is e', then value of 8e' is :

दिये गये अतिपरवलय $\left|\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2}\right| = 3$ के संयुग्मी अतिपरवलय की उत्केन्द्रता e' है, तो 8e' का मान होगा :

(A) 12 (B) 14 (C) 17 (D) 10

Ans. D

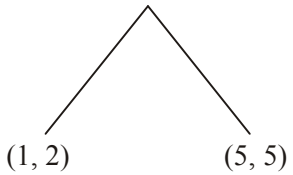
Sol. $(PS_1 - PS_2) = 2a$

$2a = 3$ $\Rightarrow a = \frac{3}{2}$

$2ae = 5$

$$2 \times \frac{3}{2} \times e = 5$$

$$e = \frac{5}{3}$$



$$\frac{1}{e^2} + \frac{1}{e'^2} = 1$$

$$\frac{9}{25} + \frac{1}{e'^2} = 1 \Rightarrow \left(\frac{1}{e'}\right)^2 = \frac{16}{25} \Rightarrow e' = \frac{5}{4}$$

$$8e' = 8 \times \frac{5}{4} = 10$$

14. A hyperbola, having the transverse axis of length 2 sin θ, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is :

एक अतिपरवलय, जिसकी अनुप्रस्थ अक्ष की लम्बाई 2 sin θ है तथा दीर्घवृत्त $3x^2 + 4y^2 = 12$ की नाभियाँ सम्पाती हैं। तब अतिपरवलय d hl ehd j . kgS

(A) $x^2 \operatorname{cosec}^2\theta - y^2 \sec^2\theta = 1$ (B) $x^2 \sec^2\theta - y^2 \operatorname{cosec}^2\theta = 1$

(C) $x^2 \sin^2\theta - y^2 \cos^2\theta = 1$ (D) $x^2 \cos^2\theta - y^2 \sin^2\theta = 1$

Ans. A

Sol. Given $3x^2 + 4y^2 = 12$ an ellipse

$\therefore$ $a^2 = 4$ $b^2 = 3$

$\therefore$ $e = \sqrt{1 - \frac{3}{4}} \Rightarrow e = \frac{1}{2}$

$\therefore$ It's focus will be $(\pm 1, 0)$

Since hyperbola is confocal to given ellipse, therefore $\pm ae = \pm 1$, but $a = \sin\theta$ given

$\therefore$ $e = \frac{1}{\sin\theta}$, Now $b^2 = a^2(e^2 - 1)$

$$\beta^2 = \sin^2\theta \frac{\cos^2\theta}{\sin^2\theta} \Rightarrow b^2 = \cos^2\theta$$

Hence required equation will be,

$$\frac{x^2}{\sin^2\theta} - \frac{y^2}{\cos^2\theta} = 1$$

$\Rightarrow$ $x^2 \operatorname{cosec}^2\theta - y^2 \sec^2\theta = 1$

15. Let a and b be non-zero real numbers. Then, the equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ represents :

(A) four straight lines, when $c = 0$ and a, b are of the same sign.

(B) two straight lines and a circle, when $a = b$, and c is of sign opposite to that of a.

(C) two straight lines and a hyperbola, when a and b are of the same sign and c is of sign opposite to that of a

(D) a circle and an ellipse, when a and b are of the same sign and c is of sign opposite to that of a.

माना a एवं b अशून्य वास्तविक संख्याएं है। समीकरण $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ प्रदर्शित करता है :

(A) चार सरल रेखाओं को जबकि $c = 0$ तथा a, b समान चिन्ह के हो

(B) दो सरल रेखाओं तथा एक वृत को यदि $a = b$, तथा c का चिन्ह a के विपरीत है।

(C) दो सरल रेखाओं तथा एक अतिपरवलय को जहाँ पर a एवं b समान चिन्ह के हो तथा c, a के विपरीत चिन्ह का है

(D) एक वृत तथा एक दीर्घवृत को जहाँ पर, a एवं b समान चिन्ह के है तथा c, a के विपरीत चिन्ह का है

Ans. B

Sol. $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$

either $x^2 - 5xy + 6y^2 = 0 \Rightarrow$ two straight lines passing through origin.

or $ax^2 + by^2 + c = 0$

(A) If c = 0, and a and b are of same sign then it will represent a point.

(B) If a = b, c is of sign opposite to a then it will represent circle.

(C) When a & b are of same sign and c is of sign opposite to a then it will represent ellipse.

(D) This is clearly incorrect.

16. Consider a branch of the hyperbola, $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A. Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A, then the area of the triangle ABC is :

माना बिन्दु A पर स्थित शीर्ष वाला अतिपरवलय $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ है। माना इसके नाभिलम्ब के शीर्षो में से एक ' kkZB है। यदि बिन्दु A के निकटतम अतिपरवलय की नाभि C है, तब ΔABC का क्षेत्रफल है :

(A) $1 - \sqrt{\frac{2}{3}}$ (B) $\sqrt{\frac{3}{2}} - 1$ (C) $1 + \sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{3}{2}} + 1$

Ans. B

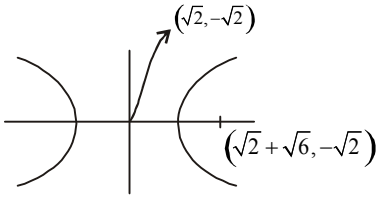
Sol. The given equation is

$(x - \sqrt{2})^2 - 2(y + \sqrt{2})^2 = 4$

$\frac{(x - \sqrt{2})^2}{4} - \frac{(y + \sqrt{2})^2}{2} = 1$

$a = 2, \quad b = \sqrt{2}$

hence eccentricity $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{\frac{3}{2}}$



$$\text{Area} = \frac{1}{2}a(e-1) \times \frac{b^2}{a}$$

$$= \left(\sqrt{\frac{3}{2}} - 1\right) \text{ sq. units.}$$

TANGENT & NORMAL

17. The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P, intersect at R, then the equation of the locus of R is :

वक्र $y^2 = 4ax$ के किसी बिन्दु P और शीर्ष को मिलाने वाली रेखा और बिन्दु P पर स्पर्श रेखा पर नाभि से लम्ब बिन्दु R पर प्रतिच्छेद करती g\$r c fcUuqR का बिन्दु पथ होगा :

(A) $x^2 + 2y^2 - ax = 0$ (B) $2x^2 + y^2 - 2ax = 0$ (C) $2x^2 + 2y^2 - ay = 0$ (D) $2x^2 + y^2 - 2ay = 0$

Ans. B

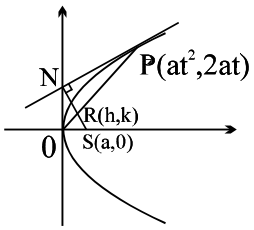
Sol. $T : ty = x + at^2$ ....(1)

line perpendicular to (1) through (a,0)

$tx + y = ta$ ....(2)

equation of OP : $y - \frac{2}{t}x = 0$ ....(3)

from (2) & (3) eleminating t we get locus



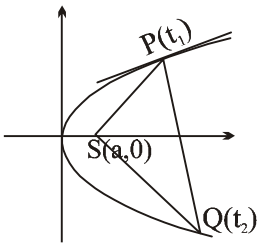
18. The normal chord of a parabola $y^2 = 4ax$ at the point whose ordinate is equal to the abscissa, then angle subtended by normal chord at the focus is :

परवलय $y^2 = 4ax$ की किसी बिन्दु, जिसके भुज और कोटि बराबर है, पर अभिलम्ब जीवा का नाभि पर अन्तरित कोण है :

(A) $\frac{\pi}{4}$ (B) $\tan^{-1}\sqrt{2}$ (C) $\tan^{-1}2$ (D) $\frac{\pi}{2}$

Ans. D

Sol. $at_1^2 = 2at_1 \quad \Rightarrow \quad t_1 = 2 \; ; P(4a, 4a)$



$$t_2 = -t_1 - \frac{2}{t_1} = -3$$

$\therefore \quad Q(9a, -6a)$

$$\therefore \quad m_{SP} = \frac{4a}{4a-a} = \frac{4}{3}$$

$$m_{SQ} = \frac{-6a}{9a-a} = -\frac{3}{4} \Bigg] \angle PSQ = 90°$$

19. PQ is a normal chord of the parabola $y^2 = 4ax$ at P, A being the vertex of the parabola. Through P a line is drawn parallel to AQ meeting the x–axis in R. Then the length of AR is :

(A) equal to the length of the latus rectum

(B) equal to the focal distance of the point P

(C) equal to twice the focal distance of the point P

(D) equal to the distance of the point P from the directrix.

परवलय $y^2 = 4ax$ की P पर अभिलम्ब जीवा है यदि P से एक रेखा AQ के समान्तर खींची जाती है जो x अक्ष को R पर मिलती है तथा A परवलय का शीर्ष है, तब AR की लम्बाई :

(A) नाभिलम्ब की लम्बाई के बराबर होगी

(B) बिन्दु P की नाभि दूरी के बराबर होगी

(C) बिन्दु P की नाभि दूरी की दुगुनी होगी

(D) नियता से बिन्दु P की दूरी के बराबर होगी

Ans. C

Sol. $t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_1t_2 + t_1^2 = -2$

Equation of the line through P parallel to AQ

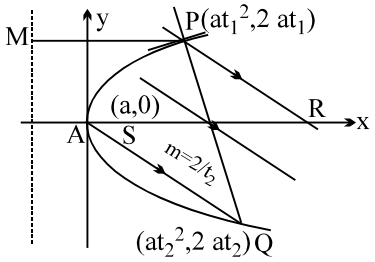
$$y - 2at_1 = \frac{2}{t_2}(x - at_1^2)$$

put $y = 0 \Rightarrow x = at_1^2 - at_1t_2$

$= at_1^2 - a(-2 - t_1^2)$

$= 2a + 2at_1^2 = 2(a + at_1^2)$

= twice the focal distance of P



20. Let BC be the latus rectum of the parabola $y^2 = 4x$ with vertex A. Minimum length of the projection of BC on a tangent drawn in the portion BAC is :

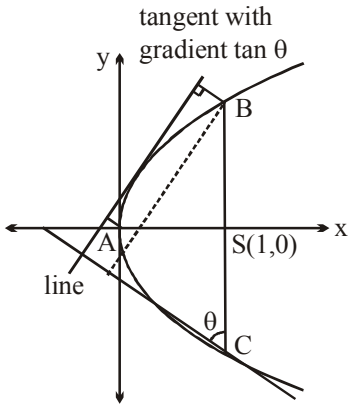
शीर्ष A वाले परवलय $y^2 = 4x$ की नाभिलम्ब BC है, तो BAC के क्षेत्र में किसी बिन्दु पर खीचें गये स्पर्श रेखा पर BC के प्रक्षेप (Projection) की न्यूनतम लम्बाई होगी :

(A) 2 (B) $2\sqrt{3}$ (C) $2\sqrt{2}$ (D) $2+\sqrt{2}$

Ans. C

Sol. Projection of BC on the tangent = | BC sin θ |

$$= 4|\sin\theta|, \theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$$



at A, $\theta \to 90° \Rightarrow$ projection = 4

at B, $\theta \to 45°$ or $135°$

Hence $(BC)_{min.} = 4\sin 45°$

$$= \frac{4}{\sqrt{2}} = 2\sqrt{2} \text{ Ans.}$$

21. The normal at the point $(bt_1^2, 2bt_1)$ on a parabola meets the parabola again in the point $\left(bt_2^2, 2bt_2\right)$, then :

परवलय के बिन्दु $(bt_1^2, 2bt_1)$ पर खींचा गया अभिलम्ब पुनः परवलय के बिन्दु $(bt_2^2, 2bt_2)$ पर मिलता है तब :

(A) $t_2 = t_1 + \frac{2}{t_1}$ (B) $t_2 = -t_1 - \frac{2}{t_1}$ (C) $t_2 = -t_1 + \frac{2}{t_1}$ (D) $t_2 = t_1 - \frac{2}{t_1}$

Ans. B

Sol. It is a fundamental theorem.

22. The tangent and normal at P(t), for all real positive t, to the parabola $y^2 = 4ax$ meet the axis of the parabola in T and G respectively, then the angle at which the tangent at P to the parabola is inclined to the tangent at P to the circle passing through the points P, T and G is :

यदि t के सभी वास्तविक धनात्मक मानों के लिए पवलय $y^2 = 4ax$ के बिन्दु P(t) से स्पर्श रेखा और अभिलम्ब परवलय की अक्ष को क्रमशः T और G पर मिलते हें तब P पर परवलय की स्पर्श रेखा और P पर वृत्त की स्पर्श रेखा के मध्य का कोण जहां वृत्त बिन्दुओं P, T और G से गुजरती है, है :

(A) $\cot^{-1}t$ (B) $\cot^{-1}t^2$ (C) $\tan^{-1}t$ (D) $\tan^{-1}t^2$

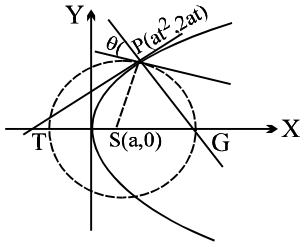
Ans. C

Sol. slope of tangent $= \frac{1}{t}$ (m_1) at P on parabola

$$\text{slope of PS} = \frac{2at}{a(t^2-1)} = \frac{2t}{t^2-1} \text{ (normal to the circle)}$$

$\therefore$ slope of tangent at P on circle $= \frac{1-t^2}{2t}(m_2)$

$$\therefore \quad \tan\theta = \frac{\frac{1}{t} - \frac{1-t^2}{2t}}{1+\frac{1-t^2}{2t^2}} = \frac{\left(2-1+t^2\right)2t^2}{2t(1+t^2)} = t$$



$\therefore$ $\theta = \tan^{-1}t \Rightarrow$ (C)

23. Equation of the other normal to the parabola $y^2 = 4x$ which passes through the intersection of those at $(4, -4)$ and $(9a, -6a)$ is :

परवलय $y^2 = 4x$ के बिन्दुओं $(4, -4)$ तथा $(9a, -6a)$ से खीचें गये अभिलम्बो के प्रतिच्छेद बिन्दु से खीचे गये अभिलम्ब का समीकरण gksxk:

(A) $5x - y + 115 = 0$ (B) $5x + y - 135 = 0$ (C) $5x - y - 115 = 0$ (D) $5x + y + 115 = 0$

Ans. B

Sol. Equation of normal at $(4, -4)$ is $y = 2x - 12$ and equation of normal at $(9, -6)$ is $y = 3x - 33a$. Point of intersection of both these normals is $(21, 30)$

Now $y = mx - 2m - m^3$

$30 = 21m - 2m - m^3 = 19m - m^3$

$\Rightarrow \quad m^3 - 19m + 30 = 0$

$m = 2, 3, -5$

24. Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the centre C of the circle, $x^2 + (y + 6)^2 = 1$. Then the equation of the circle, passing through C and having its centre at P is :

[JEE(Main)2016]

माना परवलय $y^2 = 8x$ का P का ऐसा बिन्दु है जो वृत्त $x^2 + (y + 6)^2 = 1$, के केन्द्र C से न्यूनतम दूरी पर है, तो उस वृत्त का समीकरण C से होकर जाता है तथा जिसका केन्द्र P पर है, है : **[JEE(Main)2016]**

(A) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$ (B) $x^2 + y^2 - 4x + 9y + 18 = 0$

(C) $x^2 + y^2 - 4x + 8y + 12 = 0$ (D) $x^2 + y^2 - x + 4y - 12 = 0$

Ans. C

Sol. $P(2t^2, 4t)$

Normal at P passes through $(0, -6)$

$tx + y = 4t + 2t^3$

$-6 = 4t + 2t^3$

$t^3 + 2t + 3 = 0$

$t = -1$

$P(2, -4)$

25. If the tangent at $(1, 7)$ to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ then the value of c is :

[JEE(Main)2018]

यदि वक्र $x^2 = y - 6$ के बिन्दु $(1, 7)$ पर बनी स्पर्शरेखा वृत $x^2 + y^2 + 16x + 12y + c = 0$ को स्पर्श करती है, तो c का मान है :

[JEE(Main)2018]

(A) 85 (B) 95 (C) 195 (D) 185

Ans. B

Sol. $x^2 = y - 6$

$\left(\frac{dy}{dx}\right)_{(1,7)} = 2$

$T(1, 7) = 2x - y + 5 = 0$

distance of center from tangent = radius

$$r = \left|\frac{-16+6+5}{\sqrt{5}}\right| = \sqrt{5}$$

$$\sqrt{g^2+f^2-c} = \sqrt{5}$$

$$\Rightarrow c = 95$$

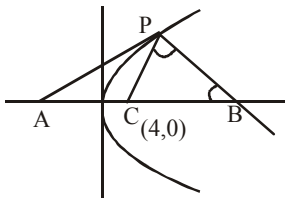
26. Tangent and normal are drawn at P(16, 16) on the parabola $y^2 = 16x$, which intersect the axis of the parabola at A and B, respectively. If $\angle CPB = \theta$, then a value of $\tan\theta$ is : **[JEE(Main)2018]**

परवलय $y^2 = 16x$ के एक बिन्दु P(16, 16) पर स्पर्श रेखा तथा अभिलम्ब खींचे जाते है तो परवलय के अक्ष को बिन्दुओं क्रमशः A तथा B पर प्रतिच्छेद करते हैं। यदि बिन्दुओं P, A तथा B से होकर जाने वाले वृत का केन्द्र C है तथा $\angle CPB = \theta$ तो $\tan\theta$ का एक मान है: **[JEE(Main)2018]**

(A) 3 (B) $\frac{4}{3}$ (C) $\frac{1}{2}$ (D) 2

Ans. D

Sol.



$$\left(-\frac{dx}{dy}\right) = -t = -2$$

Slope of normal at P = $-\tan\theta = -2$

$\tan\theta = 2$

27. The point on the ellipse $x^2 + 2y^2 = 6$, whose distance from the line $x + y = 7$ is minimum is :

(A) (2, 3) (B) (2, 1) (C) (1, 0) (D) None of these

दीर्घवृत $x^2 + 2y^2 = 6$ पर बिन्दु जिसकी रेखा $x + y = 7$ से दूरी न्यूनतम है :

(A) (2, 3) (B) (2, 1) (C) (1, 0) (D) इनमें से कोई नहीं

Ans. B

Sol. $E : \frac{x^2}{6} + \frac{y^2}{3} = 1$ $a^2 = 6$ $\Rightarrow a = \sqrt{6}$

$b^2 = 3$ $\Rightarrow b = \sqrt{3}$

Point on ellipse $\left(\sqrt{6}\cos\theta, \sqrt{3}\sin\theta\right)$

distance from line $x + y = 7$

$$d = \frac{\left(\sqrt{6}\cos\theta + \sqrt{3}\sin\theta - 7\right)}{\sqrt{2}}$$

$$\left(a\cos\theta + b\sin\theta\right) \in \left[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}\right]$$

$$d_{min.} = \frac{\left|[-3,3]-7\right|}{\sqrt{2}} = \frac{\left|[-10,-4]\right|}{\sqrt{2}}$$

$d_{min.} = 2\sqrt{2}$ $\leftarrow$ correct but we don't have to find this.

Equation of tangent at $P\left(\sqrt{6}\cos\theta, \sqrt{3}\sin\theta\right)$

$$\frac{x\sqrt{6}\cos\theta}{6} + \frac{y\sqrt{3}\sin\theta}{3} = 1 \qquad \text{....(1)}$$

it is parallel to $x + y = 7$ so slopes equal

$$\Rightarrow -\frac{\cos\theta}{\sqrt{6}} \times \frac{\sqrt{3}}{\sin\theta} = -1$$

$$\tan\theta = \frac{1}{\sqrt{2}} \left\langle \begin{matrix} \sin\theta = \frac{1}{\sqrt{3}} \\ \cos\theta = \frac{\sqrt{2}}{\sqrt{3}} \end{matrix} \right.$$

$$\therefore P\left(\sqrt{6} \times \frac{\sqrt{2}}{\sqrt{3}}, \sqrt{3} \times \frac{1}{\sqrt{3}}\right) \Rightarrow P(2,1)$$

28. The locus of the foot of perpendicular drawn from the centre of the ellipse $x^2 + 3y^2 = 6$ on any tangent to it is : **[JEE(Main)2014]**

दीर्घवृत $x^2 + 3y^2 = 6$ के केंद्र से इसकी किसी स्पर्श रेखा पर खींचे गए लंब के पाद का बिंदु पथ है : **[JEE(Main)2014]**

(A) $(x^2 + y^2)^2 = 6x^2 + 2y^2$ (B) $(x^2 + y^2)^2 = 6x^2 - 2y^2$

(C) $(x^2 - y^2)^2 = 6x^2 + 2y^2$ (D) $(x^2 - y^2)^2 = 6x^2 - 2y^2$

Ans. A

Sol. $K = mh \pm \sqrt{6m^2 + 2}$ _____(1)

$\left(\frac{K}{h}\right)m = -1$ _____(2)

Eliminating m from (1) and (2)

Locus is

$(x^2 + y^2)^2 = 6x^2 + 2y^2$

29. The locus of a point $P(\alpha, \beta)$ moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is :

(A) an ellipse (B) a circle (C) a parabola (D) a hyperbola

बिन्दु $P(\alpha, \beta)$, जो इस प्रतिबन्ध के अन्तर्गत घूम रहा है, कि $y = \alpha x + \beta$ अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ की स्पर्श रेखा है, का fcUnqiथ :

(A) एक दीर्घवृत्त है (B) एक वृत्त है (C) एक परवलय है (D) एक अतिपरवलय है

Ans. D

Sol. On comparing $y = mx + \sqrt{a^2m^2 - b^2}$ and $y = \alpha x + \beta$

$\therefore \quad \alpha = m, \beta = \sqrt{a^2m^2 - b^2}$

$\Rightarrow \beta^2 = a^2\alpha^2 - b^2$

$\Rightarrow a^2\alpha^2 - \beta^2 = b^2$

$\Rightarrow$ Hyperbola

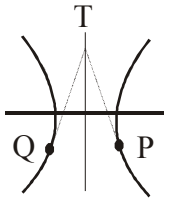
30. Tangent are drawn to the hyperbola $4x^2 - y^2 = 36$ at the points P and Q. If these tangents intersect at the point T(0, 3) then the area (in sq. units) of ΔPTQ is : **[JEE(Main)2018]**

एक अतिपरवलय $4x^2 - y^2 = 36$ के बिंदुओं P तथा Q पर स्पर्श रेखाएँ खींची जाती है। यदि यह स्पर्शरेखाएँ बिंदु T(0, 3) पर काटती g§r ksΔPTQ का क्षेत्रफल (वर्ग इकाइयों में) है : **[JEE(Main)2018]**

(A) $60\sqrt{3}$ (B) $36\sqrt{5}$ (C) $45\sqrt{5}$ (D) $54\sqrt{3}$

Ans. C

Sol.



Eqaution of tangent at P

$$\frac{x\sec\theta}{3}-\frac{y\tan\theta}{6}=1$$

it will pass though (0, 3)

$$0-\frac{3\tan\theta}{6}=1$$

$\tan\theta=-2\Rightarrow\sec\theta=\sqrt{5}$

$$\text{area}=\frac{1}{2}\left|6\sec\theta\right|\left(|6\tan\theta|+3\right)$$

$=45\sqrt{5}$

31. Let P(6, 3) be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If the normal at the point P intersects the x-axis at (9, 0), then the eccentricity of the hyperbola is : **[JEE-2011]**

माना अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ पर बिन्दु P(6, 3) स्थित है। यदि P पर अभिलम्ब x-अक्ष को (9, 0) पर काटता है तो अतिपरवलय की mRd fUz;r gksxh: **[JEE-2011]**

(A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$

Ans. B

Sol. Equation of normal at P(6, 3) on $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $\frac{a^2x}{6}+\frac{b^2y}{3}=a^2e^2$

It intersects x-axis at (9, 0)

$$\Rightarrow \quad a^2\frac{9}{6}=a^2e^2 \quad \Rightarrow \quad e=\sqrt{\frac{3}{2}}$$

COMMON TANGENTS & COMMON NORMAL

32. Minimum distance between the curves $y^2=x-1$ and $x^2=y-1$ is equal to :

वक्रों $y^2=x-1$ और $x^2=y-1$ के मध्य की न्यूनतम दूरी है :

(A) $\frac{3\sqrt{2}}{4}$ (B) $\frac{5\sqrt{2}}{4}$ (C) $\frac{7\sqrt{2}}{4}$ (D) $\frac{\sqrt{2}}{4}$

Ans. A

Sol. (A) $\left(\frac{x}{3}\right)^2+\left(\frac{y}{4}\right)^2=\cos^2 t+\sin^2 t=1 \Rightarrow$ An ellipse

(B) $x^2-2=-2\cos t=-2\left(2\cos^2\frac{t}{2}-1\right) \Rightarrow$ A parabola

(C) $\sqrt{x}=\tan x$, $\sqrt{y}=\sec t \quad (x\geq 0, y\geq 0)$

$\sec^2 t-\tan^2 t=1 \quad \Rightarrow \quad y-x=1 \Rightarrow$ a line tangent

(D) $x^2=1-\sin t=2-(1+\sin t)=2-\left(\sin\frac{t}{2}+\cos\frac{t}{2}\right)^2=2-y^2 \Rightarrow$ a circle

33. Consider two curves C_1: $(y-\sqrt{3})^2=4(x-\sqrt{2})$ and C_2: $x^2+y^2=(6+2\sqrt{2})x+2\sqrt{3}y-6(1+\sqrt{2})$, then :

(A) C_1 and C_2 touch each other only at one point.

(B) C_1 and C_2 touch each other exactly at two points.

(C) C_1 and C_2 intersect (but do not touch) at exactly two points.

(D) C_1 and C_2 neither intersect nor touch each other.

माना C_1: $(y-\sqrt{3})^2=4(x-\sqrt{2})$ और C_2: $x^2+y^2=(6+2\sqrt{2})x+2\sqrt{3}y-6(1+\sqrt{2})$ दो वक्र है, तब :

(A) C_1 और C_2 एक दूसरे को एक बिन्दु पर स्पर्श करेंगे

(B) C_1 और C_2 एक दूसरे को दो बिन्दुओं पर स्पर्श करेंगे

(C) C_1 और C_2 दो बिन्दुओं पर प्रतिच्छेद करेंगे। (स्पर्श नहीं)

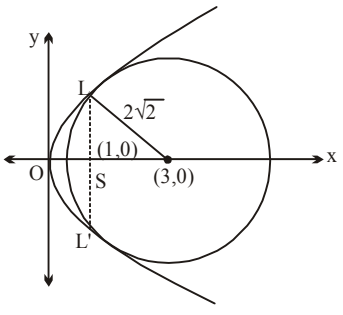
(D) C_1 और C_2 न तो प्रतिच्छेद करते हैं नहीं स्पर्श

Ans. B

Sol. Given $C_1: (y-\sqrt{3})^2=4(x-\sqrt{2})$ .....(1)

and $C_2: \left(x-(3+\sqrt{2})\right)^2+(y-\sqrt{3})^2=\left(3+\sqrt{2}\right)^2+3-6\left(1+\sqrt{2}\right)$

$\left[\left(x-\sqrt{2}\right)-3\right]^2+\left(y-\sqrt{3}\right)^2=8$ .....(2)



Let $y-\sqrt{3}=Y$ and $x-\sqrt{2}=X$

$C_1'=Y^2=4X$ .....(3)

$C_2'=(X-3)^2+Y^2=8$....(4)

Solving $(X-3)^2+4X=8$

$X^2-2X+1=0$

$(X-1)^2=0$

$X=1$

Hence the two curves touch each other exactly at two points.

34. Equation of the common tangent to the ellipses, $\frac{x^2}{a^2+b^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{a^2}+\frac{y^2}{a^2+b^2}=1$, is :

निम्न में से कौनसी सरल रेखा दोनों वक्रों $\frac{x^2}{a^2+b^2}+\frac{y^2}{b^2}=1$ तथा $\frac{x^2}{a^2}+\frac{y^2}{a^2+b^2}=1$ की उभयनिष्ठ स्पर्श रेखा है :

(A) $ay = bx + \sqrt{a^4 - a^2b^2 + b^4}$ (B) $by = ax - \sqrt{a^4 + a^2b^2 + b^4}$

(C) $ay = bx - \sqrt{a^4 + a^2b^2 + b^4}$ (D) $by = ax + \sqrt{a^4 - a^2b^2 + b^4}$

Ans. B

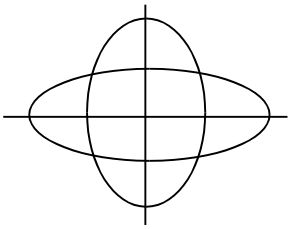
Sol. Equation of a tangent to $\frac{x^2}{a^2+b^2}+\frac{y^2}{b^2}=1$

$$y = mx \pm \sqrt{(a^2+b^2)m^2+b^2} \quad ....(1)$$

If (1) is also a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{a^2+b^2}=1$ then

$(a^2 + b^2)m^2 + b^2 = a^2m^2 + a^2 + b^2$ (using $c^2 = a^2m^2 + b^2$)

$$b^2m^2 = a^2 \Rightarrow m^2 = \frac{a^2}{b^2} \Rightarrow m = \pm\frac{a}{b}$$



$$y = \pm\frac{a}{b}x \pm \sqrt{(a^2+b^2)\frac{a^2}{b^2}+b^2}$$

$$by = \pm ax \pm \sqrt{a^4 + a^2b^2 + b^4}$$

Note : Although there can be four common tangents but only one of these appears in B

35. Two common tangents to the circle $x^2 + y^2 = 2a^2$ and parabola $y^2 = 8ax$ are :

वृत्त $x^2 + y^2 = 2a^2$ तथा परवलय $y^2 = 8ax$ की दो उभयनिष्ठ स्पर्श रेखा है होगी :

(A) $x = \pm (y + 2a)$ (B) $y = \pm (x + 2a)$ (C) $x = \pm (y + a)$ (D) $y = \pm (x + a)$

Ans. B

Sol. $C_1 : x^2 + y^2 = 2a^2$

Equation of tangent

$$y = mx \pm \sqrt{2a^2m^2 + 2a^2} \quad \text{_____(1)}$$

$P : y^2 = 8ax$

$$y = mx \pm \frac{2a}{m}$$ _____(2)

from (1) & (2)

$$\frac{4a^2}{m^2} = 2a^2m^2 + 2a^2$$

$4a^2 = 2a^2m^4 + 2a^2m^2$

$2 = m^4 + m^2$

$m^4 + m^2 - 2 = 0$

$m^4 + 2m^2 - m^2 - 2 = 0$

$m^2 (m^2 + 2) - 1(m^2 + 2) = 0$

$m = \pm 1$

$y = \pm(x + 2a) \rightarrow$ Equation of common tangents

ELLIPSE : MISCELLANEOUS

36. Two sets A and B are as under : $A = \{(a, b) \in R \times R : |a - 5| < 1$ and $|b - 5| < 1\}$: $B\{(a, b) \in R \times R: 4(a - 6)^2 + 9(b - 5)^2 \le 36\}$. Then : **[JEE(Main)2018]**

(A) $A \cap B = \phi$ (an empty set) (B) neither $A \subset B$ not $B \subset A$

(C) $B \subset A$ (D) $A \subset B$

दो समुच्चय A तथा B निम्न प्रकार के है – $A = \{(a, b) \in R \times R : |a - 5| < 1$ तथा $|b - 5| < 1\}$: $B \{(a, b) \in R \times R: 4(a - 6)^2 + 9(b - 5)^2 \le 36\}$. तो : **[JEE(Main)2018]**

(A) $A \cap B = \phi$ (एक रिक्त समुच्चय) (B) न तो $A \subset B$ और न ही $B \subset A$

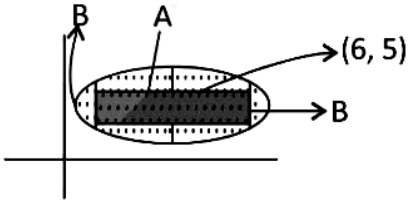
(C) $B \subset A$ (D) $A \subset B$

Ans. D

Sol. Let $a - 6 = x$ $b - 5 = y$

$|x + 1| < 1$ $|y| < 1$ $4x^2 + 9y^2 \le 36$

$-2 < x < 0$ $-1 < y < 1$



Boundaries of square are lying in ellipse

$\Rightarrow A \subset B$

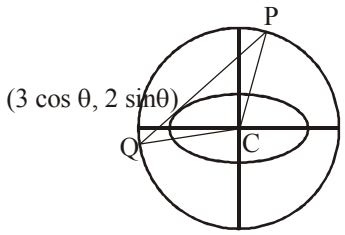
37. A tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ with centre C meets its director circle at P and Q. Then the product of the slopes of CP and CQ, is :

दीर्घवृत्त $\frac{x^2}{9}+\frac{y^2}{4}=1$, जिसका केन्द्र C है, पर स्पर्श रेखा नियामक वृत्त को P तथा Q पर काटती है तो CP तथा CQ के ढालों का xqkuQy gksxk:

(A) $\frac{9}{4}$ (B) $\frac{-4}{9}$ (C) $\frac{2}{9}$ (D) $-\frac{1}{4}$

Ans. B

Sol. The equation of the tangent at $(3\cos\theta, 2\sin\theta)$ on $\frac{x^2}{9}+\frac{y^2}{4}=1$ is



$$\frac{x}{3}\cos\theta+\frac{y}{2}\sin\theta=1 \qquad \text{... (i)}$$

The equation of the director circle is

$$x^2+y^2=9+3=13 \qquad \text{... (ii)}$$

The combined equation of CP and CQ is obtained by homogenising equation (ii) with (i). Thus combined equation is

$$x^2+y^2=13\left(\frac{x}{3}\cos\theta+\frac{y}{2}\sin\theta\right)^2$$

$$\Rightarrow \quad \left(\frac{13}{9}\cos^2\theta-1\right)x^2+\frac{13}{3}\sin\theta\cos\theta\,xy+\left(\frac{13}{4}\sin^2\theta-1\right)y^2=0$$

$\therefore$ Product of the slopes of CP and CQ

$$\frac{\text{coefficient of } x^2}{\text{coefficient of } y^2}=\frac{\frac{13}{9}\cos^2\theta-1}{\frac{13}{4}\sin^2\theta-1}=\frac{13\cos^2\theta-9}{13\sin^2\theta-4}\times\frac{4}{9}=\frac{13\cos^2\theta-9}{13-13\cos^2\theta-4}\times\frac{4}{9}=-\frac{4}{9}$$

$$\left[\text{Note: } m_1+m_2=\frac{-2h}{b};\ m_1m_2=\frac{a}{b}\right]$$

38. The normal at a variable point P on an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ of eccentricity e meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e′ such that :

दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ पर स्थित किसी बिन्दु पर खींची गयी अभिलम्ब अक्षों को क्रमशः बिन्दु Q तथा R पर मिलती है तो QR के

e/; fcUnq kfcUnqPkd hmRd fcUnq kd ekuk e′ इस प्रकार है कि :

(A) e′ is independent of e (B) e′ = 1

(C) e′ = e (D) e′ = 1/e

Ans. C

Sol. Normal at point P (a cos θ, b sin θ)

$$\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2 \qquad ...(1)$$

it meets axes of ellipse at

$$Q=\left(\frac{\left(a^2-b^2\right)\cos\theta}{a},0\right)$$

$$R=\left(0,-\frac{\left(a^2-b^2\right)\sin\theta}{a}\right)$$

mid point of Q & R is M (h, k).

$$2h=\frac{\left(a^2-b^2\right)\cos\theta}{a}$$

$$\Rightarrow \cos\theta=\frac{2ab}{a^2-b^2} \qquad ...(2)$$

$$2k=-\frac{\left(a^2-b^2\right)\sin\theta}{b}$$

$$\Rightarrow \sin\theta=-\frac{2bk}{\left(a^2-b^2\right)} \qquad ...(3)$$

from [Equation (2)]^2 + [equation (3)]^2

$$\frac{x^2}{\frac{\left(a^2-b^2\right)^2}{4a^2}}+\frac{y^2}{\frac{\left(a^2-b^2\right)^2}{4b^2}}=1$$

$$e'^2 = 1 - \frac{b^2}{a^2} = e^2 \Rightarrow e' = e$$

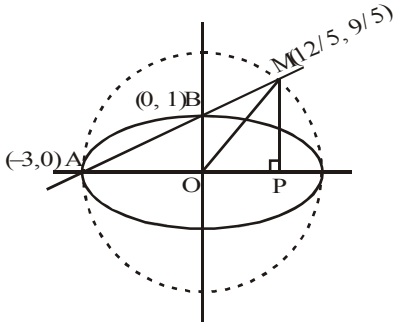
39. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M. Then the area of the triangle with vertices at A, M and the origin O is : **[JEE-2009]**

दीर्घवृत्त $x^2 + 9y^2 = 9$ के दीर्घ–अक्ष के सिरा A तथा लघु–अक्ष के सिरा B से जाने वाली रेखा सहायत–वृत्त को बिन्दु M पर काटती है। ' kkZ Uq ls A, M तथा मूलबिन्दु O से बनने वाले त्रिभुज का क्षेत्रफल होगा : **[JEE-2009]**

(A) $\frac{31}{10}$ (B) $\frac{29}{10}$ (C) $\frac{21}{10}$ (D) $\frac{27}{10}$

Ans. D

Sol. Area of triangle AOM = $\frac{1}{2}$ AO. PM



$\Rightarrow$ Eqaution of AM is $y = \frac{1}{3}(x + 3)$

$x - 3y + 3 = 0$ which is chord of auxiliary circle $x^2 + y^2 = 9$, and PM is ordinate of point M

$\Rightarrow (3y - 3)^2 + y^2 = 9 \Rightarrow y = \frac{9}{5} = PM$ $\Rightarrow$ Area of triangle = $\frac{1}{2} \cdot 3 \cdot \frac{9}{5} = \frac{27}{10}$.

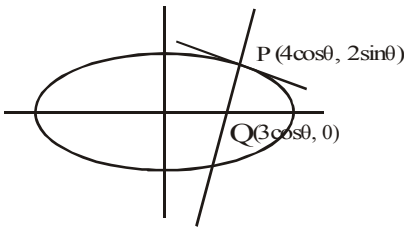
40. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x-axis at Q. If M is the mid point of the line segment PQ, then the locus of M intersects the latus rectums of the given ellipse at the point : **[JEE-2009]**

दीर्घवृत्त $x^2 + 4y^2 = 16$ के किसी बिन्दु P पर अभिलम्ब x-अक्ष से बिन्दु Q पर मिलता है। यदि रेखाखण्ड PQ का मध्य बिन्दु M है, r c M का बिन्दुपथ दिए गए दीर्घवृत्त के नाभिलम्बों को किन बिन्दुओं पर काटता है : **[JEE-2009]**

(A) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{2}{7}\right)$ (B) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{\sqrt{19}}{4}\right)$ (C) $\left(\pm 2\sqrt{3}, \pm\frac{1}{7}\right)$ (D) $\left(\pm 2\sqrt{3}, \pm\frac{4\sqrt{3}}{7}\right)$

Ans. C

Sol. $\frac{x^2}{16}+\frac{y^2}{4}=1$



$4x\sec\theta - 2y\,\text{cosec}\theta = 12$

$x = 3\cos\theta$

$Q \equiv (3\cos\theta, 0)$

$2\eta = 7\,\chi o\sigma\theta$

$2k = 2\sin\theta$

$\frac{4x^2}{49}+\frac{y^2}{1}=1$ ...(i)

L.R $\Rightarrow x = 2\sqrt{3}$

Putting in (i)

$y = \pm\frac{1}{7}$ $\therefore$ $\left(\pm 2\sqrt{3}, \pm\frac{1}{7}\right)$

AUXILLIARY CIRCLE & ECCENTERIC ANGLE OF ELLIPSE AND HYPERBOLA

41. If α & β are the eccentric angles of the extremities of a focal chord of an standard ellipse, then the eccentricity of the ellipse is :

यदि α तथा β मानक दीर्घवृत्त के नाभि जीवा के सिरों के उत्केन्द्रिय कोण हे तो दीर्घवृत्त की उत्केन्द्रियता का मान होगा :

(A) $\frac{\cos\alpha+\cos\beta}{\cos(\alpha+\beta)}$ (B) $\frac{\sin\alpha-\sin\beta}{\sin(\alpha-\beta)}$ (C) $\frac{\cos\alpha-\cos\beta}{\cos(\alpha-\beta)}$ (D) $\frac{\sin\alpha+\sin\beta}{\sin(\alpha+\beta)}$

Ans. D

Sol. $\frac{x}{a}\cos\frac{\alpha+\beta}{2}+\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$; $\frac{ae}{a}\cos\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$

$\Rightarrow$ $e=\frac{\cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha+\beta}{2}}\cdot\frac{2\sin\frac{\alpha+\beta}{2}}{2\sin\frac{\alpha+\beta}{2}}=\frac{\sin\alpha+\sin\beta}{\sin(\alpha+\beta)}$

42. If P is any point on ellipse with foci S_1 & S_2 and eccentricity is $\frac{1}{2}$ such that $\angle PS_1S_2 = \alpha$, $\angle PS_2S_1 = \beta$, $\angle S_1PS_2 = \gamma$, then $\cot\frac{\alpha}{2}$, $\cot\frac{\gamma}{2}$, $\cot\frac{\beta}{2}$ are in :

यदि P दीर्घवृत्त पर कोई बिन्दु है जिसकी नाभियां S_1 तथा S_2 है तथा उत्केन्द्रिता $\frac{1}{2}$ इस प्रकार है कि $\angle PS_1S_2=\alpha$, $\angle PS_2S_1 = \beta$, $\angle S_1PS_2 = \gamma$, तो $\cot\frac{\alpha}{2}$, $\cot\frac{\gamma}{2}$, $\cot\frac{\beta}{2}$ निम्न श्रेणी में है :

(A) A.P. (B) G.P. (C) H.P. (D) *NOT* A.P., G.P. & H.P.

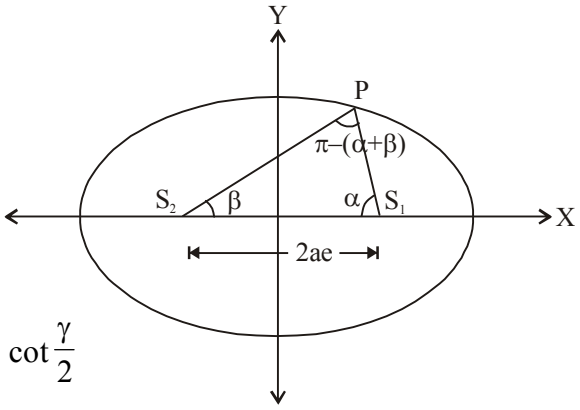
Ans. A

Sol. By sine rule in ΔPS_1S_2, we get $\frac{2ae}{\sin(\alpha+\beta)} = \frac{S_1P}{\sin\beta} = \frac{S_2P}{\sin\alpha} = \frac{2a}{\sin\alpha+\sin\beta}$

$$\Rightarrow e = \frac{\sin(\alpha+\beta)}{\sin\alpha+\sin\beta} \Rightarrow \frac{e}{1} = \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)}{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}$$

Now $\frac{1-e}{1+e} = \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1-\frac{1}{2}}{1+\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{3}$

$\therefore \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1}{3}$ ... (1)



Also we know that

$$\cot\frac{\alpha}{2}+\cot\frac{\beta}{2}+\cot\frac{\gamma}{2} = \cot\frac{\alpha}{2}\cot\frac{\beta}{2}\cot\frac{\gamma}{2} = 3\cot\frac{\gamma}{2}$$

$\Rightarrow 2\cot\frac{\gamma}{2} = \cot\frac{\alpha}{2} + \cot\frac{\beta}{2} \Rightarrow \cot\frac{\alpha}{2}, \cot\frac{\gamma}{2}, \cot\frac{\beta}{2}$ are in A.P.

RECTANGULAR HYPERBOLA

43. Locus of the middle points of the parallel chords with gradient m of the rectangular hyperbola $xy = c^2$ is :

आयताकार अतिपरवलय $xy = c^2$ के समान्तर जीवाओं (जिनके ढाल m है) के निकाय के मध्य बिन्दु का बिन्दुपथ होगा :

(A) $y + mx = 0$ (B) $y - mx = 0$ (C) $my - x = 0$ (D) $my + x = 0$

Ans. A

Sol. equation of chord with mid point (h, k) is $\frac{x}{h}+\frac{y}{k} = 2$; $m = -\frac{k}{h} \Rightarrow y + mx = 0$

44. The normal to curve $xy = 4$ at the point (1, 4) meets the curve again at :

वक्र $xy = 4$ के बिन्दु (1, 4) पर अभिलम्ब वक्र को पुनः किस बिन्दु पर मिलेगा :

(A) $(-4, -1)$ (B) $\left(-8, -\frac{1}{2}\right)$ (C) $\left(-16, -\frac{1}{4}\right)$ (D) $(-1, -4)$

Ans. C

Sol. $xy = 4,\ c^2 = 4 \Rightarrow c = 2$

$$\text{point } \left(ct, \frac{c}{t}\right) = (1, 4) \Rightarrow t = \frac{1}{2}$$

$$m_N = t^2 = \frac{1}{4}$$

Equation of normal

$$(y - 4) = \frac{1}{4}(x - 1)$$

$4y - 4 \times 4 = x - 1$

$x - 4y + 15 = 0$ ...(1)

$xy = 4$ ...(2)

from equation (1) & (2)

$$x - 4 \times \frac{4}{x} + 15 = 0$$

$x^2 + 15x - 16 = 0$

$x^2 + 16x - x - 16 = 0$

$x(x + 16) - 1(x + 16) = 0$

$x = -16,\ 1$ $\quad 1 \rightarrow$ Not possible

$$x = -16,\ y = -\frac{1}{4}$$

$$\text{Point} = \left(-16, -\frac{1}{4}\right)$$

45. Locus of a point, whose chord of contact with respect to the circle $x^2 + y^2 = 4$ is a tangent to the hyperbola $xy = 1$ is a/an :

(A) ellipse (B) circle (C) hyperbola (D) parabola

उस बिन्दु का बिन्दुपथ, जिसकी वृत $x^2 + y^2 = 4$ के सापेक्ष स्पर्श जीवा अतिपरवलय $xy = 1$ की स्पर्श रेखा है, होगा :

(A) दीर्घवृत (B) वृत (C) अतिपरवलय (D) परवलय

Ans. C

Sol. Let P(h, k) be point $\rightarrow$ C.O.C. of circle

$x^2 + y^2 = 4$

$\Rightarrow 4x + ky = 4$ ...(1) $\Rightarrow y = \frac{4 - bx}{k}$

Equation (1) is tangent to $xy = 1$.

$x\left(\frac{4 - hx}{k}\right) = 1$

$4x - hx^2 - k = 0 \Rightarrow hx^2 - 4x + k = 0$

COT $D = 0$

$16 - 4hk = 0$

$4hk = 16$

$hk = 4$

$\Rightarrow xy = 4.$ its a R.H.B.

46. If the tangent and normal at a point on rectangular hyperbola cut-off intercept a_1, a_2 on x-axis and b_1, b_2 on the y-axis, then $a_1a_2 + b_1b_2$ is equal to :

यदि आयतीय अतिपरवलय के किसी बिन्दु पर स्पर्श रेखा तथा अभिलम्ब द्वारा काटे गये अन्तःखण्ड x- अक्ष पर क्रमशः a_1 एवं a_2 तथा y-अक्ष पर b_1 एवं b_2 है, तो $a_1a_2 + b_1b_2$ का मान है :

(A) 2 (B) $\frac{1}{2}$ (C) 0 (D) –1

Ans. C

Sol. Equation of tangent at $\left(ct, \frac{c}{t}\right)$

$\frac{x}{t} + yt = 2c$ ...(1)

Equation of normal at $\left(ct, \frac{c}{t}\right)$

$\left(y - \frac{c}{t}\right) = t^2\left(x - ct\right)$ ...(2)

Put $y = 0$ in euqation (1) & (2)

$a_1 = 2ct,\ a_2 = ct - \frac{c}{t^3}$

put x = 0 in equation (1) & (2)

$$b_1 = \frac{2c}{t},\ b_2 = \frac{c}{t} - ct^3$$

$a_1a_2 + b_1b_2 = 0$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

STANDARD & SHIFTED ELLIPSE

1. Find the equation of the ellipse with its centre (1, 2), focus at (6, 2) and passing through the point (4, 6).
दीर्घवृत्त का समीकरण ज्ञात करें जिसका केन्द्र (1, 2) तथा नाभि (6, 2) है तथा बिन्दु (4, 6) से जाता है।

Ans. $20x^2 + 45y^2 - 40x - 180y - 700 = 0$

Sol. $\frac{(x-1)^2}{a^2} + \frac{(y-2)^2}{b^2} = 1$

It passes through (4, 6)

$\frac{9}{a^2} + \frac{16}{b^2} = 1$ and $ae = 5$

$\Rightarrow$ $b^2 = a^2(1 - e^2)$ $\Rightarrow$ $b^2 = a^2 - a^2e^2$ $\Rightarrow$ $b^2 = a^2 - 25$

$\Rightarrow$ $\frac{9}{a^2} + \frac{16}{a^2 - 25} = 1$ $''$ $(a^2 - 5)(a^2 - 45) = 0$ $''$ either $a^2 = 5$ or $a^2 = 45$

but $a^2 \neq 5$ $\therefore$ $a^2 = 45$ $\Rightarrow$ $b^2 = 20$

2. An ellipse passes through the points (– 3, 1) & (2, –2) & its principal axis are along the coordinate axes in order. Find its equation.
एक दीर्घवृत्त जो बिन्दुओं (– 3, 1) एवं (2, –2) से गुजरता है तथा जिसका मुख्य अक्ष क्रमवार रूप में निर्देशांक अक्षों पर है। दीर्घवृत्त का समीकरण ज्ञात कीजिए।

Ans. $3x^2 + 5y^2 = 32$

Sol. Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

it passes through (3,1) & (2,2)

$\frac{9}{a^2} + \frac{1}{b^2} = 1$ ....(1)

and $\frac{4}{a^2} + \frac{4}{b^2} = 1$ ....(2)

from equation (1)

$a^2 = \frac{32}{3}, b^2 = \frac{32}{5}$ $\Rightarrow 3x^2 + 5y^2 = 32$

3. 'O' is the origin & also the centre of two concentric circles having radii of the inner & the outer circle as 'a' & 'b' respectively. A line OPQ is drawn to cut the inner circle in P & the outer circle in Q. PR is drawn parallel to the y-axis & QR is drawn parallel to the x-axis. Prove that the locus of R is an ellipse touching the two circles. If the focii of this ellipse lie on the inner circle, find the ratio of inner : outer radii & find also the eccentricity of the ellipse.

. त्रिज्या a और b (a < b) के दो संकेन्द्रीय वृत्तों के केन्द्र O (0, 0) से होकर एक छेदक रेखा (secant) OPQ खींची गई है जो वृत्तों को P व Q पर मिलती है। P से होकर y-अक्ष के समान्तर तथा Q से होकर x-अक्ष के समान्तर रेखायें एक–दूसरे से R पर मिलती हैं। सिद्ध करो कि R का बिन्दुपथ एक दीर्घवृत है जो दोनों वृतों को स्पर्श करता है। यदि इस दीर्घवृत्त की नाभियाँ भीतर के वृत्त पर स्थित हैं, तो इसके आन्तरिक एवं बाह्य त्रिज्या का अनुपात ज्ञात करें साथ ही दीर्घवृत्त की उत्केन्द्रता ज्ञात कीजिए।

Ans. $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

Sol. $P = (b\cos\theta, b\sin\theta)$

$Q = (a\cos\theta, a\sin\theta)$

$R = (b\cos\theta, b\sin\theta)$

$\because \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is locus of R which is an ellipse.

$\therefore$ foci lie in inner circle.

$b = ae$

$$b^2 = a^2\left(1 - \frac{b^2}{a^2}\right) \Rightarrow \frac{b^2}{a^2} = \frac{1}{2} \Rightarrow \frac{b}{a} = \frac{1}{\sqrt{2}}$$

$$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$e = \frac{1}{\sqrt{2}}$$

STANDARD & SHIFTED HYPERBOLA

4. Find the equation to the hyperbola whose directrix is $2x + y = 1$, focus (1, 1) & eccentricity $\sqrt{3}$. Find also the length of its latus rectum.

अतिपरवलय का समीकरण ज्ञात करो जिसकी नियता $2x + y = 1$, नाभि (1, 1) तथा उत्केन्द्रता $\sqrt{3}$ है। इसके नाभिलम्ब की लम्बाई भी ज्ञात कीजिए।

Ans. $7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$

Sol. Let P(x,y) be any point on the hyperbola.

Draw PM perpendicular from P on the directrix.

Then by definition SP = e PM

$\Rightarrow$ $(SP)^2 = e^2 (PM)^2$

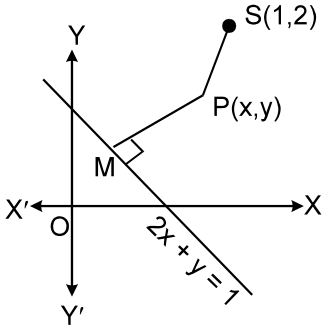
$\Rightarrow$ $(x-1)^2 + (y-2)^2 = 3\left\{\frac{2x+y-1}{\sqrt{4+1}}\right\}^2$

$\Rightarrow$ $5\{x^2 + y^2 - 2x - 4y + 5\}$

$= 3(4x^2 + y^2 + 1 + 4xy - 2y - 4x)$

$\Rightarrow$ $7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$

which is the required hyperbola.



5. The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point of intersection of the lines, $7x + 13y - 87 = 0$ and $5x - 8y + 7 = 0$ & the latus rectum is $32\sqrt{2}/5$. Find 'a' & 'b'.

अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ रेखाओं $7x + 13y - 87 = 0$ तथा $5x - 8y + 7 = 0$ के प्रतिच्छेद बिन्दु से गुजरती है तथा नाभिलम्ब $32\sqrt{2}/5$ है। 'a' एवं 'b' का मान ज्ञात कीजिए।

Ans. $a^2 = 25/2$; $b^2 = 16$

Sol. Point of intersection of lines

$7x + 13y - 87 = 0$ & $5x - 8y + 7 = 0$ is (5, 4).

Then $\frac{25}{a^2} - \frac{16}{b^2} = 1$ ...(i)

Also latus rectum $LR = \frac{2b^2}{a} = \frac{32\sqrt{2}}{5}$

$\Rightarrow$ $b^2 = \frac{16\sqrt{2}a}{5}$ ...(ii)

From (i) & (ii) $a^2 = \frac{25}{2}$, $b^2 = 16$.

6. For the hyperbola $\frac{x^2}{100} - \frac{y^2}{25} = 1$, prove that

(i) eccentricity $= \sqrt{5}/2$

(ii) $SA.\ S'A = 25$, where S & S′ are the foci & A is the vertex.

अतिपरवलय $\frac{x^2}{100}-\frac{y^2}{25}=1$ के लिये सिद्ध करो।

(i) उत्केन्द्रता $=\sqrt{5}/2$

(ii) SA. S′A = 25, जहाँ पर S तथा S′ नाभियां है तथा A शीर्ष है।

Ans.

Sol. $\frac{x^2}{10^2}-\frac{y^2}{5^2}=1 \quad \Rightarrow \quad e=\sqrt{1+\frac{25}{100}}=\frac{\sqrt{5}}{2}$

SA = ae – a

S′A = ae + a $\quad \Rightarrow \quad$ SA.S′A = $a^2(e^2-1)=25$.

7. Find the centre, the foci, the directrices, the length of the latus rectum, the length & the equations of the axes of the hyperbola $16x^2-9y^2+32x+36y-164=0$.

अतिपरवलय $16x^2-9y^2+32x+36y-164=0$ का केन्द्र, नाभि, नियता, नाभिलम्ब की लम्बाई, अक्षों की लम्बाई एवं उनका समीकरण ज्ञात कीजिए।

Ans. (–1, 2) ; (4, 2) & (–6, 2) ; 5x – 4 = 0 & 5x + 14 = 0; $\frac{32}{3}$; 6 ; 8 ; y – 2 = 0; x + 1 = 0

Sol. Given hyperbola can be written as

$$\frac{(x+1)^2}{9}-\frac{(y-2)^2}{16}=1$$

so $e=\frac{5}{3}$, centre is (–1, 2)

foci = (–1 ± 5, 2) = (–6, 2) & (4, 2)

directrix is $x+1=\pm\frac{9}{5} \Rightarrow x=-1\pm\frac{9}{5}$

L.R. = $\frac{32}{3}$, Length of axes is 8 and 6,

Equation of axis is y – 2 = 0 and x + 1 = 0.

PARAMETRIC COORDINATES OF PARABOLA

8. 'O' is the vertex of the parabola $y^2=4ax$ & L is the upper end of the latus rectum. If LH is drawn perpendicular to OL meeting OX in H, prove that the length of the double ordinate through H is $4a\sqrt{5}$.

परवलय $y^2=4ax$ का शीर्ष O है तथा उसके नाभिलम्ब का ऊपरी सिरा L है। यदि LH जो OL पर लम्बवत् है OX को H पर मिलता है, तो सिद्ध कीजिऐ H से जाने वाले द्विक्कोटि की लम्बाई $4a\sqrt{5}$ है।

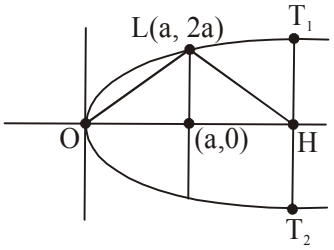
Ans.

Sol. Equation of LH

$$(y-2a)=-\frac{1}{2}(x-a)$$

$$\Rightarrow \frac{x}{2}+y=\frac{5a}{2}$$

$H = (5a, 0)$ $\Rightarrow T_1 = \left(5a, 2a\sqrt{5}\right)$



$\Downarrow$

Put this in equation of parabola.

for $y^2 = 4a(5a)$ $T_2 = \left(5a, -2a\sqrt{5}\right)$

$T_1T_2 = 4a\sqrt{5}$

AUXILLIARY CIRCLE AND ECCENTERIC ANGLE OF ELLIPSE AND HYPERBOLA

9. Suppose x and y are real numbers and that $x^2 + 9y^2 - 4x + 6y + 4 = 0$ then find the maximum value of $(4x - 9y)$.

माना x तथा y वास्तविक संख्याऐं इस प्रकार है कि $x^2 + 9y^2 - 4x + 6y + 4 = 0$ है, तो $(4x - 9y)$ का अधिकतम मान ज्ञात कीजिए।

Ans. 16

Sol. $$E: \frac{(x-2)^2}{1}+\frac{\left(y+\frac{1}{3}\right)^2}{\left(\frac{1}{9}\right)}=1$$

$$\text{centre} = \left(2, -\frac{1}{3}\right)$$

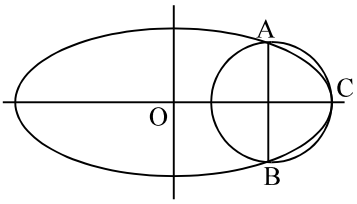
$$\text{Point on ellipse} = \left(\underbrace{2+a\cos\theta}_{\Downarrow x}, \underbrace{-\frac{1}{3}+b\sin\theta}_{\Downarrow y}\right)$$

$$4x - 9y \Rightarrow 4\left(2 + 1\cdot\cos\theta\right) - 9\cdot\left(-\frac{1}{3} + \frac{1}{3}\sin\theta\right)$$

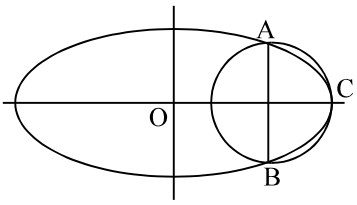
$\Rightarrow f(\theta) = 8 + 4\cos\theta + 3 - 3\sin\theta = 11 + (4\cos\theta - 3\sin\theta)$

$= 11 + [-5, 5] = 16_{max}$

10. A circle intersects an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ precisely at three points A, B, C as shown in the figure. AB is a diameter of the circle and is perpendicular to the major axis of the ellipse. If the eccentricity of the ellipse is 4/5, find the length of the diameter AB in terms of a.



एक वृत दीर्घवृत $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ को स्पष्ट रूप से तीन बिन्दुओं A, B, C पर चित्रानुसार काटता है। AB वृत का व्यास है एवं दीर्घवृत के दीर्घाक्ष पर लम्बवत् है। यदि दीर्घवृत्त की उत्केन्द्रता 4/5 है, तो व्यास AB की लम्बाई a के पदों में ज्ञात करें।

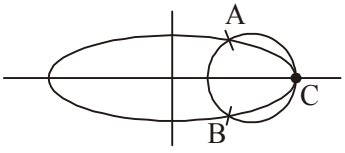


Ans. $\frac{18a}{17}$

Sol. $e^2 = 1 - \frac{b^2}{a^2}$

$\Rightarrow \frac{16}{25} = 1 - \frac{b^2}{a^2} \Rightarrow \frac{b^2}{a^2} = \frac{9}{25}$

$\Rightarrow \frac{b}{a} = \frac{3}{5}$



Let radius = r

Point A is on Ellipse so A(a cos θ, b sin θ)

M is midpoint of AB. (a cos θ, 0) A(a cos θ, –b sin θ)

$2r = 2b \sin\theta$

$(r = b \sin\theta)$ $\Rightarrow \left(r = \frac{3a}{5}\sin\theta \right)$ ...(1)

$r = OC - MO$

$\rho = \alpha\square\alpha\chi o\sigma\theta$

$r = a(1 - \cos\theta)$ ...(2)

$a\left(1-\cos\theta\right) = \frac{3a}{5}\sin\theta$

$\tan\left(\frac{\theta}{2}\right) = \frac{3}{5}$ ...(3)

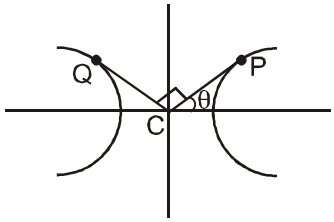
from equation (1) $r = \frac{3a}{5} \times \frac{2\cdot\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} = \frac{36a}{34} = \frac{18a}{17}$

11. If two points P & Q on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose centre is C be such that CP is perpendicular to CQ & $a < b$, then prove that $\frac{1}{CP^2} + \frac{1}{CQ^2} = \frac{1}{a^2} - \frac{1}{b^2}$.

यदि अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ का केन्द्र C हो तथा इस पर दो बिन्दु P & Q इस प्रकार है कि CP, CQ पर लम्बवत् है तथा $a < b$ सिद्ध करो कि $\frac{1}{CP^2} + \frac{1}{CQ^2} = \frac{1}{a^2} - \frac{1}{b^2}$.

Ans.

Sol.



$CP \equiv \frac{x-0}{\cos\theta} = \frac{y-0}{\sin\theta} = r_1$ where $CP = r_1$

$\therefore$ P($r_1\cos\theta, r_1\sin\theta$)

Similarly $\quad Q\left(r_2\cos\left(\frac{\pi}{2}+\theta\right), r_2\sin\left(\frac{\pi}{2}+\theta\right)\right)$

Q ($-r_2\sin\theta, r_2\cos\theta$)

P & Q lies on Hyperbola

$$\therefore \quad r_1^2\left(\frac{\cos^2\theta}{a^2}-\frac{\sin^2\theta}{b^2}\right)=1$$

$$\therefore \quad r_1^2=\frac{a^2b^2}{(b^2\cos^2\theta-a^2\sin^2\theta)} \quad \& \quad r_2^2=\frac{a^2b^2}{(b^2\sin^2\theta-a^2\cos^2\theta)}$$

$$\therefore \quad \frac{1}{r_1^2}+\frac{1}{r_2^2}=\frac{b^2-a^2}{a^2b^2}=\frac{1}{a^2}-\frac{1}{b^2} \text{ H.P.}$$

CHORD JOINING TWO POINTS

12. If any two chords be drawn through two points on the major axis of an ellipse equidistant from the centre, show that $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}\cdot\tan\frac{\gamma}{2}\cdot\tan\frac{\delta}{2}=1$, where $\alpha, \beta, \gamma, \delta$ are the eccentric angles of the extremities of the chords.

दीर्घवृत्त के दीर्घअक्ष पर स्थित केन्द्र से समान दूरी पर दो बिन्दुओं से यदि दो जीवाऐं बनायी जाती है, तो सिद्ध कीजिए $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}\cdot\tan\frac{\gamma}{2}\cdot\tan\frac{\delta}{2}=1$, जहाँ $\alpha, \beta, \gamma, \delta$ जीवाओं के सिरों के उत्केन्द्रीय कोण है।

Ans.

Sol. For ellipse if chord pass through a fixed point (d, 0)

$$\frac{d-a}{d+a}=\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2} \qquad ...(1)$$

Let the points be (d, 0) & (–d, 0)

$$\frac{-d-a}{-d+a}=\tan\frac{\gamma}{2}\times\tan\frac{\delta}{2} \qquad ...(2)$$

Equation (1) $\times$ Equation (2) $\quad \Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}\tan\frac{\delta}{2}=1$

13. If θ_1 & θ_2 are the parameters of the extremities of a chord through (ae, 0) of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, then show that $\tan\frac{\theta_1}{2}\cdot\tan\frac{\theta_2}{2}+\frac{e-1}{e+1}=0$.

यदि θ_1 एवं θ_2 अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ के (ae, 0) से गुजरने वाले किसी जीवा के सिरों के प्राचल हो तो सिद्ध करो

$\tan\frac{\theta_1}{2}\cdot\tan\frac{\theta_2}{2}+\frac{e-1}{e+1}=0.$

Ans.

Sol.

LINE AND CONIC

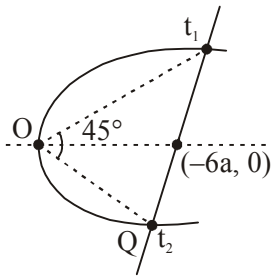
14. Find the equations of the chords of the parabola $y^2 = 4ax$ which pass through the point (–6a, 0) and which subtends an angle of 45° at the vertex.

परवलय $y^2 = 4ax$ की उन सभी जीवाओं का समीकरण ज्ञात करें जो बिन्दु (–6a, 0) से जाती है तथा शीर्ष पर 45° कोण अंकित करती है।

Ans. $7y \pm 2(x + 6a) = 0$

Sol. If chord passes through a fixed point (c, 0)

$t_1t_2=-\frac{c}{a}$



Here it passes through (–6a, 0)

$t_1t_2=-\frac{(-6a)}{a}=6$

$t_1t_2 = 6$ _____(1)

Also, $M_{OP}=\frac{2}{t_1}, M_{OQ}=\frac{2}{t_2}$

$\tan\theta=1\Rightarrow\left|\frac{\frac{2}{t_1}-\frac{2}{t_2}}{1+\frac{4}{t_1t_2}}\right|$

$\Rightarrow\frac{5}{3}=\frac{2t_2-2t_1}{6}$

$\Rightarrow t_2 - t_1 = 5$ _____(2)

$t_2-\frac{6}{t_2}=5 \quad \Rightarrow t_2^2-5t_2-6=0 \quad \rightarrow -6t_2+t_2$

$t_2 = 6, -1$

$t_1 = 1 \quad t_1 = -6$

Equation of chord = $2x + 2at_1t_2$

$y(7) = 2x + 12a$ or $-y7 = 2x + 12a$

DIRECTOR CIRCLE AND TANGENT FROM AN EXTERNAL POINT

15. Tangents drawn from the point P(2, 3) to the circle $x^2 + y^2 - 8x + 6y + 1 = 0$ touch the circle at the points A and B. The circumcircle of the ΔPAB cuts the director circle of ellipse $\frac{(x+5)^2}{9} + \frac{(y-3)^2}{b^2} = 1$ orthogonally. Find the value of b^2.

बिन्दु P(2, 3) से वृत $x^2 + y^2 - 8x + 6y + 1 = 0$ पर बनायी गई स्पर्श रेखाऐं वृत को A एवं B पर स्पर्श करती है। त्रिभुज PAB का परिवृत दीर्घवृत $\frac{(x+5)^2}{9} + \frac{(y-3)^2}{b^2} = 1$ के नियामक वृत को लाम्बिक प्रतिच्छेद करता है। b^2 का मान ज्ञात कीजिए।

Ans. 54

Sol. Equation of director circle PAB (Center of circle & point P will be ends of diameter)

$(x - 4)(x - 2) + (y + 3)(y - 3) = 0$

$x^2 + y^2 - 6x - 1 = 0$ ...(1)

Equation of director circle

$(x + 5)^2 + (y - 3)^2 = 9 + b^2$

$x^2 + 10x + 25 + y^2 - 6y + 9 = 9b^2$

$x^2 + y^2 + 10x - 6y + 25 - b^2 = 0$ ...(2)

for orthoganility

$b^2 = 54$

16. Tangents are drawn from the point (α, β) to the hyperbola $3x^2 - 2y^2 = 6$ and are inclined at angles θ and ϕ to the x –axis. If $\tan\theta . \tan\phi = 2$, prove that $\beta^2 = 2\alpha^2 - 7$.

बिन्दु (α, β) से अतिपरवलय $3x^2 - 2y^2 = 6$ पर स्पर्श रेखाएं खींची जाती है तो जो कि x-अक्ष से θ एवं ϕ पर झुकी है। यदि $\tan\theta . \tan\phi = 2$, तब सिद्ध करो $\beta^2 = 2\alpha^2 - 7$.

Ans.

Sol. $3x^2 - 2y^2 = 6$

$\frac{x^2}{2} - \frac{y^2}{3} = 1$

Let the equation of tangent

$y = mx + \sqrt{a^2m^2 - b^2}$

passes through (α, β)

$(\beta - m\alpha)^2 = a^2m^2 - b^2$

$m^2\alpha^2 + \beta^2 - 2m\alpha\beta = a^2m^2 - b^2$

$m^2(\alpha^2 - a^2) - 2m\alpha\beta + \beta^2 + b^2 = 0$

$m_1m_2 = \frac{\beta^2 + b^2}{\alpha^2 - a^2} = 2$

$2\alpha^2 - 2a^2 = \beta^2 + b^2$

or $2\alpha^2 - 4 = \beta^2 + 3$

$\beta^2 = 2\alpha^2 - 7$

TANGENT AND NORMAL

17. Show that the normals at the points (4a, 4a) & at the upper end of the latus rectum of the parabola $y^2 = 4ax$ intersect on the same parabola.

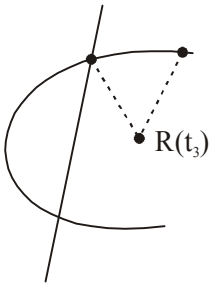
दर्शाईयें कि बिन्दुओं (4a, 4a) एवं परवलय $y^2 = 4ax$ के नाभिलम्ब के ऊपरी सिरे पर बनाये गये अभिलम्ब उसी परवलय पर प्रतिच्छेद करते है।

Ans.

Sol. Equation of normal $P(t_1)$ $(t_1 = 1)$

$y + tx = 2at + at^3$

$x + y = 3a$ ...(1)



Equation of normal at $Q(t_2)$ $(t_2 = 2)$

$y + 2x = 12a$ ...(2)

Equation (2) – Equation (1)

$x = 9a, y = -6a$ Point lie on parabola H.P.

$\Rightarrow t_3 = -3$

18. Prove that the locus of the middle point of portion of a normal to $y^2 = 4ax$ intercepted between the curve & the axis is another parabola. Find the vertex & the latus rectum of the second parabola.

सिद्ध कीजिये कि परवलय $y^2 = 4ax$ पर बनाए गए अभिलम्ब द्वारा अक्ष के मध्य काटे गये अन्तःखण्ड़ का मध्य बिन्दु का बिन्दुपथ एक परवलय होगा। उस परवलय का शीर्ष एवं नाभिलम्ब ज्ञात कीजिए?

Ans. (a, 0) ; a

Sol. Equation of normal at $(am^2, 2am)$

on $y^2 = 4ax$

$y + mx = 2am + am^3$ ... (1)

It cuts x – axis at y = 0 i.e. $(2a + am^2, 0)$

Let middle point (h, k)

$2h = am^2 + 2a + am^2$

$h = am^2 + a$ & $k = am$ ... (2)

from (1) & (2)

$h = a\frac{k^2}{a^2} + a$

Locus $y^2 = a(x - a)$

vertex (a, 0) L.R. = a

19. Find the equations of the tangents to the parabola $y^2 = 16x$, which are parallel & perpendicular respectively to the line $2x - y + 5 = 0$. Find also the coordinates of their points of contact.

परवलय $y^2 = 16x$ पर रेखा $2x - y + 5 = 0$ के समान्तर एवं लम्बवत् स्पर्श रेखाओं का समीकरण ज्ञात करें। उनके स्पर्श बिन्दुओं के निर्देशांक भी ज्ञात कीजिए।

Ans. $2x - y + 2 = 0$, (1, 4) ; $x + 2y + 16 = 0$, (16, –16)

Sol. P : $y^2 = 16x$ L : $2x - y + 5 = 0$

Equation of tangent $2x - y + \lambda = 0$

$\Rightarrow \lambda = 2$ (by COT)

$2x - y + 2 = 0$ ...(1)

$$\text{Point of contact} = \left(\frac{a}{m^2}, \frac{2a}{m}\right) = \left(\frac{4}{4}, \frac{8}{2}\right) = (1, 4)$$

Equation of perpendicular tangent $x + 2y + \mu = 0$

$\mu = 16$

$\Rightarrow x + 2y + 16 = 0$ ...(2)

$$\text{Point of contact} = \left(\frac{a}{m^2}, \frac{2a}{m}\right) = (16, -16)$$

20. Find the equations of the tangents of the parabola $y^2 = 12x$, which passes through the point (2,5). Also find the point of contact.

परवलय $y^2 = 12x$ पर बिन्दु (2,5) से डाली गई स्पर्श रेखाओं का समीकरण ज्ञात करें एवं उनकी स्पर्शी बिन्दु के निर्देशांक भी ज्ञात कीजिए।

Ans. $3x - 2y + 4 = 0$; $x - y + 3 = 0$

Sol. Equation of tangents $y = mx + \frac{a}{m}$

$$y = mx + \frac{3}{m} \qquad ...(1)$$

it passes through (2,5)

$$5 = 2m + \frac{3}{m} \Rightarrow 2m^2 - 5m + 3 = 0$$

$2m^2 - 2m - 3m + 3 = 0$

$2m(m-1) - 3(m-1) = 0$

$m = 1, \frac{3}{2}$

Equation of tangents are

$y = x + 3$ POC = (3, 6)

Equation of tangent

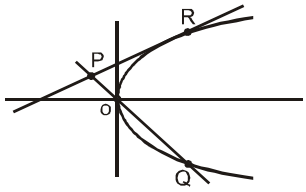
$y = \frac{3}{2}x + 2$ POC = $\left(\frac{4}{3}, 4\right)$

21. Through the vertex O of the parabola $y^2 = 4ax$, a perpendicular is drawn to any tangent meeting it at P & the parabola at Q. Show that OP · OQ = constant.

परवलय $y^2 = 4ax$ शीर्ष O से किसी भी स्पर्श रेखा पर अभिलम्ब डाला गया है जो उसे बिन्दु P पर मिलता है तथा परवलय को बिन्दु Q पर तो दशाईये OP · OQ = नियतांक।

Ans.

Sol.



$y = mx + \frac{a}{m}$ .... (i)

equation of OP is

$y = -\frac{1}{m}x$ .... (ii)

$OP = \frac{a/m}{\sqrt{1+m^2}}$

equation (ii) meets the parabola at Q

$\frac{1}{m^2}x^2 = 4ax \Rightarrow x = 4am^2, y = -4am$

$\therefore$ $OQ = 4am\sqrt{1+m^2}$, $OP.OQ = 4a^2$

22. The normal at a point P to the parabola $y^2 = 4ax$ meets its axis at G. Q is another point on the parabola such that QG is perpendicular to the axis of the parabola. Prove that $QG^2 - PG^2$ = constant.

परवलय $y^2 = 4ax$ के बिन्दु P पर बनाया गया अभिलम्ब अक्ष को G पर मिलता है। परवलय पर स्थिति एक अन्य बिन्दु Q इसप्रकार है कि QG अक्ष के लम्बवत् है। सिद्ध कीजिए $QG^2 - PG^2$ = अचर।

Ans.

Sol. Normal at $P(am^2, 2am)$ on $y^2 = 4ax$

$y + mx = 2am + am^3$ ... (1)

$G(2a + am^2, 0)$

Equation of QG is $x = 2a + am^2$

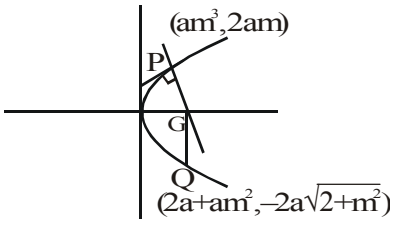
Solving with parabola we get

$y = \pm 2a\sqrt{2 + m^2}$

$QG^2 - PG^2 =$

$4a^2(2 + m^2) - (am^2 - am^2 - 2a)^2 - (2am)^2$

$= 4a^2$ which is constant



23. If the normal at P(18, 12) to the parabola $y^2 = 8x$ cuts it again at Q, show that $9PQ = 80\sqrt{10}$

यदि बिन्दु P(18, 12) पर परवलय $y^2 = 8x$ पर अभिलम्ब परवलय को फिर से Q पर काटता है तो दर्शाइये $9PQ = 80\sqrt{10}$.

Ans.

Sol.

24. Prove that, the normal to $y^2 = 12x$ at $(3, 6)$ meets the parabola again in $(27, -18)$ & circle on this normal chord as diameter is $x^2 + y^2 - 30x + 12y - 27 = 0$.

सिद्ध कीजिये कि परवलय $y^2 = 12x$ पर $(3, 6)$ पर बनाया गया अभिलम्ब फिर से परवलय को $(27, -18)$ पर मिलता है तथा उस अभिलम्ब जीवा को व्यास मानकर बनाये गये वृत्त का समीकरण $x^2 + y^2 - 30x + 12y - 27 = 0$ है।

Ans.

Sol. $P : y^2 = 12x$ $\qquad 4a = 12 \Rightarrow a = 3$

$P(t_1) = P(3,6) \Rightarrow 2at_1 = 6$

$t_1 = 1$

$$t_2 = -t_1 - \frac{2}{t_1} = -1 - 2 = -3$$

$Q(at_2^2, 2at_2) = (27, -18)$

Equation of circle diameter as PQ

$(x - 3)(x - 27) + (y - 6)(y + 18) = 0$

$x^2 - 30x + 81 + y^2 + 12y - 108 = 0$

$x^2 + y^2 - 30x + 12y - 27 = 0$

25. Find the equation of the circle which passes through the focus of the parabola $x^2 = 4y$ & touches it at the point $(6, 9)$.

परवलय $x^2 = 4y$ की नाभि से जाने वाले वृत्त का समीकरण ज्ञात करें जो परवलय को $(6, 9)$ पर स्पर्श करता है।

Ans. $x^2 + y^2 + 18x - 28y + 27 = 0$

Sol. $x\,x_1 = 2(y + y_1)$

$6x = 2(y + 9)$

$3x = y + 9$ $\qquad 3x - y - 9 = 0$

from equation of family circle is $S + \lambda L = 0$

$S \equiv (x - 6)^2 + (y - 91^2 + k(3x - y - 9) = 0$

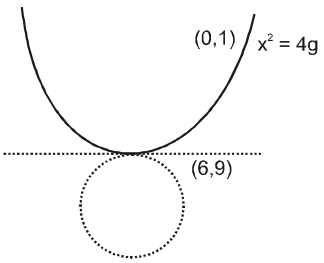
is passes through $(0, 1)$

$36 + 64 + k(-10) = 0$

$100 - 10k = 0$ $\qquad k = 10$

$x^2 + 36 - 12x + y^2 + 81 - 18y + 30x - 30y - 90 = 0$

$x^2 + y^2 + 18x - 28y + 27 = 0$

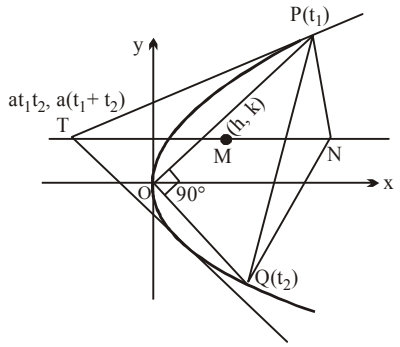


26. PQ, a variable chord of the parabola $y^2 = 4x$ subtends a right angle at the vertex. The tangents at P and Q meet at T and the normals at those points meet at N. If the locus of the mid point of TN is a parabola, then find its latus rectum.

च्फ परवलय $y^2 = 4x$ एक चर जीवा है जो शीर्ष पर समकोण अंकित करती है। बिन्दुओं P तथा Q पर बनायी गई स्पर्श रेखाऐं T पर एवं अभिलम्ब रेखाऐं N पर प्रतिच्छेद करती है। यदि TN के मध्य बिन्दु का बिन्दुपथ एक परवलय है, तो उसका नाभिलम्ब ज्ञात कीजिए।

Ans. 25/2

Sol. $\frac{2at_1}{at_1^2} \cdot \frac{2at_2}{at_2^2} = -1$



$t_1 t_2 = -4$

Now $T \equiv a\, t_1 t_2,\ a(t_1 + t_2)$

and $N \equiv a\,(t_1^2 + t_2^2 + t_1 t_2 + 2),\ -a\, t_1 t_2 (t_1 + t_2)$

Hence $2h = a((t_1 + t_2)^2 + 2)$

& $2k = a(t_1 + t_2)\,(1 - t_1 t_2)$

$2k = 5a(t_1 + t_2)$

$\therefore \quad t_1 + t_2 = \frac{2k}{5a}$

$\therefore \quad 2h = \left[\left(\frac{2k}{5a}\right)^2 + 2\right]$

$2h = \left[\frac{4k^2 + 50a^2}{25a^2}\right]$

$50ah = 4k^2 + 50a^2$

$25ah = 2k^2 + 25a^2$

$25 \cdot 2(h - 2) = 2k^2$ or $y^2 = 25(x - 2)$

Hence latus rectum = 25 Ans.

27. Let P be a point on the parabola $y^2 - 2y - 4x + 5 = 0$, such that the tangent on the parabola at P intersects the directrix at point Q. Let R be the point that divides the line segment PQ externally in the ratio $\frac{1}{2}:1$. Find the locus of R.

परवलय $y^2 - 2y - 4x + 5 = 0$ के किसी बिन्दु P पर एक स्पर्श रेखा खींची जाती है, जो नियता से बिन्दु Q पर मिलती है। बिन्दु R का बिन्दुपथ ज्ञात कीजिए, जो PQ को $\frac{1}{2}:1$. के अनुपात में बाह्य विभाजित करता है।

Ans. $2(y-1)^2(x-2) = (3x-4)^2$

Sol. The given equation of parabola is

$$y^2 - 2y - 4x + 5 = 0 \qquad \ldots (1)$$

$\Rightarrow \quad (y-1)^2 = 4(x-1)$

Any parametric point on this parabola is

$$P(t^2 + 1, 2t + 1)$$

Differentiating (1) w.r.t. x, we get

$$2y\frac{dy}{dx} - 2\frac{dy}{dx} - 4 = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{2}{y-1}$$

$\therefore$ Slope of tangent to (1) at pt.

$$P(t^2 + 1, 2t + 1) \text{ is } m = \frac{2}{2t} = \frac{1}{t}$$

$\therefore$ Equation of tangent at $P(t^2 + 1, 2t + 1)$ is

$$y - (2t + 1) = \frac{1}{t}(x - t^2 - 1)$$

$\Rightarrow \quad yt - 2t^2 - t = x - t^2 - 1$

$\Rightarrow \quad x - yt + (t^2 + t - 1) = 0 \qquad \ldots (2)$

Now direction of given parabola is

$$(x-1) = -1 \qquad \Rightarrow \quad x = 0$$

Tangent to (2) meets directrix at $Q\left(0, \frac{t^2 + t - 1}{t}\right)$

Let pt. R be (h, k)

ATQ R divides the line joining QP in the ratio $\frac{1}{2}$: 1 i.e. 1 : 2 externally.

$$\therefore (h, k) = \left[\frac{1(1 + t^2) - 0}{-1}, \frac{t + 2t^2 - 2t^2 - 2t + 2}{-t}\right]$$

$$\Rightarrow \quad h = -(1 + t^2) \text{ and } k = \frac{t-2}{t}$$

$$\Rightarrow \quad t^2 = -1 - h \text{ and } \quad t = \frac{2}{1-k}$$

Eliminating t we get $\left(\frac{2}{1-k}\right)^2 = -1 - h$

$\Rightarrow \quad 4 = -(1-k)^2 (1-h)$

$\Rightarrow \quad (h-1)(k-1)^2 + 4 = 0$

$\therefore \quad$ locus of R(h, k) is, $(x-1)(y-1)^2 + 4 = 0$

28. The tangent at any point P of a circle $x^2 + y^2 = a^2$ meets the tangent at a fixed pointA (a, 0) in T and T is joined to B, the other end of the diameter through A, prove that the locus of the intersection of AP and BT is an ellipse whose eccentricity is $1/\sqrt{2}$.

वृत्त $x^2 + y^2 = a^2$ पर स्थिति बिन्दु P पर बनायी गई स्पर्श रेखा बिन्दु A (a, 0) पर बनायी गई स्पर्श रेखा को T पर मिलती है तथा T को B से मिला दिया गया है, जोकि A से खींचे गये व्यास का दूसरा सिरा है। सिद्ध कीजिए AP एवं BT के प्रतिच्छेदन बिन्दु का बिन्दुपथ एक दीर्घवृत्त है जिसकी उत्केन्द्रता $1/\sqrt{2}$ है।

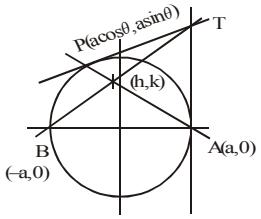
Ans.

Sol. Equation of tangent at P

$x\cos\theta + y\sin\theta = a$

which meets $x = a$ at T

$\therefore$ T $(a, a\tan\theta/2)$



Equation of AP $\rightarrow$ $y = -\cot(\theta/2)(x - a)$ ....(1)

Equation of BT $\rightarrow$ $y = \frac{\tan(\theta/2)}{2}(x + a)$ ....(2)

From (1) & (2)

$y^2 = -\frac{1}{2}(x^2 - a^2)$

$x^2 + 2y^2 = a^2$

29. If s, s' are the length of the perpendicular on a tangent from the foci, a, a' are those from the vertices, c is that from the centre and e is the eccentricity of the ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then prove that $\frac{ss' - c^2}{aa' - c^2} = e^2$

यदि s, s' एक स्पर्श रेखा की नाभियों से दूरी है एवं a, a' स्पर्श रेखा की शीर्षों से दूरी है, c केन्द्र से दूरी है एवं e दीघवृत $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

की उत्केन्द्रता है, तो सिद्ध करें $\frac{ss' - c^2}{aa' - c^2} = e^2$

Ans.

Sol.

30. Prove that the equation to the circle, having double contact with the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (with eccentricity e) at the ends of a latus rectum, is $x^2 + y^2 - 2ae^3x = a^2(1 - e^2 - e^4)$.

सिद्ध कीजिऐ की वृत का समीकरण जो दीर्घवृत $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (उत्केन्द्रियता e) को नाभिलम्ब के सिरों पर दो बार स्पर्श करता है, $x^2 + y^2 - 2ae^3x = a^2 (1 - e^2 - e^4)$ है।

Ans.

Sol. $E : \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Ends of LR : $L_1\left(ae,\frac{b^2}{a}\right), L_2\left(ae,-\frac{b^2}{a}\right)$

for double contact centre of circle should lie on normal of L_1 & L_2. By symmetry y co-ordinate of centre = 0

Equation of normal at $\left(ae,\frac{b^2}{a}\right)$

$$\frac{a^2x}{ae}-\frac{b^2y\times a}{b^2}=a^2-b^2$$

$$\frac{ax}{e}-ay=a^2-b^2 \qquad ...(1)$$

$$y=0,\ \frac{ax}{e}=\left(a^2-b^2\right)\Rightarrow x=ae^3$$

centre $(ae^3, 0)$

Equation of circle distance between L_1 & centre = r

$$\left(x-ae^3\right)+y^2=\left(ae-ae^3\right)^2+\left(\frac{b^2}{a}\right)^2$$

$x^2 - 2ae^3 + y^2 = a^2(1 - e^2 - e^4)$

31. Find the equations of the lines with equal intercepts on the axes & which touch the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$.

उन रेखाओं के समीकरण ज्ञात करें जो अक्षों पर समान अन्तःखण्ड काटती है एवं दीर्घवृत $\frac{x^2}{16}+\frac{y^2}{9}=1$ को स्पर्श करती है।

Ans. $x + y - 5 = 0, x + y + 5 = 0$

Sol. Line having equal intercepts is

$x + y = a$ ...(1)

$\Rightarrow y = a - x$

$m = -1, c = a$

COT with ellipse

$c^2 = a^2m^2 + b^2$

$a^2 = 16\cdot(-1)^2 + 9$

$a^2 = 25 \Rightarrow a = \pm 5$

Equation of line $\rightarrow x + y = 5$ or $x + y = -5$

32. A tangent having slope $-\frac{4}{3}$ to the ellipse $\frac{x^2}{18} + \frac{y^2}{32} = 1$, intersects the axis of x & y in points A & B respectively. If O is the origin, find the area of triangle OAB.

दीर्घवृत $\frac{x^2}{18} + \frac{y^2}{32} = 1$ पर एक स्पर्श रेखा जिसकी प्रवणता $-\frac{4}{3}$ है। अक्षों को क्रमशः A तथा B पर काटती है। यदि O मूल बिन्दु है, तो त्रिभुज OAB का क्षेत्रफल ज्ञात कीजिए।

Ans. 24 sq. units

Sol. Equation of ellipse $\frac{x^2}{18} + \frac{y^2}{32} = 1$ having slope $-\frac{4}{3}$

$y = mx \pm \sqrt{a^2m^2 + b^2}$

$y = -\frac{4}{3}x \pm \sqrt{18 \times \frac{16}{9} + 32}$

$y = -\frac{4}{3}x \pm 8 \qquad \Rightarrow 3y = -4x + 24$

$\frac{4x}{24} + \frac{3y}{24} = 1$

$\frac{x}{6} + \frac{y}{3} = 1$

area of $\Delta = \frac{1}{2} \times 6 \times 8 = 24$ sq. units

33. Find the equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$.

दीर्घवृत $x^2 + 4y^2 = 16$ में स्थित अधिकतम क्षेत्रफल वाले वृत्त का समीकरण ज्ञात करें जिसका केन्द्र (1, 0) है।

Ans. $(x - 1)^2 + y^2 = \frac{11}{3}$

Sol. Equation of circle

$\xi - 1 = \rho\chi o\sigma\theta$

$y - 0 = r\sin\theta$

lies in ellipse $\quad \therefore \quad S < 0$

$\Rightarrow \quad (1 + r\cos\theta)^2 + 4(r\sin\theta)^2 - 16 < 0$

$\Rightarrow \quad 3r^2\cos^2\theta - 2r\cos\theta + 15 - 4r^2 > 0$

$\because \quad a = 3r^2 > 0 \quad \therefore \quad f(\theta) > 0 \quad \forall\theta \in R$

$\therefore \quad D < 0$

$4r^2 - 4 \times 3r^2(15 - 4r^2) < 0$

$4r^2[3r^2 - 11] < 0$

$4r^2[3r^2 - 11] < 0$

$\because \quad r^2 > 0 \quad \therefore \quad 3r^2 - 11 < 0$

$r^2 < \frac{11}{3}$

Hence equation of circle

$(x - 1)^2 + y^2 = \frac{11}{3}$

34. Find the equation of the tangent to the hyperbola $x^2 - 4y^2 = 36$ which is perpendicular to the line $x - y + 4 = 0$.

अतिपरवलय $x^2 - 4y^2 = 36$ के उस स्पर्श रेखा का समीकरण ज्ञात करो जो रेखा $x - y + 4 = 0$ के लम्बवत् है।

Ans. $x + y \pm 3\sqrt{3} = 0$

Sol. Let m be the slope of the tangent. Since the tangent is perpendicular to the line $x - y = 0$

$\therefore \quad m \times 1 = -1 \quad \Rightarrow \quad m = -1$

Since $x^2 - 4y^2 = 36$ or $\frac{x^2}{36} - \frac{y^2}{9} = 1$

Comparing this with $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\therefore \quad a^2 = 36$ and $b^2 = 9$

So the equation of tangents are $y = (-1)x \pm \sqrt{36 \times (-1)^2 - 9}$

$y = -x \pm \sqrt{27} \quad \Rightarrow \quad x + y \pm 3\sqrt{3} = 0$

35. Tangents are drawn to the hyperbola $3x^2 - 2y^2 = 25$ from the point (0, 5/2). Find their equations.

अतिपरवलय $3x^2 - 2y^2 = 25$ पर बिन्दु (0, 5/2) से स्पर्श रेखा खींची जाती है। उनके समीकरण ज्ञात कीजिए।

Ans. $3x + 2y - 5 = 0$; $3x - 2y + 5 = 0$

Sol. Equation of tangents

$$y = mx + \frac{5}{2} \quad \Rightarrow m = C = \frac{5}{2}$$

COT with HB $\dfrac{x^2}{\left(\frac{25}{3}\right)} - \dfrac{y^2}{\left(\frac{25}{2}\right)} = 1$

$c^2 = a^2m^2 - b^2$

$$\frac{25}{4} = \frac{25}{3}m^2 - \frac{25}{2}$$

$$\frac{1}{4} + \frac{1}{2} = \frac{m^2}{3} \Rightarrow \frac{m^2}{3} = \frac{3}{4} \Rightarrow m = \pm\frac{3}{2}$$

Equation of tangents

$$y = \frac{3}{2}x + \frac{5}{2} \quad \text{or} \quad y = -\frac{3}{2}x + \frac{5}{2}$$

36. Locus of the feet of the perpendicular from centre of the hyperbola $x^2 - 4y^2 = 4$ upon a variable normal to it has the equation, $(x^2 + y^2)^2 (4y^2 - x^2) = \lambda x^2y^2$, find λ.

अतिपरवलय $x^2 - 4y^2 = 4$ के केन्द्र से इसके किसी चर अभिलम्ब पर डाले गये लम्बपाद का बिन्दुपथ $(x^2 + y^2)^2 (4y^2 - x^2) = \lambda x^2y^2$ है। λ ज्ञात कीजिए।

Ans. 25

Sol. Equation of N is

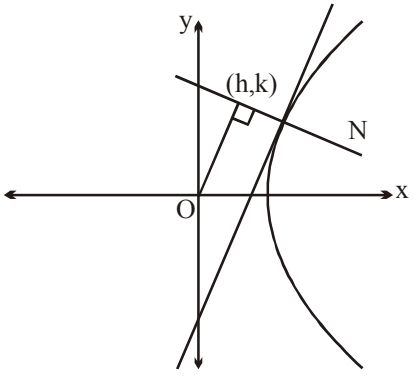
$hx + ky = h^2 + k^2$

Compair it with the normal are Q

$$\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2$$

i.e., $\dfrac{2x}{\sec\theta} + \dfrac{y}{\tan\theta} = 5$

$$\boxed{\frac{x^2}{4} - \frac{y^2}{1} = 1 \quad a = 2 \; ; \; b = 1}$$



$$\frac{h\cdot\sec\theta}{2} = k\tan\theta = \frac{h^2+k^2}{5}$$

$$\sec\theta = \frac{2(h^2+k^2)}{5h}$$

$$\tan\theta = \frac{h^2+k^2}{5k}$$

$$\therefore \quad \frac{4}{25}\,\frac{(h^2+k^2)^2}{25h^2} - \frac{(h^2+k^2)^2}{25k^2} = 1$$

$$4\,\frac{(h^2+k^2)^2}{20}\left(\frac{4}{h^2}-\frac{1}{k^2}\right) = 1 \Rightarrow k = 25 \Rightarrow \mathbf{A}.$$

37. Let P (a sec θ, b tan θ) and Q (a sec ϕ, b tan ϕ), where $\theta+\phi=\frac{\pi}{2}$, be two points on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If (h, k) is the point of intersection of the normals at P & Q, then find k.

माना P (a sec θ, b tan θ) तथा Q (a sec ϕ, b tan ϕ), जहाँ पर $\theta+\phi=\frac{\pi}{2}$ है, अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ पर दो बिन्दु है। यदि P एवं Q पर अभिलम्बो का प्रतिच्छेद बिन्दु (h, k) हो तो k ज्ञात कीजिए।

Ans. $-\left(\frac{a^2+b^2}{b}\right)$

Sol. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Equation of normal

$ax\cos\theta + by\cot\theta = a^2+b^2$ (at P) ...(1)

at Q

$ax\cos\phi + by\cot\phi = a^2+b^2$ ...(2)

Point of intersection in R(h, k)

αξ χοσθ cos ϕ + by cot θ cos ϕ = $(a^2+b^2)\cos\phi$

$ax\cos\theta\cos\phi + by\cot\phi\cos\theta = (a^2+b^2)\cos\theta$

– – –

$y[b[\cot\theta\cdot\cos\phi - \cot\phi\cdot\cos\theta]] = (a^2+b^2)(\cos\phi-\cos\theta)$

$$K=\frac{\left(a^2+b^2\right)2\cdot\sin\left(\frac{\theta+\phi}{2}\right)\cdot\sin\left(\frac{\phi-\theta}{2}\right)}{b\left[\frac{\cos\theta\cdot\cos\phi}{\sin\theta}-\frac{\cos\phi\cdot\cos\theta}{\sin\phi}\right]}$$

$$K=\frac{a^2+b^2}{b}$$

38. The line $2x+y=1$ is tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If this line passes through the point of intersection of the nearest directrix and the x-axis, then the eccentricity of the hyperbola, is. **[JEE-2010]**

सरल रेखा $2x+y=1$ अतिपरवयल $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ की स्पर्श रेखा है। यदि यह रेखा निकटतम नियता तथा x-अक्ष के कटान बिन्दु से होकर गुजरती है, तो अतिपरवलय की उत्केन्द्रता होगी। **[JEE-2010]**

Ans. 2

Sol. As directrix cut the x–axis at $(\pm a/e, 0)$

Hence, $\frac{2a}{e}+0=1$ (for nearer directrix)

$\Rightarrow$ $2a=e$ ...(i)

Now, $b^2=a^2(e^2-1)=a^2(4a^2-1)$

$\Rightarrow$ $\frac{b^2}{a^2}=4a^2-1$ ...(ii)

Given line $y=-2x+1$ is a tangent to the hyperbola condition of tangency is $c^2=a^2m^2-b^2$

$\Rightarrow$ $1=4a^2-b^2$

$\Rightarrow$ $4a^2-1=b^2$ ...(iii)

from (ii) & (iii), $a^2=1$

$\Rightarrow$ from (ii), $b^2=3$

$\Rightarrow$ $e=\sqrt{\frac{1+3}{1}}=2$

NORMAL TO CONIC FROM A POINT

39. Three normals to $y^2=4x$ pass through the point (15, 12). Show that if one of the normals is given by $y=x-3$ & find the equations of the others.

बिन्दु (15, 12) से परवलय $y^2 = 4x$ पर तीन अभिलम्ब डाले जाते है। दर्शाइये कि इनमें से एक अभिलम्ब $y = x - 3$ है तथा अन्य का समीकरण ज्ञात कीजिए।

Ans. $y = -4x + 72$, $y = 3x - 33$

Sol. $y^2 = 4x \Rightarrow 4a = 4 \Rightarrow a = 1$ it passes through (15,12)

Equation of normal

$y + xt = 2t + t^3$

$12 + 5t = 2t + t^3 \Rightarrow t^3 - 13t - 12 = 0$

$(t + 1)(t - 4)(t + 3) = 0 \Rightarrow t = -1, -3, 4$

remaining normals are

$y = -4x + 72$ $y = 3x - 33$

$(t = 4)$ $(t = -3)$

40. From the point P(h,k) three normals are drawn to the parabola $x^2 = 8y$ and m_1, m_2 and m_3 are the slops of three normals

(a) Find the algebaric sum of the slopes of these three normals.

(b) If two of the three normals are at right angles then the locus of point P is a conic, find the latus rectum of conic.

(c) If the two normals from P are such that they make complementary angles with the axis then the locus of point P is a conic, find a directrix of conic.

बिन्दु P(h, k) से परवलय $x^2 = 8y$ से तीन अभिलम्ब डाले गये है तथा m_1, m_2 तथा m_3 इन तीनों की प्रवणताऐं है।

(a) इन तीनों अभिलम्बों की प्रवणताओं का बीजगणितीय योग ज्ञात कीजिए।

(b) यदि इनमें से दो अभिलम्ब लम्बवत् है, तो बिन्दु P का बिन्दुपथ एक शंकु होगा, उस शंकु का नाभिलम्ब ज्ञात कीजिए।

(c) यदि P से दो अभिलम्ब इसप्रकार बनाये गये है कि उनके द्वारा अक्ष के साथ बनाये गये कोण पूरक है, तो बिन्दु P का बिन्दुपथ एक शंकु है, उस शंकु की नियता ज्ञात कीजिए।

Ans. (a) $\frac{k-4}{h}$; (b) 2; (c) $2y - 3 = 0$

Sol. Equation of normal (for $y^2 = 4ax \rightarrow y = mx - 2am - am^3$)

$$m \rightarrow \frac{1}{m}, x \rightarrow y, y \rightarrow x$$

$$x = \frac{y}{m} = \frac{2a}{m} - \frac{a}{m^3} \qquad 8 = 4a \Rightarrow a = 2$$

$$mx = y - 2a - \frac{a}{m^2}$$

(a) $mx = y - 4 - \frac{2}{m^2} \Rightarrow m^3x = m^2y - 4m^2 - 2$

it passes through (h, k)

$$\Rightarrow m^3h + m^2(h-k) + 2 = 0 \begin{cases} m_1 \\ m_2 \\ m_3 \end{cases}$$

$$m_1 + m_2 + m_3 = \frac{-(4-k)}{h} \Rightarrow \frac{k-4}{h}$$

(b) $\underbrace{m_1m_2}_{-1} m_3 = \frac{2}{h} \qquad \Rightarrow m_3 = -\frac{2}{h}$

Equation of normal $\quad m_3x = y - 4 - \frac{2}{m_3^2} \leftarrow (h,k) \qquad m_3 = -\frac{2}{h}$

$$-\frac{2}{h} \times h = k - 4 - \frac{2}{4} \times h^2$$

$$\frac{h^2}{2} \times k - 2 = 0 \qquad \Rightarrow h^2 + 2k - 4 = 0$$

$\Rightarrow h^2 = -2k + 4 = -2(k-2)$

$x^2 = -2(y-2)$

$\ell_{LR} = 2$

(C) if 2 normals make complementary angle

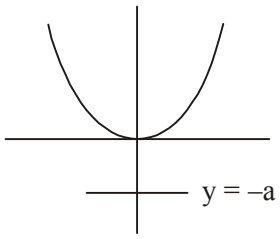
$m_1m_2 = 1$

$$\Rightarrow \underbrace{m_1m_2}_{1} m_3 = \frac{2}{h}$$

$$m_3 = \frac{2}{h}$$

$$m_3x = y - 4 - \frac{2}{m_3^2}$$

$2 = k - 4 - \frac{2}{4} \times h^2$



$\frac{h^2}{2} = k - 6$

$h^2 = 2(k-6) \Rightarrow x^2 = 2(y-6)$

Vertex = (0,6) $\qquad a = \frac{1}{2}$

directrix $\qquad y - 6 = -\frac{1}{2}$

COMMON TANGENTS & COMMON NORMAL

41. Prove that the two parabolas $y^2 = 4ax$ & $y^2 = 4c(x-b)$ cannot have a common normal, other than the axis, unless $\frac{b}{(a-c)} > 2$.

सिद्ध कीजिए कि दो परवलयों $y^2 = 4ax$ एवं $y^2 = 4c(x-b)$ में उनके अक्ष को छोड़कर कोई उभयनिष्ठ अभिलम्ब नहीं हो सकता जब तक $\frac{b}{(a-c)} > 2$ है।

Ans.

Sol. Equation of normal at $y^2 = 4ax$

$y = mx - 2am - am^3$ ....(1)

and normal to $y^2 = 4c(x-b)$ with the same slope

$y = m(x-b) - 2cm - cm^3$ ....(2)

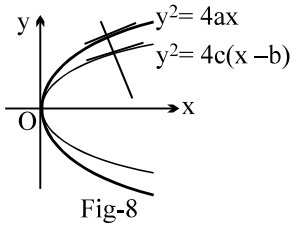


Fig-8

If (1) and (2) denotes the same line, we have

$-2am - am^3 = -bm - 2cm - cm^3$

$2a + am^2 = b + 2c + cm^2 \qquad (m \neq 0)$

$(a-c)m^2 + 2(a-c) = b$

$m^2 = \frac{b}{a-c} - 2 > 0 \Rightarrow \frac{b}{a-c} > 2$

42. Find the condition on 'a' & 'b' so that the two tangents drawn to the parabola $y^2 = 4ax$ from a point are normals to the parabola $x^2 = 4by$.

ऐसे बिन्दु के लिए शर्त ज्ञात करें जिसके लिए परवलय $y^2 = 4ax$ पर बनायी गई दो स्पर्श रेखाऐं परवलय $x^2 = 4by$ पर अभिलम्ब है।

Ans. $a^2 > 8b^2$

Sol. Equation of tangent of

$y^2 = 4ax$ in slope form at(x_1, y_1) is

$y_1 = mx_1 + \frac{a}{m}$ ... (1)

equation of normal at $(2bt_1, bt_1^2)$ on $x^2 = 4by$

$x + t_1y = 2bt_1 + bt_1^3$

It passes through (x_1, y_1)

$\therefore\ x_1 + t_1y_1 = 2bt_1 + bt_1^3$ ... (2)

(1) & (2) are same equation so compare

$\frac{1}{t_1} = -\frac{m}{1} = \frac{a}{m(2bt_1 + bt_1^3)}$

$t_1m = -1$

$-m^2t_1(2b + bt_1^2) = a$

$\Rightarrow \quad m(2b + bt_1^2) = a$ ... (3)

Put $m = -\frac{1}{t_1}$ in equation (3)

$2b + bt_1^2 = -at_1$

$bt_1^2 + at_1 + 2b = 0$

t_1 will be real

$a^2 > 8b^2$

Another Method

Equation of tangent to $y^2 = 4ax$ is

$y = mx + \frac{a}{m}$ ....(1)

Normal to $x^2 = 4by$ having slope m is

$y = mx + 2b + \frac{b}{m^2}$ ....(2)

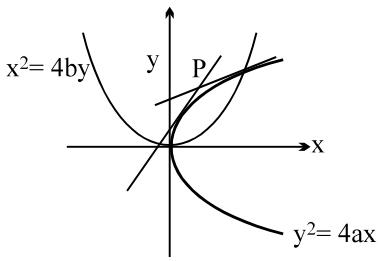
comparing (1) and (2)

$2b + \frac{b}{m^2} = \frac{a}{m}$

$2bm^2 - am + b = 0$

for two distinct m, $D > 0$

$\therefore \quad a^2 - 8b^2 > 0 \quad \Rightarrow \quad a^2 > 8b^2$



43. Find the equation of the common tangent in 1st quadrant to the circle $x^2 + y^2 = 16$ and the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$.

Also find the length of the intercept of the tangent between the coordinate axes.

प्रथम चतुर्थांश में, वृत्त $x^2 + y^2 = 16$ एवं दीर्घवृत्त $\frac{x^2}{25}+\frac{y^2}{4} = 1$ की उभयनिष्ठ स्पर्श रेखा ज्ञात कीजिये। स्पर्श रेखा के अक्षों के मध्य काटे गये अन्तःखण्ड की लम्बाई भी ज्ञात कीजिए।

Ans. $AB = \frac{14}{\sqrt{3}}$

Sol. Equation of tangent of circle

$$y = mx \pm 4\sqrt{1+m^2} \qquad ...(1)$$

Equation of tangent of ellipse

$$y = mx \pm \sqrt{25m^2+4} \qquad ...(2)$$

from equation (1) & (2)

$16 + 16m^2 = 25m^2 + 4$

$9m^2 = 12 \Rightarrow m = \pm\frac{2}{\sqrt{3}}$

Since tangent is in Ist quadrant $m < 0$

$$m = -\frac{2}{\sqrt{3}}$$

Equation of common tangent $\Rightarrow y = -\frac{2x}{\sqrt{3}} + 4\sqrt{\frac{7}{3}}$

meets coordinate axes at

$$A\left(2\sqrt{7},0\right) \text{ \& } B\left(0,4\sqrt{\frac{7}{3}}\right)$$

$$AB = \frac{14\sqrt{3}}{3}$$

COMMON CONCEPTS

44. Through the vertex O of a parabola $y^2 = 4x$, chords OP & OQ are drawn at right angles to one another. Show that for all positions of P, PQ cuts the axis of the parabola at a fixed point. Also find the locus of the middle point of PQ.

परवलय $y^2 = 4x$ के शीर्ष O से दो जीवाऐं OP तथा OQ लम्बवत् बनाई गयी है। दर्शाइये कि P की सभी स्थितियों के लिए PQ परवलय के अक्ष को सदैव निश्चित बिन्दु पर प्रतिच्छेद करेगी। साथ ही PQ के मध्य बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. $(4, 0)$; $y^2 = 2a(x - 4a)$

Sol. $P : y^2 = 4x \quad \Rightarrow 4a = 4 \Rightarrow a = 1$

$t_1t_2 = -4$ (as PQ subtend 90º at vertex)

$$\text{midpoint of PQ } M(h,k) = \left(\frac{a\left(t_1^2 + t_2^2\right)}{2}, a\left(t_1 + t_2\right) \right) \qquad a = 1$$

$2h = (t_1^2 + t_2^2)$, $t_1 + t_2 = k$

$2h = (t_1 + t_2)^2 - 2t_1t_2 \Rightarrow 2h = k^2 - 2(-4)$

$y^2 = 2x - 8$

45. From the point $(-1, 2)$ tangent lines are drawn to the parabola $y^2 = 4x$. Find the equation of the chord of contact. Also find the area of the triangle formed by the chord of contact & the tangents.

बिन्दु $(-1, 2)$ से परवलय $y^2 = 4x$ पर स्पर्श रेखाऐं बनाई गयी है। स्पर्श जीवा का समीकरण ज्ञात करें। साथ ही स्पर्श जीवा एवं स्पर्श रेखाओं द्वारा बनाये गये त्रिभुज का क्षेत्रफल ज्ञात कीजिए।

Ans. $x - y = 1$; $8\sqrt{2}$ sq. units

Sol. Equation of COC for $(-1,2)$ to parabola $y^2 = 4x$.

$(T = 0)$

$2y = 2(x + x_1) \Rightarrow y = x - 1$ _____(1)

Solving it with parabola

$(x - 1)^2 = 4x$

$x^2 - 2x + 1 - 4x = 0$

$x^2 - 6x + 1 = 0$

$x = 3 \pm 2\sqrt{2}$

$P\left(3 + 2\sqrt{2}, 2 + 2\sqrt{2}\right)$, $Q\left(3 - 2\sqrt{2}, 2 - 2\sqrt{2}\right)$

$PQ^2 = 64 \Rightarrow PQ = 8$

also perpendicular length from $(-1, 2)$ to equation (1)

$$= \frac{4}{\sqrt{2}}$$

Area of $\Delta = \frac{1}{2}\times(8)\times\frac{4}{\sqrt{2}} = 8\sqrt{2}$ sq. units

46. Consider the family of circles, $x^2 + y^2 = r^2$, $2 < r < 5$. If in the first quadrant, the common tangent to a circle of the family and the ellipse $4x^2 + 25y^2 = 100$ meets the co–ordinate axes at A & B, then find the equation of the locus of the mid–point of AB.

माना वृत का निकाय $x^2 + y^2 = r^2$, $2 < r < 5$ है। यदि प्रथम चतुर्थांश में वृत एवं दीर्घवृत $4x^2 + 25y^2 = 100$ पर उभयनिष्ठ स्पर्श रेखा, अक्षों को A तथा B पर मिलती है, तो AB के मध्य बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. $25y^2 + 4x^2 = 4x^2y^2$

Sol. Equation of tangent to circle

$x\cos\theta + y\sin\theta = r$ _____(1)

Equation of tangent to ellipse

$$\frac{x\cdot x_1}{25} + \frac{y\cdot y_1}{4} = 1 \quad \text{_____(2)}$$

from (1) & (2)

$$\frac{25\cos\theta}{x_1} = \frac{4\sin\theta}{y_1} = \frac{r}{1} \quad \text{_____(3)}$$

line (1) meet co-ordinate axes at A (r sec θ, 0), B(0, r cosec θ). Let (h,k) be mid-point of AB.

$$h = \frac{r\sec\theta}{2},\ \ k = \frac{r\,\mathrm{cosec}\theta}{2} \quad \text{_____(4)}$$

from (3) & (4)

$$x_1 = \frac{25}{2h}, y_1 = \frac{4}{2k}$$

as (x_1, y_1) is on ellipse.

$$\frac{1}{25}x_1^2 + \frac{1}{4}y_1^2 = 1$$

$$\frac{1}{25}\times\frac{625}{4h^2} + \frac{1}{4}\times\frac{16}{4k^2} = 1$$

$$\frac{25}{4h^2} + \frac{1}{k^2} = 1$$

$\Rightarrow 25k^2 + 4h^2 = 4h^2k^2$

$4x^2 + 25y^2 = 4x^2y^2$

47. Chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are tangents to the circle drawn on the line joining the foci as diameter. Find the locus of the point of intersection of tangents at the extremities of the chords.

अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ की जीवायें नाभियों को जोड़ने वाली रेखा को व्यास मानकार खीचे गये वृत पर स्पर्श रेखा है। जीवाओं के सिरों पर स्पर्श रेखाओं के प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2 + b^2}$

Sol. Equation of circle

$x^2 + y^2 = a^2e^2$ _____(1)

Equation of tangent

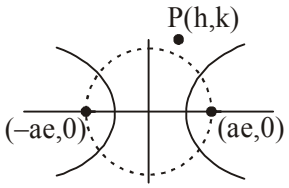
$x \cos\theta + y \sin\theta = ae$ _____(2)

Equation of chord of contact

$\frac{hx}{a^2} - \frac{ky}{b^2} = 1$ _____(3)

from (2) & (3)

$\frac{\cos\theta}{\left(\frac{h}{a^2}\right)} = \frac{\sin\theta}{\left(\frac{-k}{b^2}\right)} = \frac{ae}{1}$



$\Rightarrow \cos\theta = \frac{hae}{a^2}, \sin\theta = \frac{-ka \cdot e}{b^2}$

$\cos^2\theta + \sin^2\theta = 1 \Rightarrow \frac{h^2a^2e^2}{a^4} + \frac{k^2a^2e^2}{b^4} = 1$

$\frac{x^2}{a^4} + \frac{y^2}{b^2} = \frac{1}{c^2} = \frac{1}{a^2 + b^2}$

48. Tangents are drawn from any point on the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$ to the circle $x^2+y^2=9$. Find the locus of midpoint of the chord of contact.

अतिपरवलय $\frac{x^2}{9}-\frac{y^2}{4}=1$ के किसी बिन्दु से वृत्त $x^2+y^2=9$ पर स्पर्श रेखाऐं खींची जाती हैं। स्पर्श जीवा के मध्य बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. $\frac{x^2}{9}-\frac{y^2}{4}=\left(\frac{x^2+y^2}{9}\right)^2$

Sol. $H:\frac{x^2}{9}-\frac{y^2}{4}=1$

point on HB = $(3\sec\theta, 2\tan\theta)$

Equation of C.O.C. for circle $\rightarrow 3x\sec\theta+2y\tan\theta=9$ _____.(1)

chord in terms of mid point (h, k)

$hx+ky=h^2+k^2$ _____(2)

$$\frac{3\sec\theta}{h}=\frac{2\tan\theta}{k}=\frac{9}{h^2+k^2}$$

$$\sec\theta=\frac{9h}{3\left(h^2+k^2\right)},\tan\theta=\frac{9k}{2\left(h^2+k^2\right)}$$

$$\sec^2\theta-\tan^2\theta=1\Rightarrow\frac{9h^2}{\left(h^2+k^2\right)}-\frac{81k^2}{4\left(h^2+k^2\right)}=1$$

$$\frac{x^2}{9}-\frac{y^2}{4}=\left(\frac{x^2+y^2}{9}\right)^2$$

49. Find the number of integral values of parameter 'a' for which three chords of the ellipse $\frac{x^2}{2a^2}+\frac{y^2}{a^2}=1$ (other than its diameter) passing through the point $P\left(11a,-\frac{a^2}{4}\right)$ are bisected by the parabola $y^2=4ax$.

'a' के पूर्णांक मानों की संख्या ज्ञात कीजिए जिसके लिए दीर्घवृत्त $\frac{x^2}{2a^2} + \frac{y^2}{a^2} = 1$ की बिन्दु $P\left(11a, -\frac{a^2}{4}\right)$ से गुजरने वाली तीन जिवाऐं (व्यास के अलावा) परवलय $y^2 = 4ax$ पर समद्विभाजित होती है।

Ans. $a = \pm 1$, two value of a

Sol. Any point on the parabola $y^2 = 4ax$ is $(at^2, 2at)$. Equation of chord of the ellipse $\frac{x^2}{2a^2} + \frac{y^2}{a^2} = 1$, whose mid-point is $(at^2, 2at)$ is $\frac{x \cdot at^2}{2a^2} + \frac{y \cdot 2at}{a^2} = \frac{a^2t^4}{2a^2} + \frac{4a^2t^2}{a^2}$

$\Rightarrow tx + 4y = at^3 + 8at \ (\because t \neq 0)$

As it passes through $\left(11a, -\frac{a^2}{4}\right)$,

$\Rightarrow 11at - 4\left(\frac{a^2}{4}\right) = at^3 + 8at \Rightarrow at^3 - 3at + a^2 = 0$

$\Rightarrow t^3 - 3t + a = 0 \ (a \neq 0)$

Now, three chords of the ellipse will be bisected by the parabola if the equation (1) has three real and distinct roots.

Let $f(t) = t^3 - 3t + a$

$f'(t) = 3t^2 - 3 = 0 \Rightarrow t = \pm 1$

So, $f(1)\, f(-1) < 0$

$\Rightarrow a \in (-2, 2)$

But $a \neq 0$, so $a \in (-2, 0) \cup (0, 2)$

$\therefore$ Number of integral values of 'a' = 2

IMPORTANT HIGHLIGHTS OF PARBOLA

50. A variable circle passes through the point A(2, 1) and touches the x-axis. Locus of the other end of the diameter through A is a parabola.

(a) Find the length of the latus rectum of the parabola.

(b) Find the coordinates of the foot of the directrix of the parabola.

(c) The two tangents and two normals at the extremities of the latus rectum of the parabola constitutes a quadrilateral. Find area of quadrilateral.

एक चर वृत्त बिन्दु A(2, 1) से गुजरता हैं एवं x-अक्ष को स्पर्श करता है। बिन्दु A से व्यास के विपरीत सिरे का बिन्दुपथ एक परवलय है।

(a) परवलय के नाभिलम्ब की लम्बाई ज्ञात कीजिए।

(b) परवलय की नियता के पाद के निर्देशांक ज्ञात कीजिए।

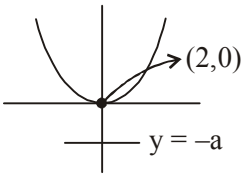
(c) परवलय के नाभिलम्ब के सिरों पर बनायी गई दो स्पर्श रेखाऐं एवं दो अभिलम्ब एक चतुर्भुज का निर्माण करते है, उस चतुर्भुज का क्षेत्रफल ज्ञात कीजिए।

Ans. (a) 4, (b) (2, – 1), (c) 8 sq. units

Sol. Equation of circle whose diameter is A(2,1) & P(h, k).

$(x-2)(x-h)+(y-1)(y-k)=0$

it touches x axis so $y=0$



$(x-2)(x-h)+(-1)(-k)=0$

$x^2-2x-hx+2h+k=0$

$x^2-x(h+2)+2h+k=0$

Discriminant should be zero so we can get only one value of x. ($\Delta=0$)

$[-(h+2)]^2-4(2h+k)=0$

$h^2+4h+4-8h-hk=0$

$h^2-4h+4=4k$

$(h-2)^2=4k$

$(x-2)^2=4y \qquad \Rightarrow 4a=4 \qquad \Rightarrow a=1$

(a) $\ell_{LR}=4$

(b) F. directrix = (2, – 1)

(C) it will be square area of square = $\frac{16a^2}{2}$

$=8a^2=8(1)^2=8$ sq. units

51. Show that the circle through three points the normals at which to the parabola $y^2=4ax$ are concurrent at the point (h, k) is $2(x^2+y^2)-2(h+2a)x-ky=0$.

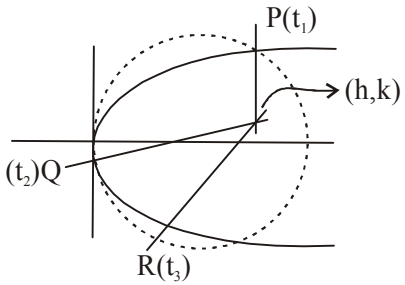
सिद्ध कीजिए कि परवलय $y^2=4ax$ पर स्थित तीन बिन्दुओं से जाने वाले वृत्त का समीकरण $2(x^2+y^2)-2(h+2a)x-ky=0$ है। जहाँ से बनाये गये अभिलम्ब (h, k) पर संगामी होते हैं।

Ans.

Sol. Equation of normal through (h, k)

$$at^3+(2a-h)t-k=0 \left\langle \begin{matrix} t_1 \\ t_2 \\ t_3 \end{matrix} \right. \qquad ...(1)$$

$$\begin{cases} t_1 + t_2 + t_3 = 0 \\ t_1t_2 + t_2t_3 + t_3t_1 = \dfrac{2a-h}{a} \\ t_1t_2t_3 = \dfrac{k}{a} \end{cases}$$



Let the circle through PQR

$x^2 + y^2 + 2gx + 2fy + c = 0$

it also passes through (0,0)

$x^2 + y^2 + 2gx + 2fy = 0$ ...(2)

solving it with parabola ($x = at^2$, $y = 2at$)

$a^2t^4 + 4a^2t^2 + 2g(at^2) + 2f(2at) = 0$

$at^3 + 4at + 2gt + 4f = 0$

$at^3 + t(4a + 2g) + 4f = 0$ ....(3)

Equation (3) & (1) are same.

$4a + 2g = 2a - 4$

$$\Rightarrow g = \frac{-h-2a}{2} = -\frac{(h+2a)}{2}$$

$$4f = -k \Rightarrow f = -\frac{k}{4}$$

From equation (2)

$$x^2 + y^2 - 2x\frac{(4+2a)}{2} - 2y\frac{k}{4} = 0$$

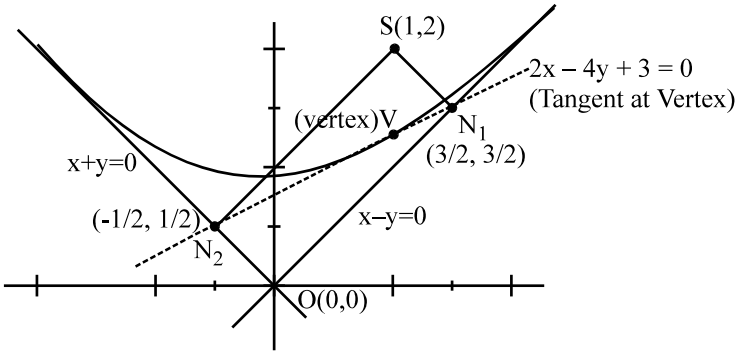
$2(x^2 + y^2) - 2x(h + 2a) - ky = 0$

52. Let $L_1 : x + y = 0$ and $L_2 : x - y = 0$ are tangent to a parabola whose focus is S(1, 2). If the length of latus-rectum of the parabola can be expressed as $\frac{m}{\sqrt{n}}$ (where m and n are coprime) then find the value of $(m + n)$.

माना $L_1 : x + y = 0$ और $L_2 : x - y = 0$ उस परवलय की दो स्पर्श रेखाऐं है जिसका नाभि S(1, 2) है यदि परवलय की नाभिलम्ब की लम्बाई $\frac{m}{\sqrt{n}}$ (m, n सह अभाज्य) द्वारा दर्शाते हैं तब (m + n) का मान होगा।

Ans. 11

Sol.



Feet of the perpendicular (N_1 and N_2) from focus upon any tangent to parabola lies on the tangent line at the vertex.

Now equation of SN_1 is $x + y = \lambda$ passing through $(1, 2)$ $\Rightarrow$ $1 = 3$

Equation of SN_1 is $x + y = 3$

Solving $x + y = 3$ and $y = x$, we get $N_1 \equiv \left(\frac{3}{2}, \frac{3}{2}\right)$

|||ly equation of SN_2 is $x - y = \lambda$ passing through $(1, 2)$ $\Rightarrow$ $1 = -1$

Equation of SN_2 is $y - x = 1$

Solving $y - x = 1$ and $y = -x$, we get $N_2 \equiv \left(\frac{-1}{2}, \frac{1}{2}\right)$

Now equation of tangent line at vertex is, $2x - 4y + 3 = 0$

Distance of S(1, 2) from tangent at vertex is

$$= \frac{|2-8+3|}{\sqrt{20}} = \frac{3}{2\sqrt{5}} = \frac{1}{4} \times \text{latus rectum}.$$

and hence length of latus rectum $= \frac{6}{\sqrt{5}} = \frac{m}{\sqrt{n}}$

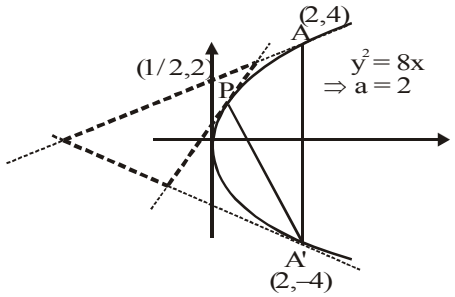
Hence $m + n = 6 + 5 = 11$ Ans.

53. Consider the parabola $y^2 = 8x$. Let Δ_1 be the area of the triangle formed by the end points of its latus rectum and the point $P\left(\frac{1}{2}, 2\right)$ on the parabola, and Δ_2 be the area of the triangle formed by drawing tangents at P and at the end points of the latus rectum. Then $\frac{\Delta_1}{\Delta_2}$ is. **[JEE-2011]**

माना परवलय $y^2 = 8x$ है। Δ_1 नाभिलम्ब के सिरों तथा परवलय पर स्थित बिन्दु $P\left(\frac{1}{2}, 2\right)$ से बने त्रिभुज का क्षेत्रफल है तथा Δ_2 इन तीन बिन्दुओं पर स्पर्श रेखाओं द्वारा बनाये गये त्रिभुज का क्षेत्रफल है तो $\frac{\Delta_1}{\Delta_2}$ होगा। **[JEE-2011]**

Ans. 2

Sol.



Δ_1 = area of Δ PAA' $= \frac{1}{2}.8.\frac{3}{2} = 6$

$\Delta_2 = \frac{1}{2}(\Delta_1)$

(Using property : Area of triangle formed by tangents is always half of original triangle)

$\Rightarrow \quad \frac{\Delta_1}{\Delta_2} = 2$

IMPORTANT HIGHLIGHTS OF ELLIPSE

54. Let d be the perpendicular distance from the centre of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ to the tangent drawn at a point P on the ellipse.If F_1 & F_2 are the two foci of the ellipse, then show that $(PF_1 - PF_2)^2 = 4a^2\left[1 - \frac{b^2}{d^2}\right]$.

माना दीर्घवृत $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ के किसी बिन्दु P पर बनायी गई स्पर्श रेखा की केन्द्र से दूरी d है। यदि F_1 एवं F_2 दीर्घवृत की नाभियाँ है,

r ksfl) d lfft , $(PF_1 - PF_2)^2 = 4a^2\left[1 - \frac{b^2}{d^2}\right]$.

Ans.

Sol. Equation of tangent at point P (acosθ, bsinθ)

on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$

foci $F_1 \equiv (ae, 0)$, $F_2 = (-ae, 0)$

and $d= \frac{1}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}}=\frac{ab}{\sqrt{a^2\sin^2\theta+b^2\cos^2\theta}}$

Now $4a^2\left(1-\frac{b^2}{d^2}\right)$

$=4a^2\left(1-\frac{b^2(a^2\sin^2\theta+b^2\cos^2\theta)}{a^2b^2}\right)$

$=4a^2\left(1-\sin^2\theta-\frac{b^2}{a^2}\cos^2\theta\right)$

$= 4\cos^2\theta\,(a^2-b^2)$

$= 4a^2e^2\cos^2\theta = (2ae\cos\theta)^2$

$= [(a-ae\cos\theta)-(a+ae\cos\theta)]^2$

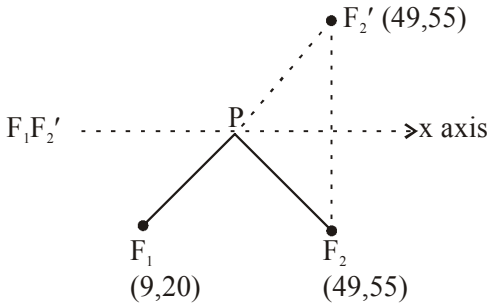
$= (PF_1 - PF_2)^2$

55. An ellipse has foci at $F_1(9, 20)$ and $F_2(49, 55)$ in the xy-plane and is tangent to the x-axis. Find the length of its major axis.

एक दीर्घवृत जिसकी नाभियाँ xy तल में $F_1(9, 20)$ तथा $F_2(49, 55)$ एवं दीर्घवृत x-अक्ष को स्पर्श करता है। इसके दीर्घाक्ष की लम्बाई ज्ञात कीजिए।

Ans. 85

Sol.



$PF_1 + PF_2 = 2a$

$\Rightarrow F_1PF_2 = 2a$

$PF_1 + PF_2 = 2a$

IMPORTANT HIGHLIGHTS OF HYPERBOLA

56. If C is the centre of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, S, S′ its foci and P a point on it. Prove that SP. S′P = $CP^2 - a^2 + b^2$.

यदि अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ का केन्द्र C , S एवं S′ इसकी नाभियां तथा P इस पर एक बिन्दु है, तो सिद्ध कीजिए SP. S′P = $CP^2 - a^2 + b^2$.

Ans.

Sol. $H:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ $\left.\begin{matrix} S=(a\ell,0) \\ S'=(-ae,0) \\ C(0,0) \end{matrix}\right] P(x_1,y_1)$

$S'P=\sqrt{(x_1+ae)^2+y_1^2}$ ...(1)

$SP=\sqrt{(x_1-ae)^2+y_1^2}$ ...(2)

$CP=\sqrt{x_1^2+y_1^2}$ ...(3)

$S'P\cdot SP=\sqrt{\left(x_1^2+y_1^2-a^2+b^2\right)^2}=x_1^2+y_1^2-a^2+b^2$

$CP^2 - a^2 + b^2$ H.P.

57. A conic C satisfies the differential equation, $(1 + y^2)\,dx - xy\,dy = 0$ and passes through the point (1, 0). An ellipse E which is confocal with C having its eccentricity equal to $\sqrt{2/3}$. (Attempt after differential equation)

(a) Find the length of the latus rectum of the conic C

(b) Find the equation of the ellipse E.

(c) Find the locus of the point of intersection of the perpendicular tangents to the ellipse E.

एक शांकव C अवकल समीकरण $(1 + y^2)\,dx - xy\,dy = 0$ को संतुष्ट करता है तथा बिन्दु (1, 0) से गुजरता है। E एक दीर्घवृत है जो C के संनाभीय है की उत्केन्द्रता $\sqrt{2/3}$ है।(Attempt after differential equation)

(a) शांकव C के नाभिलम्ब की लम्बाई ज्ञात कीजिए।

(b) दीर्घवृत E का समीकरण ज्ञात कीजिए।

(c) ₹ पर लम्बवत् स्पर्शियों का प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. (a) 2; (b) $\frac{x^2}{3}+\frac{y^2}{1}=1$; (c) $x^2+y^2=4$

Sol. Given diff^{n}. equation

$$\int\frac{dx}{x}=\int\frac{y\cdot dy}{1+y^2}$$

$$\ell nx=\frac{\ell n\left(1+y^2\right)}{2}+\ell nc$$

$$\Rightarrow \ell nx^2=\ell n\left(1+y^2\right)+\ell nc$$

$$\frac{x^2}{1+y^2}=C \qquad \Rightarrow \text{it passes } (1,0) \qquad \Rightarrow C=1$$

$x^2 = 1 + y^2$

$x^2 - y^2 = 1$ (H.B.)

$e=\sqrt{2}$, focii $=\left(1\sqrt{2},0\right)$

(a) $\ell_{LR}=\frac{2b^2}{a}=2$ units

(b) Given ellipse E confocal with C & $e=\sqrt{\frac{2}{3}}$

distance between 2 foci $2ae=2\sqrt{2}$

$\Rightarrow a=\sqrt{3}$

$$e^2=1-\frac{b^2}{a^2}\Rightarrow\frac{2}{3}==1-\frac{b^2}{3}$$

$\Rightarrow b^2 = 1$

Equation of ellipse $\frac{x^2}{3}+\frac{y^2}{1}=1$

(c) Locus of perpendicular is director circle of ellipse

$x^2 + y^2 = 4 = a^2 + b^2$

58. Let 'p' be the perpendicular distance from the centre C of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ to the tangent drawn at a point R on the hyperbola. If S & S′ are the two foci of the hyperbola, then show that $(RS + RS')^2 = 4a^2\left(1+\frac{b^2}{p^2}\right)$.

यदि अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ के केन्द्र C से बिन्दु R पर डाले गये स्पर्श रेखा की लम्बवत् दूरी 'p' हो तथा S एवं S′ दोनों नाभियाँ gksa rks fl) dhft, $(RS + RS')^2 = 4a^2\left(1+\frac{b^2}{p^2}\right)$.

Ans.

Sol. $H:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ point on HB R(a sec θ, b tan θ)

Equation of tangent

$$\frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1 \qquad ...(1)$$

$$\frac{1}{P^2}=\frac{\sec^2\theta}{a^2}+\frac{\tan^2\theta}{b^2}$$ [perpendicular distance from centre]

S(ae, 0), S′ (–ae, 0)

RS = ae sec θ – a ...(2)

RS′ = ae sec θ + a ...(3)

L.H.S.

$(P\Sigma + P\Sigma')^2 = 4a^2e^2\sec^2\theta$ ...(4)

$= 4\chi^2\sec^2\theta$

RHS

$$= 4a^2\left[1+\frac{b^2\sec^2\theta}{a^2}+\tan^2\theta\right]=4a^2\left[\sec^2\theta+\frac{b^2\sec^2\theta}{a^2}\right]$$

$$= 4a^2\times\frac{\sec^2\theta\left(a^2+b^2\right)}{a^2}$$

$= 4c^2\sec^2\theta$ H.P.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 59 to 61

Tangents are drawn from the point P(3, 4) to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B.

बिन्दु P(3, 4) से दीर्घवृत $\frac{x^2}{9} + \frac{y^2}{4} = 1$ पर स्पर्शियाँ डाली जाती हैं जो दीर्घवृत को बिन्दुओं A एवं B पर स्पर्श करती है।

59. The coordinates of A and B are : **[JEE-2010]**

(A) (3, 0) and (0, 2)

(B) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

(C) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and (0, 2)

(D) $(3, 0)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

A तथा B के निर्देशांक निम्न हैं : **[JEE-2010]**

(A) (3, 0) एवं (0, 2)

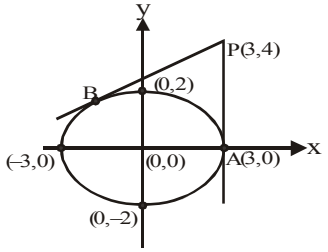
(B) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ एवं $\left(-\frac{9}{5}, \frac{8}{5}\right)$

(C) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ एवं (0, 2)

(D) $(3, 0)$ एवं $\left(-\frac{9}{5}, \frac{8}{5}\right)$

Ans. D

Sol. As shown in figure, one of the point of contact is (3,0)



Let equation of other tangent,

$y = mx + \sqrt{9m^2 + 4}$ as $c > 0$

It passes through (3,4), so

$4 = 3m + \sqrt{9m^2 + 4}$

$(4 - 3m)^2 = 9m^2 + 4$

Solving, $m = \frac{1}{2}$

As we know that point of contact for the tangent given by $\left(-\frac{a^2m}{\sqrt{a^2m^2 + b^2}}, \frac{b^2}{\sqrt{a^2m^2 + b^2}}\right)$

$\therefore$ Point of contact is $\left(-\frac{9}{5}, \frac{8}{5}\right)$

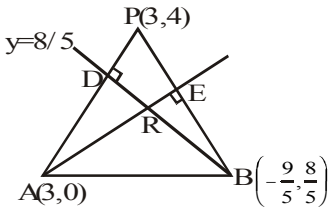
60. The orthocenter of the triangle PAB is :

त्रिभुज PAB का लम्बकेन्द्र निम्न है :

(A) $\left(5, \frac{8}{7}\right)$ (B) $\left(\frac{7}{5}, \frac{25}{8}\right)$ (C) $\left(\frac{11}{5}, \frac{8}{5}\right)$ (D) $\left(\frac{8}{25}, \frac{7}{5}\right)$

Ans. C

Sol. Equation of line BD : $y = \frac{8}{5}$



Equation of line AE : $2x + y = 6$

Now orthocentre R of ΔPAB will be intersection of line BD and line AE.

Solving for R, we get $R \equiv \left(\frac{11}{5}, \frac{8}{5}\right)$

61. The equation of the locus of the point whose distances from the point P and the line AB are equal, is :

उस बिन्दु के बिन्दुपथ का समीकरण जिसकी बिन्दु P तथा सरल रेखा AB से दूरियाँ बराबर है, निम्न है :

(A) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$ (B) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$

(C) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$ (D) $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

Ans. A

Sol. Equation of line AB is $x + 3y = 3$

Now let the point be (h,k)

According to question,

$$\left|\frac{h + 3k - 3}{\sqrt{1^2 + 3^2}}\right| = \sqrt{(h-3)^2 + (4-k)^2}$$

After solving, we get

$9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$

Paragraph for Question Nos. 62 to 63

The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B.

वृत्त $x^2 + y^2 - 8x = 0$ तथा अतिपरवलय $\frac{x^2}{9} - \frac{y^2}{4} = 1$, बिन्दुओं A तथा B पर प्रतिच्छेद करते हैं।

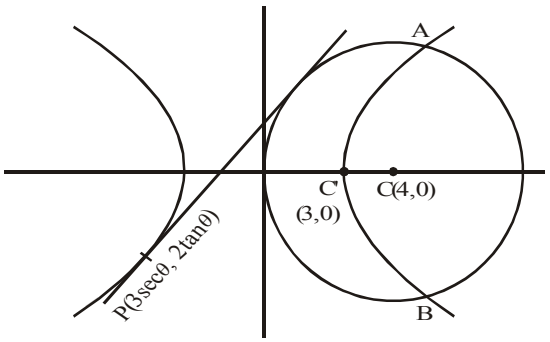
62. Equation of a common tangent with positive slope to the circle as well as to the hyperbola is : **[JEE-2010]**

धनात्मक प्रवणता वाली, वृत्त एवं अतिपरवलय की उभयनिष्ठ स्पर्श रेखा का समीकरण होगा : **[JEE-2010]**

(A) $2x - \sqrt{5}y - 20 = 0$ (B) $2x - \sqrt{5}y + 4 = 0$ (C) $3x - 4y + 8 = 0$ (D) $4x - 3y + 4 = 0$

Ans. B

Sol. Let the point on the hyperbola P(3secθ, 2tanθ)



Equation of tangent $\frac{x\sec\theta}{3} - \frac{y\tan\theta}{2} = 1$

$|p| = r$

$$\frac{\left|\frac{4}{3}\sec\theta - 1\right|}{\sqrt{\frac{\sec^2\theta}{9} + \frac{\tan^2\theta}{4}}} = 4$$

$$\Rightarrow \quad \frac{16}{9}\sec^2\theta + 1 - \frac{8}{3}\sec\theta = 16\left(\frac{4\sec^2\theta + 9\tan^2\theta}{4\times 9}\right)$$

$16\sec^2\theta + 9 - 24\sec\theta = 52\sec^2\theta - 36$

$\Rightarrow \quad 36\sec^2\theta + 24\sec\theta - 45 = 0$

$\Rightarrow \quad 12\sec^2\theta + 8\sec\theta - 15 = 0$

$\Rightarrow \quad 12\sec^2\theta + 18\sec\theta - 10\sec\theta - 15 = 0$

$\Rightarrow \quad (6\sec\theta - 5)(2\sec\theta + 3) = 0$

$\sec\theta = \frac{5}{6}$ (not possible), $\sec\theta = -\frac{3}{2}$

$$\tan\theta = \pm\sqrt{\frac{9}{4} - 1} = \pm\frac{\sqrt{5}}{2}$$

($\because$ slope is positive $\Rightarrow \tan\theta = -\frac{\sqrt{5}}{2}$)

Hence the required equation be $-\frac{3x}{2\times 3} + \frac{y\sqrt{5}}{2\times 2} = 1 \quad \Rightarrow \quad 2x - \sqrt{5}y + 4 = 0$

63. Equation of the circle with AB as its diameter is :

AB व्यास वाले वृत्त का समीकरण होगा :

(A) $x^2 + y^2 - 12x + 24 = 0$ (B) $x^2 + y^2 + 12x + 24 = 0$

(C) $x^2 + y^2 + 24x - 12 = 0$ (D) $x^2 + y^2 - 24x - 12 = 0$

Ans. A

Sol. Solving (a) & (b) for x, we get

$x = 6$

$y = \pm 2\sqrt{3}$

$(x-6)^2 + y^2 - 12 = 0$

$x^2 + y^2 - 12x + 24 = 0$

Option (A) is correct

Paragraph for Question nos. 64 to 66

Tangents are drawn to the parabola $y^2 = 4x$ from the point $P(6, 5)$ to touch the parabola at Q and R. C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R. Both the circles C_1 and C_2 pass through the focus of the parabola.

बिन्दु P(6, 5) से परवलय $y^2 = 4x$ पर डाली गयी स्पर्श रेखाऐं परवलय को Q और R पर स्पर्श करती है। यदि वृत्त C_1 परवलय को Q पर स्पर्श करता है तथा वृत्त C_2 परवलय को R पर स्पर्श करता है। दोनों वृत्त C_1 और C_2 परवलय के नाभि से होकर जाते है। तब।

64. Area of the ΔPQR equals :

ΔPQR का क्षेत्रफल बराबर :

(A) 1/2 (B) 1 (C) 2 (D) 1/4

Ans. A

Sol. Equation of common chord pass through centroid

Equation of tangents from (6,5) to $y^2 = 4x$

$$y = mx + \frac{1}{m} \leftarrow (6,5)$$

$$5 = 6m + \frac{1}{m}$$

$6m^2 - 5m + 1 = 0$

$6m^2 - 3m - 2m + 1 = 0$

$$m = \frac{1}{2}, \frac{1}{3}$$

$$\text{point of contact} = \left(\frac{a}{m^2}, \frac{2a}{m}\right)$$

Q = (9,6), R = (4,4)

Area of $\Delta PQR = = \frac{1}{2}\begin{vmatrix} 6 & 5 & 1 \\ 9 & 6 & 1 \\ 4 & 4 & 1 \end{vmatrix} = \frac{1}{2}$

Equation of circle C_2

$(x-4)^2 + (y-4)^2 + \lambda(x - 2y + 4) = 0$

it passes (1,0)

$(1-4)^2 + (-4)^2 + \lambda(1 + 4) = 0$

$\lambda = -5$

$(x-4)^2 + (y-4)^2 - 5x + 10y - 20 = 0$

$x^2 - 8x + 16 + y^2 - 8y + 16 - 5x + 10y - 20 = 0$

$x^2 + y^2 - 13x + 2y + 12 = 0$

$r = \sqrt{\frac{169}{4} + 1 - 12} = \sqrt{\frac{125}{4}} = \frac{5\sqrt{5}}{2}$

Equation of Circle C_1

$(x-9)^2 + (y-6)^2 + \lambda\,(x - 3y + 9) = 0$

it passes through (1,0)

$8^2 + 6^2 + \lambda\,(10) = 0 \Rightarrow \lambda = -10$

$x^2 - 18x + 81 + y^2 - 12y + 36 - 10x + 30y - 90 = 0$

$x^2 + y^2 - 28x + 18y + 27 = 0$

$r = \sqrt{196 + 81 - 27}$

$r = \sqrt{250}$

65. Radius of the circle C_2 is :

वृत्त C_2 की त्रिज्या है :

(A) $5\sqrt{5}$ (B) $5\sqrt{10}$ (C) $10\sqrt{2}$ (D) $\sqrt{210}$

Ans. B

Sol.

66. The common chord of the circles C_1 and C_2 passes through the :

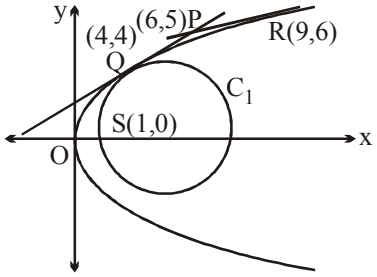
(A) incentre of the ΔPQR.
(B) circumcentre of the ΔPQR.
(C) centroid of the ΔPQR.
(D) othocentre of the ΔPQR.

C_1 और C_2 की उभयनिष्ठ जीवा होकर जाती है :

(A) ΔPQR के अन्तः केन्द्र से
(B) ΔPQR के परिकेन्द्र से
(C) ΔPQR के केन्द्रक से
(D) ΔPQR के लम्ब केन्द्र से

Ans. C

Sol. Equation of tangent of slope m to $y^2 = 4x$ is



$$y = mx + \frac{1}{m} \quad ....(1)$$

(i) As (1) passes through P(6, 5), so

$$5 = 6m + \frac{1}{m}$$

$$\Rightarrow \quad 6m^2 - 5m + 1 = 0 \quad \Rightarrow \quad m = \frac{1}{2} \text{ or } m = \frac{1}{3}$$

Points of contact are $\left(\frac{1}{m_1^2}, \frac{2}{m_1}\right)$ and $\left(\frac{1}{m_2^2}, \frac{2}{m_2}\right)$

Hence P (4, 4) and Q (9, 6)

$$\text{Area of } \Delta PQR = \frac{1}{2}\begin{vmatrix} 6 & 5 & 1 \\ 4 & 4 & 1 \\ 9 & 6 & 1 \end{vmatrix} = \frac{1}{2} \Rightarrow (A)$$

(ii) $y = \frac{1}{2}x + 2 \Rightarrow x - 2y + 4 = 0 \quad ....(2)$

and $y = \frac{1}{3}x + 3 \Rightarrow x - 3y + 9 = 0$

Now equation of circle C_2 touching $x - 3y + 9 = 0$ at (9, 6), is

$$(x - 9)^2 + (y - 6)^2 + \lambda(x - 3y + 9) = 0$$

As above circle passes through (1, 0), so

$$64 + 36 + 10\lambda = 0 \quad \Rightarrow \quad 1 = -10$$

Circle C_2 is $x^2 + y^2 - 28x + 18y + 27 = 0 \quad .....(3)$

Radius of C_2 is

$$r_2^2 = 196 + 81 - 27 = 277 - 27 = 250 \Rightarrow \quad r_2 = 5\sqrt{10} \quad \Rightarrow (B)$$

(iii) Equation of C_1

$$(x - 4)^2 + (y - 4)^2 + \lambda(x - 2y + 4) = 0$$

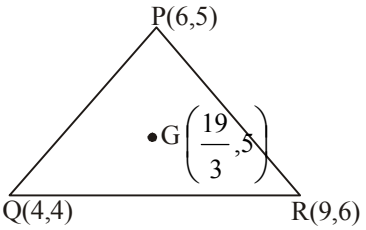
As above circle passes through (1, 0)

$9 + 16 + \lambda(5) = 0 \quad \Rightarrow \quad l = -5$

Now C_1 is $x^2 + y^2 - 13x + 2y + 12 = 0$ ....(4)

$\therefore$ Common chord of (3) and (4) is

$15x - 16y - 15 = 0$ ....(5)



Also centroid (G) of ΔPQR is $\left(\frac{19}{3}, 5\right)$

Clearly $\left(\frac{19}{3}, 5\right)$ satisfies equation (5)

Hence (C)

Paragraph for Question nos. 67 to 69

Let ABCD be a square of side length 2 units. C_2 is the circle through vertices A, B, C, D and C_1 is the circle touching all the sides of the square ABCD. L is a line through A.

माना ABCD, 2 इकाई लम्बाई की भुजा वाला एक वर्ग है। C_2 शीर्षो A, B, C एवं D से गुजरने वाला एक वृत है तथा C_1 एक ऐसा वृत है जो वर्ग ABCD की प्रत्येक भुजा को स्पर्श करता है। L, शीर्ष A से गुजरने वाली एक रेखा है।

67. If P is a point on C_1 and Q in another point on C_2, then $\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$ is equal to :

यदि बिन्दु P , वृत्त C_1 पर एवं बिन्दु Q, वृत्त C_2 पर स्थित हो, तो $\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$ का मान है :

(A) 0.75 (B) 1.25 (C) 1 (D) 0.5

Ans. A

Sol. Without loss of generality we can assume the square ABCD with its vertices A(1, 1), B(–1, 1), C(–1, –1), D(1, –1)

P to be the point (0, 1) and Q as ($\sqrt{2}$, 0)

Then, $\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$

$= \frac{1 + 1 + 5 + 5}{2[(\sqrt{2} - 1)^2 + 1] + 2((\sqrt{2} + 1)^2 + 1)} = \frac{12}{16} = 0.75$

68. A circle touches the line L and the circle C_1 externally such that both the circles are on the same side of the line, then the locus of centre of the circle is :

(A) ellipse (B) hyperbola (C) parabola (D) parts of straight line

एक वृत्त रेखा L एवं वृत्त C_1 को बाह्य स्पर्श इस प्रकार करता है कि दोनों वृत्त सरल रेखा के एक ही ओर है, तो ऐसे वृत्त के केन्द्र का fcUnqPkgS

(A) दीर्घवृत्त (B) अतिपरवलय (C) परवलय (D) सरल रेखा का भाग

Ans. C

Sol. Let C' be the said circle touching C_1 and L, so that C_1 and C' are on the same side of L. Let us draw a line T parallel to L at a distance equal to the radius of circle C_1, on opposite side of L.

Then for N, centre of circle C', MN = NO

$\Rightarrow$ N is equidistant from a line and a point

$\Rightarrow$ locus of N is a parabola.

69. A line M through A is drawn parallel to BD. Point S moves such that its distances from the line BD and the vertex A are equal. If locus of S cuts M at T_2 and T_3 and AC at T_1, then area of $\Delta T_1T_2T_3$ is :

(A) 1/2 sq. units (B) 2/3 sq. units (C) 1 sq. unit (D) 2 sq. units

एक रेखा M शीर्ष A से गुजरते हुए इस प्रकार खीचीं जाती है कि यह BD के समान्तर है। एक बिन्दु S इस प्रकार गति करता है कि इसकी रेखा BD से ओर शीर्ष A से दूरियाँ बराबर है। यदि बिन्दु S का बिन्दु पथ M को T_2 और T_3 पर तथा AC को T_1 पर काटता है तो $\Delta T_1T_2T_3$ का क्षेत्रफल होगा :

(A) 1/2 वर्ग इकाई (B) 2/3 वर्ग इकाई (C) 1 वर्ग इकाई (D) 2 वर्ग इकाई

Ans. C

Sol. Since S is equidistant from A and line BD, it traces a parabola. Clearly AC is the axis, A(1, 1) is the focus and $T_1\left(\frac{1}{2},\frac{1}{2}\right)$ is the vertex of parabola, $AT_1 = \frac{1}{\sqrt{2}}$.

T_2T_3 = latus rectum of parabola $= 4 \times \frac{1}{\sqrt{2}} = 2\sqrt{2}$

$\therefore$ Area $(\Delta T_1 T_2 T_3) = \frac{1}{2} \times 2\sqrt{2} = \frac{1}{2} = 1$ sq. units.

Paragraph for Question nos. 70 to 72

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

माना वृत्त $x^2 + y^2 = 9$ एवं परवलय $y^2 = 8x$ प्रथम चतुर्थांश एवं चतुर्थ चतुर्थांश में क्रमशः बिन्दु P एवं Q पर प्रतिच्छेद करते है। वृत्त के लिये P व Q पर स्पर्श रेखाएँ x-अक्ष को बिन्दु R पर एवं परवलय पर स्थित P व Q पर स्पर्श रेखाएँ x-को बिन्दु S पर प्रतिच्छेद करती है।

70. The ratio of the areas of the triangles PQS and PQR is :

त्रिभुज PQS एवं PQR के क्षेत्रफलों का अनुपात है :

(A) $1 : \sqrt{2}$ (B) 1 : 2 (C) 1 : 4 (D) 1 : 8

Ans. C

Sol. $$\frac{\text{Ar}\Delta PQS}{\text{Ar}\Delta PQR} = \frac{\frac{1}{2}QP \times ST}{\frac{1}{2}PQ \times TR} = \frac{ST}{TR} = \frac{2}{8} = \frac{1}{4}$$

71. The radius of the circumcircle of the triangle PRS is :

त्रिभुज PRS के परिवृत्त की त्रिज्या है :

(A) 5 (B) $3\sqrt{3}$ (C) $3\sqrt{2}$ (D) $2\sqrt{3}$

Ans. B

Sol. For ΔPRS,

$$\text{Ar}(\Delta PRS) = \Delta = \frac{1}{2} \times SR \times PT = \frac{1}{2} \times 10 \times 2\sqrt{2}$$

$\therefore$ $D = 10\sqrt{2}$, $a = PS = 2\sqrt{3}$,

$b = PR = 6\sqrt{2}$, $c = SR = 10$

$\therefore$ radius of circumference

$$= R = \frac{abc}{4\Delta} = \frac{2\sqrt{3} \times 6\sqrt{2} \times 10}{4 \times 10\sqrt{2}} 3\sqrt{3}$$

72. The radius of the incircle of the triangle PQR is :

त्रिभुज PQR के अन्तःवृत्त की त्रिज्या है :

(A) 4 (B) 3 (C) 8/3 (D) 2

Ans. D

Sol. Radius of incircle

$$= \frac{\text{area of } \Delta PQR}{\text{semiperimeter of} \Delta PQR} = \frac{\Delta}{s}$$

We have $a = PR = 6\sqrt{2}$, $b = QP = PR = 6\sqrt{2}$

$c = PQ = 4\sqrt{2}$

and $\Delta = \frac{1}{2} \times PQ \times TR = 16\sqrt{2}$

$\therefore$ $s = \frac{6\sqrt{2} + 6\sqrt{2} + 4\sqrt{2}}{2} = 8\sqrt{2}$

$\therefore \quad r = \frac{16\sqrt{2}}{8\sqrt{2}} = 2$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

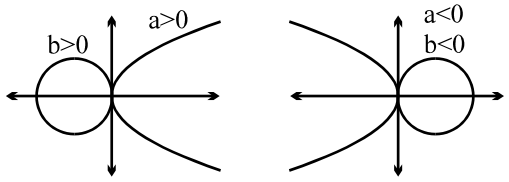
73. Let $y^2 = 4ax$ be a parabola and $x^2 + y^2 + 2bx = 0$ be a circle. If parabola and circle touch each other externally then:

यदि परवलय $y^2 = 4ax$ और वृत्त $x^2 + y^2 + 2bx = 0$ एक दूसरे को ब्राह्य स्पर्श करते हैं तब :

(A) $a > 0, b > 0$ (B) $a > 0, b < 0$ (C) $a < 0, b > 0$ (D) $a < 0, b < 0$

Ans. AD

Sol. For externally touching a & b must have the same sign



74. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter :

निम्न में से कौनसा प्राचलिक रूप अतिपरवलय को निरूपित करते हैं जहां 't' प्राचल है :

(A) $x = \frac{a}{2}\left(t + \frac{1}{t}\right)$ & $y = \frac{b}{2}\left(t - \frac{1}{t}\right)$ (B) $\frac{tx}{a} - \frac{y}{b} + t = 0$ & $\frac{x}{a} + \frac{ty}{b} - 1 = 0$

(C) $x = e^t + e^{-t}$ & $y = e^t - e^{-t}$ (D) $x^2 - 6 = 2\cos t$ & $y^2 + 2 = 4\cos^2\frac{t}{2}$

Ans. ACD

Sol. Check option using elimination $\rightarrow$ ACD

75. Let p and q be non-zero real numbers. Then the equation $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$ represents :

(A) two straight lines and a circle, when $r = 0$ and p, q are of the opposite sign.

(B) two circles, when $p = q$ and r is of sign opposite to that of p.

(C) a hyperbola and a circle, when p and q are of opposite sign and $r \neq 0$.

(D) a circle and an ellipse, when p and q are unequal but of same sign and r is of sign opposite to that of p.

माना p तथा q अशून्य वास्तविक संख्याऐं हो तो समीकरण $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$ निरूपित करती है :

(A) दो सरल रेखाओं तथा एक वृत्त जब $r = 0$ तथा p, q विपरीत चिन्ह के है

(B) दो वृत्त जब $p = q$ तथा r तथा p विपरीत चिन्ह के हैं

(C) एक अतिपरवलय तथा वृत्त, p तथा q विपरीत चिन्ह के हैं तथा $r \neq 0$

(D) एक वृत्त तथा एक दीर्घवृत्त , जब p तथा q समान चिन्ह की असमान संख्याऐं हैं तथा r तथा p विपरीत चिन्ह के हैं

Ans. ABCD

Sol. $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$

$\Rightarrow \quad 4x^2 + 4y^2 - 8x - 4 = 0 \Rightarrow (x-1)^2 + y^2 = 1$

or

$px^2 + qy^2 + r = 0$ will represents

(i) two straight lines if $r = 0$ and p, q are of opposite sign.

(ii) a circle if $p = q$ and r is of opposite sign that of p.

(iii) a hyperbola if p and q are of opposite sign & $r \neq 0$.

(iv) an ellipse if p and q are unequal but of same sign and r is of sign opposite to that of p.

76. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$, then:

यदि वृत्त $x^2 + y^2 = a^2$ अतिपरवलय $xy = c^2$ को चार बिन्दुओं $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$ पर काटता है तो :

(A) $x_1 + x_2 + x_3 + x_4 = 0$ (B) $y_1 + y_2 + y_3 + y_4 = 0$

(C) $x_1 x_2 x_3 x_4 = c^4$ (D) $y_1 y_2 y_3 y_4 = c^4$

Ans. ABCD

Sol. solving $xy = c^2$ and $x^2 + y^2 = a^2$

$$x^2 + \frac{c^4}{x^2} = a^2$$

$$x^4 - ax^3 - a^2x^2 + ax + c^4 = 0$$

$\Rightarrow \quad \sum x_i = 0 \;\; ; \;\; \sum y_i = 0$

$x_1 x_2 x_3 x_4 = c^4 \quad \Rightarrow \quad y_1 y_2 y_3 y_4 = c^4$

77. The tangent to the hyperbola, $x^2 - 3y^2 = 3$ at the point $\left(\sqrt{3}, 0\right)$ when associated with two asymptotes constitutes:

(A) isosceles triangle which is not equilateral (B) an equilateral triangle

(C) a triangles whose area is $\sqrt{3}$ sq. units (D) a right isosceles triangle

अतिपरवलय $x^2 - 3y^2 = 3$ के किसी बिन्दु $\left(\sqrt{3}, 0\right)$ पर खींची गयी स्पर्श रेखा दो अनन्त स्पर्शीयों के साथ :

(A) समद्विबाहु त्रिभुज बनाती है (B) समबाहु त्रिभुज बनाती है

(C) त्रिभुज का क्षेत्रफल $\sqrt{3}$ वर्ग इकाई है (D) समकोण समद्विबाहु त्रिभुज बनाती है

Ans.

Sol. area of the $\Delta = ab$ sq units ; H : $x^2/3 - y^2/1 = 1$

78. The tangent PT and the normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N, respectively. The locus of the centroid of the triangle PTN is a parabola whose : **[JEE-2009]**

(A) vertex is $\left(\frac{2a}{3}, 0\right)$ (B) directrix is $x = 0$ (C) latus rectum is $\frac{2a}{3}$ (D) focus is (a, 0)

परवलय $y^2 = 4ax$ के बिन्दु P पर स्पर्श रेखा PT व अभिलम्ब PN इसकी अक्ष से क्रमशः बिन्दु T व N पर मिलती है। त्रिभुज PTN के केन्द्रक का बिन्दुपथ परवलय है जिसका (जिसकी) **[JEE-2009]**

(A) शीर्ष $\left(\frac{2a}{3}, 0\right)$ है (B) नियता $x = 0$ है (C) नाभिलम्ब $\frac{2a}{3}$ है (D) नाभि (a, 0) है

Ans. AD

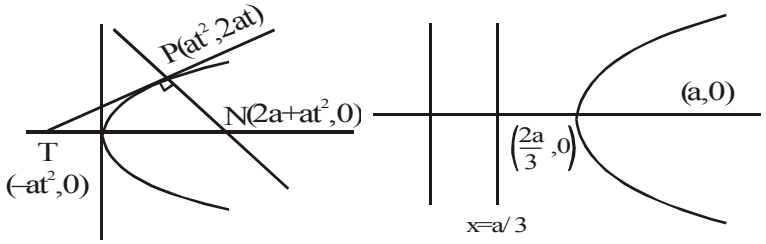
Sol. $3h = 2a + at^2$

$3k = 2at$

$3h = 2a + \frac{a \cdot 9k^2}{4a^2}$

$y^2 = \frac{4a}{9}(3x - 2a)$

$y^2 = \frac{4a}{3}\left(x - \frac{2a}{3}\right)$



79. Let A and B be two distinct point on the parabola $y^2 = 4x$. If the axis of the parabola touches a circle of radius r having AB as its diameter, then the slope of the line joining A and B can be : **[JEE-2010]**

माना कि परवलय $y^2 = 4x$ पर A तथा B दो भिन्न बिन्दु है। यदि परवलय का अक्ष, r त्रिज्या वाले उस वृत्त को स्पर्श करता है जिसका व्यास AB है तो A एवं B से होकर जाने वाली सरल रेखा की प्रवणता (slope) निम्न हो सकती है। **[JEE-2010]**

(A) $\frac{-1}{r}$ (B) $\frac{1}{r}$ (C) $\frac{2}{r}$ (D) $\frac{-2}{r}$

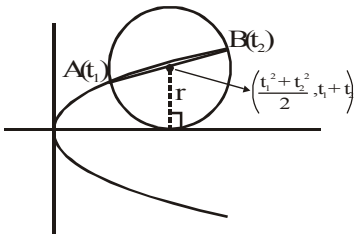
Ans. CD

Sol. $t_1 + t_2 = r$

$\frac{2}{r} = \frac{2}{t_1 + t_2}$

similarly $-\frac{2}{r}$ is

also possible



80. In a triangle ABC with fixed base BC, the vertex A moves such that $\cos B + \cos C = 4 \sin^2 \frac{A}{2}$. If a, b and c denote the lengths of the sides of the triangle opposite to the angles A, Band C, respectively, then : **[JEE-2009]**

(A) $b + c = 4a$ (B) $b + c = 2a$

(C) locus of point A is an ellipse. (D) locus of point A is a pair of straight lines.

नियत आधार BC वाले त्रिभुज ABC का शीर्ष बिन्दु A इस प्रकार गतिमान है कि $\cos B + \cos C = 4 \sin^2 \frac{A}{2}$. यदि a, b एवं c क्रमशः

f=kh1 d sd ksk.kksa A, B एवं C की सम्मुख भुजाओं की लम्बाईयों को प्रदर्शित करते है, तब : **[JEE-2009]**

(A) $b + c = 4a$ (B) $b + c = 2a$

(C) बिन्दु A का बिन्दुपथ एक दीर्घ वृत्त है (D) बिन्दु A का बिन्दु पथ एक सरल रेखा युग्म है

Ans. BC

Sol. $\cos B + \cos C = 4 \sin^2 \frac{A}{2}$

$$\Rightarrow 2\cos\left(\frac{B+C}{2}\right).\cos\left(\frac{B-C}{2}\right) = 4\sin^2\frac{A}{2}$$

$$\Rightarrow \cos\left(\frac{B-C}{2}\right) = 2\sin\left(\frac{A}{2}\right)$$

$$\Rightarrow 2\cos\frac{A}{2}\cos\frac{B-C}{2} = 4\sin\frac{A}{2}\cos\frac{A}{2}$$

$$\Rightarrow \sin B + \sin C = 2\sin A$$

$$\Rightarrow b + c = 2a$$

81. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then : **[JEE-2009]**

(A) Equation of ellipse is $x^2 + 2y^2 = 2$ (B) The foci of ellipse are $(\pm 1, 0)$

(C) Equation of ellipse is $x^2 + 2y^2 = 4$ (D) The foci of ellipse are $(\pm \sqrt{2}, 0)$

अतिपरवलय $2x^2 - 2y^2 = 1$ को एक दीर्घवृत्त लम्बवत् काटता है। दीर्घवृत्त की उत्केन्द्रता अतिपरवलय की उत्केन्द्रता की व्युत्क्रम (reciprocal) है। यदि दीर्घ वृत्त के अक्ष निर्देशाक्षों के सम्पाती हो, तब : **[JEE-2009]**

(A) दीर्घवृत्त का समीकरण $x^2 + 2y^2 = 2$ (B) दीर्घवृत्त की नाभियाँ $(\pm 1, 0)$

(C) दीर्घवृत्त का समीकरण $x^2 + 2y^2 = 4$ (D) दीर्घवृत्त की नाभियाँ $(\pm \sqrt{2}, 0)$

Ans. AB

Sol. $x^2 - y^2 = \frac{1}{2}$ ...(i) $\rightarrow$ its e = $\sqrt{2}$

e of ellipse is $\frac{1}{\sqrt{2}}$

$\frac{x^2}{2} + \frac{y^2}{1} = b^2$...(ii)

add (i) & (ii) $\frac{3x^2}{2} = \frac{1}{2} + b^2$

$3x^2 = 1 + 2b^2$

$y^2 = \frac{1}{3} + \frac{2h^2}{3} - \frac{1}{6}$

$y^2 = \frac{1}{6}(4b^2 - 1)$

$m_1 . m_2 = -1 \Rightarrow \frac{1+2b^2}{3} = \frac{2(4b^2-1)}{6}$

$b^2 = 1 \Rightarrow x^2 + 2y^2 = 2$

82. Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be reciprocal to that of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then : **[JEE-2011]**

(A) the equation of the hyperbola is $\frac{x^2}{3} - \frac{y^2}{2} = 1$ (B) a focus of the hyperbola is (2, 0)

(C) the eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$ (D) the equation of the hyperbola is $x^2 - 3y^2 = 3$

माना अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ के उत्केन्द्रियता दीर्घवृत्त $x^2 + 4y^2 = 4$ के उत्केन्द्रियता के व्युत्क्रम हैं यदि अतिपरवलय दीर्घवृत्त की

, d uklHkl sxtq jr kgSrks: **[JEE-2011]**

(A) अतिपरवलय का समीकरण $\frac{x^2}{3} - \frac{y^2}{2} = 1$ है (B) अतिपरवलय की एक नाभि (2, 0) है

(C) अतिपरवलय की उत्केन्द्रियता $\sqrt{\frac{5}{3}}$ है (D) अतिपरवलय का समीकरण $x^2 - 3y^2 = 3$ है

Ans. BD

Sol. Given hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

ellipse is $\frac{x^2}{2^2}+\frac{y^2}{1}=1$

eccentricity of ellipse $=\sqrt{1-\frac{1}{4}}=\sqrt{\frac{3}{4}}=\frac{\sqrt{3}}{2}$

eccentricity of hyperbola $=\sqrt{1+\frac{b^2}{a^2}}=\sqrt{\frac{4}{3}}$

$\Rightarrow \quad \frac{b^2}{a^2}=\frac{1}{3} \quad \Rightarrow \quad 3b^2 = a^2 \qquad .....(1)$

also hyperbola passes through foci of ellipse $(\pm\sqrt{3}, 0)$

$\frac{3}{a^2}=1 \quad \Rightarrow \quad a^2 = 3 \qquad .....(2)$

from (1) & (2)

$b^2 = 1$

equation of hyperbola is $\frac{x^2}{3}-\frac{y^2}{1}=1$

$\Rightarrow \quad x^2 - 3y^2 = 3$

eccentricity of hyperbola $=\sqrt{1+\frac{1}{3}}=\sqrt{\frac{4}{3}}$

focus of hyperbola $=\left(\pm\sqrt{3}.\frac{2}{\sqrt{3}}, 0\right)\equiv(\pm2, 0)$

83. The locus of the mid point of the focal radii of a variable point moving on the parabola, $y^2 = 8x$ is a parabola whose:

(A) Latus rectum is half the latus rectum of the original parabola

(B) Vertex is (1, 0)

(C) Directrix is y-axis

(D) Focus has the co-ordinates (2, 0)

परवलय $y^2 = 8x$ पर स्थित प्रत्येक बिन्दु के नाभिक त्रिज्या (focal radii) के मध्य बिन्दु का बिन्दुपथ एक परवलय होगा जिसका :

(A) नाभिलम्ब की लम्बाई, मूल परवलय की नाभिलम्ब की आधी होगी

(B) शीर्ष (1, 0) होगा

(C) नियता y-अक्ष होगी

(D) नाभि का निर्देशांक (2, 0) होगा

Ans. ABCD

Sol. P : $y^2 = 8x \quad \Rightarrow 4a = 8 \quad \Rightarrow a = 2$

Let any point on parabola be $(2t^2, 4t)$

focus is (2,0)

mid point (h,k) = $(1 + t^2, 2t)$

$h = 1 + t^2$, $\frac{k}{2} = t$

$h = 1 + \frac{k^2}{4} \Rightarrow y^2 = 4(x - 1)$

vertex is (1,0), $\ell_{LR} = 4$, directrix $x = 0$

focus (2,0), ℓ_{LR} of original = 8.

$\ell_{LR} = 4$

84. A variable circle is described to pass through the point (1, 0) and tangent to the curve $y = \tan(\tan^{-1} x)$. The locus of the centre of the circle is a parabola whose :

(A) length of the latus rectum is $2\sqrt{2}$ (B) axis of symmetry has the equation $x + y = 1$

(C) vertex has the co-ordinates (3/4, 1/4) (D) directrix is $x - y = 0$

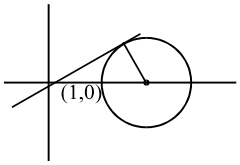
एक चर वृत्त (1, 0) से गुजरती है तथा वक्र $y = \tan(\tan^{-1} x)$ को स्पर्श करती है, तो उसके केन्द्र का बिन्दुपथ एक परवलय होगा जिसके:

(A) नाभिलम्ब की लम्बाई $2\sqrt{2}$ होगी (B) अक्ष का समीकरण $x + y = 1$

(C) शीर्ष का निर्देशांक (3/4, 1/4) (D) नियता का समीकरण $x - y = 0$

Ans. BCD

Sol.



85. Consider a circle with its centre lying on the focus of the parabola, $y^2 = 2px$ such that it touches the directrix of the parabola. Then a point of intersection of the circle & the parabola is :

माना एक वृत्त है जिसका केन्द्र परवलय $y^2 = 2px$ का नाभि है। इस प्रकार की वृत्त परवलय की नियता को स्पर्श करता है। तब परवलय vkSj oRr d ki fr PNsn fcUnq gS

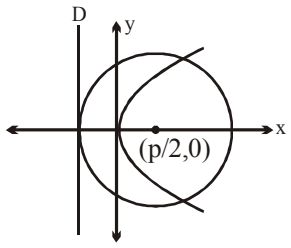
(A) $\left(\frac{p}{2}, p\right)$ (B) $\left(\frac{p}{2}, -p\right)$ (C) $\left(-\frac{p}{2}, p\right)$ (D) $\left(-\frac{p}{2}, -p\right)$

Ans. AB

Sol. Equation of circle will be $\left(x - \frac{p}{2}\right)^2 + y^2 = p^2$

which intersects $y^2 = 2px$

$\therefore \quad \left(x-\frac{p}{2}\right)^2+2px=p^2$



$x^2+px-\frac{3p^2}{4}=0$

$\left(x+\frac{3p}{4}\right)\left(x-\frac{p}{2}\right)=0$

$4x^2+4px-3p^2=0$

$4x^2+6px-2px-3p^2=0$

$2x(2x+3p)-p(2x+3p)$

$(2x+3p)(2x-p)=0$

$\therefore \quad x=\frac{p}{2}$ only

Hence $y^2=p^2 \Rightarrow y=\pm p \Rightarrow$ A, B Ans.

86. If from the vertex of a parabola $y^2=4x$ a pair of chords be drawn at right angles to one another and with these chords as adjacent sides a rectangle be made, then the locus of the further end of the rectangle is :

(A) an equal parabola
(B) a parabola with focus at (9, 0)
(C) a parabola with directrix as $x-7=0$
(D) a parabola having tangent at its vertex $x=8$

परवलय $y^2=4x$ की दो परस्पर लम्बवत् जीवा मूल बिन्दु से खीचीं जाती है जो कि एक आयत की दो संलग्न भुजा भी हो तो मूल बिन्दु d sfoi jhr vk;r d s'kh"kZd kfcUnqiFkgkrk:

(A) एक समान परवलय
(B) पर परवलय जिसकी नाभि (9, 0) है
(C) एक परवलय जिसकी नियता $x-7=0$
(D) एक परवलय जिसके शीर्ष पर स्पर्श रेखा का समीकरण $x=8$ है

Ans. ABCD

Sol. locus is $y^2=4a(x-8a)$

87. The focus of the parabola is (1, 1) and the tangent at the vertex has the equation $x+y=1$. Then :

(A) equation of the parabola is $(x-y)^2=2(x+y-1)$
(B) equation of the parabola is $(x-y)^2=4(x+y-1)$
(C) the co-ordinates of the vertex are $\left(\frac{1}{2},\frac{1}{2}\right)$
(D) length of the latus rectum is $2\sqrt{2}$

परवलय के नाभि का निर्देशांक (1, 1) है तथा इसके शीर्ष पर खीचीं गयी स्पर्श रेखा का समीकरण x + y = 1 है, तो :

(A) परवलय का समीकरण $(x-y)^2 = 2(x+y-1)$ है (B) परवलय का समीकरण $(x-y)^2 = 4(x+y-1)$ है

(C) शीर्ष का निर्देशांक $\left(\frac{1}{2}, \frac{1}{2}\right)$ है (D) नाभिलम्ब की लम्बाई $2\sqrt{2}$ है

Ans. BCD

Sol. Equation of axis $\rightarrow x - y + \lambda = 0$

$\Rightarrow x - y = 0$

Equation of parabola

$$(x-1)^2 + (y-1)^2 = \left[\frac{x+y}{\sqrt{2}}\right]^2$$

$\Rightarrow (x-y)^2 = 4(x+y-1)$

88. Consider the parabola whose equation is $y = x^2 - 4x$ and the line $y = 2x - b$. Then which of the following is/are correct?

(A) For $b = 9$ the line is a tangent to the parabola

(B) For $b = 7$ the line cuts the parabola in A and B such that the $\angle AOB$ is a right angle when 'O' is the origin

(C) For some $b \in R$ the line cuts the parabola in C and D such that x-axis bisects the $\angle COD$

(D) For $b > 9$ the line passes outside the parabola

परवलय $y = x^2 - 4x$ तथा रेखा $y = 2x - b$ के लिये सही विकल्पों का चुनो :

(A) इ ञ 9 के लिये रेखा, परवलय की स्पर्श रेखा है

(B) इ ञ 7 के लिये रेखा परवलय को A तथा B पर काटेगी जहाँ $\angle AOB = 90^\circ$ यहाँ 'O' मूल बिन्दु हो

(C) किसी $b \in R$ के लिये रेखा, परवलय को C तथा D पर काटेगी जहाँ x-अक्ष $\angle COD$ की अर्द्धक होगी

(D) इ झ 9 के लिये रेखा परवलय को कभी नही काटेगी

Ans. ABD

Sol. (A) For $b = 9$ solving line and parabola

$2x - 9 = x^2 - 4x \quad \Rightarrow \quad x^2 - 6x + 9 = 0 \quad \Rightarrow \quad (x-3)^2 = 0$

$\therefore$ line is tangent to parabola $\Rightarrow$ (A) is correct

(B) For $b = 7$, line is $y = 2x - 7$

Homogenising parabola with line

$$x^2 - 4x\left(\frac{2x-y}{7}\right) - y\left(\frac{2x-y}{7}\right) = 0$$

coefficient of x^2 + coefficient of $y^2 = 0 \Rightarrow 1 - \frac{8}{7} + \frac{1}{7} = 0$

$\therefore$ ÐAOB is 90° $\Rightarrow$ (B) is correct

(C) Homogenising the parabola with line, we get

$$x^2 - 4x\left(\frac{2x-y}{b}\right) - y\left(\frac{2x-y}{b}\right) = 0$$

for x-axis to bisect $\angle COD$

coefficient of xy should be zero

i.e. $\frac{4}{b} - \frac{2}{b} = 0$

which is not possible for any $b \in R$. $\Rightarrow$ (C) is false

(D) For $b > 9$

Solving line and parabola

$x^2 - 6x + b = 0$

$D = 36 - 4b$

If $b > 9$

then $D < 0$

$\therefore$ line passes outside the parabola $\Rightarrow$ (D) is correct.

89. PQ is a double ordinate of the parabola $y^2 = 4ax$. If the normal at P intersect the line passing through Q and parallel to axis of x at G, then locus of G is a parabola with :

(A) length of latus rectum equal to 4a. (B) vertex at (4a, 0).

(C) directrix as the line $x - 3a = 0$ (D) focus as (5a, 0)

परवलय $y^2 = 4ax$ का PQ एक द्विक् कोटि है यदि P से खीचें गये अभिलम्ब को बिन्दु Q से खीचीं गयी तथा x–अक्ष के समान्तर रेखा यदि G पर काटती हो, तो G का बिन्दुपथ एक परवलय होगा, जिसका :

(A) नाभिलम्ब की लम्बाई 4a है (B) शीर्ष (4a, 0) है

(C) नियता $x - 3a = 0$ (D) नाभि (5a, 0) है

Ans. ABCD

Sol. $N : y + tx = 2at + at^3$

$L : y = -2at$

solving together

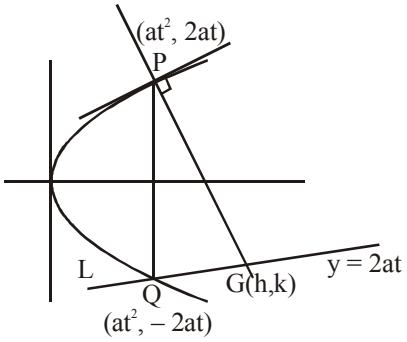
$-2at + tx = 2at + at^3$

$x = at^2 + 4a$

hence, $h = at^2 + 4a$ and $k = -2at$

$\frac{k^2}{4a^2} = \frac{h-4a}{a} \Rightarrow y^2 = 4a(x-4a)$.

Now verify all the alternatives. Ans.



90. If θ is eliminated from the equations $a\sec\theta - x\tan\theta = y$ and $b\sec\theta + y\tan\theta = x$ (a and b are constant) then the eliminant denotes the equation of :

(A) the director circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(B) auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(C) Director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(D) Director circle of the circle $x^2 + y^2 = \frac{a^2 + b^2}{2}$

यदि समीकरण $a\sec\theta - x\tan\theta = y$ तथा $b\sec\theta + y\tan\theta = x$ में θ को विलुप्त किया जाता है। (a तथा b अचर है) तो प्राप्त l ehdj.kfuEu dksfu: fir djrkgS

(A) अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ का नियामक वृत्त है (B) दीर्घवृत्त $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ का सहायक वृत्त है

(C) दीर्घवृत्त $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ का नियामक वृत्त है (D) वृत्त $x^2 + y^2 = \frac{a^2 + b^2}{2}$ का नियामक वृत्त है

Ans. CD

Sol. $a\sec\theta = y + x\tan\theta$

$b\sec\theta = x - y\tan\theta$

$(a^2 + b^2)\sec^2\theta = x^2(1 + \tan^2\theta) + y^2(1 + \tan^2\theta)$

$\Rightarrow \quad x^2 + y^2 = a^2 + b^2 \quad \Rightarrow$ (C) and (D)

91. If a hyperbola passes through the focus of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and its transverse and coincides with the major and minor axis of the ellipse, and product of their eccentricities is 1, then :

(A) equation of hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ (B) equation of hyperbola $\frac{x^2}{9} - \frac{y^2}{25} = 1$

(C) focus of hyperbola (5, 0) (D) focus of hyperbola is $\left(5\sqrt{3}, 0\right)$

एक अतिपरवलय दीर्घवृत $\frac{x^2}{25} + \frac{y^2}{16} = 1$ की नाभियों से गुजरता है उसके अनुप्रस्थ तथा संयुग्मी अक्ष क्रमशः दीर्घवृत्त के दीर्घ तथा लघु v{kkd sl kFkl Fi ir hgSA; fn v fr i joy ; r Fkk nh?kZor d hmRd sUnz k vkSj kxqkuQy 1 हो, तो :

(A) अतिपरवलय का समीकरण $\frac{x^2}{9}-\frac{y^2}{16}=1$ (B) अतिपरवलय का समीकरण $\frac{x^2}{9}-\frac{y^2}{25}=1$

(C) अतिपरवलय की नाभि (5, 0) (D) अतिपरवलय की नाभि $\left(5\sqrt{3}, 0\right)$

Ans. AC

Sol. Eccentricity of ellipse = 3/5

Eccentricity of hyperbola = 5/3 and it passes through (±3, 0)

$\Rightarrow$ its equation $\frac{x^2}{9}-\frac{y^2}{b^2}=1$

where $1+\frac{b^2}{9}=\frac{25}{9} \Rightarrow b^2 = 16$

$\Rightarrow \frac{x^2}{9}-\frac{y^2}{16}=1 \Rightarrow \frac{x^2}{9}-\frac{y^2}{16}=1$ and its foci are (±5, 0)

92. The equations of common tangents to the parabola $y = x^2$ and $y = -(x-2)^2$ is/are :

$y = x^2$ और $y = -(x-2)^2$ की उभयनिष्ठ स्पर्श रेखा का समीकरण है :

(A) $y = 4(x-1)$ (B) $y = 0$ (C) $y = -4(x-1)$ (D) $y = -30x - 50$

Ans. AB

Sol. $P_1 : x^2 = y \Rightarrow a = \frac{1}{4}$

Equation of tangent

$y = mx - am^2$

$y = mx - \frac{m^2}{4}$ ...(1)

$P_2 : (x-2)^2 = -y \qquad \Rightarrow a = \frac{1}{4}$

$y = m(x-2) + \frac{m^2}{4} = mx - 2m + \frac{m^2}{4}$ ...(2)

from equation (1) & (2)

$-2m + \frac{m^2}{4} = -\frac{m^2}{4}$

$-2m = -\frac{2m^2}{4}$ $\Rightarrow m = 0, m = 4$

$y = 4x - 4$ or $y = 0$

93. Let $P(x_1, y_1)$ and $Q(x_2, y_2)$, $y_1 < 0$, $y_2 < 0$, be the end points of the latus rectum of the ellipse $x^2 + 4y^2 = 4$. The equations of parabolas with latus rectum PQ are :

माना दीर्घवृत्त (ellipse) $x^2 + 4y^2 = 4$ के नाभिलम्ब के अन्तिम बिन्दु (end points) $P(x_1, y_1)$ व $Q(x_2, y_2)$, $y_1 < 0$, $y_2 < 0$, है। PQ नाभिलम्ब वाले परवलयों (parabolas) के समीकरण निम्न है :

(A) $x^2 + 2\sqrt{3}\, y = 3 + \sqrt{3}$ (B) $x^2 - 2\sqrt{3}\, y = 3 + \sqrt{3}$

(C) $x^2 + 2\sqrt{3}\, y = 3 - \sqrt{3}$ (D) $x^2 - 2\sqrt{3}\, y = 3 - \sqrt{3}$

Ans. BC

Sol. Equation of ellipse is $\frac{x^2}{4} + y^2 = 1$

eccentricity $e = \frac{\sqrt{3}}{2}$

so focus are $(\sqrt{3}, 0)$ & $(-\sqrt{3}, 0)$

so end points of latus rectum will be

$\left(\sqrt{3}, \frac{1}{2}\right)\left(\sqrt{3}, -1/2\right), \left(-\sqrt{3}, 1/2\right)$ & $\left(-\sqrt{3}, -1/2\right)$

$\because y_1 < 0$ & $y_2 < 0$

Hence coordinates of P & Q will be

$P\left(\sqrt{3}, -\frac{1}{2}\right)$ & $Q\left(-\sqrt{3}, -\frac{1}{2}\right)$.

So now equation of parabola taking these points as end points of latus rectum.

Focus will be $(0, -1/2)$

$4\alpha = 2\sqrt{3} \Rightarrow \quad a = \frac{\sqrt{3}}{2}$

Hence vertex of the parabolas will be

$\left(0, -\frac{1}{2} + \frac{\sqrt{3}}{2}\right), \left(0, -\frac{1}{2} - \frac{\sqrt{3}}{2}\right)$

so eq. of parabolas will be

$x^2 = -2\sqrt{3}\left(y + \frac{1}{2} - \frac{\sqrt{3}}{2}\right)$ &

$x^2 = 2\sqrt{3}\left(y + \frac{1}{2} + \frac{\sqrt{3}}{2}\right)$

$x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$

$x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$

MATCH THE COLUMN TYPE QUESTION(S)

94. Match the following column :

Column-I	**Column-II**
(A) The number of real common tangents to the circle $5x^2 + 5y^2 = 16$ and the hyperbola $3x^2 - y^2 = 48$, is	(P) 1
(B) The number of real common normals to parabola $y^2 = 4x$ and the circle $(x-1)^2 + (y-1)^2 = 1$, is	(Q) 2
(C) If P is any point on ellipse $\frac{x^2}{4} + 4y^2 = 1$ whose foci are at A and B, then the maximum value of (PA)(PB) equals	(R) 3 (S) 4
(D) The length of latus-rectum of the parabola defined parametrically by equations $x = \cos t - \sin t$ and $y = \sin 2t$ equals	(T) 5
(E) The value of a for the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a > b)$, if the extremities of the latus-rectum of the ellipse having positive ordinate lies on the parabola $x^2 = -2(y-2)$, is	

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) वृत्त $5x^2 + 5y^2 = 16$ तथा अतिपरवलय $3x^2 - y^2 = 48$ के वास्तविक उभयनिष्ठ स्पर्श रेखा की संख्या होगी	(P) 1
(B) परवलय $y^2 = 4x$ तथा वृत्त $(x-1)^2 + (y-1)^2 = 1$ के वास्तविक उभयनिष्ठ स्पर्श रेखा की संख्या होगी	(Q) 2
(C) दीर्घवृत्त $\frac{x^2}{4} + 4y^2 = 1$ पर स्थित किसी बिन्दु P का दोनों नाभियों A एवं B से दूरियों का गुणनफल (PA)(PB) का अधिकतम मान होगा	(R) 3
(D) एक परवलय जिसका प्राचलिक समीकरण $x = \cos t - \sin t$ तथा $y = \sin 2t$ है के नाभिलम्ब की लम्बाई होगी	(S) 4
(E) का क्या मान होगा जबकि दीर्घवृत्त $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a > b)$ के नाभिलम्ब के अन्तःबिन्दु (extremities) के सिरे जिसकी कोटि धनात्मक है, परवलय	(T) 5

$x^2 = -2(y-2)$ पर स्थित है

Ans. (A) S ; (B) P ; (C) S ; (D) P ; (E) Q

Sol. (A) $S : x^2 + y^2 = \frac{16}{5}$

$H : \frac{x^2}{16} - \frac{y^2}{48} = 1$ (dividing by 48)

tangent to H is, $y = mx \pm \sqrt{16m^2 - 48}$

use $p = r$

$$\left|\frac{\sqrt{16m^2-48}}{\sqrt{1+m^2}}\right| = \frac{4}{\sqrt{5}} \Rightarrow \quad 5(16m^2 - 48) = 16(1 + m^2) \quad \Rightarrow \quad 64m^2 = 16 \times 16$$

$m^2 = 4 \Rightarrow \quad m = \pm 2$

$\Rightarrow$ 4 common tangent $\Rightarrow$ (S) [for each value of m there are two parallel tangents]

(B) $P : y^2 = 4x$

$S : (x-1)^2 + (y-1)^2 = 1$

normal to P: $y = mx - 2m - m^3$ passes through the centre of the circle (1, 1)

$1 = m - 2m - m^3$

$m^3 + m + 1 = 0 \Rightarrow \quad f'(m) > 0 \quad \Rightarrow$ one real root

$\therefore$ one common normal $\Rightarrow$ (P)]

(C) We have $PA + PB = 4$

Now $\frac{PA + PB}{2} \geq \sqrt{PA \cdot PB} \Rightarrow PA \cdot PB \leq 4$ Ans.

(D) We have $x = \cos t - \sin t$, $y = \sin 2t$

$\Rightarrow x^2 = 1 - y \Rightarrow x^2 = -(y-1)$

Clearly l (L.R) = 1 Ans.

(E) $\left(\pm ae, \frac{b^2}{a}\right)$ are extremities of the latus-rectum having positive ordinates.

$\Rightarrow \quad a^2e^2 = -2\left(\frac{b^2}{a} - 2\right)$ ...(1)P-1]

But $b^2 = a^2(1 - e^2)$ ....(2)

$\therefore$ From (1) and (2), we get $a^2e^2 - 2ae^2 + 2a - 4 = 0 \Rightarrow ae^2(a-2) + 2(a-2) = 0$

$\therefore (ae^2 + 2)(a - 2) = 0$

Hence $a = 2$.

95. Identify the conic whose equations are given in column-I :

Column-I (Equation of a conic)	**Column-II (Nature of conic)**
(A) $xy + a^2 = a(x + y)$	(P) Ellipse
(B) $2x^2 - 72xy + 23y^2 - 4x - 28y - 48 = 0$	(Q) Hyperbola
(C) $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$	(R) Parabola.
(D) $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$	(S) line pair
(E) $4x^2 - 4xy + y^2 - 12x + 6y + 9 = 0$	

स्तम्भ–I में दिये गये शंकु के समीकरण का स्तम्भ–II से सुमेलित कीजिए

Column-I (शंकु का समीकरण)	**Column-II** (शंकु की प्रकृति)
(A) $xy + a^2 = a(x + y)$	(P) दीर्घवृत्त
(B) $2x^2 - 72xy + 23y^2 - 4x - 28y - 48 = 0$	(Q) अतिपरवलय
(C) $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$	(R) परवलय
(D) $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$	(S) रेखा युग्म
(E) $4x^2 - 4xy + y^2 - 12x + 6y + 9 = 0$	

Ans. (A) S, (B) Q, (C) S, (D) P, (E) R

Sol. Check Δ & h^2 related to AB

96. Match the following column :

Column-I	**Column-II**
(A) The eccentricity of the ellipse which meets the straight line $2x - 3y = 6$ $2x - 3y = 6$ on the X-axis and the straight line $4x + 5y = 20$ on the Y-axis and whose principal axes lie along the coordinate axes, is	(P) $\frac{1}{2}$
(B) A bar of length 20 units moves with its ends on two fixed straight lines at right angles. A point P marked on the bar at a distance of 8 units from one end describes a conic whose eccentricity is	(Q) $\frac{1}{\sqrt{2}}$
(C) If one extremity of the minor axis of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the foci form an equilateral triangle, then its eccentricity, is	(R) $\frac{\sqrt{5}}{3}$
(D) There are exactly two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose distance whose distance from the centre of the ellipse are greatest and	(S) $\frac{\sqrt{7}}{4}$

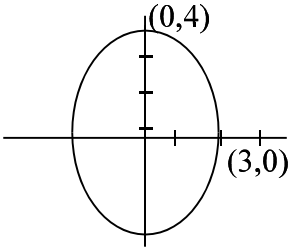
equal to $\sqrt{\frac{a^2+2b^2}{2}}$. Eccentricity of this ellipse is equal to

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) उस दीर्घवृत्त की उत्केन्द्रता बताओं जो सरल रेखाओं $2x - 3y = 6$ तथा $4x + 5y = 20$ को क्रमशः x-अक्ष तथा y-अक्ष पर काटते है तथा उसके मुख्य अक्ष (principal axes) निर्देशांक अक्षों पर है	(P) $\frac{1}{2}$
(B) 20 मात्रक लम्बाई वाले डण्डे के दोनों सिरे दो लम्बवत् रेखाओं पर खिसक रहे है, तो एक सिरे से 8 मात्रक की दूरी पर स्थित बिन्दु P का जो वक्र प्राप्त होगा उसकी उत्केन्द्रता होगी	(Q) $\frac{1}{\sqrt{2}}$
(C) यदि दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ की लघु अक्ष का एक सिरा तथा नाभियाँ यदि समबाहु त्रिभुज बनाये तो इसकी उत्केन्द्रता होगी	(R) $\frac{\sqrt{5}}{3}$
(D) दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ पर स्थित केवल दो बिन्दु ही ऐसे होगें जिसका दीर्घवृत्त के केन्द्र से दूरी अधिकतम है तथा $\sqrt{\frac{a^2+2b^2}{2}}$ है तो इसकी उत्केन्द्रता होगी	(S) $\frac{\sqrt{7}}{4}$

Ans. (A) S; (B) R ; (C) P ; (D) Q

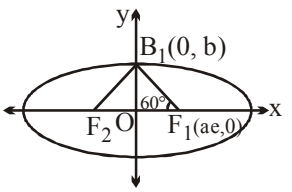
Sol. (A) $e^2 = 1 - \frac{9}{16} = \frac{7}{16}$ ellipse touch the x-axis at (3, 0) and y-axis at (0,4)



$e = \frac{\sqrt{7}}{4}$ Ans.

(B) Locus is as ellipse

(C) We have $\frac{b}{ae} = \sqrt{3} \Rightarrow a^2(1 - e^2) = 3a^2e^2$ (As $b^2 = a^2(1 - e^2)$



$\Rightarrow e^2 = \frac{1}{4}$

Hence $e = \frac{1}{2}$

(D) The given distance is clearly the length of semi major axis

Thus, $\sqrt{\frac{a^2+2b^2}{2}} = a \quad \Rightarrow \quad 2b^2 = a^2 \quad \Rightarrow \quad 2a^2(1-e^2) = a^2$

$\Rightarrow \quad e^2 = \frac{1}{2} \Rightarrow \quad e = \frac{1}{\sqrt{2}}$ Ans.

97. Match the following column :

Column-I	**Column-II**
(A) If the chord of contact of tangents from a point P to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, the locus of P is	(P) Straight line
(B) A variable circle C has the equation $x^2 + y^2 - 2(t^2 - 3t + 1)x - 2(t^2 + 2t)y + t = 0$, where t is a parameter. The locus of the centre of the circle is	(Q) Circle
(C) The locus of point of intersection of tangents to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at two points the sum of whose eccentric angles is constant is	(R) Parabola
(D) An ellipse slides between two perpendicular straight lines. Then the locus of its centre is	(S) Hyperbola

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) बिन्दु P से परवलय $y^2 = 4ax$ पर खींची गयी स्पर्श जीवा परवलय $x^2 = 4by$ को स्पर्श करता है तो P का बिन्दुपथ होगा	(P) सरल रेखा
(B) एक चर वृत्त का समीकरण $x^2 + y^2 - 2(t^2 - 3t + 1)x - 2(t^2 + 2t)y + t = 0$ है जहां t प्राचल है तो वृत्त के केन्द्र का बिन्दुपथ होगा	(Q) वृत्त
(C) दीर्घवृत्त $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ के उन बिन्दुओं पर खींची गयी स्पर्श रेखाओं के प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात कीजिए जिनके उपकेन्द्रिय कोणों का योगफल अचर है	(R) परवलय
(D) यदि एक दीर्घवृत्त दो लम्बवत रेखाओं के अनुदिश फिसलता है तो इसके केन्द्र का बिन्दुपथ होगा	(S) अतिपरवलय

Ans. (A) S; (B) R; (C) P; (D) Q

Sol. (A) $yy_1 = 2a(x + x_1)$; $x^2 = 4by = 4b[(2a/y_1)(x + x_1)] \Rightarrow y_1x^2 - 8abx - 8abx_1 = 0$;
$D = 0$ gives $xy = -2ab \Rightarrow$ Hyperbola

(B) centre is $h = t^2 - 3t + 1$ ....(1)

$k = t^2 + 2t$ ....(2)

(2) – (1) gives $-x + y = 5t - 1$

or $t = \frac{1 - x + y}{5}$

Substituting the value of t in (2)

$$y = \left(\frac{y-x+1}{5}\right)^2 + 2\left(\frac{y-x+1}{5}\right)$$

$25y = (y - x + 1)^2 + 10(y - x + 1)$

$25y = y^2 + x^2 + 1 - 2xy - 2x + 2y + 10y - 10x + 10$

$x^2 + y^2 - 2xy - 12x - 13y + 11 = 0$

which is a parabola

as $\Delta \neq 0$ and $h^2 = ab$

(C) $$h = \frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}} \; ; \; k = \frac{b\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}$$

given $\frac{\alpha+\beta}{2} = \text{constant} = C$

$$\therefore \cos\frac{\alpha-\beta}{2} = \frac{a\cos C}{h} = \frac{b\sin C}{k} \quad \Rightarrow y = \left(\frac{b}{a}\tan C\right)x$$

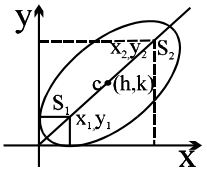
Locus of (h, k) is a straight line

(D) $y_1y_2 = x_1x_2 = b^2$....(1)

and $(x_2 - x_1)^2 + (y_2 - y_1)^2 = 4(a^2 - b^2)$ ....(2)

Also $2h = x_1 + x_2$

$2k = y_1 + y_2$



from (2) $(x_1 + x_2)^2 + (y_1+y_2)^2 - 4(x_1x_2 + y_1y_2) = 4(a^2 - b^2)$

$4(h^2 + k^2) - 4(2b^2) = 4(a^2 - b^2)$

$\therefore x^2 + y^2 = a^2 + b^2 \Rightarrow$ Circle

Alternative: Equation of director circle with centre (h, k)

$(x - h)^2 + (y - k)^2 = a^2 + b^2$

(0, 0) lies on it $\Rightarrow$ $h^2 + k^2 = a^2 + b^2$ $\Rightarrow$ locus is $x^2 + y^2 = a^2 + b^2$

98. Match the following column :

Column-I	**Column-II**
(A) For an ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with vertices A and A', tangent drawn at the point P in the first quadrant meets the y-axis in Q and the chord A'P meets the y-axis in M. If 'O' is the origin then $OQ^2 - MQ^2$ equals to	(P) 2
(B) If the product of the perpendicular distances from any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ of eccentricity $e = \sqrt{3}$ from its asymptotes is equal to 6, then the length of the transverse axis of the hyperbola is	(Q) 3

(C) The locus of the point of intersection of the lines $\sqrt{3}x-y-4\sqrt{3}t=0$ and $\sqrt{3}tx+ty-4\sqrt{3}=0$ (where t is a parameter) is a hyperbola whose eccentricity is (R) 4

(D) If F_1 and F_2 are the feet of the perpendiculars from the foci S_1 and S_2 of an ellipse $\frac{x^2}{6}+\frac{y^2}{3}=1$ on the tangent at any point P on the ellipse, then $(S_1F_1).(S_2F_2)$ is equal to (S) 6

निम्न को सुमेलित कीजिए :

Column-I	**Column-II**
(A) दीर्घवृत्त $\frac{x^2}{9}+\frac{y^2}{4}=1$ के शीर्ष A तथा A' है, पर प्रथम चर्तुथांश में स्थित किसी बिन्दु P पर खींची गयी स्पर्श रेखा y-अक्ष को बिन्दु Q तथा जीवा A'P y-अक्ष को M पर मिलती है। यदि O' मूल बिन्दु है तो $OQ^2 - MQ^2$ का मान है	(P) 2
(B) अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ जिसकी प्रवणता $e=\sqrt{3}$ है, पर स्थित किसी बिन्दु से अनन्तस्पर्शी पर खींचे गये लम्बो की लम्बाईयों का गुणनफल 6 है तो अतिपरवलय की अनुप्रस्थ अक्ष की लम्बाई का मान होगा	(Q) 3
(C) रेखाओं $\sqrt{3}x-y-4\sqrt{3}t=0$ तथा $\sqrt{3}tx+ty-4\sqrt{3}=0$ के प्रतिच्छेद बिन्दुओं का बिन्दुपथ अतिपरवलय है। जिसकी उत्केन्द्रियता का मान होगा। (जहां t प्राचल है)	(R) 4
(D) परवलय $\frac{x^2}{5}+\frac{y^2}{3}=1$ पर स्थित किसी बिन्दु P पर नाभियों S_1 तथा S_2 से खींचे गये लम्बपात F_1 तथा F_2 हो तो $(S_1F_1).(S_2F_2)$ का मान होगा।	(S) 6

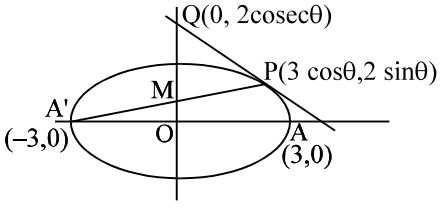
Ans. (A) R; (B) S; (C) P; (D) Q

Sol. (A) $a=3\ ;\ b=2$

$T:\ \frac{x\cos\theta}{3}+\frac{y\sin\theta}{2}=1$

$x=0\ ;\ y=2\ \text{cosec}\theta$

chord A'P, $y=\frac{2\sin\theta}{3(\cos\theta+1)}(x+3)$



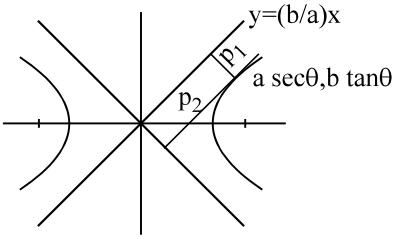
put $x=0\ \ y=\frac{2\sin\theta}{1+\cos\theta}=OM$

Now $OQ^2 - MQ^2 = OQ^2 - (OQ - OM)^2 = 2(OQ)(OM) - OM^2 = OM\{ 2(OQ) - (OM) \}$

$$= \frac{2\sin\theta}{1+\cos\theta}\left[\frac{4}{\sin\theta} - \frac{2\sin\theta}{1+\cos\theta}\right] = \frac{4\sin\theta}{1+\cos\theta}\left[\frac{2(1+\cos\theta)-(1-\cos^2\theta)}{\sin\theta(1+\cos\theta)}\right] = \frac{4(1+\cos\theta)(2-1+\cos\theta)}{(1+\cos\theta)(1+\cos\theta)} = 4$$

(B) $p_1p_2 = \dfrac{a^2b^2}{a^2+b^2} = \dfrac{a^2.a^2(e^2-1)}{a^2e^2} = 6;$

$$\frac{2a^2}{3} = 6 \Rightarrow a^2 = 9 \Rightarrow a = 3$$



hence $2a = 6$

(C) hyperbola $\dfrac{x^2}{16} - \dfrac{y^2}{48} = 1$

(D) Product of the feet of the perpendiculars is equal to the square of its semi minor axes.

99. Match the following

Normals are drawn at points P, Q and R lying on the parabola $y^2 = 4x$ which intersect at (3, 0). Then :

Column-I	**Column-II**
(A) Area of ΔPQR	(P) 2
(B) Radius of circumcircle of ΔPQR	(Q) 5/2
(C) Centroid of ΔPQR	(R) (5/2, 0)
(D) Circumcentre of ΔPQR	(S) (2/3, 0)

सुमेलित कीजिये।

i j oy ; $y^2 = 4x$ पर स्थित बिन्दुओं P, Q एवं R पर खींचे गये अभिलम्ब बिन्दु (3, 0) पर प्रतिच्छेद करते है, तो :

Column-I	**Column-II**
(A) ΔPQR का क्षेत्रफल	(P) 2
(B) ΔPQR के परिवृत की त्रिज्या	(Q) 5/2
(C) ΔPQR का केन्द्रक	(R) (5/2, 0)
(D) ΔPQR का परिकेन्द्र	(S) (2/3, 0)

Ans. (A) P, (B) Q, (C) S, (D) R

Sol.

100. Match the statements in Column I with the properties in Column II :

Column-I	**Column-II**
(A) Two intersecting circles	(P) have a common tangent
(B) Two mutually external circles	(Q) have a common normal
(C) Two circles, one strictly inside the other	(R) do not have a common tangent
(D) Two branches of a hyperbola	(S) do not have a common normal

स्तम्भ–I का स्तम्भ–II से मिलान कीजिये :

Column-I	**Column-II**
(A) दो प्रतिच्छेदी वृत्त	(P) एक उभयनिष्ठ स्पर्श रेखा
(B) दो परस्पर बाह्य वृत्त	(Q) एक उभयनिष्ठ अभिलम्ब
(C) यदि दो वृत्त, एक वृत्त दूसरे वृत्त के अंदर स्थित है	(R) कोई उभयनिष्ठ स्पर्श रेखा नहीं
(D) अतिपरवलय की दो शाखाएँ	(S) कोई उभयनिष्ठ अभिलम्ब नहीं

Ans. (A) P, Q; (B) P ,Q; (C) Q, R; (D) Q, R

Sol.

Exercise J-M

1. Let the tangent to the parabola S : $y^2 = 2x$ at the point P(2, 2) meet the x-axis at Q and normal at it meet the parabola S at the point R. Then the area (in sq. units) of the triangle PQR is equal to : **[JEE(Main)July2021]**

माना परवलय S : $y^2 = 2x$ के बिंदु P(2, 2) पर स्पर्श रेखा x – अक्ष को बिंदु Q पर मिलती है तथा इस बिंदु पर अभिलम्ब परवलय S को बिंदु R पर मिलता है। तो त्रिभुज PQR का क्षेत्रफल (वर्ग इकाईयों में) बराबर है : **[JEE(Main)July2021]**

(A) $\frac{15}{2}$ (B) $\frac{25}{2}$ (C) 25 (D) $\frac{35}{2}$

Ans. B

Sol. $y^2 = 2x \qquad a = \frac{1}{2}$

$P(2,2) \equiv P(at_1^2, 2at_1) \quad \Rightarrow t_1 = 1$

Tangent at P

$$t_1 y = x + at_1^2$$

$$Q(-at_1^2, 0) \equiv Q(-2, 0)$$

Normal at P intersects parabola at R

$$t_2 = -t_1 - \frac{2}{t_1} = -3$$

$$R\left(\frac{9}{2}, -3\right)$$

$$\text{Area of } \Delta PQR = \frac{1}{2}\begin{Vmatrix} -2 & 0 & 1 \\ 2 & 2 & 1 \\ \frac{9}{2} & -3 & 1 \end{Vmatrix} = \frac{25}{2}$$

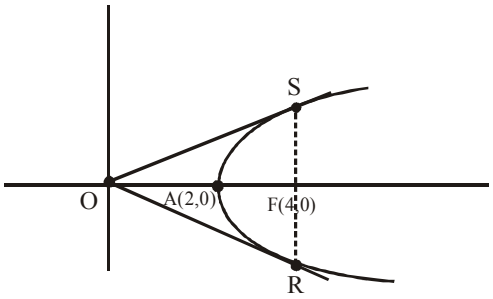
2. Let a parabola P be such that its vertex and focus lie on the positive x-axis at a distance 2 and 4 units from the origin, respectively. If tangents are drawn from O(0, 0) to the parabola P which meet P at S and R, then the area (in sq. units) of ΔSOR is equal to : **[JEE(Main)July2021]**

माना P एक परवलय है जिसका शीर्ष तथा नाभि धनात्मक x -अक्ष पर हैं और उनकी मूलबिन्दु से दूरी क्रमशः 2 तथा 4 इकाई हैं। यदि O(0, 0) से परवलय P पर दो स्पर्श रेखाऐं खींची जाती हैं, जो P को S तथा R मिलती हैं, तो ΔSOR को क्षेत्रफल (वर्ग इकाइयों में) बराबर है : **[JEE(Main)July2021]**

(A) 16 (B) 32 (C) $16\sqrt{2}$ (D) $8\sqrt{2}$

Ans. A

Sol.



Origin is foot of directrix, so tangents from 'O' meets at ends of 'LR'

Hence, points S and R are ends of LR

$S(4,4)$ $R(4,-4)$

Area of $\Delta SOR = \frac{1}{2}(8)(4) = 16$

3. Let $y = mx + c$, $m > 0$ be the focal chord $y^2 = -64x$, which is tangent to $(x + 10)^2 + y^2 = 4$. Then, the value of $4\sqrt{2}(m+c)$ is equal to _____. **[JEE(Main)July2021]**

माना $y = mx + c$, $m > 0$, परवलय $y^2 = -64x$ की नाभीय जीवा है तथा वृत $(x + 10)^2 + y^2 = 4$ की स्पर्श रेखा है। तो $4\sqrt{2}(m+c)$ का मान बराबर है _____ **[JEE(Main)July2021]**

Ans. 34

Sol. $y^2 = -64x$ $a = 16$

focus $(-16, 0)$ will lie on $y = mx + c$

$\Rightarrow c = 16m$

$y - m(x + 16) = 0$

apply COT wrt $(x + 10)^2 + y^2 = 4$

$$\left|\frac{(-10+16)m}{\sqrt{1+m^2}}\right| = 1$$

$8m^2 = 1$

$$m = \frac{1}{2\sqrt{2}}$$

$$4\sqrt{2}(m+c) = 4\sqrt{2}(m+16m)$$

$$= 68\sqrt{2}\left(\frac{1}{2\sqrt{2}}\right)$$

$= 34$

4. A ray of light through (2, 1) is reflected at a point P on the y-axis and then passes through the point (5,3). If this reflected ray is the directrix of an ellipse with eccentricity 1/3 and the distance of the nearer focus from this directrix is $\frac{8}{\sqrt{53}}$, then the equation of the other directrix can be : **[JEE(Main)July2021]**

(A) $11x + 7y + 8 = 0$ or $11x + 7y - 15 = 0$ (B) $11x - 7y - 8 = 0$ or $11x + 7y + 15 = 0$

(C) $2x - 7y - 39 = 0$ or $2x - 7y - 7 = 0$ (D) $2x - 7y + 29 = 0$ or $2x - 7y - 7 = 0$

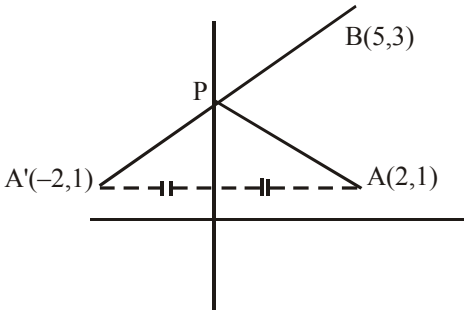
बिन्दु (2, 1) से चलकर प्रकाश की एक किरण y-अक्ष पर बिन्दु P पर परावर्तित होती है और तब बिन्दु (5, 3) से होकर जाती है। यदि यह परावर्तित किरण $\frac{1}{3}$ उत्केन्द्रता के एक दीर्घवृत्त की नियता है तथा इस नियता से पास की नाभि की दूरी $\frac{8}{\sqrt{53}}$ है, तो दूसरी नियता का l ehdj .kgksl d r kgS **[JEE(Main)July2021]**

(A) $11x + 7y + 8 = 0$ अथवा $11x + 7y - 15 = 0$ (B) $11x - 7y - 8 = 0$ अथवा $11x + 7y + 15 = 0$

(C) $2x - 7y - 39 = 0$ अथवा $2x - 7y - 7 = 0$ (D) $2x - 7y + 29 = 0$ अथवा $2x - 7y - 7 = 0$

Ans. D

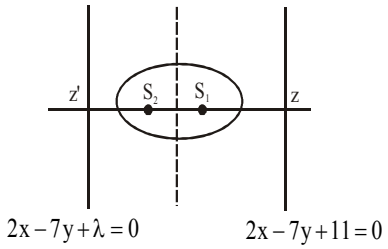
Sol.



$m_{A'B} = \frac{3-1}{5+2} = \frac{2}{7}$

Equation of $A'B \Rightarrow (y-1) = \frac{2}{7}(x+2)$

$\Rightarrow 2x - 7y + 11 = 0$



$e = \frac{1}{3}$

$S_1Z = \frac{8}{\sqrt{53}} = \frac{a}{e} - ae = 3a - \frac{a}{3} = \frac{8a}{3}$

$a = \frac{3}{\sqrt{53}}$

$ZZ' = 2\frac{a}{e} = \frac{18}{\sqrt{53}}$

$\Rightarrow \left|\frac{\lambda - 11}{\sqrt{53}}\right| = \frac{18}{\sqrt{53}}$

$\Rightarrow \lambda - 11 = \pm 18$

Þ $1 = 29$ or -7

Equation of other directrix

$2x - 7y + 29 = 0$

or $2x - 7y - 7 = 0$

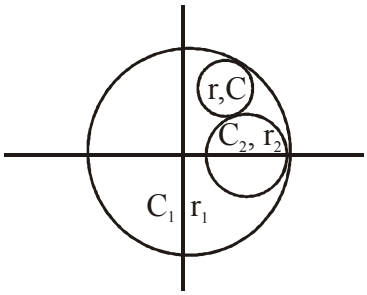
5. Let $S_1 : x^2 + y^2 = 9$ and $S_2 : (x-2)^2 + y^2 = 1$. Then the locus of center of a variable circle S which touches S_1 internally and S_2 externally always passes through the points : **[JEE(Main)Mar.2021]**

माना $S_1 : x^2 + y^2 = 9$ तथा $S_2 : (x-2)^2 + y^2 = 1$ है। तो एक चर वृत्त S, जो S_1 को अंदर से स्पर्श करता है तथा S_2 को बाहर से स्पर्श d j r kgS] sd sHkh o`Ùkksa dk fcUnq ftUgsa lnSo fLFkr fcUnq gksrk gS] gS? **[JEE(Main)Mar.2021]**

(A) $\left(2, \pm\frac{3}{2}\right)$ (B) $\left(0, \pm\sqrt{3}\right)$ (C) $\left(\frac{1}{2}, \pm\frac{\sqrt{5}}{2}\right)$ (D) $\left(1, \pm 2\right)$

Ans. A

Sol.



$CC_1 = r_1 - r$

$CC_1 = r_2 + r$

$c\,c_1 + CC_2 = r_1 + r_2$

$CC_1 + CC_2 = \text{constant}$

Locus is ellipse

$2a = 4 \Rightarrow a = 2$

$2ae = 2$

$e = 1/2$

$e^2 = 1 - \frac{b^2}{a^2}$

$\frac{1}{4} = 1 - \frac{b^2}{4}$

$$\frac{b^2}{4} = \frac{3}{4} \Rightarrow b^2 = 3$$

$$\frac{(x-1)^2}{4} + \frac{y^2}{3} = 1$$

Option 1 satisfies it.

6. Let $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$. Let E_2 be another ellipse such that it touches the end points of major axis of E_1 and the foci of E_2 are the end points of minor axis of E_1. If E_1 and E_2 have same eccentricities, then its value is :

[JEE(Main)July2021]

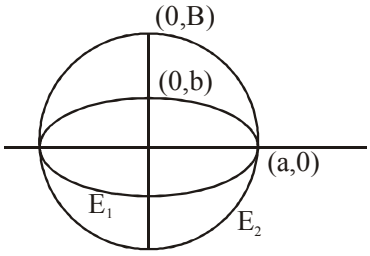
माना $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ एक दीर्घवृत्त है। माना E_2 एक और दीर्घवृत्त है, जो E_1 के दीर्घ अक्ष के छोरों को स्पर्श करता है तथा E_2 की नाभियाँ, E_1 के लघु अक्ष के छोरों पर हैं। यदि E_1 तथा E_2 की उत्केंन्द्रता बराबर है, तो उसका मान है :

[JEE(Main)July2021]

(A) $\frac{-1+\sqrt{5}}{2}$ (B) $\frac{-1+\sqrt{8}}{2}$ (C) $\frac{-1+\sqrt{6}}{2}$ (D) $\frac{-1+\sqrt{3}}{2}$

Ans. A

Sol.



E_1 :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$e = \sqrt{1 - \frac{b^2}{a^2}} \text{ ----(1)}$$

E_2:

$$\frac{x^2}{a^2} + \frac{y^2}{B^2} = 1$$

but $a^2 = B^2(1-e^2)$ ----(2)

also we know

$b = Be$

$$B = \frac{b}{e}$$

By (2)

$$a^2 = \frac{b^2}{e^2}\left(1 - e^2\right)$$

$$\frac{e^2}{1-e^2} = \frac{b^2}{a^2}$$

Now by (2) we get

$$e = \sqrt{1 - \frac{e^2}{1-e^2}}$$

$$e^2 = \frac{1-2e^2}{1-e^2}$$

$$e^2 - e^4 = 1 - 2e^2$$

$$e^4 - 2e^2 + 1 = e^2$$

$$\left(e^2 - 1\right)^2 = e^2$$

$$e^2 - 1 = \pm e$$

but $e^2 - 1 = -e$

$$e^2 + e - 1 = 0$$

$$e = \frac{-1 \pm \sqrt{5}}{2}$$

but $e = \frac{-1+\sqrt{5}}{2}$ (because $e > 0$)

7. Let an ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a^2 > b^2$, passes through $\left(\sqrt{\frac{3}{2}}, 1\right)$ and has eccentricity $\frac{1}{\sqrt{3}}$. If a circle, centered at focus $F(\alpha, 0)$, $\alpha > 0$, of E and radius $\frac{2}{\sqrt{3}}$, intersects E at two points P and Q, then PQ^2 is equal to :

[JEE(Main)July2021]

एक दीर्घवृत्त $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$, बिन्दु $\left(\sqrt{\frac{3}{2}}, 1\right)$ से होकर जाता है तथा उसकी उत्केन्द्रता $\frac{1}{\sqrt{3}}$ है। यदि एक वृत्त जिसका केन्द्र E की नाभि $F(\alpha, 0)$, $\alpha > 0$ पर और त्रिज्या $\frac{2}{\sqrt{3}}$ हैं, दीर्घवृत्त E को दो बिन्दुओं P तथा Q पर काटता हैं, तो PQ^2 बराबर है।

[JEE(Main)July2021]

(A) 3 (B) $\frac{8}{3}$ (C) $\frac{4}{3}$ (D) $\frac{16}{3}$

Ans. D

Sol. Ellipse passing through $\left(\frac{\sqrt{3}}{\sqrt{2}}, 1\right)$

$$\frac{3}{2a^2} + \frac{1}{b^2} = 1 \quad \text{.....(1)}$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

$$\frac{1}{3} = 1 - \frac{b^2}{a^2}$$

$$\frac{b^2}{a^2} = \frac{2}{3} \Rightarrow \frac{3}{2a^2} = \frac{1}{b^2} \quad \text{.....(2)}$$

from (1) & (2)

$b = \sqrt{2} \quad a = \sqrt{3}$

$F(ac, 0) \quad = F(1, 0) = \text{centre } (1, 0), \text{Radius} = \frac{2}{\sqrt{3}}$

$$E: \frac{x^2}{3} + \frac{y^2}{2} = 1$$

circle $(x-1)^2 + y^2 = \frac{4}{3}$

point of intersection

$P\left(1, \frac{2}{\sqrt{3}}\right) \quad Q\left(1, -\frac{2}{\sqrt{3}}\right)$

$$PQ = \frac{16}{3}$$

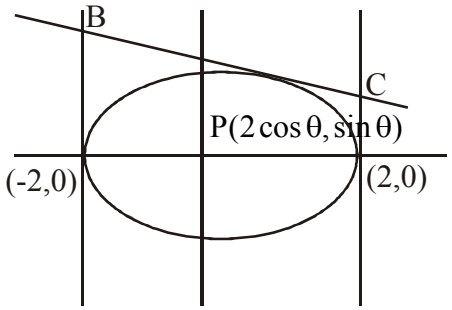
8. If a tangent to the ellipse $x^2 + 4y^2 = 4$ meets the tangents at the extremities of its major axis at B and C, then the circle with BC as diameter passes through the point : **[JEE(Main)July2021]**

यदि दीर्घवृत्त $x^2 + 4y^2 = 4$ की एक स्पर्शरेखा, इसके दीर्घ अक्ष के छोरों पर खींची गई स्पर्श रेखाओं को बिन्दुओं B तथा C पर मिलती g\$r ksBC को व्यास मान कर खींचा गया वृत्त निम्न में से किस बिन्दु से होकर जाता है : **[JEE(Main)July2021]**

(A) $(\sqrt{3}, 0)$ (B) $(1, 1)$ (C) $(-1, 1)$ (D) $(\sqrt{2}, 0)$

Ans. A

Sol.



$$\frac{2\cos\theta \times x}{4} + y \times \sin\theta = 1$$

$$x\cos\theta + 2y\sin\theta = 2$$

For B

$$x = -2$$

$$-\cos\theta + 2y\sin\theta = 2$$

ψαν θ = 1 + cos θ

$$y = \frac{1+\cos\theta}{\sin\theta}$$

For C

$$x = 2$$

$$2\cos\theta + 2y\sin\theta = 2$$

ψαν θ = 1 – cos θ

$$y = \frac{1-\cos\theta}{\sin\theta}$$

$$B\left(-2, \cot\theta/2\right)\ C\left(2, \tan\theta/2\right)$$

$$(x+2)(x-2) + \left(y - \cot\theta/2\right)\left(y - \tan\theta/2\right) = 0$$

$$x^2 - 4 + y^2 - y\left(\tan\theta/2 + \cot\theta/2\right) + 1 = 0$$

$x^2 + y^2 - y\left(\tan\theta/2 + \cot\theta/2\right) - 3 = 0$

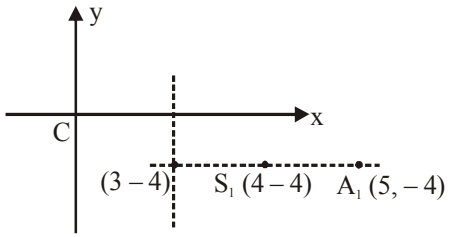
$\left(\sqrt{3}, 0\right)$ = Satisfies

9. Let E be an ellipse whose axes are parallel to the co-ordinates axes, having its center at (3, –4), one focus at (4, –4) and one vertex at (5,–4). If $mx - y = 4$, $m > 0$ is a tangent to the ellipse E, then the value of $5m^2$ is equal to_____. **[JEE(Main)July2021]**

माना E एक दीर्घवृत्त है जिसके अक्ष, निर्देशांक अक्षों के समान्तर हैं। इसका केन्द्र (3, –4) पर, एक नाभि (4, –4) तथा एक शीर्ष (5,–4) पर हैं। यदि $mx - y = 4$, $m > 0$ दीर्घवृत्त E की एक स्पर्श रेखा है, तो $5m^2$ का मान बराबर है _____ **[JEE(Main)July2021]**

Ans. 3

Sol.



$CA_1 = a = 2$

$CS_1 = ae = 1$

$\therefore e = \frac{1}{2}$

$b^2 = a^2(1 - e^2)$

$= 4\left(1 - \frac{1}{4}\right) = 3$

Equation of the given ellipse is :

$\frac{(x-3)^2}{4} + \frac{(y+4)^2}{3} = 1$ .....(1)

$\therefore mx - y = 4$ is a tangent to (1)

$m(x - 3) + 3m = y + 4$

$\Rightarrow (y + 4) = m(x - 3) + 3m$

$\therefore$ this line is tangent to the above ellipse (1)

$\therefore$ Using condition of tangency

$c^2 = a^2m^2 + b^2$

$(3m)^2 = 4.m^2 + 3$

$\Rightarrow 9m^2 = 4m^2 + 3$

$= 5m^2 = 3$

10. Let a line L : $2x + y = k$, $k > 0$ be a tangent to the hyperbola $x^2 - y^2 = 3$. If L is also a tangent to the parabola $y^2 = \alpha x$, then α is equal to : **[JEE(Main)July2021]**

माना एक रेखा L : $2x + y = k$, $k > 0$, अतिपरवलय, $x^2 - y^2 = 3$ को स्पर्श करती है। यदि रेखा L, परवलय, $y^2 = \alpha x$ को भी स्पर्श d j r hg$r ks$\alpha$ बराबर है : **[JEE(Main)July2021]**

(A) 24 (B) 12 (C) – 12 (D) –24

Ans. D

Sol. $x^2 - y^2 = 3$

$2x + y = k$ is tangent to hyperbola hence by condition of tangent

$c^2 = a^2 m^2 - b^2$

$y = -2x + k$ & $\frac{x^2}{3} - \frac{y^2}{3} = 1$

$k^2 = 3 \times 4 - 3$

$k^2 = 9$

$k = 3$

Now Line becomes

$y = -2x + 3$

Which is tangent to $y^2 = \alpha x$ by condition of tangently

$$c = \frac{a}{m}$$

$$3 = \frac{\alpha}{4(-2)}$$

$\alpha = -24$

11. The locus of the centroid of the triangle formed by any point P on the hyperbola $16x^2 - 9y^2 + 32x + 36y - 164 = 0$, and its foci is : **[JEE(Main)July2021]**

अतिपरवलय $16x^2 - 9y^2 + 32x + 36y - 164 = 0$ पर किसी बिन्दु P तथा इसकी नाभियों से बने त्रिभुज के केन्द्रक का बिन्दुपथ है: **[JEE(Main)July2021]**

(A) $16x^2 - 9y^2 + 32x + 36y - 36 = 0$ (B) $9x^2 - 16y^2 + 36x + 32y - 36 = 0$

(C) $16x^2 - 9y^2 + 32x + 36y - 144 = 0$ (D) $9x^2 - 16y^2 + 36x + 36y - 144 = 0$

Ans. A

Sol. $16 (x + 1)^2 - a (y - 2)^2 = 144$

$$\frac{(x+1)^2}{9} - \frac{(y-2)^2}{16} = 1$$

Let P $(-1 + 3 \sec\theta,\ 2 + 4 \tan\theta)$

$$e = \sqrt{1 + \frac{16}{9}} = \frac{5}{3}$$

$ae = 5$

$F_1(4, 2), F_2(-6, 2)$

Let centroid of ΔF_1F_2P is (h, k)

$$h = \frac{4-6-1+3\sec\theta}{3} \qquad k = \frac{2+2+2+4\tan\theta}{3}$$

$$h = -1+\sec\theta \qquad \left(\frac{3k-6}{4}\right) = \tan\theta$$

$\sec\theta = h + 1$

$\sec^2\theta - \tan^2\theta = 1$

$$(h+1)^2 - \left(\frac{3k-6}{4}\right)^2 = 1$$

$16(x + 1)^2 - (3y - 6)^2 = 16$

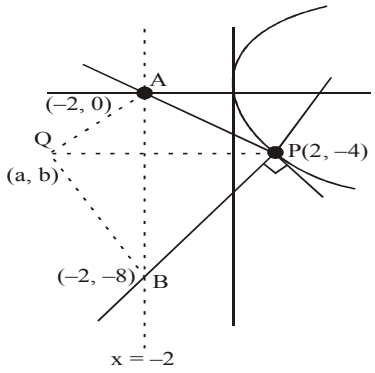
12. A tangent and a normal are drawn at the point P(2, –4) on the parabola $y^2 = 8x$, which meet the directrix of the parabola at the points A and B respectively. If Q(a, b) is a point such that AQBP is a square, then $2a + b$ is equal to : **[JEE(Main)Aug.2021]**

परवलय $y^2 = 8x$ के बिन्दु P(2, –4) पर एक स्पर्श रेखा तथा एक अभिलम्ब खींचे गए हैं, जो परवलय की नियता को क्रमशः बिन्दुओं A तथा B पर मिलते है। यदि Q(a, b) एक ऐसा बिन्दु है, जिसके लिए AQBP एक वर्ग है, तो $2a + b$ बराबर है : **[JEE(Main)Aug.2021]**

(A) – 20 (B) – 16 (C) – 12 (D) – 18

Ans. B

Sol.



$y^2 = 8x$

Tangent at $(2, -4)$

$$T = 0: \quad y(-4) = 8\left(\frac{x+2}{2}\right)$$

$x + y + 2 = 0$

at $x = -2, y = 0$

$A(-2, 0)$

$N = 0 \quad x - y = 6$

at $x = -2, y = -8$

$B(-2, -8)$

Mid point of AB and PQ must be same

$a + 2 = -2 - 2 \Rightarrow a = -6$

$b - 4 = 0 - 8 \Rightarrow b = -4$

$2a + b = -16$

13. If two tangents drawn from a point P to the parabola $y^2 = 16(x-3)$ are at right angles, then the locus of point P is:
[JEE(Main)Aug.2021]

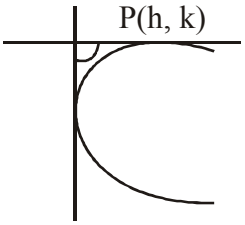
यदि एक बिन्दु P से परवलय $y^2 = 16(x-3)$ पर खींची गई दो स्पर्श रेखायें समकोण बनाती हैं,तो बिन्दु P का बिन्दुपथ है :
[JEE(Main)Aug.2021]

(A) $x + 4 = 0$ (B) $x + 2 = 0$ (C) $x + 3 = 0$ (D) $x + 1 = 0$

Ans. D

Sol. Equation of tangent

$$y = m(x-3) + \frac{4}{m}$$



$$k = m(h-3) + \frac{4}{m}$$

$$km = m^2(h-3) + 4$$

$$m^2(h-3) - km + 4 = 0 \begin{cases} m_1 \\ m_2 \end{cases}$$

$m_1 m_2 = -1$

$$\frac{4}{h-3} = -1$$

$x - 3 = 4$

$x + 1 = 0$

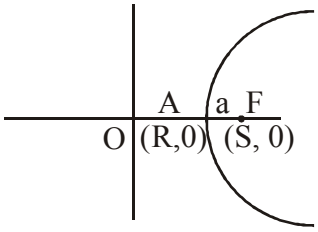
14. The length of the latus rectum of a parabola, whose vertex and focus are on the positive x-axis at a distance R and S; (S > R) respectively from the origin, is :
[JEE(Main)Aug.2021]

एक परवलय, जिसके शीर्ष तथा नाभि धनात्मक x-अक्ष पर मूल बिन्दु से क्रमशः R तथा S; (S > R) की दूरी पर हैं, कि नाभिलम्ब जीवा d hy Ecl bZ S
[JEE(Main)Aug.2021]

(A) 2 (S – R) (B) 4 (S – R) (C) 2 (S + R) (D) 4 (S + R)

Ans. B

Sol.



$a = S - R$

$\ell(LR) = 4a$

$= 4(S - R)$

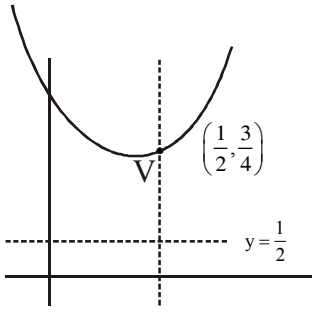
15. Consider the parabola with vertex $\left(\frac{1}{2}, \frac{3}{4}\right)$ and the directrix $y = \frac{1}{2}$. Let P be the point where the parabola meets the line $x = -\frac{1}{2}$. If the normal to the parabola at P intersects the parabola again at the point Q, then $(PQ)^2$ is equal to : **[JEE(Main)Aug.2021]**

माना एक परवलय का शीर्ष $\left(\frac{1}{2}, \frac{3}{4}\right)$ तथा नियता $y = \frac{1}{2}$ हैं। माना P एक बिन्दु है जहाँ परवलय, रेखा $x = -\frac{1}{2}$ से मिलता है। यदि P बिन्दु पर परवलय का अभिलम्ब परवलय को फिर से Q बिन्दु पर कटता है, तब $(PQ)^2$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\frac{125}{16}$ (B) $\frac{15}{2}$ (C) $\frac{25}{2}$ (D) $\frac{75}{8}$

Ans. A

Sol.



Equation of the parabola :

$\left(x - \frac{1}{2}\right)^2 = 4a\left(y - \frac{3}{4}\right)$

where $a = \frac{1}{4}$

$\Rightarrow \left(x - \frac{1}{2}\right)^2 = \left(y - \frac{3}{4}\right)$ .....(1)

when $x = -\frac{1}{2}$

$1 = y - \frac{3}{4}$ $\therefore y = \frac{7}{4}$

Coordinates of $P = \left(-\frac{1}{2}, \frac{7}{4}\right)$

For normal at P :

$$2\left(x - \frac{1}{2}\right) = \frac{dy}{dx}$$

$\therefore$ Slope of tangent (M_T) at $P = -2$

Slope of Normal at $P = \frac{1}{2}$

Equation of normal at P :

$y - \frac{7}{4} = \frac{1}{2}\left(x + \frac{1}{2}\right)$

$\Rightarrow 4y - 7 = 2x + 1$

$\Rightarrow 2x - 4y + 8 = 0$

$x - 2y + 4 = 0$ .....(2)

for Q

$\left(2y - 4 - \frac{1}{2}\right)^2 = \left(y - \frac{3}{4}\right)$

$\Rightarrow \frac{(4y-9)^2}{4} = \frac{4y-3}{4}$

$\Rightarrow 16y^2 - 72y + 81 = 4y - 3$

$\Rightarrow 16y^2 - 76y + 84 = 0$

$\Rightarrow 4y^2 - 19y + 21 = 0$

$\Rightarrow 4y^2 - 12y - 7y + 21 = 0$

$\Rightarrow 4y(y-3) - 7(y-3) = 0$

$(4y-7)(y-3)=0$

$y=\frac{7}{4}$ or 3

$\therefore$ for Q $y=3$

$\therefore x=2$

$\therefore Q=(2,3)$

$PQ=\sqrt{\left(2+\frac{1}{2}\right)^2+\left(3-\frac{7}{4}\right)^2}$

$=\sqrt{\frac{25}{4}+\frac{25}{16}}$

$PQ=\sqrt{\frac{25}{16}}$

$PQ^2=\frac{125}{16}$

16. A tangent line L is drawn at the point (2, –4) on the parabola $y^2 = 8x$. If the line L is also tangent to the circle $x^2 + y^2 = a$, then 'a' is equal to_____. **[JEE(Main)Aug.2021]**

परवलय $y^2 = 8x$ के बिन्दु (2, –4) पर एक स्पर्श रेखा L खींची गई है। यदि रेखा L वृत्त $x^2 + y^2 = a$ की भी स्पर्श रेखा है, तो 'a' बराबर gS_____ **[JEE(Main)Aug.2021]**

Ans. 2

Sol. $y\times(-4)=8\left(\frac{x+2}{2}\right)$

$-y = x + 2$

$x + y + 2 = 0$

$\left|\frac{\sqrt{D}}{a}\right| = \frac{C}{a}$

$|\sqrt{2}| = \sqrt{a}$

$a = 2$

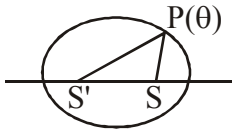
17. On the ellipse $\frac{x^2}{8}+\frac{y^2}{4}=1$ let P be a point in the second quadrant such that the tangent at P to ellipse is perpendicular to the line $x + 2y = 0$. Let S and S' be the foci of the ellipse and e be its eccentricity. If A is the area of the triangle SPS' then, the value of $(5 – e^2)$. A is : **[JEE(Main)Aug.2021]**

माना दीर्घवृत्त $\frac{x^2}{8}+\frac{y^2}{4}=1$ पर दूसरे चतुर्थांश में एक बिन्दु P इस प्रकार है कि P पर दीर्घवृत्त की स्पर्श रेखा, रेखा $x + 2y = 0$ के y Ecor ~gSAeku knh?kZUkd hukdHk, k¡S तथा S' है तथा इसकी उत्केन्द्रता e है। यदि त्रिभुज SPS' का क्षेत्रफल A है, तो $A(5 - e^2)$ का मान है : **[JEE(Main)Aug.2021]**

(A) 24 (B) 14 (C) 12 (D) 6

Ans. D

Sol.



E $\frac{x^2}{8}+\frac{y^2}{4}=1$

$e^2=1-\frac{b^2}{a^2}=1-\frac{4}{8}=\frac{1}{2}$

$P(2\sqrt{2}\cos\theta, 2\sin\theta)$

$A=\left|\frac{1}{2}(2ae)(b\sin\theta)\right|$

$=|abe\sin\theta|$

$=|4\sin\theta|$

Tangent at P

$\frac{x\cos\theta}{2\sqrt{2}}+\frac{y\sin\theta}{2}=1$

slope of tangent = 2

$-\frac{\cos\theta}{\sin\theta\sqrt{2}}=2$

$\cot\theta=-2\sqrt{2}$

$\Rightarrow|\sin\theta|=\frac{1}{3}$

$A=|4\sin\theta|=\frac{4}{3}$

$(5-e^2)A=\left(5-\frac{1}{2}\right)\frac{4}{3}$

$$= \frac{9}{2} + \frac{4}{3} = 6$$

18. The line $12x\cos\theta + 5y\sin\theta = 60$ is tangent to which of the following curves? **[JEE(Main)Aug.2021]**

रेखा $12x\cos\theta + 5y\sin\theta = 60$ निम्न में से किस वक्र की स्पर्श रेखा है ? **[JEE(Main)Aug.2021]**

(A) $x^2 + y^2 = 169$ (B) $25x^2 + 12y^2 = 3600$

(C) $144x^2 + 25y^2 = 3600$ (D) $x^2 + y^2 = 60$

Ans. C

Sol. $\frac{x\cos\theta}{5} + \frac{y\sin\theta}{12} = 1$

is tangent to the ellipse

$$\frac{x^2}{25} + \frac{y^2}{144} = 1$$

$$\Rightarrow 144x^2 + 252 = 3600$$

19. The locus of mid-points of the line segments joining (–3, –5) and the points on the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is : **[JEE(Main)Aug.2021]**

बिन्दु (–3, –5) को दीर्घवृत्त $\frac{x^2}{4} + \frac{y^2}{9} = 1$ के बिन्दुओं से मिलाने वाले रेखाखण्डों के मध्य–बिन्दुओं का बिन्दुपथ है : **[JEE(Main)Aug.2021]**

(A) $36x^2 + 16y^2 + 90x + 56y + 145 = 0$ (B) $9x^2 + 4y^2 + 18x + 8y + 145 = 0$

(C) $36x^2 + 16y^2 + 72x + 32y + 145 = 0$ (D) $36x^2 + 16y^2 + 108x + 80y + 145 = 0$

Ans. D

Sol. $\frac{x^2}{4} + \frac{y^2}{9} = 1$

$P(2\cos\theta, 3\sin\theta)$ $Q(-3,-5)$

$$\text{mid point} = \left(\frac{2\cos\theta - 3}{2}, \frac{3\sin\theta - 5}{2}\right) = (h, k)$$

$$2h = 2\cos\theta - 3$$

$$\frac{2h+3}{2} = \cos\theta \qquad \ldots.(1)$$

$$2k = 3\sin\theta - 5$$

$$\frac{2k+5}{3}=\sin\theta \qquad \text{.....(2)}$$

$$(1)^2+(2)^2$$

$$\left(\frac{2h+3}{2}\right)^2+\left(\frac{2k+5}{3}\right)^2=1$$

$$\frac{4h^2+9+12h}{4}+\frac{4k^2+25+20k}{9}=1$$

$$36h^2+81+108h+16k^2+100+80k=36$$

$$36h^2+16k^2+108h+80k+145=0$$

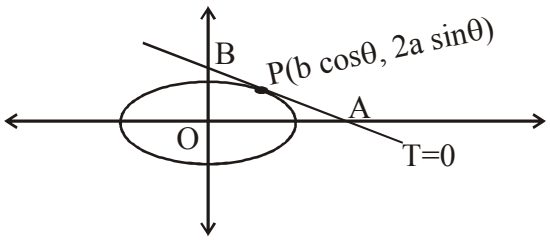
$$36x^2+16y^2+108x+80y+145=0$$

20. If the minimum area of the triangle formed by a tangent to the ellipse $\frac{x^2}{b^2}+\frac{y^2}{4a^2}=1$ and the co-ordinate axis is kab, then k is equal to_____. **[JEE(Main)Aug.2021]**

यदि दीर्घवृत्त $\frac{x^2}{b^2}+\frac{y^2}{4a^2}=1$ की एक स्पर्श रेखा तथा निर्देशांक अक्षों द्वारा बने त्रिभुज का न्यूनतम क्षेत्रफल kab है, तो k बराबर gS_____ **[JEE(Main)Aug.2021]**

Ans. 2

Sol.



$$E:\frac{x^2}{b^2}+\frac{y^2}{4a^2}=1$$

$$\underline{T=0}\quad \frac{x\cos\theta}{b}+\frac{y\sin\theta}{2a}=1$$

$$A\left(\frac{b}{\cos\theta},0\right),B\left(0,\frac{2a}{\sin\theta}\right)$$

$$\therefore \text{ Area } \Delta OAB=\frac{1}{2}.\frac{b}{\cos\theta}\frac{2a}{\sin\theta}=\frac{2ab}{\sin 2\theta}$$

$$A_{Min}=2ab=Kab$$

$$K=2$$

21. The locus of the mid points of the chord of the hyperbola $x^2 - y^2 = 4$, which touch the parabola $y^2 = 8x$, is :

[JEE(Main)Aug.2021]

अतिपरवलय $x^2 - y^2 = 4$ की उन जीवाओं, जो परवलय $y^2 = 8x$ को स्पर्श करती है, के मध्य बिन्दुओं का बिन्दुपथ है :

[JEE(Main)Aug.2021]

(A) $x^3(x-2) = y^2$ (B) $y^3(x-2) = x^2$ (C) $x^2(x-2) = y^3$ (D) $y^2(x-2) = x^3$

Ans. D

Sol. Tangent to $y^2 = 8x$ is $y = mx + \frac{2}{m}$

$y = \left(\frac{h}{k}\right)x + \frac{k^2 - h^2}{k}$ is the equation of chord with mid point (h, k)

Comparing above equations we get

$m = \frac{h}{k}$ and $\frac{2}{m} = \frac{k^2 - h^2}{k}$

$h^3 + 2k^2 - k^2h = 0$

Equation of locus is $x^3 + 2y^2 - y^2x = 0$

22. The point $P(-2\sqrt{6}, \sqrt{3})$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ having eccentricity $\frac{\sqrt{5}}{2}$. If the tangent and normal at P to the hyperbola intersect its conjugate axis at points Q and R respectively, then QR is equal to :

[JEE(Main)Aug.2021]

अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ जिसकी उत्केन्द्रता $\frac{\sqrt{5}}{2}$ है, एक बिन्दु $P(-2\sqrt{6}, \sqrt{3})$ है। यदि इस अतिपरवलय के बिन्दु P पर स्पर्श रेखा तथा अभिलम्ब अतिपरवलय के संयुग्मी अक्ष को क्रमशः बिन्दुओं Q तथा R पर काटते हैं, तो QR बराबर है :

[JEE(Main)Aug.2021]

(A) 6 (B) $3\sqrt{6}$ (C) $4\sqrt{3}$ (D) $6\sqrt{3}$

Ans. D

Sol. Let $P(a\sec\theta, b\tan\theta)$

$T : \frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1 \Rightarrow Q\left(0, -\frac{b}{\tan\theta}\right)$

$N : \frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2 \Rightarrow R\left(0, \frac{a^2 + b^2}{b}\tan\theta\right)$

$$QR = \left|\left(\frac{a^2+b^2}{b}\right)\tan\theta + \frac{b}{\tan\theta}\right|$$

$\because \frac{b^2}{a^2} = e^2 - 1 = \frac{1}{4} \Rightarrow 4b^2 = a^2$, so $b^2 = 3$ and $a^2 = 12$

$\sec\theta = -\sqrt{2}$ and $\tan\theta = 1$

Now, $QR = \left|\left(\frac{12+3}{\sqrt{3}}\right)1 + \frac{\sqrt{3}}{1}\right| = 6\sqrt{3}$

23. Let $x^2 + y^2 + Ax + By + C = 0$ be a circle passing through (0, 6) and touching the parabola $y = x^2$ at (2, 4). Then A + C is equal to _____. **[JEE(Main)June2022]**

माना बिन्दु (0, 6) से होकर जाने वाला तथा परवलय $y = x^2$ को (2, 4) पर स्पर्श करने वाला एक वृत्त $x^2 + y^2 + Ax + By + C = 0$ है, तो A + C बराबर है _____ **[JEE(Main)June2022]**

(A) 16 (B) $\frac{88}{5}$ (C) 72 (D) –8

Ans. A

Sol. Tangent to $y = x^2$ at (2, 4)

$$\frac{dy}{dx} = 2x$$

$$\left(\frac{dy}{dx}\right)_{x=2} = 4$$

$y - 4 = 4(x - 2)$

$4x - y - 4 = 0$

equation of circle touching $y = x^2$ at (2, 4)

$(x - 2)^2 + (y - 4)^2 + \lambda(4x - y - 4) = 0$

(0, 6) lies on it $\Rightarrow \lambda = \frac{4}{5}$

$$(x - 2)^2 + (y - 4)^2 + \frac{4}{5}(4x - y - 4) = 0$$

$$A = -4 + \frac{16}{5} \qquad c = 20 - \frac{16}{5}$$

$\Rightarrow A + C = 16$

24. If $y = m_1x + c_1$ and $y = m_2x + c_2$, $m_1 \neq m_2$ are two common tangents of circle $x^2 + y^2 = 2$ and parabola $y^2 = x$, then the value of $8|m_1m_2|$ is equal to : **[JEE(Main)June2022]**

यदि $x^2 + y^2 = 2$ तथा परवलय $y^2 = x$ की दो उभयनिष्ठ स्पर्श रेखाएँ $y = m_1x + c_1$ तथा $y = m_2x + c_2$, $m_1 \neq m_2$ हैं तो $8|m_1m_2|$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $3+4\sqrt{2}$ (B) $-5+6\sqrt{2}$ (C) $-4+3\sqrt{2}$ (D) $7+6\sqrt{2}$

Ans. C

Sol. Let equation of tangent $y = mx + \frac{1}{4m}$

then is also tangent to curve $\left|\frac{0-0+\frac{1}{4m}}{\sqrt{m^2+1}}\right| = \sqrt{2}$

$\frac{1}{16m^2} = 2\left(m^2+1\right)$

$32m^4 + 32m^2 - 1 = 0$

$m^2 = \frac{-32 \pm \sqrt{(32)^2 + 4\times 32}}{64}$

then the value of $8\left|m_1.m_2\right| = 8m^2$

$= 8\times\left(\frac{-32+\sqrt{(32)^2+4\times 32}}{64}\right)$

$= -4+\sqrt{18} = 3\sqrt{2}-4$

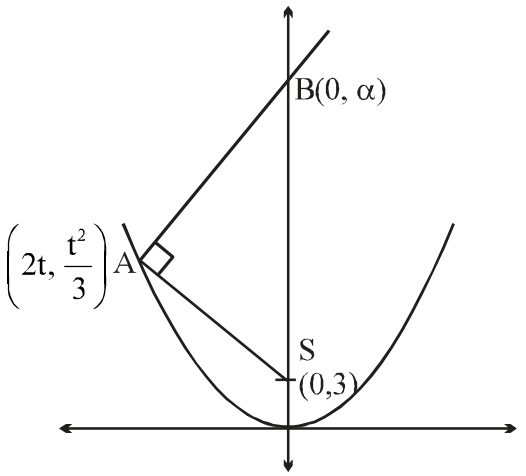
25. Let $x = 2t$, $y = \frac{t^2}{3}$ be conic. Let S be the focus and B be the point on the axis of the conic such that $SA \perp BA$, where A is any point on the conic. If k is the ordinate of the centroid of the ΔSAB, then $\lim\limits_{t\to 1} k$ is equal to : **[JEE(Main)June2022]**

माना $x = 2t$, $y = \frac{t^2}{3}$ एक शांकव है। माना शांकव की नाभि S है तथा शांकव के अक्ष पर बिन्दु B इस प्रकार है कि $SA \perp BA$, जहाँ A शांकव पर बिन्दु है। यदि ΔSAB के केन्द्रक की कोटि k है, तो $\lim\limits_{t\to 1} k$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{17}{18}$ (B) $\frac{19}{18}$ (C) $\frac{11}{18}$ (D) $\frac{13}{18}$

Ans. D

Sol.



parabola $x^2 = 12y$

$SA \perp SB$

so, $m_{AS} \cdot m_{AB} = -1$

$$\frac{\left(3-\frac{t^2}{3}\right)}{(0-2t)} \cdot \frac{\left(\alpha-\frac{t^2}{3}\right)}{(0-2t)} = -1$$

by solving $3\alpha = \frac{27t^2 + t^4}{t^2 - 9}$

ordinate of centriod of $\Delta SAB = K = \frac{\alpha + \frac{t^2}{3} + 3}{3}$

$$\lim_{t \to 1} k = \lim_{t \to 1} \frac{1}{9}\left(9 + t^2 + \frac{27t^2 + t^4}{\left(t^2 - 9\right)}\right) = \frac{13}{18}$$

26. If the line $y = 4 + kx$, $k > 0$, is the tangent to the parabola $y = x - x^2$ at the point P and V is the vertex of the parabola, then the slope of the line through P and V is : **[JEE(Main)June2022]**

यदि परवलय $y = x - x^2$ के बिन्दु P पर स्पर्श रेखा का समीकरण $y = 4 + kx$, $k > 0$ है तथा परवलय का शीर्ष V है, तो P तथा V से gkdj t kusokyh js[kk dh izo.krk gksxh **[JEE(Main)June2022]**

(A) $\frac{3}{2}$ (B) $\frac{26}{9}$ (C) $\frac{5}{2}$ (D) $\frac{23}{6}$

Ans. C

Sol. $L : y = 4 + kx$

$P : y = x - x^2$

$4 + kx = x - x^2$

$x^2 + (k-1)x + 4 = 0$

$D = 0$

$(k-1)^2 - 16 = 0 \Rightarrow k = -3, 5\ (k > 0)$

$k = 5$

$x^2 + 4x + 4 = 0 \Rightarrow x = 2$

$y = 2 - 4 = -2$

$P(2, -2)$

$P : y = -\left(x^2 - x + \frac{1}{4} - \frac{1}{4}\right)$

$y = \frac{1}{4} - \left(\frac{x-1}{2}\right)^2 \Rightarrow \left(x - \frac{1}{2}\right)^2 = -\left(y - \frac{1}{4}\right)$

$V = \left(\frac{1}{2}, \frac{1}{4}\right)$

$m_{PV} = \frac{5}{2}$

27. Let the normal at the point P on the parabola $y^2 = 6x$ pass through the point $(5, -8)$. If the tangent at P to the parabola intersects its directrix at the point Q, then the ordinate of the point Q is :

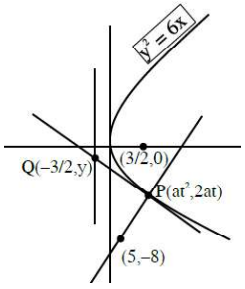
[JEE(Main)June2022]

माना परवलय $y^2 = 6x$ के बिंदु P पर अभिलंब, बिंदु $(5, -8)$ से होकर जाता है। यदि P पर परवलय की स्पर्श रेखा इसकी नियता को Q पर काटती है, तो बिंदु Q की कोटी है : **[JEE(Main)June2022]**

(A) -3 (B) $-\frac{9}{4}$ (C) $-\frac{5}{2}$ (D) -2

Ans. B

Sol.



Equation of normal : $y = -tx + 2at + at^3 \quad \left(a = \frac{3}{2}\right)$

since passing through $(5, -8)$, we get $t = -2$

Co-ordinate of P : $(6, -6)$

Equation of tangent at P : $x + 2y + 6 = 0$

Put $x = \frac{-3}{2}$ to get $Q\left(\frac{-3}{2}, \frac{-9}{4}\right)$

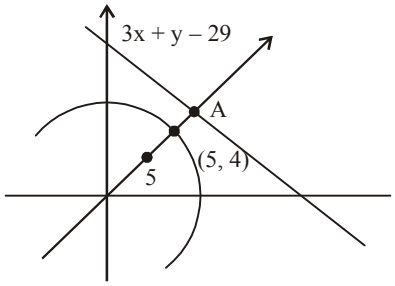
28. If the equation of the parabola, whose vertex is at (5, 4) and the directtrix is $3x + y - 29 = 0$, is $x^2 + ay^2 + bxy + cx + dy + k = 0$, then $a + b + c + d + k$ is equal to : **[JEE(Main)June2022]**

एक परवलय का शीर्ष (5, 4) तथा नियता $3x + y - 29 = 0$ हैं। यदि इसका समीकरण $x^2 + ay^2 + bxy + cx + dy + k = 0$ है, तो $a + b + c + d + k$ बराबर है : **[JEE(Main)June2022]**

(A) 575 (B) – 575 (C) 576 (D) – 576

Ans. D

Sol.



$$\frac{x-5}{3} = \frac{y-4}{1} = \frac{-(15+4-29)}{3^2+1^2}$$

$$\frac{x-5}{3} = \frac{y-4}{1} = 1$$

$x = 8, y = 5$

A(8, 5)

$S(x, y)$ $\frac{x+8}{2} = 5$ $\frac{y+5}{2} = 4$

$x = 2$ $y = 3$

S(2, 3) PS = PM

$\Rightarrow PS^2 = PM^2$

$$(x-2)^2 + (y-3)^2 = \left(\frac{3x+y-29}{\sqrt{10}}\right)^2$$

$\Rightarrow x^2 + 9y^2 - 6xy + 134x - 2y - 711 = 0$

$\Rightarrow a + b + c + d + k = -576$

29. If vertex of a parabola is (2, –1) and the equation of its directrix is $4x - 3y = 21$, then the length of its latus rectum is : **[JEE(Main)June2022]**

यदि एक परवलय का शीर्ष (2, –1) है इसकी नियता की समीकरण 4x – 3y = 21 है, तो इसकी नाभिलंब जीवा की लंबाई है :

[JEE(Main)June2022]

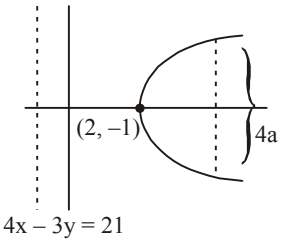
(A) 2 (B) 8 (C) 12 (D) 16

Ans. B

Sol. a = distance between vertex and direct x

$$a = \frac{|4\times 2 - 3\times 1 - 21|}{\sqrt{4^2+3^2}}$$

$$a = 2$$



Now, length of latus rectum = 4a

$= 4 \times 2 = 8$ units

30. Let P : $y^2 = 4ax$, $a > 0$ be a parabola with focus S. Let the tangents to the parabola P make an angle of $\frac{\pi}{4}$ with the line y = 3x + 5 touch the parabola P at A and B. Then the value of a for which A, B and S are collinear is :

[JEE(Main)June2022]

(A) 8 only (B) 2 only (C) $\frac{1}{4}$ only (D) any $a > 0$

माना P : $y^2 = 4ax$, $a > 0$ एक परवलय है, जिसकी नाभि S है। माना परवलय P की स्पर्श रेखाऐं, रेखा y = 3x + 5 से $\frac{\pi}{4}$ का कोण cukr hgas Hkki joy ; P को बिन्दुओं A तथा B पर स्पर्श करती हैं। तो बिन्दुओं A, B तथा a का मान है :

[JEE(Main)June2022]

(A) केवल 8 (B) केवल 2 (C) केवल $\frac{1}{4}$ (D) कोई भी $a > 0$

Ans. D

Sol. Given P : $y^2 = 4ax$, $(a > 0)$ and focus $S \equiv (a, 0)$

equation of tangent on parabola $y = mn + \frac{a}{m}$

Given line, y = 3x + 5

Angle between given line and tangent is $\frac{\pi}{4}$

$$\tan\frac{\pi}{4} = \left|\frac{m-3}{1+3m}\right| \Rightarrow m - 3 = \pm(1+3m)$$

$\Rightarrow m – 3 = 1 + 3m$

$\boxed{m = -2}$

$m – 3 = – 1 – 3m$

$\boxed{m = \frac{1}{2}}$

So, equation of one tangent : $y = -2 \times \frac{a}{2}$

equation of three tangent : $y = \frac{x}{z} + 29$

Point of contant are $\left(\frac{a}{(-2)^2}, \frac{-2a}{(-2)}\right)$ and $\left(\frac{a}{\left(\frac{1}{2}\right)^2}, \frac{-29}{\frac{1}{2}}\right)$

$\therefore A\left(\frac{a}{4}, a\right)$ and B(4a, –4a)

So, A, B and S are collineal

then $\boxed{\Delta ABS = 0}$

$$\frac{1}{2}\begin{vmatrix} \frac{a}{4} & a & 1 \\ 4a & -4a & 1 \\ a & 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{a}{4}(-4a - 0) - a(4a - a) + \left(0 - \left(-4a^2\right)\right) = 0$$

$$\Rightarrow -a^2 - 3a^2 + 4a^2 = 0$$

$$\Rightarrow 0 = 0$$

It is not depend on a.

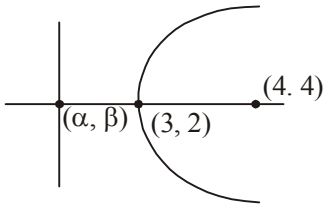
So, any a > 0

31. Let P_1 be a parabola with vertex (3, 2) and focus (4, 4) and P_2 be its mirror image with respect to the line x + 2y = 6. Then the directrix of P_2 is x + 2y = _____. **[JEE(Main)June2022]**

माना P_1 एक परवलय है जिसका शीर्ष (3, 2) है तथा नाभि (4, 4) है, तथा रेखा x + 2y = 6 के सापेक्ष P_1 का दर्पण प्रतिबिंब P_2 है। तो P_2 की नियता x + 2y = _____ है। **[JEE(Main)June2022]**

Ans. 10

Sol.



We know $\frac{\alpha+4}{2}=3$

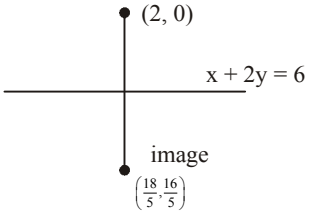
$\alpha = 2$

$\frac{\alpha+4}{2}=3$

$\beta = 0$

Foot of directrix = (2, 0)

image of (2, 0) with respect to line $x + 2y = 6$



Directrix of image of parabola is

$\xi + 2\psi = \lambda$

$\frac{18}{5}+\frac{32}{5}=\lambda$

$\lambda = 10$

$x + 2y = 10$

32. Let the common tangents to the curves $4(x^2+y^2) = 9$ and $y^2 = 4x$ intersect at the point Q. Let an ellipse, centrered *at the origin O, has lengths of semi-minor and semi-major axes equal to OQ and 6, respectively. If e and l* respectively denote the eccentricity and the length of the latus rectum of this ellipse, then l/e^2 is equal to _____.

[JEE(Main)June2022]

माना वक्रों $4(x^2+y^2) = 9$ तथा $y^2 = 4x$ की उभयनिष्ठ स्पर्श रेखाएँ, बिन्दु Q पर मिलती हैं। माना एक दीर्घवृत्त, जिसका केन्द्र मूलबिंदु O है, के अर्ध लघु अक्ष तथा अर्ध दीर्घ अक्ष की लंबाईयाँ क्रमशः OQ तथा 6 हैं। यदि इस दीर्घवृत्त की उत्तकेन्द्रता तथा नाभिलंब जीवा की y alkbZ Øe' k% e तथा l हैं, तो l/e^2 बराबर है _____ **[JEE(Main)June2022]**

Ans. D

Sol. $x^2 + y^2 = \frac{9}{4}$ $y^2 = 4x$

Equation tangent in slope form

$y = mx \pm \frac{3}{2}\sqrt{\left(1+m^2\right)}$ _____(1)

$y = mx + \frac{1}{m}$ _____(2)

compare (1) & (2)

$\pm\frac{3}{2}\sqrt{\left(1+m^2\right)} = \frac{1}{m}$

$9m^2(1 + m^2) = 4$

$9m^4 + 9m^2 - 4 = 0$

$9m^4 + 12m^2 - 3m^2 - 4 = 0$

$3m^2(3m^2 + 4) - (3m^2 + 4) = 0$

$m^2 = -\frac{4}{3}$ (Rejected)

$m^2 = \frac{1}{3} \Rightarrow m = \pm\frac{1}{\sqrt{3}}$

Equation of common tangent

$y = \frac{1}{\sqrt{3}}x + \sqrt{3}$

on X axis $y = 0$

$OQ = -3$

$b = |OQ| = 3$

$a = 6$

$b^2 = a^2(1 - e^2) \Rightarrow e^2 = 1 - \frac{9}{36} = \frac{3}{4}$

$\ell = \frac{2b^2}{a} = \frac{2\times 9}{6} = 3$

$\frac{\ell}{e^2} = \frac{3}{3/4} = 4$

33. A circle of radius 2 unit passes through the vertex and the focus of the parabola $y^2 = 2x$ and touches the parabola $y = \left(x - \frac{1}{4}\right)^2 + \alpha$, where $\alpha > 0$. Then $(4\alpha - 8)^2$ is equal to _____. **[JEE(Main)June2022]**

इकाई त्रिज्या का वृत्त, परवलय $y^2 = 2x$ के शीर्ष तथा नाभि से होकर जाता है तथा परवलय $y = \left(x - \frac{1}{4}\right)^2 + \alpha$, जहाँ $\alpha > 0$ है, को स्पर्श करता है। तो $(4\alpha - 8)^2$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 63

Sol. Vertex and focus of parabola $y^2 = 2x$

are V (0, 0) and $S\left(\frac{1}{2}, 0\right)$ resp. Let equation of circle be $(x - h)^2 + (y - k)^2 = 4$

$\therefore$ Circle passes through (0, 0)

$\Rightarrow h^2 + k^2 = 4$ _____(1)

$\therefore$ Circle passes through $\left(\frac{1}{2},0\right)$

$$\left(\frac{1}{2}-h\right)^2+k^2=4$$

$$\Rightarrow h^2+k^2-h=\frac{15}{4} \quad \text{_____(2)}$$

On solving (1) and (2)

$$4-h=\frac{15}{4}$$

$$h=4-\frac{15}{4}=\frac{1}{4}$$

$$k=+\frac{\sqrt{63}}{4}$$

$k=-\frac{\sqrt{63}}{4}$ is rejected as circle with centre $\left(\frac{1}{4},-\frac{\sqrt{63}}{4}\right)$ can't touch given parabola. Equatioin of circle is

$$\left(x-\frac{1}{4}\right)^2+\left(k-\frac{\sqrt{63}}{4}\right)^2=4$$

from figure

$$\alpha=2+\frac{\sqrt{63}}{4}=\frac{8+\sqrt{63}}{4}$$

$$4\alpha-8=\sqrt{63}$$

$$(4\alpha-8)^2=63$$

34. The line $y=x+1$ meets the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$ at two points P and Q. If r is the radius of the circle with PQ as diameter then $(3r)^2$ is equal to : **[JEE(Main)June2022]**

रेखा $y=x+1$, दीर्घवृत्त $\frac{x^2}{4}+\frac{y^2}{2}=1$ को दो बिन्दुओं P तथा Q पर मिलती है। यदि वृत्त, जिसका एक व्यास PQ है, की त्रिज्या r है, तो $(3r)^2$ बराबर है : **[JEE(Main)June2022]**

(A) 20 (B) 12 (C) 11 (D) 8

Ans. A

Sol. $L : y=x+1$

$E: \frac{x^2}{4}+\frac{y^2}{2}=1$

$x^2 + 2y^2 = 4$

Quadratic in x

$x^2 + 2(x + 1)^2 = 4$

$3x^2 + 4x - 2 = 0$

Quadratic in y

$(y - 1)^2 + 2y^2 = 4$

$3y^2 - 2y - 3 = 0$

Eq. of circle

$(3x^2 + 4x - 2) + (3y^2 - 2y - 3) = 0$

$x^2+y^2+\frac{4}{3}x-\frac{2}{3}y-\frac{5}{3}=0$

$r=\sqrt{\frac{4}{9}+\frac{1}{9}-\frac{5}{3}}=\frac{\sqrt{20}}{3}$

$(3r)^2 = 20$

35. If m is the slope of a common tangent to the curves $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2 + y^2 = 12$, then $12m^2$ is equal to:

[JEE(Main)June2022]

यदि वक्रों $\frac{x^2}{16}+\frac{y^2}{9}=1$ तथा $x^2 + y^2 = 12$ की एक उभ्यनिष्ठ स्पर्श रेखा की प्रवणता m है, तो $12m^2$ बराबर है :

[JEE(Main)June2022]

(A) 6 (B) 9 (C) 10 (D) 12

Ans. B

Sol. $\frac{x^2}{16}+\frac{y^2}{9}=1$ $x^2 + y^2 = 12$

$y = mx \pm \sqrt{a^2m^2 + b^2}$

$y = mx \pm \sqrt{16m^2 + 9}$

Apply COT

$mn - 9 \pm \sqrt{16m^2 + 9} = 0$

$\left|\frac{\pm\sqrt{16m^2+9}}{\sqrt{1+m^2}}\right| = 2\sqrt{3}$

$16m^2 + 9 = 12(1 + m^2)$

$16m^2 + 9 = 12 + 12m^2$

$4m^2 = 3$

$$m = \pm\frac{\sqrt{3}}{2}$$

$$m^2 = \frac{3}{4}$$

$$12m^2 = 12\left(\frac{3}{4}\right) = 9$$

36. The locus of the mid point of the line segment joining the point (4, 3) and the points on the ellipse $x^2 + 2y^2 = 4$ is an ellipse with ecentricity : **[JEE(Main)June2022]**

दीर्घवृत्त $x^2 + 2y^2 = 4$ के बिन्दुओं को बिन्दु (4, 3) से मिलाने वाले रेखाखण्ड के मध्यबिन्दु का बिन्दुपथ एक दीर्घवृत्त है, जिसकी mRd Hz kgS **[JEE(Main)June2022]**

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Ans. C

Sol. $\frac{x^2}{4} + \frac{y^2}{2} = 1$

$a = 2$ $b = \sqrt{2}$

$p(\theta) = \left(2\cos\theta, \sqrt{2}\sin\theta\right)$ A(4, 3)

$$S(h,k) = \left(\frac{2\cos\theta + 4}{2}, \frac{\sqrt{2}\sin\theta + 3}{2}\right)$$

$$\frac{2h-4}{2} = \cos\theta \quad \text{———(1)}$$

$$\frac{2k-3}{\sqrt{2}} = \sin\theta \quad \text{———(2)}$$

$(1)^2 + (2)^2$

$$1 = \left(\frac{2h-4}{2}\right)^2 + \left(\frac{2k-3}{\sqrt{2}}\right)^2$$

$$1 = (h-2)^2 + \frac{(2k-3)^2}{2}$$

$$\frac{(x-2)^2}{1} + \frac{(2y-3)^2}{2} = 1$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

$e^2 = 1 - \frac{1}{2}$

$e^2 = \frac{1}{2}$ $\Rightarrow e = \frac{1}{\sqrt{2}}$

37. Let the ecentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b, e = \frac{1}{4}$. If this ellipse passes through the point $\left(-4\sqrt{\frac{2}{5}}, 3\right)$, then $a^2 + b^2$ is qual to : **[JEE(Main)June2022]**

माना दीर्घवृत्त $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ की उत्केन्द्रता $\frac{1}{4}$ है। यदि यह दीर्घवृत्त बिन्दु $\left(-4\sqrt{\frac{2}{5}}, 3\right)$ से होकर जाता है, तो $a^2 + b^2$ बराबर gS **[JEE(Main)June2022]**

(A) 29 (B) 31 (C) 32 (D) 34

Ans. B

Sol. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$e = \frac{1}{4} = \sqrt{1 - \frac{b^2}{a^2}}$

$\frac{1}{16} = \frac{a^2 - b^2}{a^2}$

$b^2 = \frac{15}{16}a^2$

Now point $\left(-4\sqrt{\frac{2}{5}}, 3\right)$

$\frac{16 \times \frac{2}{5}}{a^2} + \frac{9}{b^2} = 1$

$\frac{32}{5a^2} + \frac{9 \times 16}{15a^2} = 1$

$5a^2 = 80$

$a^2 = 16$

$b^2 = \frac{15}{16} \times 16 = 15$

$a^2 + b^2 = 31$

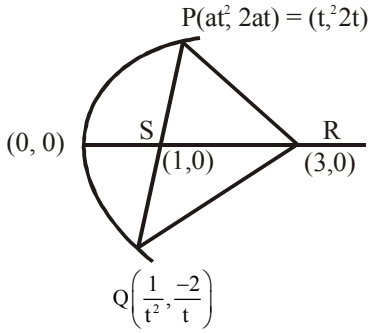
38. Let PQ be a focal chord of the parabola $y^2 = 4x$ such that it subtends an angle of $\frac{\pi}{2}$ at the point (3, 0). Let the line segment PQ be also a focal chord of the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$. If e is the eccentricity of the ellipse E, then the value of $\frac{1}{e^2}$ is equal to : **[JEE(Main)June2022]**

माना परवलय $y^2 = 4x$ की एक नाभीय जीवा PQ, बिंदु (3, 0) पर $\frac{\pi}{2}$ कोण बनाती है माना रेखाखण्ड PQ, दीर्घवृत्त $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$, की भी एक नाभीय जीवा है। यदि दीर्घवृत्त E की उत्तकेन्द्रता e है, तो $\frac{1}{e^2}$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $1+\sqrt{2}$ (B) $3+2\sqrt{2}$ (C) $1+2\sqrt{3}$ (D) $4+5\sqrt{3}$

Ans. B

Sol.



$M_{PR} . M_{QR} = -1$

$$\frac{2t-0}{t^2-3} \cdot \frac{\frac{-2}{t}-0}{\frac{1}{t^2}-3} = -1$$

$$\frac{4t^2}{\left(t^2-3\right)\left(1-3t^2\right)} = 1$$

$t = +1$

P(1, 2) Q (1, – 2)

PQ also focal chord of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a > b)$

$\frac{1}{a^2} + \frac{4}{b^2} = 1$, $ae = 1$

$$\frac{1}{a^2}+\frac{4}{a^2\left(1-e^2\right)}=1$$

$$\frac{1}{1}+\frac{4}{1-e^2}=\frac{1}{e^2} \Rightarrow e^4-6e^2-1=0$$

$$\Rightarrow \frac{1}{e^2}=3+2\sqrt{2}$$

39. If two tangents drawn from a point (α, β) lying on the ellipse $25x^2+4y^2=1$ to the parabola $y^2=4x$ are such that the slope of one tangent is four times the other, then the value of $(10\alpha+5)^2+(16\beta^2+50)^2$ equals _____.

[JEE(Main)June2022]

दीर्घवृत्त $25x^2+4y^2=1$ पर एक बिन्दु (α, β) से परवलय $y^2=4x$ पर दो स्पर्श रेखाएँ खीची जाती हैं। यदि एक स्पर्श रेखा की प्रवणता nwj sd hpkj xqkgSr ks $(10\alpha+5)^2+(16\beta^2+50)^2$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 2929

Sol. $\alpha=\frac{1}{5}\cos\theta, \beta=\frac{1}{2}\sin\theta$

Equation of tangent to $y^2=4x$

$$y=mx+\frac{1}{m}$$

It passes through (α, β)

$$\frac{1}{2}\sin\theta=m\frac{1}{5}\cos\theta+\frac{1}{m}$$

$$m^2\left(\frac{\cos\theta}{5}\right)-m\left(\frac{1}{2}\sin\theta\right)+1=0$$

It has two rootss m_1 and m_2 where $m_1=4m_2$

$$m_1+m_2=\frac{\frac{1}{2}\sin\theta}{\frac{\cos\theta}{5}}$$

$$m_1m_2=\frac{5}{\cos\theta}$$

After eliminating m_1 and m_2

$$\cos\theta=\frac{-5\pm\sqrt{29}}{2}$$

$$\alpha=\frac{-5\pm\sqrt{29}}{10} \Rightarrow 10\alpha+5=\pm\sqrt{29}$$

$$\beta^2=\frac{1}{4}\sin^2\theta \Rightarrow 16\beta^2=-50\pm10\sqrt{29}$$

$(10\alpha+5)^2+(16\beta^2+50)^2=2929$

40. Let $\lambda x-2y=\mu$ be a tangent to the hyperbola $a^2x^2-y^2=b^2$. then $\left(\frac{\lambda}{a}\right)^2-\left(\frac{\mu}{b}\right)^2$ is equal to :

[JEE(Main)June2022]

माना अतिपरवलय $a^2x^2-y^2=b^2$ की एक स्पर्श रेखा $\lambda x-2y=\mu$ है। तो $\left(\frac{\lambda}{a}\right)^2-\left(\frac{\mu}{b}\right)^2$ बराबर है :

[JEE(Main)June2022]

(A) -2 (B) -4 (C) 2 (D) 4

Ans. D

Sol. Hyperbola $a^2x^2-y^2=b^2$

$$\frac{x^2}{\left(\frac{b^2}{a^2}\right)}-\frac{y^2}{b^2}=1$$

Apply cot

$c^2=a^2m^2-b^2$

$$\left(-\frac{\mu}{2}\right)^2=\left(\frac{b^2}{a^2}\right)\left(\frac{\lambda}{2}\right)^2-b^2$$

$$\frac{\lambda^2}{a^2}-\frac{\mu^2}{b^2}=4$$

41. The normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{9}=1$ at the point $\left(8,3\sqrt{3}\right)$ on it passes through the point :

[JEE(Main)June2022]

अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{9}=1$ के बिन्दु $\left(8,3\sqrt{3}\right)$ पर अभिलंब किस बिन्दु से होकर जाता है? **[JEE(Main)June2022]**

(A) $\left(15,-2\sqrt{3}\right)$ (B) $\left(9,2\sqrt{3}\right)$ (3) $\left(-1,9\sqrt{3}\right)$ (D) $\left(-1,6\sqrt{3}\right)$

Ans. C

Sol. $\frac{x^2}{a^2}-\frac{y^2}{9}=1$ $\left(8,3\sqrt{3}\right)$

$$\frac{64}{a^2}-3=1$$

$$\frac{64}{a^2}=4 \quad \Rightarrow \frac{16}{a^2}=1$$

$\frac{x^2}{16}-\frac{y^2}{9}=1$ $\quad a^2 = 16$

$\frac{16x}{8}+\frac{9y}{3\sqrt{3}}=25$

$2y+\sqrt{3}y=25$

$\left(-1,9\sqrt{3}\right)$ $\quad \Rightarrow -2+27=25$

42. Let the eccentricity of the hyperbola $H:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be $\sqrt{\frac{5}{2}}$ and length of its latus retum be $6\sqrt{2}$, If $y=2x+c$ is a tangent to the hyperbola H, then the value of c^2 is equal to : **[JEE(Main)June2022]**

माना अतिपरवलय $H:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ की उत्केन्द्रता $\sqrt{\frac{5}{2}}$ है तथा इसकी नाभिलंब जीवा की लंबाई $6\sqrt{2}$ है। यदि अतिपरलय H की एक स्पर्श रेखा $y = 2x + c$ है, तो c^2 का मान है : **[JEE(Main)June2022]**

(A) 18 (B) 20 (C) 24 (D) 32

Ans. B

Sol. $1+\frac{b^2}{a^2}=\frac{5}{2}\Rightarrow\frac{b^2}{a^2}=\frac{3}{2}$ .....(1)

and $\frac{2b^2}{a}=6\sqrt{2}$ .....(2)

from (1) and (2) $a=2\sqrt{2},\ b=2\sqrt{3}$

Now condition of tangency

$c^2 = a^2m^2 - b^2$

$\Rightarrow$ $c^2 = 8 \times 4 - 12 = 20$

43. Let $a > 0, b > 0$. Let e and l repsectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. Let e′ and l' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2=\frac{11}{14}l$ and $(e')^2=\frac{11}{8}l'$, then the value of $77a + 44b$ is equal to : **[JEE(Main)June2022]**

माना $a > 0, b > 0$ हैं। माना अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ की उत्केन्द्रता तथा नाभिलंब जीवा की लंबाई क्रमशः e तथा l है। माना इसके संयुग्मी अतिपरवलय की उत्केन्द्रता तथा नाभिलंब जीवा की लंबाई क्रमशः e′ तथा l' हैं। यदि $e^2=\frac{11}{14}l$ तथा $(e')^2=\frac{11}{8}l'$ हैं, तो $77a + 44b$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 100 (B) 110 (C) 120 (D) 130

Ans. D

Sol. Given $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ ————(1)

$e=\sqrt{1+\frac{b^2}{a^2}}$ and $\ell=\frac{2b^2}{a}$

$\Rightarrow e^2=\frac{11}{14}\ell$ $\Rightarrow \frac{a^2+b^2}{a^2}=\frac{11}{4}\times\frac{2b^2}{a}$

$\Rightarrow a^2+b^2=\frac{11}{7}b^2a$ ————(2)

In conjugate hyperbola

$$-\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

$e'=\sqrt{1+\frac{a^2}{b^2}}$, $\ell'=\frac{2a^2}{b}$

given $(e')^2=\frac{11}{8}\ell'$ $\Rightarrow \frac{b^2+a^2}{b^2}=\frac{11}{8}\times\frac{2a^2}{b}$

$\Rightarrow a^2+b^2=\frac{11}{4}a^2b$ ————(3)

from eq. (2) and eq. (3)

$$7a=4b \Rightarrow a=\frac{4}{7}b$$

from eq. (2)

$44b = 65$ and $77a = 65$

$\therefore\ 77a + 44b = 65 + 65 = 130$

44. Let the hyperbola H: $\frac{x^2}{a^2}-y^2=1$ and the ellipse E : $3x^2+4y^2=12$ be such that the length of latus rectum of H is equal to the length of latus rectum of E. If e_H and e_E are the ecentricities of H and E respectively, then the value of $12\left(e_H^2+e_E^2\right)$ is equal to _____. **[JEE(Main)June2022]**

माना अतिपरवलय H : $\frac{x^2}{a^2}-y^2=1$ की नाभिलंब जीवा की लंबाई, दीर्घवृत्त E : $3x^2+4y^2+4y^2=12$ की नाभिलंब जीवा की लम्बाई d scj kcj gSA; fn H तथा E की उत्केन्द्रताएँ क्रमशः e_H तथा e_E है, तो $12\left(e_H^2+e_E^2\right)$ का मान बराबर है _____

[JEE(Main)June2022]

Ans. 42

Sol. $\frac{x^2}{a^2}-\frac{y^2}{1}=1$ & $\frac{x^2}{4}+\frac{y^2}{3}=1$

By equations length of lates-reactum

$$\frac{2\times1}{a}=\frac{2\times3}{2}$$

$$a=\frac{2}{3}$$

$$e_H=\sqrt{1+\frac{b^2}{a^2}}=\sqrt{1+\frac{1\times9}{9}}=\sqrt{\frac{13}{4}}$$

$$e_E=\sqrt{1-\frac{3}{4}}=\frac{1}{2}$$

value = $12(e_H^2+e_E^2)$

$$=12\left(\frac{13}{4}+\frac{1}{4}\right)$$

$$=42$$

45. Let the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be $\frac{5}{4}$. If the equation of the normal at the point $\left(\frac{8}{\sqrt{5}},\frac{12}{5}\right)$ on the hyperbola is $8\sqrt{5}x+\beta y=\lambda$, then $\lambda-\beta$ is equal to _____. **[JEE(Main)June2022]**

माना अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ की उत्केन्द्रता $\frac{5}{4}$ है। यदि अतिपरवलय के बिन्दु $\left(\frac{8}{\sqrt{5}},\frac{12}{5}\right)$ पर अभिलम्ब का समीकरण $8\sqrt{5}x+\beta y=\lambda$ है, तो $\lambda-\beta$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 85

Sol. H: $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

$$e^2=1+\frac{b^2}{a^2}=\frac{25}{16}\Rightarrow\frac{b^2}{a^2}=\frac{9}{16}$$

eq. of normal at (x_1, y_1)

$$\frac{a^2x}{x_1}+\frac{b^2y}{y_1}=a^2+b^2$$

$$\frac{x}{x_1}+\frac{\frac{b^2}{a^2}}{y_1}y=1+\frac{b^2}{a^2}$$

$$\frac{x}{x_1}+\frac{9y}{16y_1}=\frac{25}{16}$$

$$\frac{x}{\frac{8}{\sqrt{5}}}+\frac{9y}{16.\frac{12}{5}}=\frac{25}{16}$$

$$8\sqrt{5}x+15y=100$$

$$\beta=15,\ \gamma=100$$

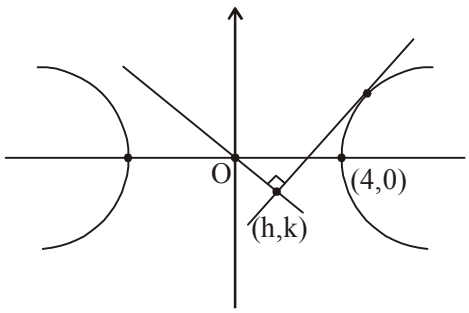
$$\gamma-\beta=85$$

46. Let a line L_1 be tangent to the hyperbola $\frac{x^2}{16}-\frac{y^2}{4}=1$ and let L_2 be the line passing through the origin and perpendicular to L_1. If the locus of the point of intersection of L_1 and L_2 is $(x^2+y^2)^2=\alpha x^2+\beta y^2$, then $\alpha+\beta$ is equal to _____. **[JEE(Main)June2022]**

माना L_1 अतिपरवलय $\frac{x^2}{16}-\frac{y^2}{4}=1$ की एक स्पर्श रेखा है तथा रेखा L_2 मूलबिन्दु से होकर जाती है और L_1 के लम्बवत है। यदि L_1 तथा L_2 के प्रतिच्छेदन बिन्दु का बिन्दुपथ $(x^2+y^2)^2=\alpha x^2+\beta y^2$ है, तो $\alpha+\beta$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 12

Sol.



tangent :

$$\frac{x\sec\theta}{4}-\frac{y\tan\theta}{2}=1$$

$$m_1=\frac{\sec\theta}{4}\times\frac{2}{\tan\theta}=\frac{\sec\theta}{2\tan\theta}$$

$$m_2=\frac{k}{h}$$

$\because$ $m_1\times m_2=-1$

$$\frac{k}{h}\times\frac{\sec\theta}{2\tan\theta}=-1$$

$$\frac{k}{2h\sin\theta}=-1$$

$$\sin\theta = -\frac{k}{2h}$$

$$\cos\theta = \frac{\sqrt{4h^2 - k^2}}{2h}$$

$$\frac{h}{4\cos\theta} - \frac{k\sin\theta}{2\cos\theta} = 1$$

$$\frac{h.2h}{4\sqrt{4h^2 - k^2}} + \frac{k \times \frac{k}{2h}}{2 \times \frac{\sqrt{4h^2 - k^2}}{2h}} = 1$$

$$\frac{h^2}{2\sqrt{4h^2 - k^2}} + \frac{k^2}{2\sqrt{4h^2 - k^2}} = 1$$

$$\left(h^2 + k^2\right) = 2\sqrt{4h^2 - k^2}$$

$$\left(x^2 + y^2\right) = 2\sqrt{4x^2 - y^2}$$

$$\left(x^2 + y^2\right)^2 = 4\left(4x^2 - y^2\right)$$

$\alpha = 16, \beta = -4$

$\alpha + \beta = 16 - 4 = 12$

47. Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a > 0, b > 0$, be a hyperbola such that the sum of lengths of the transverse and the conjugate axes is $4\left(2\sqrt{2} + \sqrt{14}\right)$. If the eccentricity H is $\frac{\sqrt{11}}{2}$, then the value of $a^2 + b^2$ is equal to _____.

[JEE(Main)June2022]

माना अतिपरवलय $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a > 0, b > 0$, के लिए अनुप्रस्थ तथा संयुग्मी अक्षों की लंबाईयों का योग $4\left(2\sqrt{2} + \sqrt{14}\right)$ हैं ; fn H की उत्केन्द्रता $\frac{\sqrt{11}}{2}$ है, तो $a^2 + b^2$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 88

Sol. $2a + 2b = 4\left(2\sqrt{2} + \sqrt{14}\right)$ $\quad H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, a > 0,\ b > 0$

$a + b = 4\sqrt{2} + 2\sqrt{14}$ .....(1)

$e^2 = 1 + \frac{b^2}{a^2} = \frac{11}{4}$

$$\frac{b^2}{a^2}=\frac{7}{4}$$

$$\frac{b}{a}=\frac{\sqrt{7}}{2}$$

$$\frac{a}{b}=\frac{2}{\sqrt{7}}\Rightarrow\frac{a+b}{a-b}=\frac{2+\sqrt{7}}{2-\sqrt{7}}\Rightarrow\frac{a+b}{2+\sqrt{7}}=\frac{a-b}{2-\sqrt{7}}$$

$$a-b=2\sqrt{2}\left(2-\sqrt{7}\right)$$

$$b-a=2\sqrt{14}-4\sqrt{2}$$

$$2b=4\sqrt{14}\Rightarrow b=2\sqrt{14}$$

$$\Rightarrow a^2+b^2=32+56=88$$

48. The equation of a common tangent to the parabolas $y=x^2$ and $y=-(x-2)^2$ is :

[JEE(Main)July2022]

परवलयों $y=x^2$ तथा $y=-(x-2)^2$ की एक उभयनिष्ठ स्पर्श रेखा का समीकरण है : **[JEE(Main)July2022]**

(A) $y=4(x-2)$ (B) $y=4(x-1)$ (C) $y=4(x+1)$ (D) $y=4(x+2)$

Ans. B

Sol. Equation of tangent of $y=x^2$ be

$$tx=y+at^2 \qquad ...(1)$$

$$y=tx-\frac{t^2}{4}$$

Solve with $y=-(x-2)^2$

$$tx=\frac{t^2}{4}=-\left(x-2\right)^2$$

$$x^2+x\left(t-4\right)-\frac{t^2}{4}+4=0$$

$D=0$

$$\left(t-4\right)^2-4.\left(4-\frac{t^2}{4}\right)=0$$

$t^2-4t=0$

$t=0$ or $t=4$

From eq. (1), required common tangent is

$y=4(x-1)$

49. Let P(a, b) be a point on the parabola $y^2 = 8x$ such that the tangent at P passes through the centre of the circle $x^2 + y^2 - 10x - 14y + 65 = 0$. Let A be the product of all possible values of a and B be the product of all possible values of b. Then the value of A + B is equal to : **[JEE(Main)July2022]**

माना परवलय $y^2 = 8x$ के बिन्दु P (a, b) पर स्पर्श रेखा, वृत्त $x^2 + y^2 - 10x - 14y + 65 = 0$ के केन्द्र से होकर जाती है। माना a के l Hkh l Hkh laHko eku ksa dk xq.kuQy A है तथा b के सभी संभव मानों का गुणनफल B है। तो A + B का मान है: **[JEE(Main)July2022]**

(A) 0 (B) 25 (C) 40 (D) 65

Ans. D

Sol. P(a, b) is point on $y^2 = 8x$, such that tangent at P

pass through centre of $x^2 + y^2 - 10x - 14y + 65 = 0$

i.e. (5, 7)

Tangent at P(at^2, 2at) is $ty = x + at^2$

A = 2 & it pass through (5, 7)

$7t = 5 + 2t^2$

$\Rightarrow t = 1, t = \frac{5}{2}$

$\therefore$ P (at^2, 2at) $\Rightarrow$ (2, 4) when t = 1

$\& \left(\frac{25}{2}, 10\right)$ when $t = \frac{5}{2}$

$\therefore A = 2 \times \frac{25}{2} = 25$

$B = 4 \times 10 = 40$ $\quad \therefore A + B = 65$

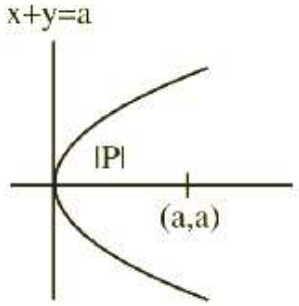
50. If the length of the latus rectum of a parabola, whose focus is (a, a) and the tangent at its vertex is x + y = a, is 16, then |a| is equal to : **[JEE(Main)July2022]**

यदि एक परवलय, जिसकी नाभि (a,a) है तथा जिसके शीर्ष पर स्पर्श रेखा का समीकरण x + y = a है, की नाभिलंब जीवा की लंबाई 16 है, तो |a| बराबर है : **[JEE(Main)July2022]**

(A) $2\sqrt{2}$ (B) $2\sqrt{3}$ (C) $4\sqrt{2}$ (D) 4

Ans. C

Sol.



$$|P| = \left|\frac{a}{\sqrt{2}}\right| = \frac{16}{4} = 4$$

$$|a| = 4\sqrt{2}$$

51. If the tangents drawn at the points P and Q on the parabola $y^2 = 2x - 3$ intersect at the point R(0, 1), then the orthocentre of the triangle PQR is : **[JEE(Main)July2022]**

यदि परवलय $y^2 = 2x - 3$ के बिन्दुओं P तथा Q पर खींची गई स्पर्श रेखाऐं बिन्दु R(0, 1), पर मिलती हैं। तो त्रिभुज PQR का लंब d sit gS **[JEE(Main)July2022]**

(A) (0, 1) (B) (2, –1) (C) (6, 3) (D) (2, 1)

Ans. B

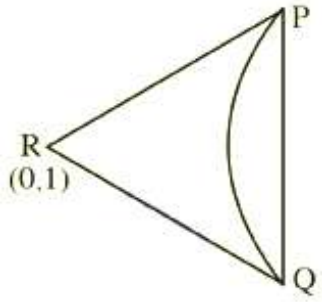
Sol. $y^2 = 2x - 3$ _____(1)

Equation of chord of contact

PQ : r = 0

yx1 = (x + 0) – 3

y = x – 3 _____(2)



from (1) and (2)

$(x . 3)^2 = 2x - 3$

$x^2 - 8x + 12 = 0$

(x – 2)(x – 6) = 0

x = 2or 6

$y = -1 \text{or } 3$

$MPQ = \frac{1}{4} = 1$

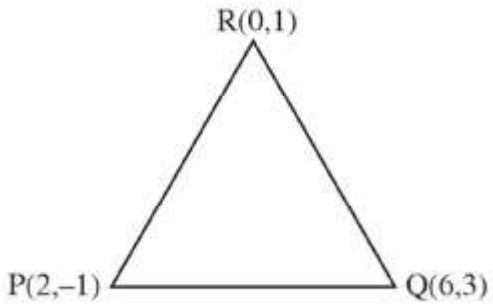
$MQR = \frac{2}{6} = \frac{1}{3}$

$MPR = \frac{2}{6} = \frac{1}{3}$

$MPR = \frac{2}{-2} = -1$

$MPQ \times MPR = - \Rightarrow PQ \perp PR$

Orthocentre = P (2, –1)



52. The sum of diameters of the circles that touch (i) the parabola $75x^2 = 64(5y - 3)$ at the point $\left(\frac{8}{5}, \frac{6}{5}\right)$ and (ii) the y-axis, is equal to _____. **[JEE(Main)July2022]**

वृत्तों, जो (i) परवलय $75x^2 = 64(5y - 3)$ को बिन्दु $\left(\frac{8}{5}, \frac{6}{5}\right)$ पर तथा (ii) y-अक्ष को स्पर्श करते हैं, के व्यासों का योग है _____

[JEE(Main)July2022]

Ans. 10

Sol. $x^2 = \frac{64.5}{75}\left(y - \frac{3}{5}\right)$

equation of tangent at $\left(\frac{8}{5}, \frac{6}{5}\right)$

$$x - \frac{8}{5} = \frac{64}{15}\left(\frac{y + \frac{6}{5}}{2} - \frac{3}{5}\right)$$

$3x - 4y = 0$

equation of family of circle is

$$\left(x - \frac{8}{5}\right)^2 + \left(y - \frac{6}{5}\right)^2 + \lambda(3x - 4y) = 0$$

It touches y axis so $f^2 = c$

$$x^2 + y^2 + x\left(3\lambda - \frac{16}{5}\right) + y\left(-4\lambda - \frac{12}{5}\right) + 4 = 0$$

$$\frac{\left(4\lambda+\frac{12}{5}\right)^2}{4}=4$$

$$\lambda=\frac{2}{5} \text{ or } \lambda=-\frac{8}{5}$$

$$\lambda=\frac{2}{5}, \quad r=1$$

$$\lambda=-\frac{8}{5}, \ r=4$$

$d_1 + d_2 = 10$

53. Two tangent lines l_1 and l_2 are drawn from the point (2, 0) to the parabola $2y^2 = -x$. If the lines l_1 and l_2 are also tangent to the circle $(x-5)^2 + y^2 = r$, then 17r is equal to _____. **[JEE(Main)July2022]**

बिन्दु (2, 0) से परवलय $2y^2 = -x$ पर दो स्पर्श रेखाएँ l_1 तथा l_2 खींची गई हैं। यदि रेखाएँ l_1 तथा l_2 वृत्त $(x-5)^2 + y^2 = r$ की 17r बराबर है _____ **[JEE(Main)July2022]**

Ans. 9

Sol. $y^2 = -\frac{x}{2}$

$$y = mx - \frac{1}{8m}$$

this tangent pass through (2, 0)

$m = \pm\frac{1}{4}$ i.e., one tangent is $x - 4y - 2 = 0$

$17r = 9$

54. If the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the line $\frac{x}{7}+\frac{y}{2\sqrt{6}}=1$ on the x-axis and the line $\frac{x}{7}-\frac{y}{2\sqrt{6}}=1$ on the y-axis, then the eccentricity of the ellipse is : **[JEE(Main)July2022]**

यदि दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, रेखा $\frac{x}{7}+\frac{y}{2\sqrt{6}}=1$ को x-अक्ष पर तथा रेखा $\frac{x}{7}-\frac{y}{2\sqrt{6}}=1$ को y-अक्ष पर मिलता है, तो दीर्घवृत्त की mRlHiz kgS **[JEE(Main)July2022]**

(A) $\frac{5}{7}$ (B) $\frac{2\sqrt{6}}{7}$ (C) $\frac{3}{7}$ (D) $\frac{2\sqrt{5}}{7}$

Ans. A

Sol. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meet the line $\frac{x}{7}+\frac{y}{2\sqrt{6}}=1$ on the x-axis

So, $a=7$ and $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meet the line $\frac{x}{7}-\frac{y}{2\sqrt{6}}=1$ y-axis

So, $b=2\sqrt{6}$

Therefore, $e^2=1-\frac{b^2}{a^2}=1-\frac{24}{49}$

$e=\frac{5}{7}$

55. The acute angle between the pair of tangents drawn to the ellipse $2x^2+3y^2=5$ from the point (1, 3) is:

[JEE(Main)July2022]

बिन्दु (1, 3) से दीर्घवृत्त $2x^2+3y^2=5$ पर डाली गई दो स्पर्श रेखाओं के बीच न्यून कोण हैं : **[JEE(Main)July2022]**

(A) $\tan^{-1}\left(\frac{16}{7\sqrt{5}}\right)$ (B) $\tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$ (C) $\tan^{-1}\left(\frac{32}{7\sqrt{5}}\right)$ (D) $\tan^{-1}\left(\frac{3+8\sqrt{5}}{35}\right)$

Ans. B

Sol. Equation of tangent to the ellipse $2x2+3y^2=5$ is

$y=mx\pm\sqrt{\frac{5}{2}m^2+\frac{5}{3}}$

It pass through (1, 3)

$3=m\pm\sqrt{\frac{5}{2}m^2+\frac{5}{3}}$

$3m^2+12m-\frac{44}{3}=0$

Let θ be the angle between the tangents

$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$

$\tan\theta=\left|\frac{3\sqrt{320}}{-35}\right|$

$\theta=\tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$

56. Let a line L pass through the point of intersection of the lines $bx + 10y - 8 = 0$ and $2x - 3y = 0$, $b \in R - \left\{\frac{4}{3}\right\}$. If the line L also passes through the point (1,1) and touches the circle $17(x^2 + y^2) = 16$, then the eccentricity of all ellipse $\frac{x^2}{5} + \frac{y^2}{b^2} = 1$ is : **[JEE(Main)July2022]**

माना एक रेखा L, रेखाओं $bx + 10y - 8 = 0$ तथा $2x - 3y = 0$, $b \in R - \left\{\frac{4}{3}\right\}$ के प्रतिच्छेदन बिन्दु से होकर जाती है। यदि रेखा L, बिन्दु (1,1) से भी होकर जाती है तथा वृत्त $17(x^2 + y^2) = 16$ को स्पर्श करती है, तो दीर्घवृत्त $\frac{x^2}{5} + \frac{y^2}{b^2} = 1$ की उत्केन्द्रता है : **[JEE(Main)July2022]**

(A) $\frac{2}{\sqrt{5}}$ (B) $\sqrt{\frac{3}{5}}$ (C) $\frac{1}{\sqrt{5}}$ (D) $\sqrt{\frac{2}{5}}$

Ans. B

Sol. Line is passing through intersection of
$bx + 10y - 8 = 0$ and $2x - 3y = 0$ is
$(bx + 10y - 8) + \lambda(2x - 3y) = 0$. As line is
passing through (1, 1) so $\lambda = b + 2$
Now line $(3b + 4)x - (3b - 4)y - 8 = 0$ is
tangent to circle $17(x^2 + y^2) = 16$

So $\frac{8}{\sqrt{(3b+4)^2 + (3b-4)^2}} = \frac{4}{\sqrt{17}}$

$\Rightarrow b^2 = 2 \Rightarrow e = \sqrt{\frac{3}{5}}$

57. If the length of the latus rectum of the ellipse $x^2 + 4y^2 + 2x + 8y - \lambda = 0$ is 4, and l is the length of its major axis, then $\lambda + l$ is equal to _____. **[JEE(Main)July2022]**

यदि दीर्घवृत्त $x^2 + 4y^2 + 2x + 8y - \lambda = 0$ की नाभिलंब जीवा की लम्बाई 4 है तथा इसके दीर्घअक्ष की लंबाई l है, तो $\lambda + l$ बराबर है _____. **[JEE(Main)July2022]**

Ans. 75

Sol. $\lambda + \ell = 75$
$x^2 + 4y^2 + 2x + 8y - \lambda = 0$

$$\frac{(x+1)^2}{\lambda+5}+\frac{(y+1)^2}{\frac{\lambda+5}{4}}=1$$

$$\because \frac{2b^2}{a}=4$$

$$\Rightarrow \lambda = 59$$

$$\lambda \neq -5$$

$$l = 2a = 2\sqrt{\lambda+5} = 2\sqrt{65} = 16$$

$$\Rightarrow \lambda + \ell = 59+16 = 75$$

58. Let the tangents at the points P and Q on the ellipse $\frac{x^2}{2}+\frac{y^2}{4}=1$ meet at the point $R\left(\sqrt{2}, 2\sqrt{2}-2\right)$. If S is the focus of the ellipse on its negative major axis, then $SP^2 + SQ^2$ is equal to _____.

[JEE(Main)July2022]

माना दीर्घवृत्त $\frac{x^2}{2}+\frac{y^2}{4}=1$ के बिन्दुओं P तथा Q पर स्पर्श रेखाएँ बिन्दु $R\left(\sqrt{2}, 2\sqrt{2}-2\right)$ पर मिलती हैं। यदि दीर्घवृत्त के _ .kkeed nh?kZv {ki j ukfHkki j S है, तो $SP^2 + SQ^2$ बराबर है _____ **[JEE(Main)July2022]**

Ans. 13

Sol. Ellipse is

$$\frac{x^2}{2}+\frac{y^2}{4}=1;\ e=\frac{1}{\sqrt{2}};S\equiv\left(0,-\sqrt{2}\right)$$

Chord of contact is

$$\frac{x}{\sqrt{2}}+\frac{\left(2\sqrt{2}-2\right)y}{4}=1$$

$$\Rightarrow \frac{x}{\sqrt{2}}=1-\frac{\left(\sqrt{2}-1\right)y}{2} \text{ solving with ellipse}$$

$$\Rightarrow y = 0, \sqrt{2} \quad \therefore x = \sqrt{2}, 1$$

$$P\equiv\left(1,\sqrt{2}\right)\ Q\equiv\left(\sqrt{2},0\right)$$

$$\therefore (SP)^2 + (SQ)^2 = 13$$

59. Let the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{7}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{\alpha}=\frac{1}{25}$ coincide. Then the length of the latus rectum of the hyperbola is : **[JEE(Main)July2022]**

माना दीर्घवृत्त $\frac{x^2}{16}+\frac{y^2}{7}=1$ तथा अतिपरवलय $\frac{x^2}{144}-\frac{y^2}{\alpha}=\frac{1}{25}$ की नाभियाँ सम्पाती हैं। तो अतिपरवलय की नाभिलंब जीवा की y alb Z gS **[JEE(Main)July2022]**

(A) $\frac{32}{9}$ (B) $\frac{18}{5}$ (C) $\frac{27}{4}$ (D) $\frac{27}{10}$

Ans. D

Sol. Ellipse : $\frac{x^2}{16}+\frac{y^2}{7}=1$

Eccentricity $=\sqrt{1-\frac{7}{16}}=\frac{3}{4}$

Foci $\equiv(\pm a\,e,0)\equiv(\pm 3,0)$

Hyperbola : $\frac{x^2}{\left(\frac{144}{25}\right)}-\frac{y^2}{\left(\frac{\alpha}{25}\right)}=1$

Eccentricity $=\sqrt{1+\frac{\alpha}{144}}=\frac{1}{12}\sqrt{144+\alpha}$

Foci $\equiv(\pm ae,0)\equiv\left(\pm\frac{12}{5}.\frac{1}{12}\sqrt{144+\alpha},0\right)$

If foci coincide then $3=\frac{1}{5}\sqrt{144+\alpha}\Rightarrow\alpha=81$

Hence, hyperbola is $\frac{x^2}{\left(\frac{12}{5}\right)^2}-\frac{y^2}{\left(\frac{9}{5}\right)^2}=1$

Length of latus rectum $=2.\frac{\frac{81}{25}}{\frac{12}{5}}=\frac{27}{10}$

60. Let the tangent drawn to the parabola $y^2 = 24x$ at the point (α, β) is perpendicular to the line $2x + 2y = 5$. Then the normal to the hyperbola $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ at the point $(\alpha + 4, \beta + 4)$ does NOT pass through the point : **[JEE(Main)July2022]**

माना परवलय $y^2 = 24x$ के बिन्दु (α, β) पर स्पर्शरेखा, $2x + 2y = 5$ के लंबवत है। तो अतिपरवलय $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ के बिन्दु $(\alpha + 4, \beta + 4)$ पर अभिलंब किस बिन्दु से होकर नहीं जाता है? **[JEE(Main)July2022]**

(A) (25, 10) (B) (20, 12) (C) (30, 8) (D) (15, 13)

Ans. D

Sol. Tangent at (α, β) has slope 1

$\beta^2 = 24\alpha$

Equation of tangent $y\beta = 12(x + \alpha), \frac{12}{\beta} = 1$

$\Rightarrow \alpha = 6, \beta = 12$

$\therefore (\alpha + 4, \beta + 4) = (10,16)$

Normal at (10, 16) to $\frac{x^2}{36} - \frac{y^2}{144} = 1$ is

$2x + 5y = 100$

61. If the line $x - 1 = 0$ is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point : **[JEE(Main)July2022]**

यदि रेखा $x - 1 = 0$, अतिपरवलय $kx^2 - y^2 = 6$ की एक नियता है, तो यह अतिपरवलय किस बिन्दु से होकर जाता है: **[JEE(Main)July2022]**

(A) $\left(-2\sqrt{5}, 6\right)$ (B) $\left(-\sqrt{5}, 3\right)$ (C) $\left(\sqrt{5}, -2\right)$ (D) $\left(2\sqrt{5}, 3\sqrt{6}\right)$

Ans. C

Sol. $\frac{x^2}{6/k} - \frac{y^2}{6} = 1$ .....(1)

$e^2 = 1 + \frac{6}{6/k}$

$e = \sqrt{1+k}$

$a = \sqrt{\frac{6}{k}}$

Equation of directrix $x = \frac{a}{e} \Rightarrow x = \sqrt{\frac{6}{k(k+1)}}$

$\frac{6}{k(k+1)} = 1$

$k = 2$

From eq. (1) ,we get $2x^2 - y^2 = 6$

Check options

62. Let the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ pass through the point $\left(2\sqrt{2}, -2\sqrt{2}\right)$. A parabola is drawn whose focus is same as the focus of H with positive abscissa and the directrix of the parabola passes through the other focus of H. If the length of the latus rectum of the parabola is e times the length of the latus rectum of H, where e is the eccentricity of H, then which of the following points lies on the parabola? **[JEE(Main)July2022]**

माना अतिपरवलय $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, बिन्दु $\left(2\sqrt{2}, -2\sqrt{2}\right)$ से होकर जाता है। एक परवलय खींचा जाता है जिसकी नाभि, H की धनात्मक भुज वाली नाभि पर है तथा परवलय की नियता H की दूसरी नाभि से होकर जाती है। यदि परवलय की नाभि लंब जीवा की y alkbZ H की नाभि लंब की जीवा की लंबाई e गुना है, जहाँ e, H की उत्केन्द्रता है, तो निम्न में से कौनसा बिन्दु परवलय पर है ?

[JEE(Main)July2022]

(A) $\left(2\sqrt{3}, 3\sqrt{2}\right)$ (B) $\left(3\sqrt{3}, -6\sqrt{2}\right)$ (C) $\left(\sqrt{3}, -\sqrt{6}\right)$ (D) $\left(3\sqrt{6}, 6\sqrt{2}\right)$

Ans. B

Sol. $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Foci : S (ae, 0), S' (–ae, 0)

Foot of directrix of parabola is (–ae, 0)

Focus of parabola is (ae, 0)

Now, semi latus rectum of parabola = |SS'| = 2ae

Given, $4ae = e\left(\frac{2b^2}{a}\right)$

$\Rightarrow b^2 = 2a^2$ _____(1)

Given, $\left(2\sqrt{2}, -2\sqrt{2}\right)$ lies on H

$\Rightarrow \frac{1}{a^2} - \frac{1}{b^2} = \frac{1}{8}$ _____(2)

From (1) and (2)

$a^2 = 4, b^2 = 8$

$\because b^2 = a^2\left(e^2 - 1\right)$

$\therefore e = \sqrt{3}$

$\Rightarrow$ Equation of parabola is $y^2 = 8\sqrt{3}x$

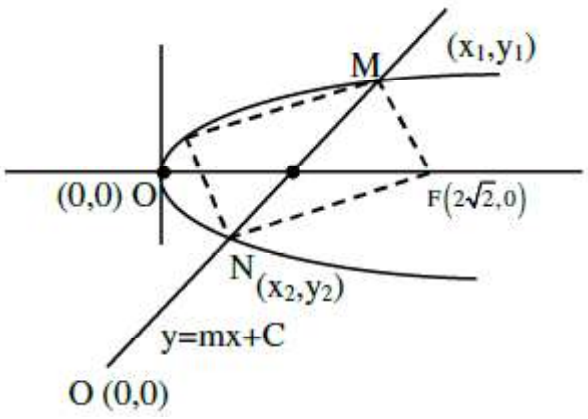
63. Let the focal chord of the parabola P : $y^2 = 4x$ along the line L : $y = mx + c$, $m > 0$ meet the parabola at the points M and N. Let the line L be a tangent to the hyperbola H : $x^2 - y^2 = 4$. If O is the vertex of P and F is the focus of H on the positive x-axis, then the area of the quadrilateral OMFN is : **[JEE(Main)July2022]**

माना रेखा L : $y = mx + c$, $m > 0$ के अनुदिश परवलय P : $y^2 = 4x$ की नाभिलंब जीवा परवलय को बिन्दुओं M तथा N पर मिलती है। माना रेखा L अतिपरवलय H : $x^2 - y^2 = 4$ की एक स्पर्श रेखा है। यदि P का शीर्ष O है तथा H की धनात्मक x-अक्ष पर नाभि F है, r ksprqHkqt OMFN का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $2\sqrt{6}$ (B) $2\sqrt{14}$ (C) $4\sqrt{6}$ (D) $4\sqrt{14}$

Ans. B

Sol.



$H : \frac{x^2}{4} - \frac{y^2}{4} = 1$

Focus (ae, 0)

$F(2\sqrt{2}, 0)$

Line L : $y = mx + c$ pass (1, 0)

$o = m + C$ .....(1)

List L is tangent to Hyperbola. $\frac{x^2}{4} - \frac{y^2}{4} = 1$

$C = \pm\sqrt{a^2 m^2 - \ell^2}$

$C = \pm\sqrt{4m^2 - 4}$

From (1)

$-m = \pm\sqrt{4m^2 - 4}$

Squaring

$m^2 = 4m^2 - 4$

$4 = 3m^2$

$\frac{2}{\sqrt{3}} = m$ (as $m > 0$)

$C = -m$

$C = \frac{-2}{\sqrt{3}}$

$y = \frac{2x}{\sqrt{3}} - \frac{2}{\sqrt{3}}$

$y^2 = 4x$

$\Rightarrow \left(\frac{2x-2}{\sqrt{3}}\right)^2 = 4x$

$\Rightarrow x^2 + 1 - 2x = 3x$

$\Rightarrow x^2 - 5x + 1 = 0$

$y^2 = 4\left(\frac{\sqrt{3}y + 2}{2}\right)$

$y^2 = 2\sqrt{3}y + 4$

$\Rightarrow y^2 - 2\sqrt{3}y - 4 = 0$

Area

$$\left|\frac{1}{2}\right| \begin{matrix} 0 & x_1 & 2\sqrt{2} & x_2 & 0 \\ 0 & y_1 & 0 & y_2 & 0 \end{matrix} \Bigg\|$$

$$= \left|\frac{1}{2}\left[-2\sqrt{2}y_1 + 2\sqrt{2}y_2\right]\right|$$

$$= \sqrt{2}\left|y_2 - y_1\right| = \frac{(\sqrt{2})\sqrt{12+16}}{111}$$

$$= \sqrt{56}$$

$$= 2\sqrt{14}$$

64. Let the equation of two diameters of a circle $x^2 + y^2 - 2x + 2fy + 1 = 0$ be $2px - y = 1$ and $2x + py = 4p$. Then then slope $m \in (0, \infty)$ of the tangent to the hyperbola $3x^2 - y^2 = 3$ passing through the centre of the circle is equal to _____. **[JEE(Main)July2022]**

माना एक वृत्त $x^2 + y^2 - 2x + 2fy + 1 = 0$ के दो व्यासों के समीकरण $2px - y = 1$ तथा $2x + py = 4p$ हैं। तो अतिपरवलय $3x^2 - y^2 = 3$ की एक स्पर्श रेखा, जो वृत्त के केन्द्र होकर जाती है, की प्रवणता $m \in (0, \infty)$ बराबर है _____. **[JEE(Main)July2022]**

Ans. B

Sol. $2p + f - 1 = 0$ _____(1)

$2 - pf - 4p = 0$ _____(2)

$2 = p(f + 4)$

$$p = \frac{2}{f+4}$$

$2p = 1 - f$

$$\frac{4}{f+4} = 1 - f$$

$f^2 + 3f = 0$

$f = 0$ or -3

Hyperbola $3x^2 - y^2 = 3$, $x^2 - \frac{y^2}{3} = 1$

$y = mx \pm\sqrt{m^2 - 3}$

It passes (1, 0)

$o = m \pm \sqrt{m^2 - 3}$

m tends ∞

It passes (1, 3)

$3 = m \pm \sqrt{m^2 - 3}$

$(3 - m)^2 = m^2 - 3$

$m = 2$

65. An ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the vertices of the hyperbola $H : \frac{x^2}{49} - \frac{y^2}{64} = -1$. Let the major and minor axes of the ellipse E coincide with the transverse and conjugate axes of the hyperbola H, . Let the product of the eccentricities of E and H be $\frac{1}{2}$. if l is the length of the latus rectum of the ellipse E, then the value of $113l$ is equal to _____. **[JEE(Main)July2022]**

एक दीर्घवृत्त $E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, अतिपरवलय $H:\frac{x^2}{49}-\frac{y^2}{64}=-1$ के शीर्षों से होकर जाता है। माना दीर्घवृत्त E के दीर्घ तथा लघु अक्ष क्रमशः अतिपरवलय H के अनुप्रस्थ तथा संयुग्मी अक्ष के सम्पाती हैं। माना E तथा H की उत्केन्द्रताओं का गुणनफल $\frac{1}{2}$ है। यदि दीर्घवृत्त E की नाभिलंब जीवा की लंबाई l है, तो $113l$ का मान है _____ **[JEE(Main)July2022]**

Ans. 1552

Sol. $\text{Hyp}:\frac{y^2}{64}-\frac{x^2}{49}=1$

An ellipse $E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through the vertices of the hyperbola $H:\frac{x^2}{49}-\frac{y^2}{64}=-1$.

So $b^2 = 64$

$$e_H=\sqrt{1+\frac{a^2}{b^2}}=\sqrt{1+\frac{49}{64}}$$

Ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$$e_H=\sqrt{1-\frac{a^2}{b^2}}=\sqrt{1-\frac{a^2}{64}}$$

$$b=8,\sqrt{\frac{1-a^2}{64}}\times\frac{\sqrt{113}}{8}=\frac{1}{2}\Rightarrow\sqrt{64-a^2}\times\sqrt{113}=32$$

$$\left(64-a^2\right)=\frac{32^2}{113}$$

$$\Rightarrow a^2=64-\frac{32^2}{113}$$

$$l=\frac{2a^2}{b}=\frac{2}{8}\left(64-\frac{32^2}{113}\right)=\frac{1552}{113}$$

$113l = 1552$

66. A common tangent T to the curve $C_1\ \frac{x^2}{4}+\frac{y^2}{9}=1$ and $C_2:\frac{x^2}{42}-\frac{y^2}{143}=1$ does not pass through the fourth quadrant. If T touches C_1 at (x_1, y_1) and C_2 at (x_2, y_2), then $|2x_1 + x_2|$ is equal to _____.

[JEE(Main)July2022]

वक्रों $C_1 \frac{x^2}{4}+\frac{y^2}{9}=1$ तथा $C_2 : \frac{x^2}{42}-\frac{y^2}{143}=1$ की एक उभयनिष्ठ स्पर्श रेखा T चतुर्थ चतुर्थांश से होकर नहीं जाती है। यदि T, वक्र C_1 को (x_1, y_1) पर तथा वक्र C_2 को (x_2, y_2) को स्पर्श करती है, तो $|2x_1 + x_2|$ बराबर है _____

[JEE(Main)July2022]

Ans. 20

Sol. Let common tangents are

$T_1 : y = mx \pm \sqrt{4m^2+9}$

& $T_2 : y = mx \pm \sqrt{42m^2-13}$

So, $4m^2 + 9 = 42m^2 - 143$

$\Rightarrow 38m^2 = 152$

$\Rightarrow m = \pm 2$

& $c = \pm 5$

For given tangent not pass through 4th quadrant

$T : y = 2x + 5$

Now, comparing with $\frac{xx_1}{4}+\frac{yy_1}{9}=1$

We get, $\frac{x_1}{8}=\frac{1}{5} \Rightarrow x_1 = -\frac{8}{5}$

$\frac{xx_2}{42}-\frac{yy_2}{143}=1$

2x /0 y = –5 we have

$x_2 = -\frac{84}{5}$

So, $|2x_1 + x_2| = \left|\frac{-100}{5}\right| = 20$

67. For the hyperbola $H : x^2 - y^2 = 1$ and the ellipse $E : \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a > b > 0$, let the

(1) eccentricity of E be reciprocal of the eccentricity of H, and

(2) the line $y = \sqrt{\frac{5}{2}}x + K$ be a common tangent of E and H.

Then $4(a^2 + b^2)$ is equal to _____.

[JEE(Main)July2022]

अतिपरवलय $H : x^2 - y^2 = 1$ तथा दीर्घवृत्त $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$ के लिए, माना

(1) E की उत्केन्द्रता, H की उत्केन्द्रता की व्युत्क्रमणीय हैं, तथा

(2) रेखा $y = \sqrt{\frac{5}{2}}x + K$, E तथा H की एक उभयनिष्ठ स्पर्श रेखा है।

r ks4$(a^2 + b^2)$ बराबर है _____. **[JEE(Main)July2022]**

Ans. 3

Sol. $e_E = \sqrt{1 - \frac{b^2}{a^2}}, e_H = \sqrt{2}$

$$\text{If} \Rightarrow e_E = \frac{1}{e_H}$$

$$\Rightarrow \frac{a^2 - b^2}{a^2} = \frac{1}{2}$$

$2a^{2-2b} = a^2$

$a^2 = 2b^2$

and $y = \sqrt{\frac{5}{2}}x + k$ is tangent to ellipse then

$$K^2 = a^2 \times \frac{5}{2} + b^2 = \frac{3}{2}$$

$$6b^2 = \frac{3}{2} \Rightarrow b^2 = \frac{1}{4} \text{ and } a^2 = \frac{1}{2}$$

$$\therefore \ 4.\left(a^2 + b^2\right) = 3$$

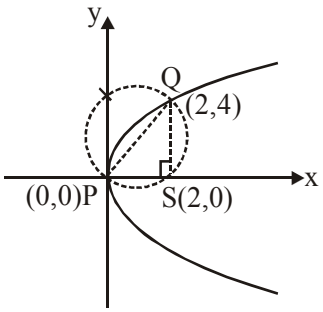
Exercise J - A

1. Let S be the focus of the parabola $y^2 = 8x$ and let PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the triangle PQS is : **[JEE-2012]**

   माना परवलय $y^2 = 8x$ की नाभि S है तथा PQ वृत्त $x^2 + y^2 - 2x - 4y = 0$ तथा दिये गये परवलय की उभयनिष्ठ जीवा है तो त्रिभुज PQS का क्षेत्रफल होगा : **[JEE-2012]**

Ans. 4

Sol. Focus of parabola S(2,0) points of intersection of given curves : (0,0) and (2,4).



Area $(\Delta PSQ) = \frac{1}{2}.2.4 = 4$ sq. units

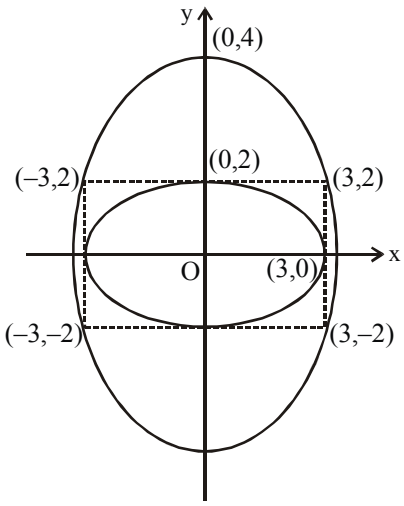
2. The ellipse $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ is inscribed in a rectangle R whose sides are parallel to the coordinate axes. Another ellipse E_2 passing through the point (0, 4) circumscribes the rectangle R. The eccentricity of the ellipse E_2 is : **[JEE-2012]**

   दीर्घवृत्त $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ एक आयत R के अन्तर्गत परिबद्ध है जिसकी भुजाऐं निर्देशी अक्षों के समान्तर है। दूसरा दीर्घवृत्त E_2 बिन्दु (0, 4) से गुजरता है तथा आयत R को परिगत करता है तो E_2 की उत्केन्द्रियता होगी : **[JEE-2012]**

   (A) $\frac{\sqrt{2}}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

Ans. C

Sol. Let equation of E_2 be



$\frac{x^2}{a^2}+\frac{y^2}{16}=1$ ($\because$ E_2 passes through (0, 4))

$\because$ E_2 passes through (3,2)

$\therefore$ $\frac{9}{a^2}+\frac{4}{16}=1$

$\Rightarrow$ $a^2 = 12$

$\therefore$ $e^2 = 1-\frac{a^2}{16}=1-\frac{3}{4}\Rightarrow e=\frac{1}{2}$

3. Tangents are drawn to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$, parallel to the straight line $2x-y=1$. The points of contact of the tangents on the hyperbola are : **[JEE-2012]**

अतिपरवलय $\frac{x^2}{9}-\frac{y^2}{4}=1$ पर रेखा $2x - y = 1$ के समान्तर स्पर्श रेखाऐं खींची जाती हैं। स्पर्श रेखाओं का अतिपरवलय पर स्पर्श fबिंदु है **[JEE-2012]**

(A) $\left(\frac{9}{2\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ (B) $\left(-\frac{9}{2\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$ (C) $\left(3\sqrt{3},-2\sqrt{2}\right)$ (D) $\left(-3\sqrt{3},2\sqrt{2}\right)$

Ans. AB

Sol. Let parametric coordinates be P(3secθ, 2 tanθ)

Equation of tangent at point P will be

$\frac{x\sec\theta}{3}-\frac{y\tan\theta}{2}=1$

$\because$ tangent is parallel to $2x - y = 1$

$\Rightarrow \frac{2\sec\theta}{3\tan\theta} = 2 \quad \Rightarrow \sin\theta = \frac{1}{3}$

$\therefore$ coordinates are

$\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and $\left(-\frac{9}{2\sqrt{2}}, -\frac{1}{2}\right)$

Paragraph for Question No. 4 to 5

Let PQ be a focal chord of the parabola $y^2 = 4ax$. The tangents to the parabola at P and Q meet at a point lying on the line $y = 2x + a$, $a > 0$.

माना कि PQ परवलय $y^2 = 4ax$ की एक नाभीय जीवा है। बिन्दुओं P तथा Q पर परवलय की स्पर्श रेखाएँ एक बिन्दु पर मिलती है जो कि रेखा $y = 2x + a$, $a > 0$ पर स्थित है।

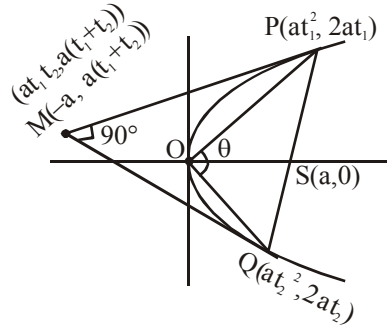
4. Length of chord PQ is : **[JEE-2013]**

जीवा PQ की लम्बाई है : **[JEE-2013]**

(A) 7a (B) 5a (C) 2a (D) 3a

Ans. B

Sol. Single tangent at the extrimities of a focal



chord will intersect on directrix.

$\therefore \quad M(-a, a(t_1 + t_2))$

lies on $y = 2x + a$

$a(t_1 + t_2) = -2a + a \Rightarrow \quad t_1 + t_2 = -1$

$\& \quad t_1t_2 = -1$

$$\tan\theta = \left(\frac{\frac{2}{t_1} - \frac{2}{t_2}}{1 + \frac{4}{t_1t_2}}\right) = \left(\frac{2(t_2 - t_1)}{3}\right)$$

$\because \quad (t_2 - t_1)^2 = (t_2 + t_1)^2 - 4t_1t_2 = 5$

$t_2 - t_1 = \pm\sqrt{5}$

$\therefore \quad \tan\theta = \pm\frac{2\sqrt{5}}{3}$

but θ is obtuse because O is the interior point of the circle for which PQ is diameter.

$\therefore \quad \tan\theta = \frac{-2\sqrt{5}}{3}$

5. If chord PQ subtends an angle θ at the vertex of $y^2 = 4ax$, then $\tan\theta =$: **[JEE-2013]**

यदि जीवा PQ, $y^2 = 4ax$ के शीर्ष पर कोण θ अंतरित करती है, तब $\tan\theta =$: **[JEE-2013]**

(A) $\frac{2}{3}\sqrt{7}$ (B) $\frac{-2}{3}\sqrt{7}$ (C) $\frac{2}{3}\sqrt{5}$ (D) $\frac{-2}{3}\sqrt{5}$

Ans. D

Sol. Length of focal chord

PQ $= a(t_1 - t_2)^2$

$= a[(t_1 + t_2)^2 - 4t_1t_2]$

$= a[1 + 4] = 5a$

6. A line L : $y = mx + 3$ meets y–axis at E(0, 3) and the arc of the parabola $y^2 = 16x$, $0 \le y \le 6$ at the point $F(x_0, y_0)$. The tangent to the parabola at $F(x_0, y_0)$ intersects the y-axis at $G(0, y_1)$. The slope m of the line L is chosen such that the area of the triangle EFG has a local maximum.

Match Column I with Column II and select the correct answer using the code given below the column :

[JEE(Adv.)2013]

Column-I	**Column-II**
P. m =	1. 1/2
Q. Maximum area of ΔEFG is	2. 4
R. y_0 =	3. 2
S. y_1 =	4. 1

एक रेखा L : $y = mx + 3$, y–अक्ष के बिन्दु E(0, 3) तथा परवलय के चाप $y^2 = 16x$, $0 \le y \le 6$ के बिन्दु $F(x_0, y_0)$ पर मिलती है। परवलय की बिन्दु $F(x_0, y_0)$ पर स्पर्शी y-अक्ष को बिन्दु $G(0, y_1)$ पर काटती है। रेखा L की प्रवणता m ऐसी चुनी जाती है कि त्रिभुज EFG के क्षेत्रफल का एक स्थानीय अधिकतम है।

d kyWe I को कॉलम II से सुमेलित कीजिए तथा कॉलम के नीचे दिए गए कोड का प्रयोग करके सही उतर चुनिये :

[JEE(Adv.)2013]

Column-I	**Column-II**
P. m =	1. 1/2
फण ΔEFG का महत्म क्षेत्रफल है	2. 4
R. y_0 =	3. 2
S. y_1 =	4. 1

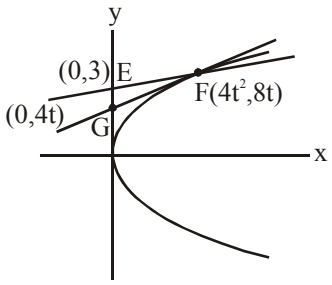
कोड :

(A) P-4; Q-1; R-2; S-3 (B) P-3; Q-4; R-1; S-2

(C) P-1; Q-3; R-2; S-4 (D) P-1; Q-3; R-4; S-2

Ans. A

Sol. Let $F(4t^2, 8t)$



where $0 \le 8t \le 6 \quad \Rightarrow \quad 0 \le t \le \frac{3}{4}$

$$\Delta EFG = \frac{1}{2}(3-4t)4t^2$$

$$\Delta = (6t^2 - 8t^3)$$

$$\frac{d\Delta}{dt} = 12t - 24t^2 = 0 \quad \begin{cases} t = 0 \text{ (minima)} \\ t = \frac{1}{2} \text{ (maxima)} \end{cases}$$

$$\underset{0}{-} \quad \underset{1/2}{+} \quad -$$

$$\Rightarrow \quad m = \frac{8t-3}{4t^2-0} = \frac{4-3}{1} = 1$$

$$(\Delta EFG)_{max} = \frac{6}{4} - 1 = \frac{1}{2}$$

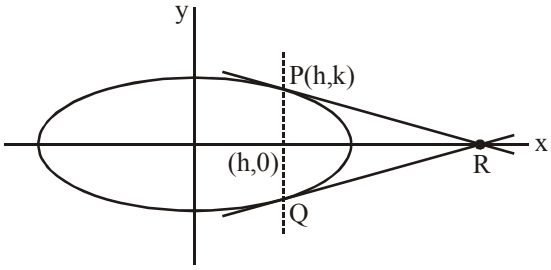
$y_0 = 8t = 4$ & $y_1 = 4t = 2$

7. A vertical line passing through the point (h, 0) intersects the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ at the point P and Q. Let the tangents to the ellipse at P and Q meet at the point R. If $\Delta(h)$ = area of the triangle PQR, $\Delta_1 = \max_{1/2 \le h \le 1} \Delta(h)$ and $\Delta_2 = \min_{1/2 \le h \le 1} \Delta(h)$, then $\frac{8}{\sqrt{5}}\Delta_1 - 8\Delta_2 =$ **[JEE(Adv.)2013]**

बिन्दु (h, 0) से गुजरने वाली एक ऊर्ध्वाधर रेखा दीर्घवृत $\frac{x^2}{4} + \frac{y^2}{3} = 1$ को बिन्दुओं P तथा Q पर काटती है। माना कि बिन्दुओं P तथा Q पर दीर्घवृत की स्पर्श रेखाएँ बिन्दु R पर मिलती है। यदि $\Delta(h)$ = त्रिभुज PQR का क्षेत्रफल, $\Delta_1 = \max_{1/2 \le h \le 1} \Delta(h)$ और $\Delta_2 = \min_{1/2 \le h \le 1} \Delta(h)$ है, तब $\frac{8}{\sqrt{5}}\Delta_1 - 8\Delta_2 =$ **[JEE(Adv.)2013]**

Ans. 9

Sol.



Tangent at P(h, k) is $\frac{xh}{4}+\frac{ky}{3}=1$

$\Rightarrow \quad R\left(\frac{4}{h},0\right)$

$\Delta PQR = k\left(\frac{4}{h}-h\right)$

$=\sqrt{3\left(1-\frac{h^2}{4}\right)}\left(\frac{4}{h}-h\right)$

which is a decreasing function in $\left[\frac{1}{2},1\right]$

$\Rightarrow \quad \Delta_1=\sqrt{3\left(1-\frac{1}{16}\right)}\left(8-\frac{1}{2}\right)=\frac{45\sqrt{5}}{8}$

& $\quad \Delta_2=\sqrt{3\left(1-\frac{1}{4}\right)}(4-1)=\frac{9}{2}$

$\frac{8}{\sqrt{5}}\Delta_1-8\Delta_2=45-36=9$

8. The common tangents to the circle $x^2+y^2=2$ and the parabola $y^2=8x$ touch the circle at the points , Q and the parabola at the points R, S. Then the area of the quadrilateral PQRS is : **[JEE(Adv.)2014]**

वृत्त $x^2+y^2=2$ तथा परवलय $y^2=8x$ की उभयनिष्ठ स्पर्श रेखाऐं वृत्त को P, Q पर स्पर्श करती है तथा परवलय को R, S पर स्पर्श d j r hgS kspr djrh gSA PQRS का क्षेत्रफल होगा : **[JEE(Adv.)2014]**

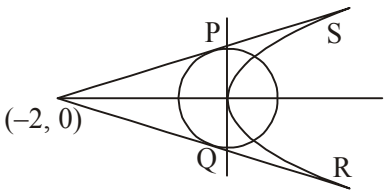
(A) 3 (B) 6 (C) 9 (D) 15

Ans. D

Sol. $y=mx+\frac{2}{m}$

if it is tangent to $x^2 + y^2 = 2$

Then,



$$\left|\frac{\frac{2}{m}}{\sqrt{1+m^2}}\right| = \sqrt{2} \qquad \Rightarrow \frac{4}{m^2\left(1+m^2\right)} = 2 \Rightarrow m = \pm 1.$$

Hence equation of tangent is $y = x + 2$ and $y = -x - 2$.

Chord of contact PQ is $-2x = 2 \Rightarrow x = -1$

Chord of contact RS is $y.\ 0 = 4\ (x - 2) \Rightarrow x = 2$

Hence co-ordinates of P, Q, R, S are $(-1, 1); (-1, -1); (2, -4)$ and $(2, 4)$

Area of trapezium is $= \frac{1}{2}(PQ + RS) \times \text{Height} = \frac{1}{2}(10) \times 3 = 15$

Paragraph For Questions 9 and 10

Let a, r, s t be nonzero real numbers. Let $P(at^2, 2at)$, Q, R $(ar^2, 2ar)$ and $S(as^2, 2as)$ be distinct points on the parabola $y^2 = 4ax$. Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point $(2a, 0)$.

माना a, r, s t अशून्य वास्तविक संख्याऐं है तथा $P(at^2, 2at)$, Q, R $(ar^2, 2ar)$ तथा $S(as^2, 2as)$ परवलय $y^2 = 4ax$ पर स्थित भिन्न बिन्दु हैं। PQ नाभि जीवा है तथा रेखाऐं QR तथा PK समान्तर हैं जहाँ बिन्दु K $(2a, 0)$ है।

9. The value of r is : **[JEE(Adv.)2014]**

r का मान होगा : **[JEE(Adv.)2014]**

(A) $-\frac{1}{t}$ (B) $\frac{t^2+1}{t}$ (C) $\frac{1}{t}$ (D) $\frac{t^2-1}{t}$

Ans. D

Sol. $m_{PK} = m_{QR}$

$$\frac{2at - 0}{at^2 - 2a} = \frac{2at' - 2ar}{a(t')^2 - ar^2}$$

$$\frac{t}{t^2 - 2} = \frac{t' - r}{(t')^2 - r^2}$$

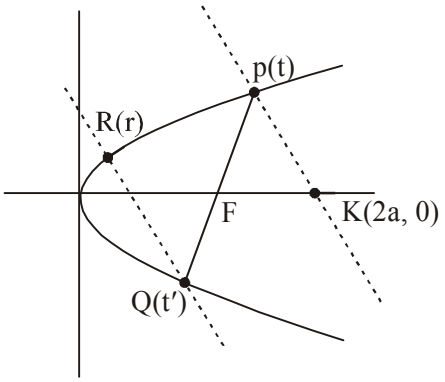
$$-t' - tr^2 = -t - rt^2 - 2t' + 2r,\ tt' = -1$$

$$t' - tr^2 = -t + 2r - rt^2$$

$$-tr^2 + r\left(t^2-2\right) + t' + t = 0$$

$$\lambda = \frac{\left(2-t^2\right) \pm \sqrt{\left(t^2-2\right)^2 + \left(-1+t^2\right)}}{-2t}$$

$$= \frac{\left(2-t^2\right) \pm \sqrt{t^4}}{-2t} = \frac{2-t^2 \pm t^2}{-2t}$$



$r = -\frac{1}{t}$ It is not possible as the R and Q will be one same.

or $r = \frac{t^2-1}{t}$

10. If st =1, then the tangent at P and the normal at S to the parabola meet at a point whose ordinate is :

[JEE(Adv.)2014]

यदि st = 1 तो P पर स्पर्श रेखा तथा S पर अभिलम्ब परवलय को एक बिन्दु पर काटेंगे जिसकी कोटि होगी :

[JEE(Adv.)2014]

(A) $\frac{(t^2+1)^2}{2t^3}$ (B) $\frac{a(t^2+1)^2}{2t^3}$ (C) $\frac{a(t^2+1)^2}{t^3}$ (D) $\frac{a(t^2+2)^2}{t^3}$

Ans. B

Sol. Tangent at P is $ty = x + at^2$

Normal at S is $y + sx = 2as + as^2$

$$ty + x = 2a + \frac{a}{t^2}$$

$$ty = 2a + \frac{a}{t^2} - ty + at^2$$

$$2t^3y = at^4 + 2at^2 + a$$

$$y = \frac{a\left(t^2+1\right)^2}{2t^3}$$

11. Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is.

[JEE(Adv.)2015]

माना कि वक्र C, रेखा $x + y + 4 = 0$ के सापेक्ष में, परवलय $y^2 = 4x$ का दर्पण प्रतिबिम्ब है। यदि A और B वक्र C और रेखा $y = -5$ के प्रतिच्छेद बिन्दु है, तब A और B के बीच की दूरी है। **[JEE(Adv.)2015]**

Ans. 4

Sol. Let $P(t^2, 2t)$ be a point on the curve $y^2 = 4x$ and $Q(h, k)$ be it's image in $x + y + 4 = 0$

$$\frac{h - t^2}{1} = \frac{k - 2t}{1} = \frac{2\left(t^2 + 2t + 4\right)}{2} \quad \Rightarrow \quad h = -(2t + 4)$$

$k = -(t^2 + 4)$

Now $k = -5$

so $t = \pm 1$

hence $h = -2, -6$

so A, B are $(-2, -5)$ and $(-6, -5)$

Hence $AB = 4$

12. If the normals of the parabola $y^2 = 4x$ drawn at the end point of its latus rectum are tangents to the circle $(x-3)^2+(y+2)^2 = r^2$, then the value of r^2 is. **[JEE(Adv.)2015]**

यदि परवलय $y^2 = 4x$ के नाभिलम्ब जीवा (latus rectum) के शिखर बिन्दुओं पर खींचे गए अभिलम्ब वृत $(x-3)^2+(y+2)^2 = r^2$ की स्पर्श रेखाएँ हैं, तब r^2 का मान है। **[JEE(Adv.)2015]**

Ans. 2

Sol. Equation of normals at points $(1, \pm 2)$ are

$y = -x + 3$ and $y = x - 3$

$\Rightarrow$ $x + y - 3 = 0$ and $x - y - 3 = 0$

Now $\left|\frac{3 - 2 - 3}{\sqrt{1+1}}\right| = r \quad \Rightarrow \quad r^2 = 2$

13. Let P and Q be distinct points on the parabola $y^2 = 2x$ such that a circle with PQ as diameter passes through the vertex O of the parabola. If P lies in the first quadrant and the area of the triangle DOPQ is $3\sqrt{2}$, then which of the following is (are) the coordinates of P ? **[JEE(Adv.)2015]**

माना कि विभिन्न बिन्दु P और Q परवलय $y^2 = 2x$ पर इस प्रकार लिए गए है कि एक वृत, जिसका व्यास PQ है, इस परवलय के शीर्ष O से जाता है। यदि P प्रथम चतुर्थांश में स्थित है तथा त्रिभुज ΔOPQ का क्षेत्रफल $3\sqrt{2}$ है, तो निम्न में से कौन कौन से बिन्दु P के fun;kd gS **[JEE(Adv.)2015]**

(A) $\left(4, 2\sqrt{2}\right)$ (B) $\left(9, 3\sqrt{2}\right)$ (C) $\left(\frac{1}{4}, \frac{1}{\sqrt{2}}\right)$ (D) $\left(1, \sqrt{2}\right)$

Ans. AD

Sol. $OP \perp OQ \quad \Rightarrow \quad t_1 t_2 = -4$

Now $\frac{1}{2}.OP.OQ = 3\sqrt{2}$

$\Rightarrow \quad \left|\frac{1}{2}\sqrt{\frac{t_1^4}{4}+t_1^2}\cdot\sqrt{\frac{t_2^4}{4}+t_2^2}\right| = 3\sqrt{2}$

$\Rightarrow \quad \frac{1}{2}.4.\sqrt{\frac{\left(t_1^2+4\right)}{4}\frac{\left(t_2^2+4\right)}{4}} = 3\sqrt{2}$

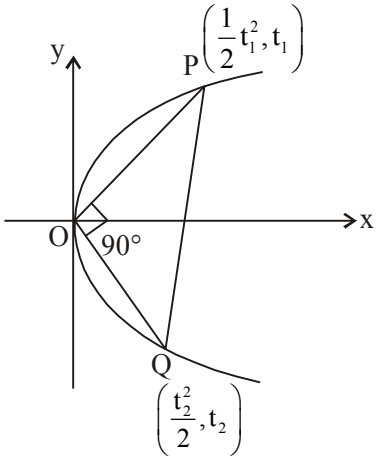
$\Rightarrow \quad 4.\frac{\left(16+4\left(t_1^2+t_2^2\right)+16\right)}{16} = 9\times 2$

$\Rightarrow \quad 8+t_1^2+t_2^2 = 18$

$\Rightarrow \quad t_1^2+t_2^2-10 = 0$

$\Rightarrow \quad t_1^4-10t_1^2+16 = 0$

$\Rightarrow \quad t_1^2 = 2, 8$



14. Suppose that the foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ are $(f_1, 0)$ and $(f_2, 0)$ where $f_1>0$ and $f_2<0$. Let P_1 and P_2 be two parabolas with a common vertex at (0,0) and with foci at $(f_1, 0)$ and $(2f_2, 0)$ respectively. Let T_1 be a tangent to P_1 which passes through $(2f_2, 0)$ and T_2 be a tangent to P_2 which passes through $(f_1, 0)$. If m_1 is the slope of T_1 and m_2 is the slope of T_2, then the value of $\left(\frac{1}{m_1^2}+m_2^2\right)$ is : **[JEE(Adv.)2015]**

माना कि दीर्घ वृत्त $\frac{x^2}{9}+\frac{y^2}{5}=1$ की नाभियाँ (foci) $(f_1, 0)$ और $(f_2, 0)$ है, जहाँ $f_1>0$ और $f_2<0$ हैं। माना कि P_1 एवं P_2 दो परवलय (parabola) हैं जिनकी नाभियाँ क्रमशः $(f_1, 0)$ एवं $(2f_2, 0)$ हैं तथा दोनों के शीर्ष (vertex) (0, 0) है। माना कि P_1 की स्पर्श रेखा T_1 बिन्दु $(2f_2, 0)$ एवं P_2 की स्पर्श रेखा T_2 बिन्दु $(f_1, 0)$ से गुजरती हैं। यदि T_1 की प्रवणता (slope) m_1 हो और T_2 की प्रवणता m_2 हो, तब $\left(\frac{1}{m_1^2}+m_2^2\right)$ का मान है : **[JEE(Adv.)2015]**

Ans. 4

Sol. $e^2 = 1-\frac{b^2}{a^2} = 1-\frac{5}{9} = \frac{4}{9}$

$e = \frac{2}{3}$ focii (2, 0) (–2, 0)

$p_1 : y^2 = 8x,$

$y = m_1 x + \frac{2}{m_1}$

$0 = -4m_1 + \frac{2}{m_1}$

$\Rightarrow 4m_1^2 = 2$

$\Rightarrow m_1 = \pm\frac{1}{\sqrt{2}}$

$p_2 : y^2 = -16x$

$\Rightarrow y = m_2 x - \frac{4}{m_2}$

$\Rightarrow 0 = 2m_2 - \frac{4}{m_2}$

$\Rightarrow 2m_2^2 = 4$

$\frac{1}{m_1^2} + m_2^2 = 2 + 2 = 4$

15. Let E_1 and E_2 be two ellipses whose centers are at the origin. The major axes of E_1 and E_2 lie along the x-axis and the y-axis, respectively. Let S be the circle $x^2 + (y-1)^2 = 2$. The straight line $x + y = 3$ touches the curces S, E_1 and E_2 at P, Q and R, respectively. Suppose that PQ = PR = $\frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2 respectively, then the correct expression (s) is (are) : **[JEE(Adv.)2015]**

माना कि E_1 और E_2 दो दीर्घवृत्त है जिनके केन्द्र मूलबिन्दु हैं। E_1 और E_2 की दीर्घ अक्षायें क्रमशः x-अक्ष और y-अक्ष पर स्थित है। माना कि S : $x^2 + (y-1)^2 = 2$ एक वृत्त है। सरल रेखा $x + y = 3$, वक्रों S, E_1 और E_2 को क्रमशः P, Q और R पर स्पर्श करती है। माना fd PQ = PR = $\frac{2\sqrt{2}}{3}$ है। यदि e_1 और e_2 क्रमशः E_1 और E_2 की उत्केन्द्रता (eccentricities) हैं, तब सही कथन है (हैं) :

[JEE(Adv.)2015]

(A) $e_1^2 + e_2^2 = \frac{43}{40}$ (B) $e_1 e_2 = \frac{\sqrt{7}}{2\sqrt{10}}$ (C) $\left|e_1^2 - e_2^2\right| = \frac{5}{8}$ (D) $e_1 e_2 = \frac{\sqrt{3}}{4}$

Ans. AB

Sol. $E_1 \to \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$E_2 \to \frac{x^2}{A^2}\frac{y^2}{B^2}=1$

Now as $x + y = 3$ is a tangent

$a^2 + b^2 = A^2 + B^2 = 9$

Now point P is

$x^2 + (2 - x)^2 = 2$

$2x^2 - 4x + 2 = 0$

$x = 1$

so P is (1, 2)

points Q and R are $\left(\frac{5}{3},\frac{4}{3}\right)$ and $\left(\frac{1}{3},\frac{8}{3}\right)$

Now $\left(\frac{5}{3},\frac{4}{3}\right)$ lies on E_1 so $\frac{25}{9a^2}+\frac{16}{9\left(9-a^2\right)}=1$

$\Rightarrow \quad 225 - 25a^2 + 16a^2 = 9a^2(9 - a^2) \quad \Rightarrow \quad 225 - 9a^2 = 9a^2(9 - a^2)$

$\Rightarrow \quad 25 - a^2 = a^2(9 - a^2)$

$\Rightarrow \quad a^4 - 10a^2 + 25 = 0 \quad \Rightarrow \quad a^2 = 5$ so $b^2 = 4$

$e_1^2 = \frac{1}{5}$

Now $\left(\frac{1}{3},\frac{8}{3}\right)$ lies on E_2

$\frac{1}{A^2}+\frac{64}{\left(9-A^2\right)}=9$

$9 - A^2 + 64A^2 = 9A^2(9 - A^2)$

$1 + 7A^2 = A^2 = 9A^2 - A^4 \quad \Rightarrow \quad A^4 - 2A^2 + 1 = 10 \quad \Rightarrow \quad A^2 = 1$ so $B^2 = 8$

$e_2^2 = \frac{7}{8}$

16. Consider the hyperbola H : $x^2 - y^2 = 1$ and a circle S with center N $(x_2, 0)$. Suppose that H and S touch each other at a point P (x_1, y_1) with $x_1 > 1$ and $y_1 > 0$. The common tangent to H and S at P intersects the x-axis at point M. IF (l, m) is the centroid of the triangle ΔPMN, then the correct expression (s) is (are) : **[JEE(Adv.)2015]**

माना कि H : $x^2 - y^2 = 1$ एक अतिपरवलय (hyperbola) है और S एक वृत्त है जिसका केन्द्र N $(x_2, 0)$ है। माना कि H और S एक दूसरे को बिन्दु P (x_1, y_1) पर स्पर्श करते है, जहाँ $x_1 > 1$ और $y_1 > 0$ है। बिन्दु P पर, H और S की सामान्य स्पर्श रेखा x-अक्ष को बिन्दु M पर प्रतिच्छेद करती है। यदि (l, m) त्रिभुज ΔPMN का केन्द्रक (centroid) है, तब सही कथन है (हैं) :

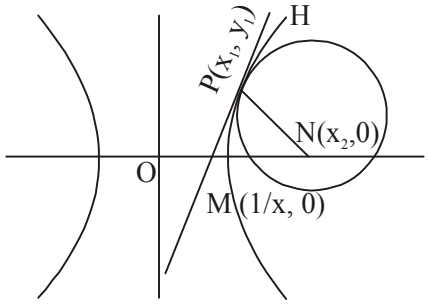
[JEE(Adv.)2015]

(A) $\frac{dl}{dx_1} = 1 - \frac{1}{3x_1^2}$ for $x_1 > 1$(B) $\frac{dm}{dx_1} = \frac{x_1}{3\left(\sqrt{x_1^2 - 1}\right)}$ for $x_1 > 1$

(C) $\frac{dl}{dx_1} = 1 + \frac{1}{3x_1^2}$ for $x_1 > 1$ (D) $\frac{dm}{dy_1} = \frac{1}{3}$ for $y_1 > 1$

Ans. ABD

Sol.



Equation tangent to H at P is $xx_1 - yy_1 = 1$

$$\ell = \frac{x_1 + x_2 + \frac{1}{x_1}}{3}, \qquad m = \frac{y_1}{3} = \frac{\sqrt{x_1^2 - 1}}{3}$$

$$\text{now, } \left.\frac{dy}{dx}\right|_{H \text{ at } P} = \left.\frac{dy}{dx}\right|_{S \text{ at } P} \qquad \Rightarrow \frac{x_1}{y_1} = \frac{x_2 - x_1}{y_1} \qquad \Rightarrow x_2 = 2x$$

$$\text{So, } \quad \ell = x_1 + \frac{1}{3x_1}$$

$$\frac{d\ell}{dx_1} = 1 - \frac{1}{3x_1^2}, \ \frac{dm}{dy_1} = \frac{1}{3}, \frac{dm}{dx_1} = \frac{1}{3} \cdot \frac{x_1}{\sqrt{x_1^2 - 3}}$$

17. The circle $C_1 : x^2 + y^2 = 3$, with centre at O, intersects the parabola $x^2 = 2y$ at the point P in the first quadrant. Let the tangent to the circle C_1 at P touches other two circles C_2 and C_3 at R_2 and R_3, respectively. Suppose C_2 and C_3 have equal radii $2\sqrt{3}$ and centres Q_2 and Q_3, respectively. If Q_2 and Q_3 lie on the y-axis, then :

[JEE(Adv.)2016]

(A) $Q_2Q_3 = 12$ (B) $R_2R_3 = 4\sqrt{6}$

(C) area of the triangle OR_2R_3 is $6\sqrt{2}$ (D) area of thje triangle PQ_2Q_3 is $4\sqrt{2}$

वृत $C_1 : x^2 + y^2 = 3$ जिसका केन्द्रबिन्दु O है, परवलय $x^2 = 2y$ को प्रथम चमुर्थांश में बिन्दु P पर प्रतिच्छेदित करता है। माना कि वृत C_1 के बिन्दु P पर खींची गई स्पर्शरेखा अन्य दो वृतों C_2 और C_3 को क्रमशः बिन्दुओं R_2 और R_3 पर स्पर्श करती हैं। मान लीजिये कि C_2 और C_3 दोनों की त्रिज्याएँ $2\sqrt{3}$ के बराबर हैं और उनके केन्द्रबिन्दु क्रमशः Q_2 और Q_3 है। यदि Q_2 और Q_3, y-अक्ष पर स्थित हैं, तब:

[JEE(Adv.)2016]

(A) $Q_2Q_3 = 12$ (B) $R_2R_3 = 4\sqrt{6}$

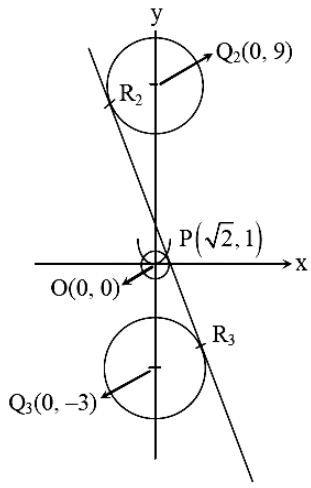
(C) त्रिभुज OR_2R_3 का क्षेत्रफल $6\sqrt{2}$ है (D) त्रिभुज PQ_2Q_3 का क्षेत्रफल $4\sqrt{2}$ है

Ans. ABC

Sol. Equation of tangent at $P\left(\sqrt{2},1\right)$ is $\sqrt{2}x + y - 3 = 0$

If centre of C_2 at $(0, \alpha)$ and radius equal to $2\sqrt{3}$

$$2\sqrt{3} = \left|\frac{\alpha - 3}{\sqrt{3}}\right| \Rightarrow \alpha = -3, 9$$



(A) $Q_2Q_3 = 12$

(B) R_2R_3 = length of transverse common tangent

$$= \sqrt{\left(Q_2Q_3\right)^2 - \left(r_1 + r_2\right)^2} = \sqrt{(12)^2 - \left(2\sqrt{3} + 2\sqrt{3}\right)^2} = 4\sqrt{6}$$

(C) Area of ΔOR_2R_3

$$= \frac{1}{2} \times R_2R_3 \times \text{perpendicular dis tan ce of O from line}$$

$$= \frac{1}{2} \times 4\sqrt{6} \times \sqrt{3} = 6\sqrt{2}$$

(D) Area of $\Delta PQ_2Q_3 = \frac{1}{2} \times 12 \times \sqrt{2} = 6\sqrt{2}$

18. Let P be the point on the parabola $y^2 = 4x$ which is at the shortest distance from the center S of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$. Let Q be the point on the circle dividing the line segment SP internally. Then : **[JEE(Adv.)2016]**

(A) $SP = 2\sqrt{5}$

(B) $SQ : QP = \left(\sqrt{5} + 1\right) : 2$

(C) the x-intercept of the normal to the parabola at P is 6

(D) the slope of the tangent to the circle at Q is $\frac{1}{2}$

माना कि परवलय $y^2 = 4x$ पर P एक ऐसा बिन्दु है जो वृत $x^2 + y^2 - 4x - 16y + 64 = 0$ के केन्द्र बिन्दु S से न्यूनतम दूरी पर है। माना fd or` ij fcUnq Q ऐसा है कि वह रेखाखंड SP को आंतरिक विभाजित करता है। तब : **[JEE(Adv.)2016]**

(A) $SP = 2\sqrt{5}$

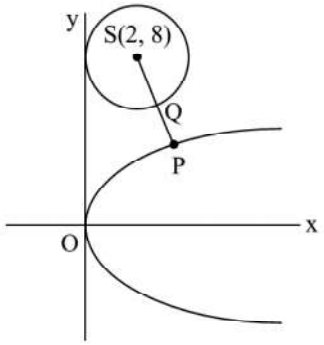
(B) $SQ : QP = \left(\sqrt{5} + 1\right) : 2$

(C) परवलय के बिंदु P पर अभिलम्ब का x अंतः खंड 6 है

(D) वृत के बिन्दु Q पर स्पर्श रेखा की ढाल $\frac{1}{2}$ है

Ans. ACD

Sol.



Equation of normal of parabola

$y + tx = 2t + t^3$

Normal passes through S(2, 8)

$8 + 2t = 2t + t^3$

$\Rightarrow \quad t = 2$

Hence P ≡ (4, 4) and SQ = radius = 2

Paragraph for Question Nos. 19 to 20

Let $F_1(x_1, 0)$ and $F_2(x_2, 0)$ for $x_2 < 0$ and $x_2 > 0$, be the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{8} = 1$. Suppose a parabola having vertex at the origin and focus at F_2 intersects the ellipse at point M in the first quadrant and at point N in the fourth quadrant.

माना कि $F_1(x_1, 0)$ और $F_2(x_2, 0)$ (जिसमें $x_1 < 0, x_2 > 0$) दीर्घवृत्त (ellipse) $\frac{x^2}{9} + \frac{y^2}{8} = 1$ की नाभियाँ (foci) हैं। माना कि एक परवलय (parabola) जिसका शीर्ष (vertex) मूलबिन्दु (origin) पर और नाभि (focus) F_2 पर है, दीर्घवृत्त को प्रथम चतुर्थांश (first quadrant) में M पर और चतुर्थ चतुर्थांश (fourth quadrant) में N पर प्रतिच्छेदित करता है।

19. The orthocentre of the triangle F_1MN is : **[JEE(Adv.)2016]**

त्रिभुज F_1MN का लम्बकेन्द्र (orthocentre) है : **[JEE(Adv.)2016]**

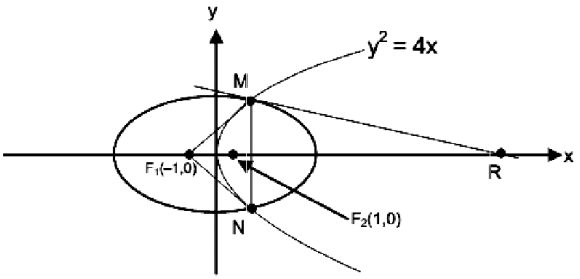
(A) $\left(-\frac{9}{10}, 0\right)$ (B) $\left(\frac{2}{3}, 0\right)$ (C) $\left(\frac{9}{10}, 0\right)$ (D) $\left(\frac{2}{3}, \sqrt{6}\right)$

Ans. A

Sol. Ellipse : $\frac{x^2}{9} + \frac{y^2}{8} = 1$ _____(1)

foci of ellipse are (±1, 0)

Equation of parabola having vertex (0, 0) and focus (1, 0) is $y^2 = 4x$ _____(2)



from equation (1) & (2) $\frac{x^2}{9}+\frac{4x}{8}=1 \Rightarrow 2x^2+9x-18=0 \Rightarrow x=\frac{3}{2},-6$ (rejected)

$\therefore M\left(\frac{3}{2},\sqrt{6}\right)$ and $N\left(\frac{3}{2},-\sqrt{6}\right)$

Equation of altitude from vertex $M\left(\frac{3}{2},\sqrt{6}\right)$ is $\left(y-\sqrt{6}\right)=\frac{5}{2\sqrt{6}}\left(x-\frac{3}{2}\right)$

put $y=0$ we get $\Rightarrow \frac{-12}{5}=x-\frac{3}{2}\Rightarrow x=\frac{-9}{10}$

$\Rightarrow$ orthocener of $\Delta F_1 MN$ is $\left(\frac{-9}{10},0\right)$

20. If the tangents to the ellipse at M and N meet at R and the normal to the parabola at M meets the x-axis at Q, then the ratio of area of the triangle MQR to area of the quadrilateral MF_1NF_2 is : **[JEE(Adv.)2016]**

यदि दीर्घवृत्त के बिन्दुओं M और N पर स्पर्श रेखाएँ (tangents) R पर मिलती हैं और परवलय के बिन्दु M पर अभिलम्ब x-अक्ष को Q पर मिलता है, तब त्रिभुज MQR के क्षेत्रफल और चतुर्भुज (quadrilateral) MF_1NF_2 के क्षेत्रफल का अनुपात (ratio) है : **[JEE(Adv.)2016]**

(A) 3 : 4 (B) 4 : 5 (C) 5 : 8 (D) 2 : 3

Ans. C

Sol. Equation of tangent at point $M\left(\frac{3}{2},\sqrt{6}\right)$ to the ellipse is $\frac{x(3/2)}{9}+\frac{y\sqrt{6}}{8}=1$

put $y=0 \Rightarrow$ R is (6, 0)

Equation of normal to the parabola at point $M\left(\frac{3}{2},\sqrt{6}\right)$ is $\left(y-\sqrt{6}\right)=-\frac{\sqrt{6}}{2}\left(x-\frac{3}{2}\right)$

put $y=0 \Rightarrow$ Q is $\left(\frac{7}{2},0\right)$

$$\text{Now}\ \frac{\text{Area of } \Delta MQR}{\text{Area of quadrilateral } MF_1NF_2} = \frac{\frac{1}{2}.\frac{5}{2}.\sqrt{6}}{\frac{1}{2}.2.2\sqrt{6}} = \frac{5}{8}$$

21. If a chord, which is not a tangent, of the parabola $y^2 = 16x$ has the quation $2x + y = p$, and midpoint (h, k), then which of the following is(are) possible value(s) of p, h and k? **[JEE(Adv.)2017]**

यदि परवलय $y^2 = 16x$ की एक जीवा जो स्पर्शरेखा नहीं है, का समीकरण $2x + y = p$ तथा मध्यबिन्दु (h, k) है, तो निम्न में से p, h तथा k के संभावित मान है/हैं? **[JEE(Adv.)2017]**

(A) $p = -2, h = 2, k = -4$ (B) $p = 2, h = 3, k = -4$

(C) $p = -1, h = 1, k = -3$ (D) $p = 5, h = 4, k = -3$

Ans. B

Sol. $T = S_1$

$k^2 - 16h = ky - 8(x + h)$

$ky - 8x = k^2 - 8h$

compare with $2x + y = p$

$$-\frac{8}{2} = \frac{k}{1} = \frac{k^2 - 8h}{p}$$

Check options.

22. If $2x - y + 1 = 0$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$, then which of the following CANNOT be sides of a right angled triangle? **[JEE(Adv.)2017]**

यदि $2x - y + 1 = 0$ अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$ की स्पर्शरेखा है, तो निम्न में से कौनसी समकोणीय त्रिभुज की भुजायें नहीं हो सकती हैं? **[JEE(Adv.)2017]**

(A) a, 4, 2 (B) a, 4, 1 (C) 2a, 8, 1 (D) 2a, 4, 1

Ans. ABC

Sol. For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Condition of tanjency for $y = mx + c$ is

$c^2 = a^2m^2 - b^2$

$1 = a^2(4) - 16$

$4a^2 = 17$

$2a = \pm\sqrt{17}$

Paragraph for Question 23 to 25

By appropriately matching the information given in the three columns of the following table. Columns 1, 2 and 3 contain conics, equations of tangents to the conics and points of contact, respectively.

नीचे दी गयी टेबल के तीन कॉलमों में उपलब्ध सूचना का उपयुक्त ढंग से सुमेल कर प्रश्नों 47 से 49 के उत्तर दीजिये। कॉलम 1, 2 तथा 3 में क्रमशः कॉनिक (conics), कॉनिक पर स्पर्शरेखा (tangents) का समीकरण तथा स्पर्शबिन्दु दिये गये हैं।

Column-I	**Column-II**	**Column-III**
(I) $x^2 + y^2 = a^2$	(i) $my = m^2x + a$	(P) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$
(II) $x^2 + a^2y^2 = a^2$	(ii) $y = mx + a\sqrt{m^2+1}$	(Q) $\left(\frac{-ma}{\sqrt{m^2+1}}, \frac{a}{\sqrt{m^2+1}}\right)$
(III) $y^2 = 4ax$	(iii) $y = mx + \sqrt{a^2m^2-1}$	(R) $\left(\frac{-a^2m}{\sqrt{a^2m^2+1}}, \frac{1}{\sqrt{a^2m^2+1}}\right)$
(IV) $x^2 - a^2y^2 = a^2$	(iv) $y = mx + \sqrt{a^2m^2+1}$	(S) $\left(\frac{-a^2m}{\sqrt{a^2m^2-1}}, \frac{-1}{\sqrt{a^2m^2-1}}\right)$

23. The tangent to a suitable conic (Column 1) at $(\sqrt{3}, \frac{1}{2})$ is found to be $\sqrt{3}x + 2y = 4$, then which of the following options is the only CORRECT combinaton? **[JEE(Adv.)2017]**

यदि उपयुक्त कॉनिक (कॉलम 1) के बिन्दु $(\sqrt{3}, \frac{1}{2})$ पर स्पर्शरेखा $\sqrt{3}x + 2y = 4$ है, तब निम्न में से कौनसा विकल्प केवल सही l a;ks u gS **[JEE(Adv.)2017]**

(A) (II) (iv) (R) (B) (IV) (iii) (S) (C) (II) (iii) (R) (D) (IV) (iv) (S)

Ans. A

Sol.

24. For $a = \sqrt{2}$, if a tangent is drawn to suitable conic (Column 1) at the point of contact (–1, 1), then which of the following options is the only CORRECT combination for obtaining its equation : **[JEE(Adv.)2017]**

$a = \sqrt{2}$ के लिये उपयुक्त कॉनिक (Column 1) पर एक स्पर्शरेखा खींची जाती है, जिसका स्पर्शबिन्दु (–1, 1) है, तब निम्न में से कौनसा fod Yi bl Li 'kZ{ks kkd kl eldj .ki kir d jusd kd sy l ghl a;ks u gS **[JEE(Adv.)2017]**

(A) (I) (i) (P) (B) (III) (i) (P) (C) (II) (ii) (Q) (D) (I) (ii) (Q)

Ans. D

Sol. (–1, 1) will satisfy curve I for $a = \sqrt{2}$ I $\rightarrow$ (ii) $\rightarrow$ Q

25. If a tangent to a suitable conic (Column 1) is found to be $y = x + 8$ and its point of contact is (8, 16), then which of the following options is the only CORRECT combination? **[JEE(Adv.)2017]**

यदि उपयुक्त कॉनिक (कॉलम 1) के स्पर्शबिन्दु (8, 16) पर स्पर्शरेखा $y = x + 8$ है, तब निम्न में से कौनसा विकल्प केवल सही संयोजन g8 **[JEE(Adv.)2017]**

(A) (III) (i) (P) (B) (I) (ii) (Q) (C) (II) (iv) (R) (D) (III) (ii) (Q)

Ans. A

Sol. $m = 1, \quad C = 8$

for $y^2 = 4ax \Rightarrow (8, 16)$ will satisfying for $a = 8$

$4 - mx + \frac{8}{m}$

$y = x + 8$

III $\rightarrow$ (i) $\rightarrow$ P

26. Consider two straight lines, each of which is tangent to both the circle $x^2 + y^2 = \frac{1}{2}$ and the parabola $y^2 = 4x$. Let these lines intersect at the point Q. Consider the ellipse whose center is at the origin O(0, 0) and whose semi-major axis is OQ. If the length of the minor axis of this ellipse is $\sqrt{2}$, then which of the following statement(s) is (are) TRUE : **[JEE(Adv.)2018]**

(A) For the ellipse, the eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is 1

(B) For the ellipse, the eccentricity is $\frac{1}{2}$ and the length of the latus rectum is $\frac{1}{2}$

(C) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$ is $\frac{1}{4\sqrt{2}}(\pi - 2)$

(D) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$ is $\frac{1}{16}(\pi - 2)$

ऐसी दो सरल रेखाओं पर विचार कीजिये, जिनमें से प्रत्येक, वृत्त $x^2 + y^2 = \frac{1}{2}$ और परवलय दोनों पर ही स्पर्शी है। माना कि ये रेखाऐं बिन्दु Q पर प्रतिच्छेद करती है। एक ऐसे दीर्घवृत्त पर विचार कीजिये जिसका केन्द्र मूलबिन्दु पर है और जिसका अर्ध–दीर्घाक्ष OQ है। ; fn bl nh?kZoÙkkd sy ?kq {kd hy Ecbz $\sqrt{2}$ है, तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं) : **[JEE(Adv.)2018]**

(A) दीर्घवृत्त की उत्केन्द्रता $\frac{1}{\sqrt{2}}$ है और नाभिलम्ब जीवा की लम्बाई 1 है

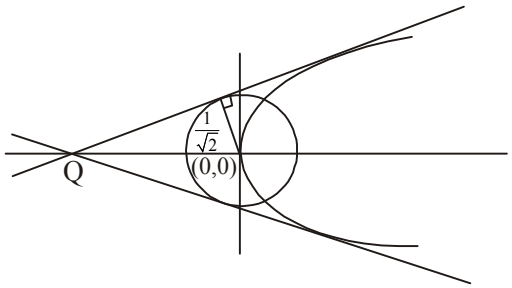
(B) दीर्घवृत्त की उत्केन्द्रता $\frac{1}{2}$ है और नाभिलम्ब जीवा की लम्बाई $\frac{1}{2}$ है

(C) रेखाओं $x = \frac{1}{\sqrt{2}}$ व $x = 1$ के बीच दीर्घवृत्त द्वारा परिबद्ध क्षेत्र का क्षेत्रफल $\frac{1}{4\sqrt{2}}(\pi - 2)$ है

(D) रेखाओं $x = \frac{1}{\sqrt{2}}$ व $x = 1$ के बीच दीर्घवृत्त द्वारा परिबद्ध क्षेत्र का क्षेत्रफल $\frac{1}{16}(\pi - 2)$ है

Ans. AC

Sol.



Let equation of common tangent is $y = mx + \frac{1}{m}$

$$\therefore \left|\frac{0+0+\frac{1}{m}}{\sqrt{1+m^2}}\right| = \frac{1}{\sqrt{2}}$$

$\Rightarrow m^4 + m^2 - 2 = 0$

$\Rightarrow m = \pm 1$

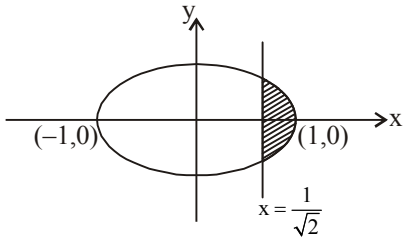
Equation of common tangents are $y = x + 1$ and $y = -x - 1$

point Q is $(-1, 0)$

$\therefore$ Equation of ellipse is $\frac{x^2}{1} + \frac{y^2}{\frac{1}{2}} = 1$

(A) $e = \sqrt{1 - \frac{1}{2}} = \frac{1}{\sqrt{2}}$ & $LR = \frac{2b^2}{a} = 1$

(C)



Area $2.\int_{\frac{1}{\sqrt{2}}}^{1} \frac{1}{\sqrt{2}}.\sqrt{1-x^2}dx$

$$= \sqrt{2}\left[\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}x\right]^1$$

$$= \sqrt{2}\left[\frac{\pi}{4}-\left(\frac{1}{4}+\frac{\pi}{8}\right)\right] = \sqrt{2}\left(\frac{\pi}{8}-\frac{1}{4}\right) = \frac{\pi-2}{4\sqrt{2}}$$

27. Let $H: \frac{x^2}{\alpha^2}-\frac{y^2}{b^2}=1$, where $\alpha > b > 0$, be a hyperbola in the xy-plane whose conjugate axis LM subtends an angle of 60º at one of its vertices N. Let the area of the triangle LMN be $4\sqrt{3}$: **[JEE(Adv.)2018]**

Column-I	**Column-II**
P. The length of the conjugate axis of H is	1. 8
Q. The eccentricity of H is	2. $\frac{4}{\sqrt{3}}$
R. The distance between the foci of H is	3. $\frac{2}{\sqrt{3}}$
S. The length of the latus rectum of H is	4. 4

The correct option is :

माना कि $H: \frac{x^2}{\alpha^2}-\frac{y^2}{b^2}=1$, जहाँ $\alpha > b > 0$, xy-समतल में एक ऐसा अतिपरवलय है जिसका संयुग्मी अक्ष LM उसके एक शीर्ष N पर 60º का कोण अंतरित करता है। माना कि त्रिभुज LMN का क्षेत्रफल $4\sqrt{3}$ है : **[JEE(Adv.)2018]**

Column-I	**Column-II**
P. ℍ के संयुग्मी अक्ष की लम्बाई है	1. 8
फ०ℍ की उत्केन्द्रता है	2. $\frac{4}{\sqrt{3}}$
R. H की नभियों के बीच की दूरी है	3. $\frac{2}{\sqrt{3}}$
S. H के नाभिलम्ब जीवा की लम्बाई है	4. 4

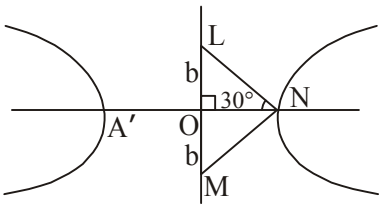
दिए हुए विकल्पों में से सही विकल्प है :

(A) $P \to 4; Q \to 2; R \to 1; S \to 3$ (B) $P \to 4; Q \to 3; R \to 1; S \to 2$

(C) $P \to 4; Q \to 1; R \to 3; S \to 2$ (D) $P \to 3; Q \to 4; R \to 2; S \to 1$

Ans. B

Sol.



Area of LMN = $4\sqrt{3}$

$\frac{1}{2}(2b)\left(\sqrt{3}b\right) = 4\sqrt{3} \quad \Rightarrow b^2 = 4 \qquad \Rightarrow b = 2 \qquad \Rightarrow 2b = 4$

Here $\frac{a}{b} = \cot 30° \qquad \Rightarrow a = \sqrt{3b} \quad \Rightarrow a = 2\sqrt{3}$

$b^2 = a^2(e^2 - 1)$

$4 = 12(e^2 - 1)$

$e^2 = 1 + \frac{1}{3} = \frac{4}{3} \Rightarrow e = \frac{2}{\sqrt{3}}$ and $2ae = 2 \times 2\sqrt{3} \times \frac{2}{\sqrt{3}} = 8$

and length of latus rectum $= \frac{2b^2}{a} = \frac{2 \times 4}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$

This section contains TWO (02) Column-Match sets.

Let the circle $C_1: x^2 + y^2 = 9$ and $C_2: (x-3)^2 + (y-4)^2 = 16$ intersect at the points X and Y. Suppose that another circle $C_3: (x-h)^2 + (y-k)^2 = r^2$ satisfies the following conditions.
(i) Centre of C_3 is collinear with the centres of C_1 & C_2,
(ii) C_1 & C_2 both lie inside C_3 and
(iii) C_3 touches C_1 at M and C_2 at N
Let the line through X and Y intersect C_3 at Z and W and let a common tangent of C_1 & C_3 be a tangent to the parabola $x^2 = 8\alpha y$
There are some expressions given in the column-I, whose values are given in column-II below :

Column-I	**Column-I**
(I) $2h + k$	(P) 6
(II) $\frac{\text{Length of ZW}}{\text{Length of XY}}$	(Q) $\sqrt{6}$
(III) $\frac{\text{Area of triange MZN}}{\text{Area of triangle ZMW}}$	(R) $\frac{5}{4}$
(IV) α	(S) $\frac{21}{5}$
	(T) $2\sqrt{6}$
	(U) $\frac{10}{3}$

This section contains TWO (02) Column-Match sets.

माना कि वृत्त (circle) C_1: $x^2 + y^2 = 9$ और वृत्त C_2: $(x-3)^2 + (y-4)^2 = 16$, एक दूसरे को बिन्दुओं X और Y पर काटते हैं। मान y hft ; s, d vkSj oÙkk C_3 : $(x-h)^2 + (y-k)^2 = r^2$ निम्नलिखित शर्तों को सन्तुष्ट करता है :

; पद्ध C_3 का केन्द्र (centre), C_1 और C_2 के केन्द्रों के संरेख (collinear) है

(ii) C_1 और C_2 दोनों C_3 के अन्दर है और

(iii) C_3, C_1 को M और C_2 को N पर स्पर्श करता है।

माना कि X और Y से होकर जाने वाली रेखा C_3 को Z और W पर काटती है तथा C_1 और C_3 की एक उभयनिष्ठ स्पर्श–रेखा (common tangent), परवलय $x^2 = 8\alpha y$ की स्पर्श–रेखा है।

कॉलम-I में कुछ व्यंजक (expression) हैं जिनका मान नीचे दी गयी कॉलम-II में है।

Column-I	**Column-I**
(I) $2h + k$	(P) 6
(II) $\frac{\text{ZW की लम्बाई}}{\text{XY की लम्बाई}}$	(Q) $\sqrt{6}$
(III) $\frac{\text{त्रिभुज MZN का क्षेत्रफल}}{\text{त्रिभुज ZMW का क्षेत्रफल}}$	(R) $\frac{5}{4}$
(IV) α	(S) $\frac{21}{5}$
	(T) $2\sqrt{6}$
	(U) $\frac{10}{3}$

28. Which of the following is the only incorrect combination : **[JEE(Adv.)2019]**

निम्न में से कौन सा एकमात्र संयोजन गलत है : **[JEE(Adv.)2019]**

(A) (III), (R) (B) (IV), (S) (C) (I), (P) (D) (IV), (U)

Ans. B

Sol.

29. Which of the following is the only correct combination : **[JEE(Adv.)2019]**

निम्न में से कौन सा एकमात्र संयोजन सही है : **[JEE(Adv.)2019]**

(A) (I), (U) (B) (II), (Q) (C) (I), (S) (D) (II), (T)

Ans. B

Sol. $C_1 : x^2 + y^2 = 9$ $C_1 (0, 0)$ $r_1 = 3$

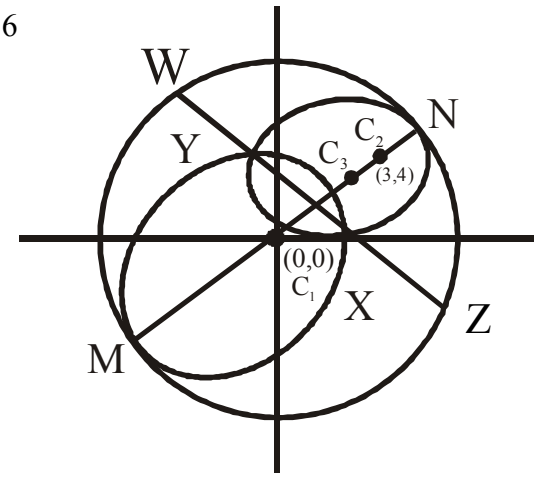
$C_2 : (x-3)^2 + (y-4)^2 = 16$ $C_2 : (3, 4)$ $r_2 = 4$

$C_3 : (x-h)^2 + (y-k)^2 = r^2$ $C_3 : (h, k)$ $r_3 = r$

$MN = MC_1 + C_1C_2 + C_2N$

$MN = r_1 + \sqrt{9+16} + r_2$

$MN = 3 + 5 + 4 = 12 = 2r \qquad \Rightarrow r = 6$



C_3C_1 and C_2 are colinear

$C_1(0, 0)$ $C_2(3, 4)$

$\frac{4}{3} = \frac{y}{x} \qquad \Rightarrow y = \frac{4x}{3}$

$C_3(h, k) \qquad k = \frac{4h}{3}$

$C_3 : \left(h, \frac{4h}{3}\right)$

$C_1C_3 = MC_3 - C_1M$

$\sqrt{h^2 + \frac{16h^2}{9}} = 6 - 3 = 3$

$\frac{5|h|}{3} = 3 \Rightarrow |h| = \frac{9}{5} \Rightarrow h = \frac{9}{5}$

$k = \frac{4h}{3} = \frac{4}{3} \times \frac{9}{5} = \frac{12}{5}$

(I) $\qquad 2h + k = \frac{18}{5} + \frac{12}{5} = \frac{30}{5} = 6$

(II) Equation of line ZW

$C_1 - C_2 = 0$

$x^2 + y^2 - 9 - x^2 - 9 + 6x - y^2 + 8y - 16 + 16 = 0$

$6x + 8y = 18$

$3x + 4y = 9$

Distance of ZW from $C_1(0, 0)$

$\left|\frac{-9}{\sqrt{25}}\right| = \frac{9}{5}$

Length $XY = 2\sqrt{3^2 - \left(\frac{9}{5}\right)^2}$

$= 2\sqrt{9 - \frac{81}{25}} = \sqrt{\frac{225 - 81}{25}}$

$= 2\sqrt{\frac{144}{25}} = \frac{12 \times 2}{5} = \frac{24}{5}$

$C_3 : \left(\frac{9}{5}, \frac{12}{5}\right)$

Distance of C_3 from ZW

$$\left|\frac{3\times\frac{9}{5}+4\times\frac{12}{5}-9}{5}\right|$$

$$=\left|\frac{\frac{27}{5}+\frac{48}{5}}{5}-9\right| = \left|\frac{27+48-45}{25}\right| = \left|\frac{30}{25}\right| = \frac{6}{5}$$

Length of $WZ = \sqrt{36-\frac{36}{25}}$

$$=12\sqrt{\frac{24}{25}} = \frac{12}{5}\times 2\sqrt{6} = \frac{24}{5}\sqrt{6}$$

$$\frac{\text{Length of ZW}}{\text{Length of XY}} = \frac{\frac{24\sqrt{6}}{5}}{\frac{12}{5}\times 2} = \sqrt{6}$$

(III) Area of $\Delta MZN = \frac{1}{2}(MN)\left(\frac{1}{2}ZW\right)$

$$= \frac{1}{4}\times 12\times\frac{24}{5}\times\sqrt{6}$$

$$= \frac{72\sqrt{6}}{5}$$

Area of $\Delta ZMW = \frac{1}{2}(ZW)$ (OM + Distance of C_1 from ZW)

$$= \frac{1}{2}\times\frac{24\sqrt{6}}{5}\times\left(3+\frac{9}{5}\right)$$

$$= \frac{12\sqrt{6}}{5}\times\left(\frac{24}{5}\right) = \frac{288\sqrt{6}}{25}$$

$$\frac{\text{Area of triangle MZN}}{\text{Area of triangle ZMW}} = \frac{\frac{72\sqrt{6}}{5}}{\frac{288\sqrt{6}}{25}}$$

$$= \frac{72\sqrt{6}}{5} \times \frac{25}{288\sqrt{6}} = \frac{5}{4}$$

(IV) Slope of tangent to C_1 at $M = \frac{-3}{4}$

Equation of tangent

$$y = mx - 3\sqrt{1+m^2}$$

$$y = \frac{-3x}{4} - 3\sqrt{1+\frac{9}{16}}$$

$$y = \frac{-3x}{4} - \frac{3\times 5}{4}$$

$4y = -3x - 15$

$4y + 3x + 15 = 0$ ..... (i)

$$T \Rightarrow x = my + \frac{2\alpha}{m}$$

$3x + 4y + 15 = 0$

$mx - m^2y - 2\alpha = 0$

$$\frac{3}{m} = \frac{-4}{m^2} = \frac{-15}{2\alpha}$$

$3m^2 = -4m$

$3m^2 + 4m = 0$

$m(3m + 4) = 0$

$$m = \frac{-4}{3}$$

$$\frac{3\times 3}{-4} = \frac{-15}{2\alpha} \Rightarrow \alpha = \frac{10}{3}$$

30. Define the collections $\{E_1, E_2, E_3 .....\}$ of ellipses and $\{R_1, R_2, R_3 .....\}$ of rectangles as follows

$$E_1 = \frac{x^2}{9} + \frac{y^2}{4} = 1;$$

R_1 : rectangles of largest area, with sides parallel to the axes, inscribed in E_1

E_n : ellipse $\frac{x^2}{a_n^2}+\frac{y^2}{b_n^2}=1$ of largest area inscribed in R_{n-1}, $n>1$;

R_n : rectangle of largest area, with sides parallel to the axes, inscribed in E_n, $n>1$
Then which of the following options is/are correct : **[JEE(Adv.)2019]**

(A) The distance of a focus from the centre in E_9 is $\frac{\sqrt{5}}{32}$

(B) The length of latus rectum of E_9 is $\frac{1}{6}$

(C) The eccentricities of E_{18} and E_{19} are not equal

(D) $\sum_{n=1}^{N}(\text{area of } R_n)<24$ for each positive integer N

दीर्घवृतो (ellipses) $\{E_1,E_2,E_3.....\}$ और आयतों (rectangles) $\{R_1,R_2,R_3.....\}$ के संग्रहो को निम्न प्रकार से परिभाषित करें

$E_1=\frac{x^2}{9}+\frac{y^2}{4}=1$;

R_1 : अधिकतम क्षेत्र (largest area) का आयत जिसकी भुजाएं अक्षों (axes) के समान्तर है और जो E_1 में अंतर्स्थित (inscribed) है

E_n : अधिकतम क्षेत्र वाला दीर्घवृत $\frac{x^2}{a_n^2}+\frac{y^2}{b_n^2}=1$ जो R_{n-1}, $n>1$ में अंतर्स्थित है

R_n : अधिकतम क्षेत्र का आयत जिसकी भुजाएं अक्षों के समान्तर है और जो E_n, $n>1$ में अंतर्स्थित है।
r c fuEkkfudksa esa lgh fodYi ¼ls½ fodYi gS ¼gSa½ : **[JEE(Adv.)2019]**

(A) E_9 में केन्द्र से एक नाभि (focus) की दूरी $\frac{\sqrt{5}}{32}$ है

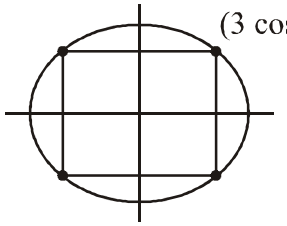
(B) E_9 के नाभिलम्ब (latus rectum) की लम्बाई $\frac{1}{6}$ है

(C) E_{18} और E_{19} की उत्केंन्द्रतायें (eccentricities) समान नहीं है

(D) प्रत्येक पूर्णांक N के लिए $\sum_{n=1}^{N}(R_n \text{ का क्षेत्रफल})<24$ है

Ans. BD

Sol. $E_1 \quad \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$



Area of rectangle = (2a cos)(2b sin)

= 2 ab sin2

area of rectangle will be maximum when $= \frac{\pi}{4}$

$$E_2 = \frac{x^2}{(a\cos\theta)^2} + \frac{y^2}{(b\sin\theta)^2} = 1$$

$$\Rightarrow a_n = \frac{a_{n-1}}{\sqrt{2}} \ \& \ b_n = \frac{b_{n-1}}{\sqrt{2}}$$

$\Rightarrow$ Forming G.P.

for E_9 : $a_9 = 3\left(\frac{1}{\sqrt{2}}\right)^8 = \frac{3}{16}$

$$b_9 = 2\left(\frac{1}{\sqrt{2}}\right)^8 = \frac{2}{16}$$

eccentricity of all ellipses will remain same as $\frac{b_n}{a_n}$ is same for all ellipse

(1) distance of focus from centre in $E_9 = a_9 e = \frac{\sqrt{5}}{3} \times \frac{3}{16} = \frac{\sqrt{5}}{16}$

(2) $l_{LR} = \frac{2b_9}{a_9} = \frac{2\left(\frac{1}{8}\right)^2}{3/16} = \frac{2\times 16}{3\times 64} = \frac{1}{6}$

(4) area of $R_n = \frac{1}{2}(\text{area of } R_{n-1})$

area of $R_1 = 12 = A_1$

$$\sum_{n=1}^{N} A_n = A_1 + A_2 + ... + A_n < \lim_{n\to\infty} \sum_{r=1}^{N} A_n$$

$$< \left(12 + \frac{12}{2} + ...\infty\right)$$

$$< \frac{12}{1-\frac{1}{2}} = 24$$

$$\sum_{n=1}^{N} (\text{area of } R_n) < 24$$

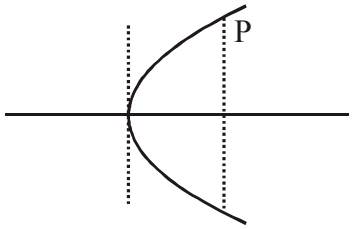
31. Let a, b and स be positive real numbers. Suppose P is an end point of the latus rectum of the parabola $y^2 = 4$सx, and suppose the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the point P. If the tangents to the parabola and the ellipse at the point P are perpendicular to each other, then the eccentricity of the ellipse is : **[JEE(Adv.)2020]**

माना कि a, b एवं स धनात्मक वास्तविक संख्याएँ (positive real numbers) हैं। मान लिजिये कि परवलय $y^2 = 4$सx के नाभिलम्ब जीवा का एक अंत्य बिन्दु P है, एवं मान लीजिये कि दीर्घवृत्त $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ बिन्दु P से गुजरता है। यदि बिन्दु P पर परवलय एवं दीर्घवृत्त d hLi ' kqfdks, i, d nwljs d sy Eor gSa r c nh?kZo`Ùkd hmd Rdsan zkgS **[JEE(Adv.)2020]**

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$

Ans. A

Sol.



p (d, 2λ)

slope of tangent at P

$m_1 = 1$

for ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{dy}{dx} = -\frac{xb^2}{ya^2}$$

$$m)_{(\lambda, 2\lambda)} = -1$$

$$-\frac{\lambda b^2}{2\lambda a^2} = -1$$

$$\frac{b^2}{a^2} = 2$$

$\because$ ellipse is vertical

$$e = \sqrt{1-\frac{a^2}{b^2}} = \sqrt{1-\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

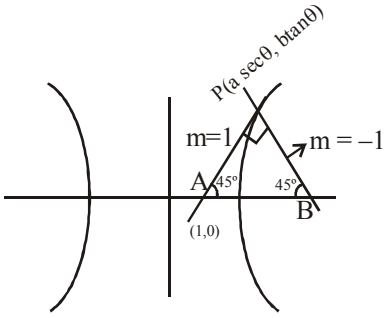
32. Let a and b be positive real numbers such that $a > 1$ and $b < a$. Let P be a point in the first quadrant that lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Suppose the tangent to the hyperbola at P passes through the point $(1, 0)$, and suppose the normal to the hyperbola at P cuts off equal intercepts on the coordinate axes. Let Δ denote the area of the triangle formed by the tangent at P, the normal at P and the x-axis. If e denotes the eccentricity of the hyperbola, then which of the following statements is/are TRUE : **[JEE(Adv.)2020]**

माना कि a एवं b इस प्रकार की धनात्मक वास्तविक संख्याएँ हैं जिनके लिये $a > 1$ और $b < a$ हैं। माना कि प्रथम चतुर्थांश का एक बिन्दु P अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ पर स्थित है। मान लीजिये कि बिन्दु P पर डाली गयी अतिपरवलय की स्पर्श रेखा बिंदु $(1, 0)$ से गुजरती है, एवं मान लीजिये कि बिन्दु P पर डाले गये अतिपरवलय का अभिलम्ब निर्देशक अक्षों पर बराबर अन्तःखण्ड काटता है। माना कि P पर डाली गयी स्पर्शरेखा, P पर डाले गये अभिलम्ब एवं x- अक्ष द्वारा बनाये गए त्रिभुज के क्षेत्रफल को Δ से दर्शाया जाता है। यदि vfr ijoy ; d hmRd Hkqt dks se से दर्शाया जाता है, तब निम्न में से कौन सा (से) कथन सही है (हैं) : **[JEE(Adv.)2020]**

(A) $1 < e < \sqrt{2}$ (B) $\sqrt{2} < e < 2$ (C) $\Delta = a^4$ (D) $\Delta = b^4$

Ans. AD

Sol.



equation of tangent $\Rightarrow T = 0$

$$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$$

passing through $(1, 0)$

$\alpha = \sec\theta$

$$m = \frac{b\sec\theta}{a\tan\theta} = 1$$

$b = \tan\theta$

$p(a^2, b^2)$ or $p(\sec^2\theta, \tan^2\theta)$

$b^2 = a^2(e^2 - 1)$

$\tan^2\theta = \sec^2\theta\,(e^2 - 1)$

$\varepsilon^2 - 1 = \sin^2\theta$

$0 < \sin^2\theta < 1$

$1 < 1 + \sin^2\theta < 2$

$1 < e^2 < \sqrt{2}$

$1 < e < \sqrt{2}$

$$\text{Area of } \Delta PAB = \frac{1}{2}(AP)(PB)$$

$$= \frac{1}{2}(AP)^2 \qquad \{\because PA = PB\}$$

$$= \frac{1}{2}\left\{(\sec^2\theta - 1)^2 + (\tan^2\theta - 0)^2\right\}$$

$= \tan^4\theta = b^4$

33. Let E denote the parabola $y^2 = 8x$. Let $P = (-2, 4)$, and let Q and Q' be two distinct points on E such that the lines PQ and PQ' are tangents to E. Let F be the focus of E. Then which of the following statements is (are) TRUE ?

[JEE(Adv.)2021]

(A) The triangle PFQ is a right-angled triangle

(B) The triangle QPQ' is a right-angled triangle

(C) The distance between P and F is $5\sqrt{2}$

(D) F lies on the line joining Q and Q'

माना E परवलय $y^2 = 8x$ को निरूपित करता है। माना $P = (-2, 4)$ तथा माना Q व Q' , E पर दो भिन्न बिन्दु इस प्रकार है कि रेखाएँ PQ व PQ', E पर स्पर्श रेखाएँ हैं। F, E की नाभि है। तब निम्न में से कौनसा/कौनसे कथन सही है/ हैं?

[JEE(Adv.)2021]

(A) त्रिभुज PFQ एक समकाणीय त्रिभुज है

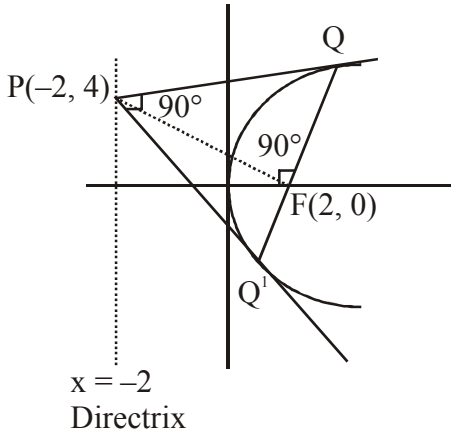
(B) त्रिभुज QPQ' एक समकोणीय त्रिभुज है

(C) च व F के मध्य दूरी $5\sqrt{2}$ है

(B) थएफ व Q' को मिलाने वाली रेखा पर स्थित है

Ans. ABD

Sol.



P lies on the directrix of the parabola.

$\Rightarrow$ QQ' is focal chord.

$\Rightarrow \angle QPQ' = 90°$ (Directrix is also director circle of the parabola)

$\Rightarrow \angle PFQ = 90°$ (Portion of tangent intercepted between point of contact and directrix subtends right angle at focus.)

$\Rightarrow$ $PF = \sqrt{4^2 + 4^2} = 4\sqrt{2}$

As (ABD)

Question Stem for Question Nos. 34 and 35

Consider the region $R = \{(x,y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ and } y^2 \leq 4 - x\}$. Let F be the family of all circles that are contained in R and have centers on the x-axis. Let C be the circle that has largest radius among the circles in F. Let (α, β) be a point where the circle C meets the curve $y^2 = 4 - x$.

क्षेत्र $R = \{(x,y) \in \mathbb{R} \times \mathbb{R} : x \geq 0$ तथा $y^2 \leq 4 - x\}$ पर विचार कीजिए। माना F उन सभी वृत्तों का कुल है जो R में स्थित हैं तथा जिनके केन्द्र x-अक्ष पर हैं। माना C वह वृत्त है जिसकी F के वृत्तों में से बड़ी त्रिज्या है। माना (α, β) बिन्दु है जहाँ वृत्त C, वक्र $y^2 = 4 - x$ से मिलता है।

34. The radius of the circle C is ___ . **[JEE(Adv.)2021]**

वृत्त C की त्रिज्या ___ है। **[JEE(Adv.)2021]**

Ans. 1.5

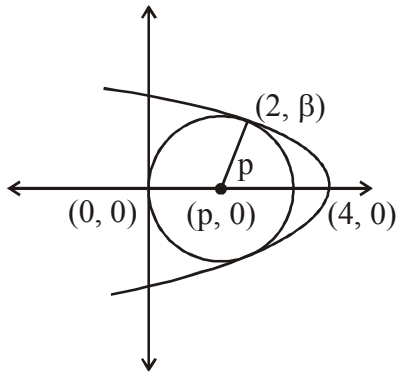
Sol.

35. The value of α is ___ . **[JEE(Adv.)2021]**

α का मान ___है। **[JEE(Adv.)2021]**

Ans. 2

Sol.



$C : x^2 + y^2 - 2px = 0$

$P : y^2 = 4 - x$

By solving both the curves

$x^2 - (2p + 1)x + 4 = 0$

$D = 0$ (Both curves touch each other)

$(2p + 1)^2 - 4^2 = 0$

$(2p + 5)(2p - 3) = 0 \Rightarrow p = 3/2 =$ radius of circle

$\phi o \rho\ p = \frac{3}{2} \Rightarrow x^2 - 4x + 4 = 0 \Rightarrow x = 2 = \alpha$

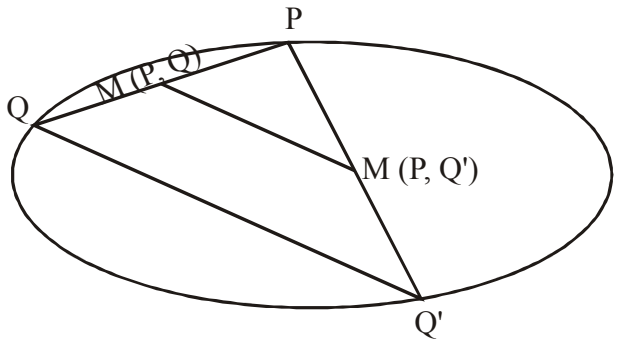
36. Let E be the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. For any three distinct points P, Q and Q' On E, let M (P, Q) be the mid-point of the line segment joining P and Q, and M (P, Q') be the mid-point of the line segment joining P and Q'. Then the maximum possible value of the distance between M (P, Q) and M(P, Q'), as P, Q and Q' vary on E, is_____.

[JEE(Adv.)2021]

माना E, दीर्घवृत्त $\frac{x^2}{16} + \frac{y^2}{9} = 1$ है। E पर किन्हीं तीन भिन्न बिन्दुओं P, Q व Q' के लिए माना M (P, Q), P व Q को मिलाने वाले रेखाखण्ड का मध्य बिन्दु है तथा M (P, Q'), P व Q' को मिलाने वाले रेखाखण्ड का मध्य बिन्दु है। E पर P, Q व Q' के परिवर्तन के अनुसार M (P, Q) तथा M(P, Q') के मध्य दूरी का अधिकतम संभावित मान ज्ञात कीजिए। **[JEE(Adv.)2021]**

Ans. 4

Sol.



We know from triangle geometry

distance between M(P, Q) and M(P, Q') = $\frac{1}{2}$ QQ'

hence distance between M(P, Q) and M(P, Q') will be maximum if QQ' is maximum and in given ellipse maximum value of QQ' will be 2a

then maximum distance between M(P, Q) and M(P, Q') = $\frac{1}{2}$ (2a) = $\frac{8}{2}$ = 4

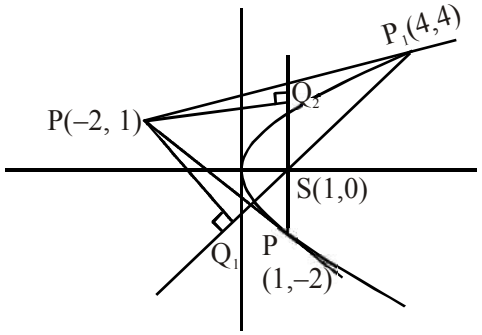
37. Consider the parabola $y^2 = 4x$. Let S be the focus of the parabola. A pair of tangents drawn to the parabola from the point P = (– 2, 1) meet the parabola at P_1 and P_2. Let Q_1 and Q_2 be points on the lines SP_1 and SP_2 respectively such that PQ_1 is perpendicular to SP_1 and PQ_2 is perpendicular to SP_2. Then, which of the following is/are TRUE? **[JEE(Adv.)2022]**

परवलय $y^2 = 4x$ पर विचार कीजिए। माना कि S इस परवलय की नाभि है। बिंदु P = (– 2, 1) से परवलय पर डाली गई दो स्पर्श रेखायें परवलय को बिंदुओं P_1 एवं P_2 पर मिलती है। माना कि Q_1 एवं Q_2 क्रमशः रेखाओं SP_1 एवं SP_2 पर ऐसे बिंदु है कि PQ_1 , SP_1 पर लंब है, और PQ_2 , SP_2 पर लंब है। तब निम्न में से कौन सा सत्य हैं ? **[JEE(Adv.)2022]**

(A) $SQ_1 = 2$ (B) $Q_1Q_2 = \frac{3\sqrt{10}}{5}$ (C) $PQ_1 = 3$ (D) $SQ_2 = 1$

Ans. BCD

Sol. Let equation of tangent with slope 'm' be



$T : y = mx + \frac{1}{m}$

T : passes through (–2, 1) so

$1 = -2m + \frac{1}{m}$

$\Rightarrow m = -1$ or $m = \frac{1}{2}$

Points are given by $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

So, one point will be (1, –2) and (4, 4)

$P_1S : 4x - 3y - 4 = 0$

$P_2S : x - 1 = 0$

$PQ_1 = \left|\frac{4(-2) - 3(1) - 4}{5}\right| = 3$

$SP = \sqrt{10}; PQ_2 = 3; SQ_1 = 1 = SQ_2$

$$\frac{1}{2}\left(\frac{Q_1Q_2}{2}\right)\times\sqrt{10}=\frac{1}{2}\times3\times1 \qquad \text{(comparing Areas)}$$

$$\Rightarrow Q_1Q_2=\frac{2\times3}{\sqrt{10}}=\frac{3\sqrt{10}}{5}$$

38. Consider the ellipse

$$\frac{x^2}{4}+\frac{y^2}{3}=1.$$

Let H(α, 0), $0<\alpha<2$, be a point. A straight line drawn through H parallel to the y-axis crosses the ellipse and its auxiliary circle at points E and F respectively, in the first quadrant. The tangent to the ellipse at the point E intersects the positive x-axis at a point G. Suppose the straight line joining F and the origin makes an angle ϕ with the positive x-axis. **[JEE(Adv.)2022]**

Column -I	**Column-II**
(I) If $\phi=\frac{\pi}{4}$, then the area of the triangle FGH is	(P) $\frac{\left(\sqrt{3}-1\right)^4}{8}$
(II) If $\phi=\frac{\pi}{3}$, than the area of the triangle FGH is	(Q) 1
(III) If $\phi=\frac{\pi}{6}$, than the area of the triangle FGH is	(R) $\frac{3}{4}$
(IV) If $\phi=\frac{\pi}{12}$, than the area of the triangle FGH is	(S) $\frac{1}{2\sqrt{3}}$
	(T) $\frac{3\sqrt{3}}{2}$

The correct option is:

दीर्घवृत्त $\frac{x^2}{4}+\frac{y^2}{3}=1.$

पर विचार कीजिए। माना कि H(α, 0), $0<\alpha<2$, एक बिंदु है। बिंदु H से होती हुई एवं y-अक्ष के समांतर एक सरल रेखा दीर्घवृत्त , oabl d sl gksÙkd ksi Zhe pr qkZkesa Oe' k¼f cnq lsE एवं F पर प्रतिच्छेदित करती है। बिंदु E पर दीर्घवृत्त की स्पर्श रेखा धनात्मक x-अक्ष को एक बिंदु G पर प्रतिच्छेदित करती है। मान लीजिए कि F एवं मूलबिंदु को जोड़ने वाली सरल रेखा, धनात्मक x-अक्ष के साथ एक कोण ϕ बनाती है। **[JEE(Adv.)2022]**

Column -I	**Column-II**
(I) यदि $\phi = \frac{\pi}{4}$ है, तब त्रिभुज FGH का क्षेत्रफल	(P) $\frac{\left(\sqrt{3}-1\right)^4}{8}$
(II) यदि $\phi = \frac{\pi}{3}$ है, तब त्रिभुज FGH का क्षेत्रफल	(Q) 1
(III) यदि $\phi = \frac{\pi}{6}$ है, तब त्रिभुज FGH का क्षेत्रफल	(R) $\frac{3}{4}$
(IV) यदि $\phi = \frac{\pi}{12}$ है, तब त्रिभुज FGH का क्षेत्रफल	(S) $\frac{1}{2\sqrt{3}}$
	(T) $\frac{3\sqrt{3}}{2}$

सही विकल्प है :

(A) (I) $\rightarrow$ (R); (II) $\rightarrow$ (S); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (P)

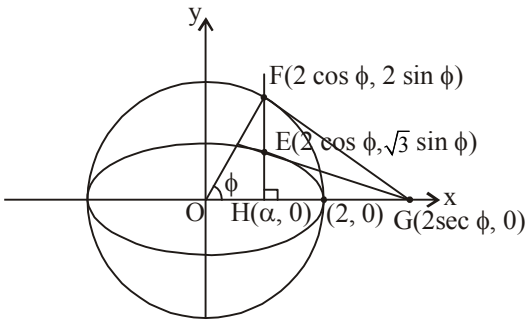
(B) (I) $\rightarrow$ (R); (II) $\rightarrow$ (T); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (P)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (T); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (P)

(D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (P

Ans. C

Sol.



$\alpha \equiv 2\cos\phi$

Tangent at $E(2\cos\phi, \sqrt{3}\sin\phi)$ to ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$

i.e. $\frac{x\cos\phi}{2}+\frac{y\sin\phi}{\sqrt{3}}=1$ intersect x-axis at G(2sec ϕ, 0)

Area of triangle $FGH = \frac{1}{2}(2\sec\phi - 2\cos\phi)2\sin\phi$

$\Delta = 2\sin^2\phi.\tan\phi$

$\Delta = (1-\cos 2\phi)\cdot\tan\phi$

I. If $\phi = \frac{\pi}{4}, \Delta = 1 \rightarrow (Q)$

II. $\phi = \frac{\pi}{3}, \Delta = 2 \cdot \left(\frac{\sqrt{3}}{2}\right)^2 \cdot \sqrt{3} = \frac{3\sqrt{3}}{2} \rightarrow (T)$

III. If $\phi = \frac{\pi}{6}, \Delta = 2 \cdot \left(\frac{1}{2}\right)^2 \cdot \frac{1}{\sqrt{3}} = \frac{1}{2\sqrt{3}} \rightarrow (S)$

IV. If $\phi = \frac{\pi}{12}, \Delta = \left(1 - \frac{\sqrt{3}}{2}\right) \cdot (2 - \sqrt{3}) = \frac{(2-\sqrt{3})^2}{2} \rightarrow (P)$

39. Consider the hyperbola $\frac{x^2}{100} - \frac{y^2}{64} = 1$ with foci at S and S_1, where S lies on the positive x-axis. Let P be a point on the hyperbola, in the first quadrant. Let $\angle SPS_1 = \alpha$, with $\alpha < \frac{\pi}{2}$. The straight line passing through the point S and parallel to the straight line passing through the point P, intersects the straight line S_1P at P_1. Let δ be the distance of P from the straight line SP_1, and $\beta = S_1P$. Then the greatest integer less than or equal to $\frac{\beta\delta}{9}\sin\frac{\alpha}{2}$ is _____. **[JEE(Adv.)2022]**

अतिपरवलय $\frac{x^2}{100} - \frac{y^2}{64} = 1$ पर विचार कीजिए जिसकी नाभियाँ S एवं S_1 है, जहाँ S धनात्मक x-अक्ष पर स्थित है। माना कि P प्रथम चतुर्थांश में अतिपरवलय पर एक बिंदु है। माना कि $\angle SPS_1 = \alpha$ है, जहाँ $\alpha < \frac{\pi}{2}$ है। बिंदु S से जाने वाली सरल रेखा, जिसकी ढाल अतिपरवलय के बिंदु P पर स्पर्श रेखा के ढाल के बराबर है, सरल रेखा S_1P को P_1 पर प्रतिच्छेदित करती है। माना कि P की सरल रेखा SP_1 से दूरी δ है, एवं $\beta = S_1P$ है। तब $\frac{\beta\delta}{9}\sin\frac{\alpha}{2}$ से कम या बराबर महत्तम पूर्णांक _____ है। **[JEE(Adv.)2022]**

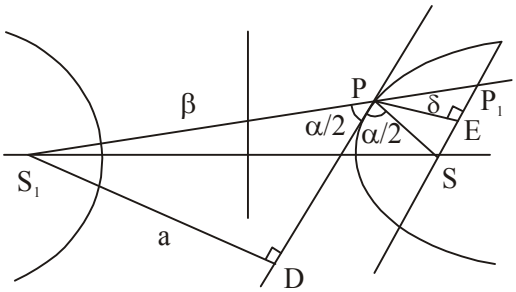
Ans. 7

Sol. Let $S_1D = a$ in S_1PD $\sin\left(\frac{\alpha}{2}\right) = \frac{a}{\beta}$

Now $\frac{1}{9}\beta\delta\sin\left(\frac{\alpha}{2}\right) = \frac{1}{9}\beta\delta \cdot \frac{a}{\beta} = \frac{a\delta}{9} = \frac{b^2}{9} = \frac{64}{9}$

we know product of perpendiculars from foci upon any tangent to Hyperbola = b^2, ($a\delta = b^2$)

Hence $\left[\frac{64}{9}\right] = 7$



40. Let P be a point on the parabola $y^2 = 4ax$, where $a > 0$. The normal to the parabola at P meets the x-axis at a point Q. The area of the triangle PFQ, where F is the focus of the parabola, is 120. If the slope m of the normal and a are both positive integers, then the pair (a, m) is : **[JEE(Adv.)2023]**

माना कि परवलय (parabola) $y^2 = 4ax$, जहाँ $a > 0$ है, पर P एक बिन्दु है। बिन्दु P पर परवलय का अभिलम्ब (normal) x-अक्ष से fcUnq Q पर मिलता है। त्रिभुज PFQ, जहाँ F इस परवलय कि नाभि (focus) है, का क्षेत्रफल 120 है। यदि अभिलम्ब की ढाल (slope) m एवं a दोनों धनात्मक पूर्णांक (positive integers) हैं, तब युग्म (pair) (a, m) है : **[JEE(Adv.)2023]**

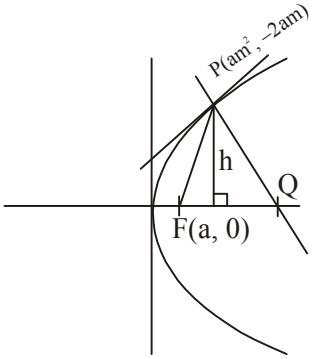
(A) (2,3) (B) (1,3) (C) (2,4) (D) (3,4)

Ans. A

Sol. Equation of normal

$y = mx - 2am - am^3$

$Q(2a + am^2, 0)$



$$\Delta PFQ = \frac{1}{2} \times h \times FQ = 120$$

$$\frac{1}{2}|-2am|\left(a + am^2\right) = 120$$

$a^2 |m(1 + m^2)| = 120$

Now check options

$a = 2$ $m = 3$

41. Let T_1 and T_2 be two distinct common tangents to the ellipse $E : \frac{x^2}{6} + \frac{y^2}{3} = 1$ and the parabola $P : y^2 = 12x$.

Suppose that the tangent T_1 touches P and E at the points A_1 and A_2, respectively and the tangent T_2 touches P and E at the points A_4 and A_3, respectively. Then which of the following statements is(are) true?

[JEE(Adv.)2023]

(A) The area of the quadrilateral $A_1A_2A_3A_4$ is 35 square units

(B) The area of the quadrilateral $A_1A_2A_3A_4$ is 36 square units

(C) The tangents T_1 and T_2 meet the x-axis at the point $(-3,0)$

(D) The tangents T_1 and T_2 meet the x-axis at the point $(-6,0)$

माना कि T_1 एवं T_2 दीर्घवृत (ellipse) $E: \frac{x^2}{6}+\frac{y^2}{3}=1$ एवं परवलय (parabola) $P: y^2=12x$ की दो भिन्न उभयनिष्ठ स्पर्श रेखाएं (distinct common tangents) हैं। माना कि स्पर्श रेखा T_1, P एवं E को क्रमशः बिन्दुओं A_1 एवं A_2 पर स्पर्श करती है और स्पर्श j§ kk T_2, P एवं E को क्रमशः बिन्दुओं A_4 एवं A_3 पर स्पर्श करती है। तब निम्न में से कौनसा(से) कथन सत्य है(हैं)?

[JEE(Adv.)2023]

(A) चतुर्भुज $A_1A_2A_3A_4$ का क्षेत्रफल 35 वर्ग इकाई है

(B) चतुर्भुज $A_1A_2A_3A_4$ का क्षेत्रफल 36 वर्ग इकाई है

(C) स्पर्श रेखाएं T_1 एवं T_2, x-अक्ष को बिन्दु $(-3,0)$ पर मिलती हैं

(D) स्पर्श रेखाएं T_1 एवं T_2, x-अक्ष को बिन्दु $(-6,0)$ पर मिलती हैं

Ans. AC

Sol. $E: \frac{x^2}{6}+\frac{y^2}{3}=1,\ \text{Tangent}: y=m_1x\pm\sqrt{6m_1^2+3}$

$P: y^2=12x,\ \text{Tangent}: y=m_2x+\frac{3}{m_2}$

For common tangent $m=m_1=m_2$

$\pm\sqrt{6m^2+3}=\frac{3}{m}$

$\Rightarrow m=\pm1$

$\Rightarrow$ equation of common tangents are $y=x+3$ and $y=-x-3$

point of contact for parabola is $\left(\frac{a}{m^2},\frac{2a}{m}\right)$

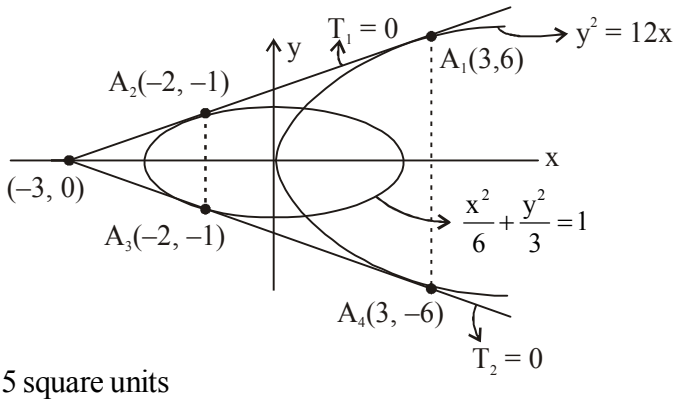
$\Rightarrow A_1\equiv(3,6),\ A_4(3-6)$

Let $A_2(x_1,y_1)\Rightarrow$ tangent to E is $\frac{xx_1}{6}+\frac{yy_1}{3}=1$

$A_2(x_1,y_1)=A_2(-2,1)$

A_3 is mirror image of A_2 in x–axis $\Rightarrow A_3(-2,-1)$

Intersection point of $T_1=0$ and $T_2=0$ is $(-3,0)$



Area of quadrilateral $A_1A_2A_3A_4=\frac{1}{2}(12+2)\times5=35$ square units

42. Consider the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$. Let S(p, q) be a point in the first quadrant such that $\frac{p^2}{9}+\frac{q^2}{4}>1$. Two tangents are drawn from S to the ellipse, of which one meets the ellipse at one end point of the minor axis and the other meets the ellipse at a point T in the fourth quadrant. Let R be the vertex of the ellipse with positive

x-coordinate and O be the center of the ellipse. If the area of the triangle ΔORT is $\frac{3}{2}$, then which of the following options is correct ? **[JEE(Adv.)2024]**

दीर्घवृत्त (ellipse) $\frac{x^2}{9}+\frac{y^2}{4}=1$ पर विचार कीजिए। माना कि S(p, q) प्रथम चतुर्थांश (first quadrant) में एक इस प्रकार का बिन्दु है कि $\frac{p^2}{9}+\frac{q^2}{4}>1$ है। बिन्दु S से दीर्घवृत्त के लिए दो स्पर्श रेखाएँ (tangents) खींची गयी हैं, जिनमें से एक रेखा दीर्घवृत्त पर लघु अक्ष (minor axis) के एक अंत्य बिन्दु (end point) पर मिलती है तथा दूसरी रेखा चौथे चतुर्थांश (fourth quadrant) में दीर्घवृत्त के एक बिन्दु T पर मिलती है। माना कि R दीर्घवृत्त का वह शीर्ष (vertex) है जिसका x-निर्देशांक (x-coordinate) धनात्मक (positive) है, और nhZkd kd lsaz O है। यदि त्रिभुज ΔORT का क्षेत्रफल $\frac{3}{2}$ है, तब निम्नलिखित विकल्पों में से कौनसा सही है ?

[JEE(Adv.)2024]

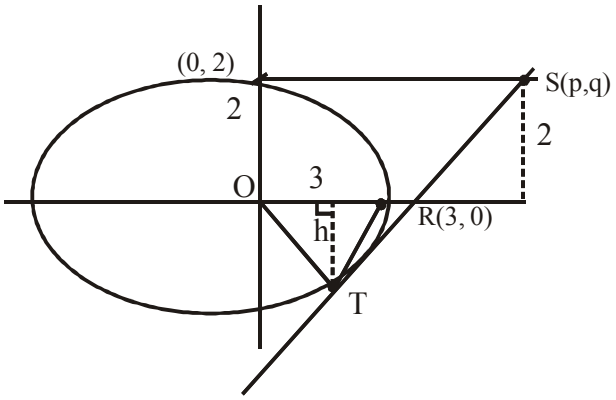
(A) $q=2,\ p=3\sqrt{3}$ (B) $q=2,\ p=4\sqrt{3}$ (C) $q=1,\ p=5\sqrt{3}$ (D) $q=1,\ p=6\sqrt{3}$

Ans. A

Sol. $\frac{x^2}{9}+\frac{y^2}{4}=1$, S(p, q)

$\frac{p^2}{9}+\frac{q^2}{4}>1$

Tthen s gives outside ellipse in 1st quadrant.



So $q=2$ Let $T=(3\cos\theta, 2\sin\theta)$

Area $(\Delta ORT)=\frac{3}{2}$

$\frac{1}{2}\times OR\times h=\frac{3}{2}$

$$\frac{1}{2}\times 3\times |2\sin\theta| = \frac{3}{2}$$

$$|\sin\theta| = \frac{1}{2}$$

$$\sin\theta = \frac{-1}{2} \text{ (IV Quadrant)}$$

$$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$$

$$T\left(\frac{3\sqrt{3}}{2}, 2\times\left(\frac{-1}{2}\right)\right) = \left(\frac{3\sqrt{3}}{2}, -1\right)$$

Now equation of tangent at point T

$T = 0$

$$\frac{x\sqrt{3}}{2\times 9} - \frac{y\times 1}{4} = 1$$

S (P, 2)

$$\frac{P\times 3\sqrt{3}}{2\times 9} - \frac{2}{4} = 1$$

$$\frac{P\times 3\sqrt{3}}{2\times 9} = \frac{3}{2}$$

43. Let A_1, B_1, C_1 be three points in the xy-plane. Suppose that the lines A_1C_1 and B_1C_1 are tangents to the curve $y^2 = 8x$ at A_1 and B_1, respectively. If O = (0, 0) and C_1 = (– 4, 0), then which of the following statements is (are) TRUE? **[JEE(Adv.)2024]**

(A) The length of the line segment OA_1 is $4\sqrt{3}$

(B) The length of the line segment A_1B_1 is 16

(C) The orthocenter of the triangle $A_1B_1C_1$ is (0, 0)

(D) The orthocenter of the triangle $A_1B_1C_1$ is (1, 0)

माना कि A_1, B_1, C_1 xy-तल (xy-plane) में स्थित तीन बिंदु हैं। मान लीजिये कि रेखाएं A_1C_1 और B_1C_1, वक्र (curve) $y^2 = 8x$ के लिए क्रमशः A_1 और B_1 पर स्पर्श रेखाएं (tangents) हैं। यदि O = (0, 0) और C_1 = (– 4, 0), तब निम्नलिखित कथनों में से कौनसा (से) सत्य है (हैं)? **[JEE(Adv.)2024]**

(A) रेखाखण्ड (line segment) OA_1 की लम्बाई $4\sqrt{3}$ है

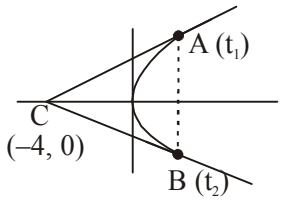
(B) रेखाखण्ड A_1B_1 की लम्बाई 16 है

(C) त्रिभुज (triangle) $A_1B_1C_1$ का लंबकेन्द्र (orthocenter) (0, 0) है

(D) त्रिभुज $A_1B_1C_1$ का लंबकेन्द्र (1, 0) है

Ans. AC

Sol.



tangent

$ty = x + at^2$

$ty = x + 2t^2$

(–4,0) lies on it.

$0 = -4 + 2t^2$

$\Rightarrow t^2 = 2 \Rightarrow t = \pm\sqrt{2}$

$A_1\left(at_1^2, 2at_1\right) \equiv A_1\left(4, 4\sqrt{2}\right)$

$B_1\left(at_2^2, 2at_2\right) \equiv B_1\left(4, -4\sqrt{2}\right)$

(A) $OA_1 = 4\sqrt{3}$

(B) $A_1B_1 = 8\sqrt{2}$

(C) Orthocenter.

equation of altitude through A

$y - 4\sqrt{2} = \sqrt{2}\left(x - 4\right)$

$\Rightarrow y = 4\sqrt{2}x$

equation of altitude through C $\quad y = 0$

$\Rightarrow$ orthocenter $\equiv (0,0)$

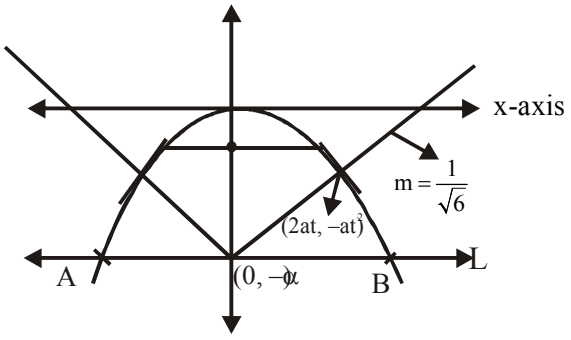
44. A normal with slope $\frac{1}{\sqrt{6}}$ is drawn from the point $(0, -\alpha)$ to the parabola $x^2 = -4ay$, where $a > 0$. Let L be the line passing through $(0, -\alpha)$ and parallel to the directrix of the parabola. Suppose that L intersects the parabola at two points A and B. Let r denote the length of the latus rectum and s denote the square of the length of the line segment AB. If $r : s = 1 : 16$, then the value of 24a is _____. **[JEE(Adv.)2024]**

बिंदु $(0, -\alpha)$ से परवलय (parabola) $x^2 = -4ay$, जहाँ $a > 0$ है, के लिए $\frac{1}{\sqrt{6}}$ ढाल (slope) का एक अभिलंब (normal) खींचा गया है। माना कि L, परवलय की नियता (directrix) के समान्तर $(0, -\alpha)$ से होकर जाने वाली रेखा है। मान लीजिये कि L, परवलय को

दो बिन्दुओं A और B पर प्रतिच्छेदित (intersect) करती है। माना कि r, परवलय की नाभिलंब जीवा (latus rectum) की लंबाई को n' kZkgS vkSj s, रेखाखण्ड (line segment) AB की लंबाई के वर्ग को दर्शाता है। यदि r : s = 1 : 16 है, तब 24a का मान _____ है। **[JEE(Adv.)2024]**

Ans. 12

Sol. $x^2 = -4ay\ (a > 0)$



$r = 4a$

$S = |AB|^2$

$\frac{r}{s} = \frac{1}{16}$

$\Rightarrow 16r = s$

Now $x^2 = -4ay$

$2x = -4a\frac{dy}{dx}$

$\left.\frac{dy}{dx}\right| = -\frac{x}{2a}$

$(2at, -at^2)$

$= -\frac{2at}{2a}$

$= -t$

$m_{tangent} = -t$

$m_{Normal} = \frac{1}{t}$

Given : $\frac{1}{t} = \frac{1}{\sqrt{6}} \Rightarrow t = \sqrt{6}$

Equation of Normal

$\left(y + at^2\right) = \frac{1}{\sqrt{6}}\left(x - 2at\right)$

$\left(y + 6a\right) = \frac{1}{\sqrt{6}}\left(x - 2\sqrt{6}a\right)$

$y\sqrt{6} + 6\sqrt{6}a = x - 2\sqrt{6}a$

$x - y\sqrt{6} = 8\sqrt{6}a$

Cuts y-axis at $(0, -\alpha)$

$0 + \alpha\sqrt{6} = 8\sqrt{6}a$

$\alpha = 8a$

Now equation of L $\Rightarrow y = -\alpha$

$y = -8a$

For coordinates of A & B

$x^2 = (4a)(8a)$

$x^2 = 32a^2$

$A\left(-4\sqrt{2}a, -8a\right) \& B\left(4\sqrt{2}a, -8a\right)$

$s = |AB| = 8\sqrt{2}a$

$s = |AB|^2 = 128$

Now, $16.4a = 128a^2$

$a = \frac{1}{2}$

$\therefore 24a \Rightarrow 24 \times \frac{1}{2}$

$= 12$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. If on a given base, a triangle be described such that the sum of the tangents of the base angles is a constant, then the locus of the vertex is :

(A) a circle (B) a parabola (C) an ellipse (D a hyperbola

यदि दिये गये आधार पर एक त्रिभुज इस प्रकार परिभाषित है कि उनके आधार कोणों के स्पर्शज्याओं का योग एक स्थिरांक हो तो शीर्ष d kfcUnq Hkgksk:

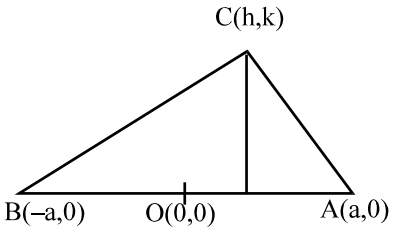
(A) एक वृत्त (B) एक परवलय (C) एक दीर्घवृत्त (D एक अतिपरवलय

Ans. B

Sol. $\tan A = \frac{k}{a-h}$; $\tan B = \frac{k}{a+h}$

$\tan A + \tan B = 'C'$ (a constant)

$$\frac{k}{a-h} + \frac{k}{a+h} = C \quad \Rightarrow \quad \frac{k(a+h)+k(a-h)}{a^2-h^2} = C$$



$\therefore \quad a^2 - h^2 = \frac{2a}{C} \cdot k \quad \Rightarrow$ locus of (h, k) will be

$x^2 = a^2 - \frac{2a}{C} \cdot y \quad \Rightarrow$ A parabola Ans.

2. Through the vertex O of the parabola, $y^2 = 4ax$ two chords OP & OQ are drawn and the circles on OP & OQ as diameters intersect in R. If θ_1, θ_2 & ϕ are the angles made with the axis by the tangents at P & Q on the parabola & by OR then the value of, $\cot\theta_1 + \cot\theta_2 =$

परवलय $y^2 = 4ax$ के शीर्ष O से दो जीवाऐं OP तथा OQ खींची जाती है तथा OP तथा OQ को व्यास मानकर बनाया गया वृत्त R पर प्रतिच्छेद करता है तथा θ_1, θ_2 क्रमशः परवलय के अक्ष से बिन्दुओं P और Q पर स्पर्श रेखा द्वारा निर्मित कोण तथा OR द्वारा निर्मित d kskϕ है तब $\cot\theta_1 + \cot\theta_2 =$

(A) $-2\tan\phi$ (B) $-2\tan(\pi-\phi)$ (C) 0 (D) $2\cot\phi$

Ans. A

Sol. Slope of tangant at P is $\frac{1}{t_1}$

and at Q = $\frac{1}{t_2}$

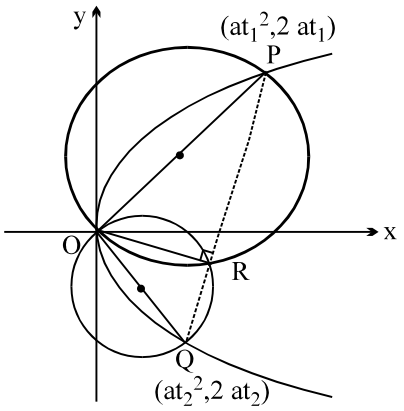
$\Rightarrow \cot\theta_1 = t_1$ and $\cot\theta_2 = t_2$

Slope of PQ $= \dfrac{2}{t_1 + t_2}$

$\Rightarrow$ Slope of OR is $\left(-\dfrac{t_1 + t_2}{2}\right) = \tan\phi$

(Note angle in a semicircle is 90º)

$\Rightarrow \tan\phi = -\dfrac{1}{2}(\cot\theta_1 + \cot\theta_2) \Rightarrow \cot\theta_1 + \cot\theta_2 = -2\tan\phi$



3. If a normal to a parabola $y^2 = 4ax$ makes an angle ϕ with its axis, then it will cut the curve again at an angle :

यदि परवलय $y^2 = 4ax$ के किसी बिन्दु पर अभिलम्ब अक्ष से ϕ कोण बनाती है जब यह वक्र को पुनः किस कोण पर काटेगा :

(A) $\tan^{-1}(2\tan\phi)$ (B) $\tan^{-1}\left(\dfrac{1}{2}\tan\phi\right)$ (C) $\cot^{-1}\left(\dfrac{1}{2}\tan\phi\right)$ (D) none

Ans. B

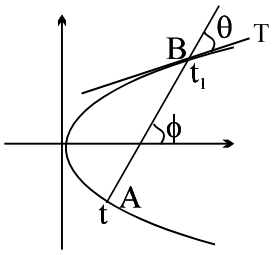
Sol. equation of normal at t : $y + tx = 2at + at^3$

$\therefore$ m_N at $A = -t = \tan\phi$

$t = -\tan\phi = m_1$

Now tangent at B $t_1 y = x + at_1^2$ with $m_2 = \dfrac{1}{\tan\theta}$

also $t_1 = -t - \dfrac{2}{t}$



$\therefore$ $\tan\theta = \left|\dfrac{\frac{1}{t_1} + t}{1 - \frac{t}{t_1}}\right| = \left|\dfrac{1 + tt_1}{t - t_1}\right| = \left|\dfrac{1 + t^2}{2\left(t + \frac{1}{t}\right)}\right| = \left|\dfrac{\sec^2\phi . \tan\phi}{2(\sec^2\phi)}\right|$ [As $t\,t_1 = -t^2 - 2$]

Hence $\tan\theta = \dfrac{\tan\phi}{2} \Rightarrow \theta = \tan^{-1}\left(\dfrac{\tan\phi}{2}\right)$

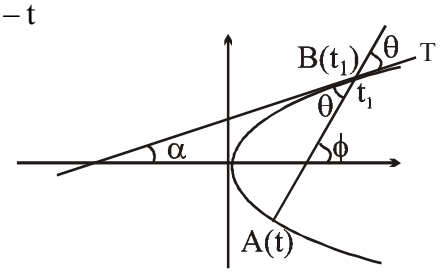
Alterntively: Equation of normal at A

$y + tx = 2at + at^3$

$\therefore$ Slope of normal at A, $m_A = -t \Rightarrow \tan\phi = -t$

Equation of tangent at B: $t_1 y = xt + at_1^2$

slope, $\tan a = \dfrac{1}{t_1}$ where $t_1 = -t - \dfrac{2}{t}$



$$\Rightarrow \qquad \frac{1}{\tan\alpha} = \tan\phi + \frac{2}{\tan\phi} \quad \Rightarrow \quad \tan\alpha = \frac{\tan\phi}{2+\tan^2\phi}$$

$$\therefore \qquad \tan\theta = \left|\frac{\tan\alpha - \tan\phi}{1+\tan\alpha\tan\phi}\right| = \left|\frac{\frac{\tan\phi}{2+\tan^2\phi} - \tan\phi}{1+\frac{\tan\phi}{2+\tan^2\phi}\tan\phi}\right| = \left|\tan\phi\left(\frac{1-2-\tan^2\phi}{2+\tan^2\phi+\tan^2\phi}\right)\right|$$

$$= \frac{\tan\phi}{2} \qquad \therefore \qquad q = \tan^{-1}\left(\frac{\tan\phi}{2}\right)$$

4. If $x + iy = \sqrt{\varphi + i\psi}$ where $i = \sqrt{-1}$ and ϕ and ψ are non zero real parameters then ϕ = constant and ψ = constant, represents two systems of rectangular hyperbola which intersect at an angle of :

यदि $x + iy = \sqrt{\varphi + i\psi}$ जहां $i = \sqrt{-1}$ तथा ϕ तथा ψ अशून्य वास्तविक प्राचल है। तो ϕ = अचर तथा ψ = अचर, दो आयताकार vfrijoy; kadksfu: fir djrkgSAt ks, d nwljs dksfuEu dksdkskij ifjPNsn djrsgSa

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. D

Sol. $x^2 - y^2 + 2xyi = \phi + i\psi$; $x^2 - y^2 = \phi$ and $xy = \psi$; which intersects at $\frac{\pi}{2} \Rightarrow$ (D)

5. For each positive integer *n*, consider the point P with abscissa *n* on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$ then $\lim\limits_{n\to\infty}(n \cdot d_n)$ has the value equal to :

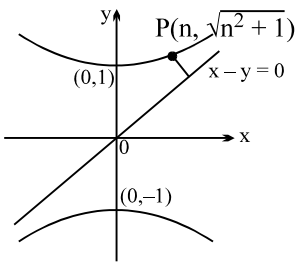
n के धनात्मक पूर्णांक के लिए माना बिन्दु P वक्र $y^2 - x^2 = 1$ पर स्थित है जिसकी भुज *n* है। यदि d_n बिन्दु P तथा $y = x$ मध्य न्यूनतम nwjh gSA rks $\lim\limits_{n\to\infty}(n \cdot d_n)$ का मान होगा :

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) 0

Ans. A

Sol. Curve is rectangular hyperbola.

perpendicular distance, $d_n = \left|\frac{n - \sqrt{n^2+1}}{\sqrt{2}}\right|$



$$\lim_{n\to\infty}(n\cdot d_n) = \lim_{n\to\infty}\frac{n}{\sqrt{2}}\left|\left(\sqrt{n^2+1}-n\right)\right|$$

$$= \lim_{n\to\infty}\frac{n}{\sqrt{2}}\left|\frac{1}{\sqrt{n^2+1}+n}\right| = \frac{1}{2\sqrt{2}} \text{ Ans.}$$

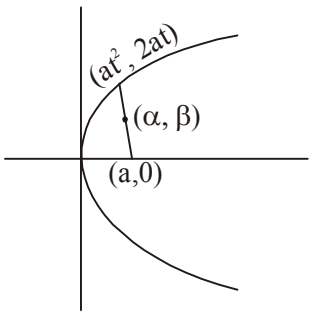
6. The locus of the mid-point of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with directrix :

परवलय $y^2 = 4ax$ की नाभि को परवलय पर स्थित एक चर बिन्दु से जोड़ने वाली रेखा के मध्य बिन्दु का बिन्दुपथ एक अन्य परवलय gkxk] ft l d hfu; r kgS

(A) $x = -a$ (B) $x = -a/2$ (C) $x = 0$ (D) $x = a/2$

Ans. C

Sol. Let (α, β) is mid point



$$\alpha = \frac{at^2+a}{2},\ \beta = \frac{2at+0}{2} \Rightarrow 2\alpha = at^2 + a,\ at = \beta$$

$$\therefore\quad 2\alpha = a.\frac{\beta^2}{a^2} + a \text{ or } 2a\alpha = \beta^2 + a^2$$

For locus $\alpha \to x$, $\beta \to y$

$\therefore$ The locus is $y^2 = \frac{4a}{2}\left(x-\frac{a}{2}\right) = 4b(x-b),\ \left(b=\frac{a}{2}\right)$

$\therefore$ Directrix is $(x-b)+b=0$ or $x=0$

7. The equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ is :

वक्रों $y^2 = 8x$ और $xy = -1$ की उभयनिष्ठ स्पर्शी का समीकरण है :

(A) $3y = 9x + 2$ (B) $y = 2x + 1$ (C) $2y = x + 8$ (D) $y = x + 2$

Ans. D

Sol. The given curves are

$$y^2 = 8x \quad \ldots (1)$$

and $$xy = -1 \quad \ldots (2)$$

If m is the slope of tangent to (1), then equation of tangent is

$$y = mx + 2/m.$$

If this tangent is also a tangent to (2), then

$$x\left(mx + \frac{2}{m}\right) = -1$$

$\Rightarrow \quad mx^2 + \frac{2}{m} x + 1 = 0$

$\therefore \quad m^2 x^2 + 2x + m = 0$

We should get repeated roots for this equation (conditions of tangency)

$\Rightarrow \quad D = 0$

$\therefore \quad (2)^2 - 4m^2 \cdot m = 0$

$\Rightarrow \quad m^3 = 1$

$\Rightarrow \quad m = 1$

Hence required tangent is $y = x + 2$.

8. The slope of the focal chords of the parabola $y^2 = 16x$ which are tangents to the circle $(x - 6)^2 + y^2 = 2$ are :

परवलय $y^2 = 16x$ की नाभिय जीवा (focal chord), वृत्त $(x - 6)^2 + y^2 = 2$ की स्पर्श रेखा है, तो जीवा की प्रवणता के सम्भव मान है:

(A) ± 2 (B) $-1/2, 2$ (C) ± 1 (D) $-2, 1/2$

Ans. C

Sol. Focus of parabola $y^2 = 6x$ ...(1) is (4,0)

Thus equation of focal chord of (1) with slope say m is,

$(y - 0) = m(x - 4) \Rightarrow y = m(x - 4)$ ...(2)

Now since line (2) touches circle $(x - 6)^2 + y^2 = 2$

So, its distance from centre of the circle is equal to the radius

$$\Rightarrow \left|\frac{m(6-4)}{\sqrt{1+m^2}}\right| = \sqrt{2} \rightarrow 2m^2 = 1 + m^2 \Rightarrow m = \pm 1$$

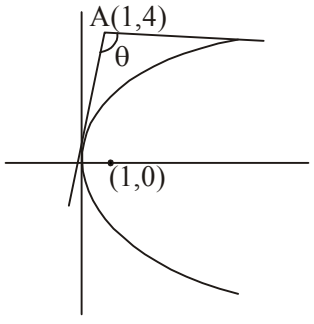
9. The angle between the tangents drawn from the point (1, 4) to the parabola $y^2 = 4x$ is :

बिन्दु (1, 4) से परवलय $y^2 = 4x$ पर खींची गई स्पर्श रेखाओं के मध्य कोण है :

(A) $\pi/2$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/6$

Ans. B

Sol.



Equation of tangent is $y = mx + 1m$.

Since it passes through (1,4)

$\therefore\ m^2 - 4m + 1 = 0$

$\Rightarrow m_1 + m_2 = 4,\ m_1 m_2 = 1$

$\Rightarrow |m_1 - m_2| = 2\sqrt{3}$

$\therefore \tan\theta = \dfrac{2\sqrt{3}}{2} = \sqrt{3} \Rightarrow \theta = \dfrac{\pi}{3}$

10. The axis of parabola is along the line $y = x$ and the distance of vertex from origin is $\sqrt{2}$ and that of origin from its focus is $2\sqrt{2}$. If vertex and focus both lie in the 1st quadrant, then the equation of the parabola is :

एक परवलय का अक्ष $y = x$ है और शीर्ष और नाभि मूल बिन्दु से क्रमशः $\sqrt{2}$ और $2\sqrt{2}$. की दूरी पर है। परवलय का समीकरण , यदि ' kkZ Fkk kFkk Fke pr qkZ ke gS gS gk:

(A) $(x + y)^2 = (x - y - 2)$ (B) $(x - y)^2 = (x + y - 2)$

(C) $(x - y)^2 = 4(x + y - 2)$ (D) $(x - y)^2 = 8(x + y - 2)$

Ans. D

Sol. Since, distance of vertex from origin is $\sqrt{2}$ and focus is $2\sqrt{2}$

$\therefore$ V(1,1) and f(2,2) (i.e. lying on $y = x$)

where, the length of latus rectum $= 4a = 4\sqrt{2}$

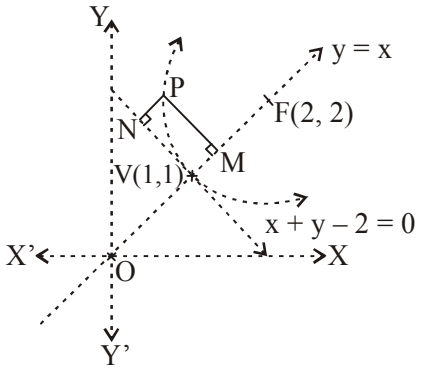
$\therefore$ By definition of parabola

$PM^2 = (4a)PN$

Where, PN is length of perpendicular upon $x + y - 2 = 0$, i.e., tangent at vertex

$$\Rightarrow \frac{(x - y)^2}{2} = 4\sqrt{2}\left(\frac{x + y - 2}{\sqrt{2}}\right)$$

$\Rightarrow (x - y)^2 = 8(x + y - 2)$



11. The locus of the middle point of the intercept of the tangents drawn from an external point to the ellipse $x^2 + 2y^2 = 2$, between the coordinates axes, is :

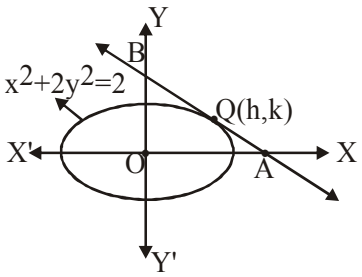
किसी बाह्य बिन्दु से दीर्घवृत्त $x^2 + 2y^2 = 2$ पर खींची गई स्पर्श रेखा द्वारा अक्षों पर काटे गये अन्तःखण्ड के मध्य बिन्दु का बिन्दुपथ है:

(A) $\frac{1}{x^2}+\frac{1}{2y^2}=1$ (B) $\frac{1}{4x^2}+\frac{1}{2y^2}=1$ (C) $\frac{1}{2x^2}+\frac{1}{4y^2}=1$ (D) $\frac{1}{2x^2}+\frac{1}{y^2}=1$

Ans. C

Sol. Let the point of contact be

$R \equiv (\sqrt{2}\cos\theta, \sin\theta)$



Equation of tangent AB is

$\frac{x}{\sqrt{2}}\cos\theta + y\sin\theta = 1$

$\Rightarrow A \equiv (\sqrt{2}\sec\theta, 0); B \equiv (0, \text{cosec}\,\theta)$

Let the middle point Q of AB be (h, k)

$h = \frac{\sqrt{2}\sec\theta + 0}{2}, \; k = \frac{\text{cosec}\,\theta + 0}{2}$

$\Rightarrow h = \frac{\sec\theta}{\sqrt{2}}, k = \frac{\text{cosec}\,\theta}{2}$

$\Rightarrow \cos\theta = \frac{1}{h\sqrt{2}}, \sin\theta = \frac{1}{2k} \Rightarrow \frac{1}{2h^2}+\frac{1}{4k^2}=1,$

$\therefore$ Required locus is $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$

Trick : The locus of mid-points of the portion of tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intercepted between axes is $a^2y^2 + b^2x^2 = 4x^2y^2$

i.e., $\frac{a^2}{4x^2} + \frac{b^2}{4y^2} = 1$ or $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$

12. The minimum area of triangle formed by the tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and coordinate axes is :

(A) ab sq. units (B) $\frac{a^2+b^2}{2}$ sq. units (C) $\frac{(a+b)^2}{2}$ sq. units (D) $\frac{a^2+ab+b^2}{3}$ sq. units

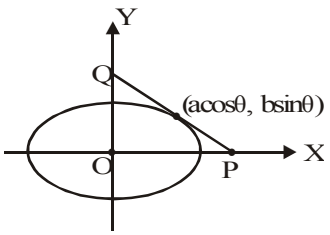
दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ की स्पर्श रेखा एवं अक्षों से निर्मित त्रिभुज का न्यूनतम क्षेत्रफल है :

(A) इ वर्ग इकाई (B) $\frac{a^2+b^2}{2}$ वर्ग इकाई (C) $\frac{(a+b)^2}{2}$ वर्ग इकाई (D) $\frac{a^2+ab+b^2}{3}$ वर्ग इकाई

Ans. A

Sol. Equation of tangent at (acosθ, bsinθ) is

$\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$



$P = \left(\frac{a}{\cos\theta}, 0\right)$

$Q = \left(0, \frac{b}{\sin\theta}\right)$

Area of $OPQ = \frac{1}{2}\left|\left(\frac{a}{\cos\theta}\right)\left(\frac{b}{\sin\theta}\right)\right| = \frac{ab}{|\sin 2\theta|}$

$\therefore\ (\text{Area})_{min} = ab$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

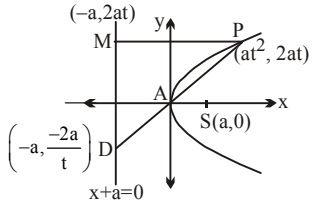
13. P is a point on the parabola $y^2 = 4ax$ $(a > 0)$ whose vertex is A. PA is produced to meet the directrix in D and M is the foot of the perpendicular from P on the directrix. If a circle is described on MD as a diameter then it intersects the x–axis at a point whose co–ordinates are :

परवलय $y^2 = 4ax$ $(a > 0)$ पर एक बिन्दु P है तथा इसका शीर्ष A है यदि PA को बढ़ाने पर यह नियता को D पर मिलता है तथा P से fu; r ki j y Ec i kn M है यदि MD को व्यास मानकर खींचे गये वृत्त जहाँ पर x–अक्ष को काटे, तो उसका निर्देशांक होगा :

(A) $(-3a, 0)$ (B) $(-a, 0)$ (C) $(-2a, 0)$ (D) $(a, 0)$

Ans. AD

Sol. Circle : $(x + a)^2 + (y - 2at)\left(y + \frac{2a}{t}\right) = 0$



from $y = 0$, $x^2 + 2ax - 3a^2 = 0 \Rightarrow x = a$ or $-3a$

14. The parabola $x = y^2 + ay + b$ intersect the parabola $x^2 = y$ at $(1, 1)$ at right angle. Which of the following is/are correct?

(A) $a = 4,\ b = -4$

(B) $a = 2,\ b = -2$

(C) Equation of the director circle for the parabola $x = y^2 + ay + b$ is $4x + 1 = 0$.

(D) Area enclosed by the parabola $x = y^2 + ay + b$ and its latus rectum is $\frac{1}{6}$.

परवलय $x = y^2 + ay + b$ एक दूसरे परवलय $x^2 = y$ को $(1, 1)$ पर लम्बवत् काटता है, तो निम्न में से सही विकल्प चुनो?

(A) $a = 4,\ b = -4$

(B) $a = 2,\ b = -2$

(C) परवलय $x = y^2 + ay + b$ की नियामक वृत्त का समीकरण $4x + 1 = 0$ है

(D) परवलय $x = y^2 + ay + b$ तथा इसकी नाभिलम्ब से घिरे क्षेत्र का क्षेत्रफल $\frac{1}{6}$ है

Ans. CD

Sol. We have $x = y^2 + ay + b$ ....(1) and $x^2 = y$ ....(2)

As (1) passing through (1, 1)

$\Rightarrow \quad 1 = 1 + a + b \quad \Rightarrow \quad a + b = 0$

Now $1 = 2y\frac{dy}{dx} + a\frac{dy}{dx}$

$\left.\frac{dy}{dx}\right]_{(1,1)} = \frac{1}{2+a}$ and for $y = x^2$, $\frac{dy}{dx} = 2x \Rightarrow \left.\frac{dy}{dx}\right]_{(1,1)} = 2$

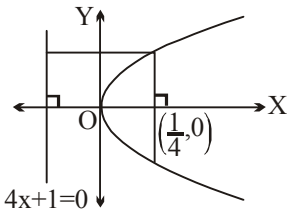
$\therefore \quad \frac{2}{2+a} = -1$

$\Rightarrow \quad -2 = 2 + a \quad \Rightarrow \quad a = -4$ and $b = 4$

Hence parabola is $y^2 - 4y + 4 = x \Rightarrow (y-2)^2 = x$

Let $y - 2 = Y$

$\therefore \quad Y^2 = X$



Hence Required area $= 2\int_0^{\frac{1}{4}} \sqrt{x}\,dx = (2)\left(\frac{2}{3}\right)\left(x^{\frac{3}{2}}\right)_0^{\frac{1}{4}} = \frac{1}{6}$ sq. units Ans. $\Rightarrow$ (D)

Equation of directrix is $x = \frac{-1}{4} \quad \Rightarrow \quad 4x + 1 = 0$ Ans. $\Rightarrow$ (C)

15. A circle 'S' is described on the focal chord of the parabola $y^2 = 4x$ as diameter. If the focal chord is inclined at an angle of 45° with axis of x, then which of the following is/are true?

(A) Radius of the circle is 4

(B) Centre of the circle is (3, 2)

(C) The line $x + 1 = 0$ touches the circle

(D) The circle $x^2 + y^2 + 2x - 6y + 3 = 0$ is orthogonal to 'S'

परवलय $y^2 = 4x$ की नाभि जीवा x–अक्ष से 45° कोण बनाती है तथा इस जीवा को व्यास मानकर एक वृत्त S खींचा जाता है, तो सही fod Yi lkad ksp ; u d hft , :

(A) वृत्त की त्रिज्या 4

(B) वृत्त का केन्द्र (3, 2) है

(C) रेखा $x + 1 = 0$ वृत्त को स्पर्श करती है

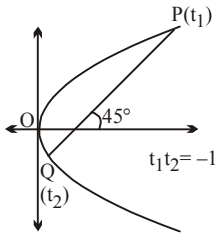
(D) वृत्त $x^2 + y^2 + 2x - 6y + 3 = 0$ वृत्त 'S' को लम्बकोणीय काटता है

Ans. ABC

Sol. Circle on PQ as diameter is

$$(x - t_1^2)(x - t_2^2) + (y - 2t_1)(y - 2t_2) = 0$$

$$x^2 + y^2 - (t_1^2 + t_2^2)x - 2(t_1 + t_2)y + t_1^2 t_2^2 + 4t_1t_2 = 0$$



But $\frac{t_1 + t_2}{2} = 1 \Rightarrow t_1 + t_2 = 2$ and $t_1t_2 = -1$

Hence equation of the circle is

$$x^2 + y^2 - \left((t_1 + t_2)^2 - 2t_1t_2\right)x - 2(t_1 + t_2)y + t_1^2 t_2^2 + 4t_1t_2 = 0$$

$x^2 + y^2 - 6x - 4y - 3 = 0$ .....(1) with centre (3, 2) and radius = 4

(1) is not orthogonal to $x^2 + y^2 + 2x - 6y + 3 = 0$

16. TP and TQ are tangents to parabola $y^2 = 4x$ and normals at P and Q intersect at a point R on the curve. The locus of the centre of the circle circumscribing ΔTPQ is a parabola whose :

(A) vertex is (1, 0). (B) foot of directrix is $\left(\frac{7}{8}, 0\right)$

(C) length of latus-rectum is $\frac{1}{4}$ (D) focus is $\left(\frac{9}{8}, 0\right)$

TP और TQ परवलय $y^2 = 4x$ की स्पर्श रेखाऐं हैं और P तथा Q पर अभिलम्ब वक्र का R पर प्रतिच्छेद करते है। तब ΔTPQ के i fj o Ùkd sd sUnz d kfc UnqH k, d i j oy ; gk s kft l d sfy ; s:

(A) शीर्ष (1, 0) (B) नियता का पाद $\left(\frac{7}{8}, 0\right)$

(C) नाभिलम्ब की लम्बाई $\frac{1}{4}$ (D) नाभि $\left(\frac{9}{8}, 0\right)$

Ans. ABD

Sol. We have $2h = t_3^2 + 2$ ....(1)

$2k = t_3$ ....(2)

$\therefore\ 2h = 4k^2 + 2$

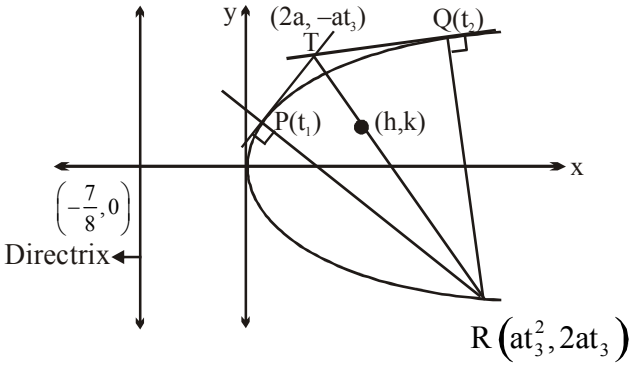
Here $a = 1$

$t_1t_2 = 2$

$t_1 + t_2 + t_3 = 0$

$\therefore\ 2y^2 = x - 1$

$y^2 = \frac{1}{2}(x - 1)$ (Parabola)



$y^2 = 4\cdot\frac{1}{8}(x-1)$

Directrix $x-1=-\frac{1}{8} \Rightarrow x=\frac{7}{8}$

Vertex (1, 0)

Focus $= 1+\frac{1}{8}=\frac{9}{8}$

Now interpret.

17. Through a point P (– 2, 0), tangents PQ and PR are drawn to the parabola $y^2 = 8x$. Two circles each passing through the focus of the parabola and one touching at Q and other at R are drawn. Which of the following point(s) with respect to the triangle PQR lie(s) on the radical axis of the two circles?

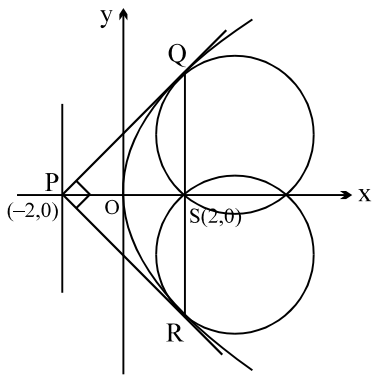
(A) centroid (B) orthocentre (C) incentre (D) circumcentre

बिन्दु P (– 2, 0) से परवलय $y^2 = 8x$ पर PQ और PR स्पर्श रेखाऐं खिचीं जाती है। तथा दो वृत्त जो प्रत्येक परवलय के नाभि से जाते हैं और एक वृत्त Q पर और दूसरा R पर स्पर्श करता है तब निम्न में से कौनसे बिन्दु ΔPQR के सापेक्ष दोनों वृत्तों के मूलाक्ष (radical axis) पर होंगे?

(A) केन्द्रक (B) लम्बकेन्द्र (C) अन्तःकेन्द्र (D) परिकेन्द्र

Ans. ABCD

Sol. (–2, 0) is the foot of directrix.



Hence Q and R are the extremities of the latus rectum and angle ∠QPR = 90° with ΔPQR as right isosceles.

Hence by symmetric the common chord of the two circles will be the x-axis which will be the median, altitude, angle bisector and also the perpendicular bisector.

Hence centroid, orthocentre, incentre and circumcentre all will lie on it.

18. If a number of ellipse be described having the same major axis 2a but a variable minor axis then the tangents at the ends of their latus rectum pass through fixed points which can be :

यदि दीर्घवृत्त जिसकी दीर्घअक्ष का मान 2a तथा लघुअक्ष चर है तो इसके नाभिलम्ब के सिरों पर खींची गयी स्पर्श रेखाऐं सदैव किस fu; r fcUnq sxtq jsxh:

(A) (0, a) (B) (0, 0) (C) (0, – a) (D) (a, a)

Ans. AC

Sol. e is a variable quantity

$$\frac{xae}{a^2}+\frac{yb^2}{ab^2}=1 \Rightarrow ex+y=a \qquad \Rightarrow \qquad y-a+ex=0$$

it passes through (0, a).

|||ly other point is (0, – a)

19. The locus of the point of intersection of those normals to the parabola $x^2=8y$ which are at right angles to each other, is a parabola. Which of the following hold(s) good in respect of the locus?

(A) Length of the latus rectum is 2 (B) Coordinates of focus are $\left(0,\frac{11}{2}\right)$

(C) Equation of a director circle is $2y-11=0$ (D) Equation of axis of symmetry is $y=0$

परवलय $x^2=8y$ के दो लम्बवत अभिलम्बों के प्रतिच्छेद बिन्दुओं का बिन्दुपथ एक परवलय है। तो बिन्दुपथ के लिए निम्न में से कौनसा d Hkq l R; gS

(A) नाभिलम्ब की लम्बाई 2 है (B) नाभि के निर्देशांक $\left(0,\frac{11}{2}\right)$

(C) नियामक वृत्त का समीकरण $2y-11=0$ (D) सममित अक्ष का समीकरण $y=0$.

Ans. AC

Sol. Locus is $x^2-2y+12=0$

20. Solutions of the differential equation $(1-x^2)\frac{dy}{dx}+xy=ax$ where $a\in R$, is : (Attempt after differential equation)

(A) a conic which is an ellipse or a hyperbola with principal axes parallel to coordinates axes.
(B) centre of the conic is (0, a)
(C) length of one of the principal axes is 1
(D) length of one of the principal axes is equal to 2

अवकलन समीकरण $(1-x^2)\frac{dy}{dx}+xy=ax$ जहां $a\in R$ का हल है : (Attempt after differential equation)

(A) एक शांकव जो दीर्घवृत्त या अतिपरवलय है। जिसके मुख्य अक्ष निर्देशों अक्षों के समान्तर है
(B) शांकव का केन्द्र (0, a) है
(C) किसी एक मुख्य अक्ष की लम्बाई 1 है
(D) किसी एक मुख्य अक्ष की लम्बाई 2 है

Ans. ABD

Sol. $\frac{dy}{dx} + \frac{x}{1-x^2}y = \frac{ax}{1-x^2}$

I.F. $e^{\int \frac{x}{1-x^2}dx} = e^{-\frac{1}{2}\log|1-x^2|} = \frac{1}{\sqrt{|1-x^2|}}$

$$\frac{y}{\sqrt{|1-x^2|}} = a\int \frac{x\,dx}{|1-x^2|^{3/2}} + C$$

let $|1-x^2| = v^2$; $\quad -2x\,dx = 2v\,dv$; $\quad x\,dx = -v\,dv$

hence $\frac{y}{v} = -a\int \frac{v\,dv}{v^3} = +\frac{a}{v} + C$

$y = a + Cv$

$y = a + C\sqrt{|1-x^2|}$

$(y-a)^2 = C^2|1-x^2| = C^2(1-x^2)$ or $C^2(x^2-1)$

$(y-a)^2 + C^2x^2 = C^2$ or $(y-a)^2 - C^2x^2 = -C^2$

$\frac{(y-a)^2}{C^2} + \frac{x^2}{1} = 1$ or $\frac{(y-a)^2}{C^2} - \frac{x^2}{1} = -1 \Rightarrow$ centre $(0, a)$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. The tangent at the point α on a standard ellipse meets the auxiliary circle in two points which subtends a right angle at the centre. Show that the eccentricity of the ellipse is $(1 + \sin^2\alpha)^{-1/2}$.

   मानक दीर्घवृत्त पर बिन्दु $P(\alpha)$ पर बनायी गई स्पर्श रेखा सहायक वृत्त को दो बिन्दुओं पर काटती है जो केन्द्र पर समकोण अंकित करते gSA' kkZ; sfd nh?kZo`Ùkd hmRdsUnz k$(1 + \sin^2\alpha)^{-1/2}$ है।

Ans.

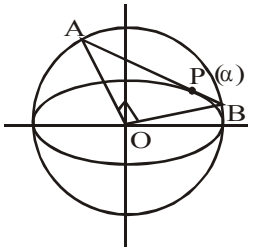
Sol. Equation of auxiliary circle is $x^2 + y^2 = a^2$ ....(1)

Equation of tangent at point P $(a\cos\alpha, b\sin\alpha)$

is $\frac{x}{a}\cos\alpha + \frac{y}{b}\sin\alpha = 1$ ....(2)

Equation of pair of lines OA, OB is obtained by homogenous equation of (1) with the help of (2)

$$\therefore \quad x^2 + y^2 = a^2\left(\frac{x}{a}\cos\alpha + \frac{y}{b}\sin\alpha\right)^2$$



$$\Rightarrow (1-\cos^2\alpha)x^2 - \frac{2xya\sin\alpha\cos\alpha}{b} + y^2\left(1 - \frac{a^2}{b^2}\sin\alpha\right)$$

But $\angle AOB = 90°$

$\therefore$ Coeff. of x^2 + coeff. of $y^2 = 0$

$$1-\cos^2\alpha + 1 - \frac{a^2}{b^2}\sin^2\alpha = 0$$

$$\Rightarrow \quad 1 = \frac{a^2 - b^2}{b^2}\sin^2\alpha \quad \Rightarrow 1 = \frac{a^2e^2}{a^2(1-e^2)}\sin^2\alpha$$

$$\Rightarrow \quad e = (1 + \sin^2\alpha)^{-1/2}$$

2. If the normal at the point $P(\theta)$ to the ellipse $\frac{x^2}{14} + \frac{y^2}{5} = 1$, intersects it again at the point $Q(2\theta)$, show that $\cos\theta = -(2/3)$.

   यदि बिन्दु $P(\theta)$ से दीर्घवृत्त $\frac{x^2}{14} + \frac{y^2}{5} = 1$ पर अभिलम्ब दीर्घवृत्त को पुनः $Q(2\theta)$ पर प्रतिच्छेद करता है, तो दर्शाइये $\cos\theta = -(2/3)$.

Ans.

Sol. Normal at $P(\theta)$

$$\frac{\sqrt{14}x}{\cos\theta} - \frac{\sqrt{5}y}{\sin\theta} = 14 - 5$$

Passes through Q(2θ)

$$\frac{14\cdot\cos 2\theta}{\cos\theta} - \frac{5\cdot\sin 2\theta}{\sin\theta} = 9$$

Let $\cos\theta = t$

$14(2t^2 - 1) - 10 \cdot t^2 = 9t$

$18t^2 - 9t - 14 = 0$

$$\cos\theta = -\frac{2}{3}, \frac{7}{6}$$

$\because \cos\theta \neq \frac{7}{6}$ for any θ.

3. Let circles C_1 and C_2 on Argand plane be given by $|z + 1| = 3$ and $|z - 2| = 7$ respectively. If a variable circle $|z - z_0| = r$ be inside circle C_2 such that it touches C_1 externally and C_2 internally then locus of 'z_0' describes a conic E. If eccentricity of E can be written in simplest form as $\frac{p}{q}$ where $p, q \in N$, then find the value of $(p + q)$.

आर्गड तल में स्थित वृत C_1 एवं C_2 क्रमशः $|z + 1| = 3$ तथा $|z - 2| = 7$ द्वारा बताये जाते है। यदि वृत C_2 के अन्तर्निष्ठ एक चर oर $|z - z_0| = r$ है, जो C_1 को बाह्य एवं C_2 को अन्तः स्पर्श कर रहा है, तो 'z_0' का बिन्दुपथ एक शंकु E है। यदि E की उत्केन्द्रता $\frac{p}{q}$ के रूप में व्यक्त की जा सकती है। जहाँ $p, q \in N$, तो $(p + q)$ का मान ज्ञात कीजिए।

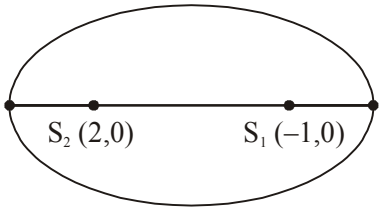
Ans. 13

Sol. $PS_1 + PS_2 = r_1 + r + r_2 - r$

$2a = 10$

$2ae = 3$

$$e = \frac{3}{10}$$



4. If the tangent at any point of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ makes an angle α with the major axis and an angle β with the focal radius of the point of contact then show that the eccentricity 'e' of the ellipse is given by the absolute value of $\frac{\cos\beta}{\cos\alpha}$.

दीर्घवृत $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ के किसी बिन्दु पर स्पर्श रेखा यदि दीर्घ अक्ष के साथ α कोण बनाती है तथा स्पर्शी बिन्दु की नाभीय त्रिज्या के l kFk β कोण बनाती है, तो दर्शाइये दीर्घवृत की उत्केन्द्रता 'e' $= \left|\frac{\cos\beta}{\cos\alpha}\right|$.

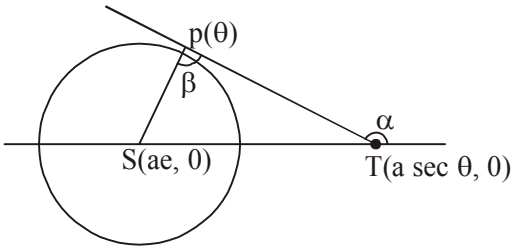
Ans.

Sol. ΔPST, sine rule

$$\frac{PS}{\sin(\pi-\alpha)}=\frac{ST}{\sin\beta}$$

$$\frac{a(1-e\cos\theta)}{\sin\alpha}=\frac{a\sec\theta-ae}{\sin\beta}$$

$$\cos\theta=\frac{\sin\alpha}{\sin\beta} \quad ...(1)$$



slope of tangent

$$-\frac{b}{a}\cot\theta=\tan\alpha$$

$$\frac{b^2}{a^2}=\tan^2\alpha\left(\sec^2\theta-1\right)=\frac{\sin^2\beta-\sin^2\alpha}{\cos^2\alpha}$$

$$e=\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1-\frac{\sin^2\beta-\sin^2\alpha}{\cos^2\alpha}}=\left|\frac{\cos\beta}{\cos\alpha}\right|$$

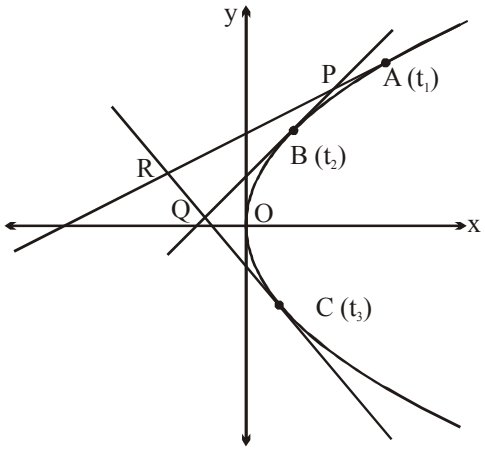
5. Tangents are drawn to the parabola $y^2=12x$ at the points A, B and C such that the three tangents form a triangle PQR. If θ_1, θ_2 and θ_3 be the inclinations of these tangents with the axis of x such that their cotangents form an arithmetical progression (in the same order) with common difference 2. Find the area of the triangle PQR.

परवलय $y^2 = 12x$ के बिन्दुओं A, B तथा C पर बनायी गई स्पर्श रेखाऐं इस प्रकार है कि वे त्रिभुज PQR की संरचना करती है। यदि θ_1, θ_2 तथा θ_3 स्पर्श रेखाओं द्वारा x अक्ष के साथ बनाये गए कोण है, तो उनके कोटस्पर्शज्या एक समानान्तर श्रेणी (उसी क्रम में) बनाते g\$ft ud kl okfLrj 2 है। त्रिभुज PQR का क्षेत्रफल ज्ञात कीजिए।

Ans. 72

Sol. $y^2 = 12x$; $a = 3$

Slopes of the tangents at A, B and C are $\frac{1}{t_1}, \frac{1}{t_2}, \frac{1}{t_3}$



$\Rightarrow \tan\theta_1 = \frac{1}{t_1}$; $\tan\theta_2 = \frac{1}{t_2}$; $\tan\theta_3 = \frac{1}{t_3}$

$\therefore \cot\theta_1 = t_1$, $\cot\theta_2 = t_2$ and $\cot\theta_3 = t_3$

$\therefore$ t_1, t_2, t_3 are in A.P. with

$d = 2$

$t_2 - t_1 = 2$, $t_3 - t_1 = 4$

Now area of $\Delta PQR = \frac{1}{2}\begin{vmatrix} at_1t_2 & a(t_1+t_2) & 1 \\ at_2t_3 & a(t_2+t_3) & 1 \\ at_3t_1 & a(t_3+t_1) & 1 \end{vmatrix} = a^2d^3 = 9 \times 8 = 72$ Ans.

6. In the parabola $y^2 = 4ax$, the tangent at the point P, whose abscissa is equal to the latus rectum meets the axis in T & the normal at P cuts the parabola again in Q. Prove that PT : PQ = 4 : 5.

परवलय $y^2 = 4ax$ में बिन्दु P, जिसका भुज नाभिलम्ब के बराबर है, पर स्पर्श रेखा अक्ष को T पर मिलती है एवं P पर अभिलम्ब परवलय d ksQj l sQ पर मिलता है, तो सिद्ध कीजिए PT : PQ = 4 : 5 है।

Ans.

Sol. Let point on parabola $y^2 = 4ax$ is $P(at^2, 2t)$ Given $at^2 = 4a \Rightarrow t = \pm 2$

taking positive $t = 2$

P(4a, 4a)

equation of tangent at P is $2y = x + 4a$

If intersect x-axis at T then T(–4a, 0)

Normal at (4a, 4a) meet again parabola at

$Q(at_2^2, 2at_2)$ (using $t_2 = -t_1 - \frac{2}{t_1} = -3$)

$\therefore$ Q(9a, – 6a)

Now P(4a, 4a), T(–4a, 0), Q(9a, –6a)

$PT = \sqrt{(4a+4a)^2 + (4a)^2} = \sqrt{80a^2}$

$$PQ = \sqrt{(4a-9a)^2 + (4a+6a)^2} = \sqrt{125a^2}$$

$$\frac{PT}{PQ} = \sqrt{\frac{80a^2}{125a^2}} = \frac{4}{5}$$

7. Two tangents to the parabola $y^2 = 8x$ meet the tangent at its vertex in the points P & Q. If PQ = 4 units, prove that the locus of the point of the intersection of the two tangents is $y^2 = 8(x+2)$.

परवलय $y^2 = 8x$ पर दो स्पर्श रेखाऐं, शीर्ष पर बनायी गई स्पर्श रेखाओं को P तथा Q पर मिलती है। यदि PQ = 4 ईकाई है, तो सिद्ध d lhft , nksksLi ' kZ[ks kvksad hi fr PNsu fcUnqiq kfcUnqiFky $y^2 = 8(x+2)$ है।

Ans.

Sol. tangent at $P(t_1)$

$y t_1 = x + 2t_1^2$

$x = 0$

$y = 2t_1$

$PQ = 2|t_1 - t_2| = 4$ ...(1)

now $h = at_1t_2$, $k = a(t_1 + t_2)$

from (1)

$(t_1 + t_2)^2 - 4t_1t_2 = 4$

$$\frac{k^2}{a^2} - \frac{4h}{a} = 4$$

$k^2 = 4a^2 + 4ah$

$y^2 = 8(x+2)$

8. Two straight lines one being a tangent to $y^2 = 4ax$ and the other to $x^2 = 4by$ are right angles. Find the locus of their point of intersection.

दो लम्बवत रेखाऐं जिनमें से एक $y^2 = 4ax$ को एवं अन्य $x^2 = 4by$ को स्पर्श करती है। दोनों के प्रतिच्छेदन बिन्दु का बिन्दुपथ ज्ञात d lhft , A

Ans. $(ax + by)(x^2 + y^2) + (bx - ay)^2 = 0$

Sol. $y^2 = 4ax$ $x^2 = 4by$

tangent tangent

$y = mx + \frac{a}{m}$ _____(1) $x = m_1 y + \frac{b}{m_1}$ _____(2)

$$m \cdot \frac{1}{m_1} = -1$$

$$m_1 = -m$$

Let P(h, k) & $k = mh + \frac{a}{m}$

$m^2h - mk + a = 0$

$m^2k + mh + b = 0$

$$\frac{m^2}{-kb - ah} = \frac{m}{ak - bh} = \frac{1}{h^2 + k^2}$$

$-(h^2 + k^2)(kb + ah) = (bh - ak)^2$

$(x^2 + y^2)(ax + by) + (bx - ay)^2 = 0$

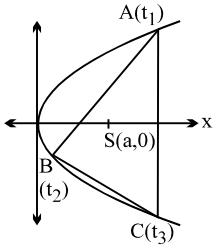
9. Let $A(at_1^2, 2at_1)$; $B(at_2^2, 2at_2)$; $C(at_3^2, 2at_3)$ be three points on the parabola $y^2 = 4ax$. If the orthocentre of the triangle ABC is at the focus, then find $\left| t_1 + t_2 + t_3 + t_1t_2t_3 + t_1t_2 + t_2t_3 + t_3t_1 \right|$.

माना तीन बिन्दु $A(at_1^2, 2at_1)$; $B(at_2^2, 2at_2)$; $C(at_3^2, 2at_3)$ परवलय $y^2 = 4ax$ पर स्थित है। यदि त्रिभुज ABC का लम्बकेन्द्र i j oy ; d huk fLFkr gS rks $\left| t_1 + t_2 + t_3 + t_1t_2t_3 + t_1t_2 + t_2t_3 + t_3t_1 \right|$ ज्ञात कीजिए।

Ans. 5

Sol. Slope of AB × slope of BC = –1

$$\frac{2}{t_1 + t_2} \times \frac{2at_3}{a(t_3^2 - 1)} = -1$$



$$\frac{2}{t_1 + t_2} = \frac{1 - t_3^2}{2t_3}$$

$$t_1 + t_2 = \frac{4t_3}{1 - t_3^2}$$

$$t_1 + t_2 + t_3 = \frac{4t_3}{1 - t_3^2} + t_3 = \frac{4t_3 + t_3 - t_3^3}{1 - t_3^2} = \frac{5t_3 - t_3^3}{1 - t_3^2}$$

If $t_1 + t_2 + t_3 = u$ then $u = \dfrac{5t_3 - t_3^3}{1 - t_3^2}$ ....(1)

The Equation (1) suggests that t_3 is a root of

$t^3 - ut^2 - 5t + u = 0$ ....(2)

by symmetry t_1 and t_2 are also the root of (2). Hence

$t_1 + t_2 + t_3 = u$

and $t_1 t_2 t_3 = -u$

Hence $t_1 + t_2 + t_3 + t_1 t_2 t_3 = 0$

also $t_1t_2 + t_2t_3 + t_3t_1 = -5$

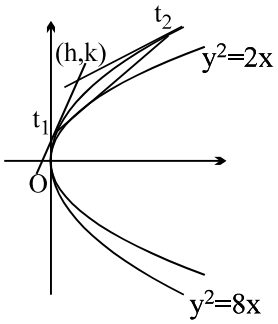
Hence $\left| t_1 + t_2 + t_3 + t_1t_2t_3 + t_1t_2 + t_2t_3 + t_3t_1 \right| = 5$ Ans.

10. A variable chords of the parabola $y^2 = 8x$ touches the parabola $y^2 = 2x$. The the locus of the point of intersection of the tangent at the end of the chord is a parabola. Find its latus rectum.

परवलय $y^2 = 8x$ की चर जीवाऐं परवलय $y^2 = 2x$ को स्पर्श करती है। उनके सिरे पर बनायी गई स्पर्श रेखाओं के प्रतिच्छेदन बिन्दु का fcUnqQk, d ijoy ; g\$ft l dkukfHkyEc Kkr dhft , A

Ans. 32

Sol. $\left.\begin{matrix} h = 2t_1t_2 \\ k = 2(t_1 + t_2) \end{matrix}\right\}$ ....(1)



equation of the chord joining t_1 and t_2

$2x - (t_1 + t_2)y + 4t_1t_2 = 0$

$y = \dfrac{2}{t_1 + t_2}x + \dfrac{4t_1t_2}{t_1 + t_2}$ ....(2)

(2) touches $y^2 = 2x$ $\left(a = \dfrac{1}{2}\right)$

$\therefore$ $c = \dfrac{a}{m}$; $\dfrac{4t_1t_2}{t_1 + t_2} = \dfrac{t_1 + t_2}{2 \cdot 2}$ $\Rightarrow$ $16t_1t_2 = (t_1 + t_2)^2$

also from (1) $(t_1 + t_2)^2 = \dfrac{k^2}{4}$; $t_1t_2 = \dfrac{h}{2}$; $\therefore$ $\dfrac{(t_1 + t_2)^2}{t_1t_2} = \dfrac{k^2}{4} \cdot \dfrac{2}{h} = 16$

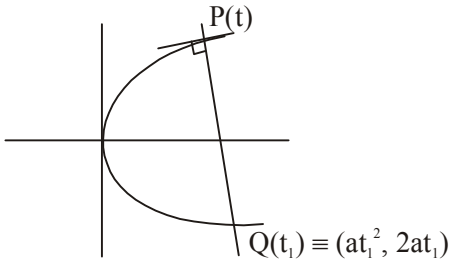
$\therefore$ $k^2 = 32h$; Locus is $y^2 = 32x$ which is a parabola.

11. Show that the normals at two suitable distinct real points on the parabola $y^2 = 4ax$ $(a > 0)$ intersect at a point on the parabola whose abscissa > 8a.

दर्शाइये कि परवलय $y^2 = 4ax$ $(a > 0)$ पर स्थित दो भिन्न वास्तविक बिन्दुओं पर बनाये गये अभिलम्बों का प्रतिच्छेदन बिन्दु परवलय पर fLFkr gksxk ftldk Hkqt > 8a है।

Ans.

Sol. $t_1 = -t - \frac{2}{t}$



$t^2 + t_1 t + 2 = 0$

For distinct roots, $D > 0$

$t_1^2 - 8 > 0$

$t_1^2 > 8$

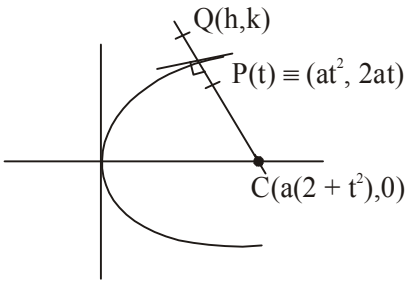
$at_1^2 > 8a$

12. PC is the normal at P to the parabola $y^2 = 4ax$, C being on the axis. CP is produced outwards to Q so that PQ = CP; show that the locus of Q is a parabola.

परवलय $y^2 = 4ax$ पर बिन्दु P पर अभिलम्ब है, C अक्ष पर स्थित है। CP को बाहर की ओर Q तक इस प्रकार आगे बढ़ाया जाता gSSd PQ = CP, दर्शाइये की Q का बिन्दुपथ एक परवलय होगा।

Ans.

Sol. PC : $y = -tx + 2at + at^3$



Put $y = 0$

$x = a(2 + t^2)$

Now, P is mid point of QC

$$\frac{h + a\left(2 + t^2\right)}{2} = at^2 \qquad ...(1)$$

$$\frac{k + 0}{2} = 2at$$

$$t = \frac{k}{4a}$$

Put in equation (1)

$$h + 2a + a\left(\frac{k}{4a}\right)^2 = 2a\left(\frac{k}{4a}\right)^2$$

parabola

13. A quadrilateral is inscribed in a parabola $y^2 = 4ax$ and three of its sides pass through fixed points on the axis. Show that the fourth side also passes through fixed point on the axis of the parabola.

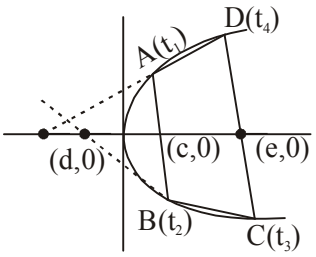
परवलय $y^2 = 4ax$ के अर्न्तनिष्ठ में स्थित एक चतुर्भुज है। जिसकी तीन भुजाऐं अक्ष पर स्थित नियत बिन्दु से होकर गुजरती है। सिद्ध d lft , pkSFkh Hkqtk kHkhv {ki j fLFkr fu; r fcUnq sgksdj xqtjrh gSA

Ans.

Sol. equation of AB

$2x - (t_1 + t_2)y + 2at_1t_2 = 0$

Passes through (c,0)



$$t_1t_2 = -\frac{c}{a} \qquad \text{_____(1)}$$

Similarly

$$t_2t_3 = -\frac{d}{a} \qquad \text{_____(2)}$$

$$\& \ t_3t_4 = -\frac{e}{a} \qquad \text{_____(3)}$$

(1) × (3)

$$t_1t_2t_3t_4 = \left(-\frac{c}{a}\right)\left(-\frac{e}{a}\right) = \frac{ec}{a^2}$$

$$t_1 \cdot t_4\left(-\frac{d}{a}\right) = \frac{ec}{a^2}$$

$$t_1t_4 = -\frac{ec}{ad}$$

Hence chord AD passes through fixed point.

14. Let P (a, b) and Q (c, d) are the two points on the parabola $y^2 = 8x$ such that the normals at them meet in (18, 12). Find the product (abcd).

माना P (a, b) तथा Q (c, d) परवलय $y^2 = 8x$ पर इस प्रकार है कि उनपर बनाये गये अभिलम्ब (18, 12) पर मिलते है। गुणन (abcd) का मान ज्ञात कीजिए।

Ans. 512

Sol. $y^2 = 8x \rightarrow a = 2$

normal at P and Q meet on R (18, 12) on the parabola

Hence $t_1 t_2 = 2$

Also point of intersection of normals is

$a(t_1^2 + t_2^2 + t_1 t_2 + 2) = 18$ .....(1)

and $-a t_1 t_2 (t_1 + t_2) = 12$ ..... (2)

From (1), $2(t_1^2 + t_2^2 + 4) = 18$

from (2), $\boxed{t_1^2 + t_2^2 = 5}$ and

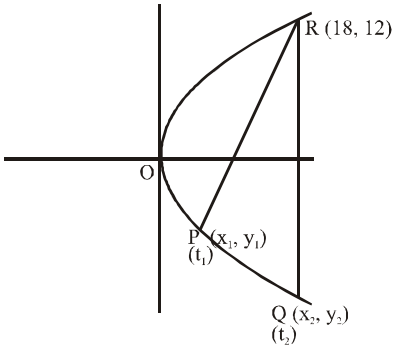
$-4(t_1 + t_2) = 12 \Rightarrow t_1 + t_2 = -3$

Hence $t_1 = -1$ and $t_2 = -2$

Hence $P \equiv 2(-1)^2, 2(2)(-1) \Rightarrow P \equiv (2, -4)$

and $Q \equiv 2(4), 2(2)(-2) \Rightarrow Q \equiv (8, -8)$

Hence, $(abcd) = (2)(-4)(8)(-8) = 512$ Ans.



15. A fixed parabola $y^2 = 4ax$ touches a variable parabola. Find the equation to the locus of the vertex of the variable parabola. Assume that the two parabolas are equal and the axis of the variable parabola remains parallel to the x-axis.

एक नियत परवलय $y^2 = 4ax$, एक चर परवलय को स्पर्श करता है। चर परवलय के शीर्ष का बिन्दुपथ ज्ञात करें। यह दिया गया है कि nksuksa ijoy; l eku gSa rFkk pj ijoy; d kv {kl nksuksa x-अक्ष के समानान्तर रहता है।

Ans. $y^2 = 8ax$

Sol. Given parabola $y^2 = 4ax$ ...(1)

Let variable parabola $(y - k)^2 = -4a(x - h)$ ...(2)

Solving equation (1) & (2)

$2y^2 - 2yk + k^2 - 4ah = 0$

For tangency, $D = 0$

$4k^2 - 8(k^2 - 4ah) = 0$

$k^2 = 8ah$

16. Prove that the locus of the centre of the circle, which passes through the vertex of the parabola $y^2 = 4ax$ & through its intersection with a normal chord is $2y^2 = ax - a^2$.

सिद्ध कीजिए परवलय $y^2 = 4ax$ के शीर्ष एवं इसके अभिलम्ब जीवा के प्रतिच्छेदन से जाने वाले वृत्त के केन्द्र का बिन्दुपथ $2y^2 = ax - a^2$ है।

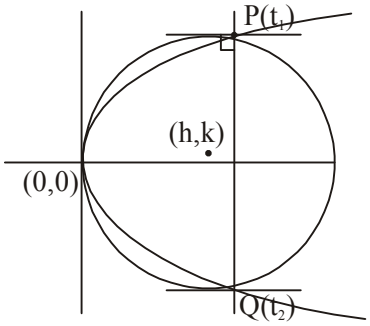
Ans.

Sol. where $t_2 = -t_1 - \frac{2}{t_1}$ or $t_1^2 + t_1 t_2 = -2$

Let equation of circle

$x^2 + y^2 - 2hx - 2ky = 0$

It passes through P & Q



$at_1^3 + 4at_1 - 2ht_1 - 4k = 0$ _____(1)

$at_2^3 + 4at_2 - 2ht_2 - 4k = 0$ _____(2)

(1) – (2)

$a\left(t_1^3 - t_2^3\right) + 4a\left(t_1 - t_2\right) - 2h\left(t_1 - t_2\right) = 0$

or $a\left(t_1^2 + t_2^2 + t_1t_2\right) + 4a - 2h = 0$

or $a\left(t_2^2 - 2\right) + 4a - 2h = 0$

or $at_2^2 + 2a - 2h = 0$ _____(3)

Now $(1) \times t_2 - (2) \times t_1$

$at_1t_2\left(t_1^2 - t_2^2\right) + 4k\left(t_1 - t_2\right) = 0$

$at_1t_2\left(t_1 + t_2\right) + 4k = 0$

$-2at_2 + 4f = 0$

$t_2 = -\dfrac{2k}{a}$

Put in equation (3)

$a\left(\dfrac{4k^2}{a^2}\right) + 2a - 2h = 0$ or $2y^2 + a^2 - ax = 0$

17. The equation of two tangents to a parabola are $L_1 : y + 3 = \dfrac{-1}{3}(x+1)$ and $L_2 : y - 4 = \dfrac{1}{4}(x+1)$ and one more tangent at the vertex is $L_3 : x + 1 = 0$. The straight line $L : y - 3 = m\left(x - \dfrac{3}{m}\right)$, $m \neq 0$ is a variable chord of the parabola.

(a) Find the minimum length of the focal chord.

(b) Find the locus of the point of intersection of the tangents drawn at the extremities of the variable chord L.

(c) If L' is the family of lines perpendicular to the variable chord L which are passing through the point of concurrency of L, then find the locus of the middle point of point of intersection of L' with directrix of the parabola and point of concurrency of L'.

एक परवलय पर दो स्पर्श रेखाओं का समीकरण $L_1 : y + 3 = \frac{-1}{3}(x + 1)$ तथा $L_2 : y - 4 = \frac{1}{4}(x + 1)$ है एवं एक अन्य स्पर्श रेखा जो शीर्ष पर स्पर्श करती है का समीकरण $L_3 : x + 1 = 0$ है। एक रेखा $L : y - 3 = m\left(x - \frac{3}{m}\right),\ m \neq 0$ परवलय की चर जीवा है।

(a) नाभि जीवा की न्यूनतम लम्बाई ज्ञात कीजिए।

(b) चर जीवा L के सिरों पर बनायी गई स्पर्श रेखाओं के प्रतिच्छेदन बिन्दु का बिन्दुपथ ज्ञात कीजिए।

(c) यदि L' चर जीवा L के लम्बवत् रेखाओं के निकाय को दर्शाता है जो L के संगामी बिन्दु से होकर जाती है। तो L' के परवलय की fu; r kl sc fr PNsu fcUnq oa L' के संगामी बिन्दु के मध्य बिन्दु का बिन्दुपथ ज्ञात कीजिए।

Ans. (a) 4, (b) $x + 2 = 0$, (c) $x + 1 = 0$

Sol. We have $L_1 : y + 3 = \frac{-1}{3}(x + 1)$; $L_2 : y - 4 = \frac{1}{4}(x + 1)$

tangent at the vertex is

$L_3 : x + 1 = 0$

Point of intersection of L_1 and L_3 is $(-1, -3)$ and point of intersection of L_2 and L_3 is $(-1, 4)$

Now, straight line perpendicular to L_1 and passing through $(-1, -3)$ is

$y + 3 = 3(x + 1)$

$\Rightarrow \quad y = 3x \qquad \ldots.(i)$

Straight line perpendicular to L_2 and passing through $(-1, 4)$ is

$y - 4 = -4(x + 1)$

$\Rightarrow \quad y = -4x \qquad \ldots.(ii)$

Point of intersection of (i) and (ii) which is the focus.

Minimum length of the focal chord is the length of the latus rectum.

$\therefore$ Latus rectum $= 4 \times$ (perpendicular distance from the focus on the tangent at the vertex)

$= 4 \times 1 = 4$ Ans.(i)

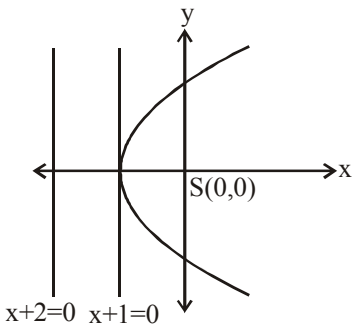
Variable chord L is $y - 3 = m\left(x - \frac{3}{m}\right)$

$y = mx$

Which passes through $(0, 0)$ i.e. focus

$\therefore$ variable chord is a focal chord.

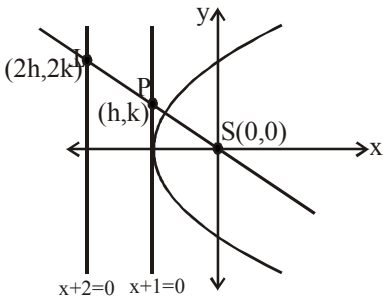
Locus of point of intersection of tangents drawn at extremities of focal chord is the directrix



$\therefore$ Equation of directrix is

$x + 2 = 0$ Ans.(ii)

Equation of family of lines L' perpendicular given family of lines L passing through the point of concurrency of L is the L it self.



Let P(h, k) be the middle point of ST.

$\therefore$ T = (2h, 2k)

$2h + 2 = 0 \Rightarrow x + 1 = 0$ is the required locus. Ans.(iii)

18. TP and TQ are two tangents to the parabola $y^2 = 8x$ such that the normals at the points P and Q intersect the curve at R.

(a) Find the locus of the middle points of the chord PQ.

(b) The chord PQ always passes through a fixed point F. Find the distance of F from the focus of the given parabola.

(c) Find the locus of the centre of the circle circumscribing the trianle TPQ.

ज्च तथा TQ परवलय $y^2 = 8x$ पर दो स्पर्श रेखाऐं इस प्रकार है कि P तथा Q पर अभिलम्ब वक्र पर R पर प्रतिच्छेद करते है।

(a) जीवा PQ के मध्य बिन्दु का बिन्दुपथ ज्ञात कीजिए।

(b) जीवा PQ सदैव एक नियत बिन्दु F से होकर जाती है। बिन्दु F की परवलय की नाभि से दूरी ज्ञात कीजिए।

(c) त्रिभुज TPQ के परिवृत्त के केन्द्र का बिन्दुपथ ज्ञात कीजिए।

Ans. (a) $y^2 = 4(x + 4)$; (b) 6

Sol. $y^2 = 8x \Rightarrow a = 2$

Given $t_1t_2 = 2$

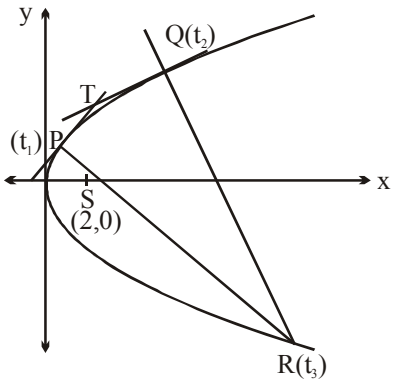
Now $2h = a(t_1^2 + t_2^2)$

$2k = 2a(t_1 + t_2)$

$$k^2 = a^2\left[\frac{2h}{a} + 4\right]$$

$k^2 = 2ah + 4a^2 \Rightarrow y^2 = 2a(x + 2a)$

$y^2 = 4(x + 4)$ Ans.(i)



Again line PQ is

$2x - (t_1 + t_2)y + 2at_1t_2 = 0$

$2x - (t_1 + t_2)y + 8 = 0 \Rightarrow (x + 4) + \lambda y = 0$

which passes through $(-4, 0) \Rightarrow$ fixed point.

Also focus is $(2, 0) \Rightarrow$ distance of 'S' from the fixed point = 6. Ans.(ii)

(iii) Circle through TPQ will be the circle on TR as diameter as T, P, R, Q are concyclic.

Let the centre of the variable circle be (x_1, y_1)

Now, $2x_1 = at_1t_2 + at_2^2$

$2x_1 = 4a + at_3^2$ $\quad (t_1t_2 = 2)$ $\quad$ .....(1)

Also $t_1 + t_2 + t_3 = 0$

and $2y_1 = a(t_1 + t_2) + 2at_3$

$$= a\underbrace{(t_1 + t_2 + t_3)}_{\text{zero}} + at_3$$

$2y_1 = at_3$

$$t_3 = \frac{2y_1}{a}\ \frac{2y_1}{a}$$

Substituting the value of $t_3 = \frac{2y_1}{a}$ in (1)

$$2x_1 = 4a + a \cdot \frac{4y_1^2}{a^2}$$

$ax_1 = 2a^2 + 2y_1^2$

$2y_1^2 = a(x_1 - 2a)$

put $a = 2$

$2y^2 = 2(x - 4) \Rightarrow \quad y^2 - x + 4 = 0$ Ans.

19. Two equal parabolas P_1 and P_2 have their vertices at $V_1(0, 4)$ and $V_2(6, 0)$ respectively. P_1 and P_2 are tangent to each other and have vertical axes of symmetry.

(a) Find the sum of the abscissa and ordinate of their point of contact.

(b) Find the length of latus rectum.

(c) Find the area of the region enclosed by P_1, P_2 and the x-axis.

P_1 तथा P_2 दो परवलय के शीर्ष क्रमशः $V_1(0, 4)$ एवं $V_2(6, 0)$ है। P_1 तथा P_2 एक दूसरे को स्पर्श करते हैं तथा उनका सममित अक्ष उर्ध्वाधर है

(a) स्पर्शीय बिन्दु के भुज एवं कोटि का योग ज्ञात कीजिए।

(b) नाभिलम्ब की लम्बाई ज्ञात कीजिए।

(c) P_1, P_2 द्वारा x-अक्ष के साथ परिबद्ध क्षेत्रफल ज्ञात कीजिए।

Ans. (a) 5, (b) 9/2, (c) $4\left(3-2\sqrt{2}\right)$

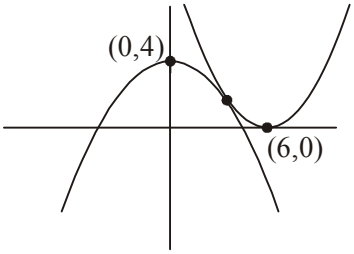
Sol. Parabola (I)

$x^2 = -4a(y-4)$

differentiating

$2\xi = \square 4\alpha\psi$

$$y' = -\frac{x}{2a} \qquad ...(1)$$



Parabola (II)

$(x-6)^2 = 4ay$

differentiating

$2(\xi \square 6) = 4\alpha\psi$

$$y' = \frac{x-6}{2a} \qquad ...(2)$$

For tangency (1) = (2) $\quad x = 3$

Put x = 3 in both parabolas

$-4a(y-4) = 4ay \Rightarrow y = 2$

Put (3,2) in parabola (II) $\quad a = \frac{9}{8}$

(a) $3 + 2 = 5$

(b) = LR = $4a = \frac{9}{2}$

(c) area = $4\left(3-2\sqrt{2}\right)$

20. Find the locus of the points of intersection of tangents drawn at the ends of all normal chords of the parabola $y^2 = 8(x-1)$.

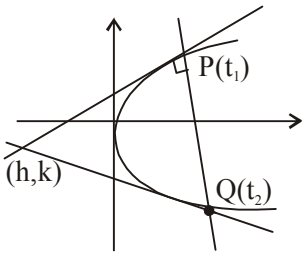
परवलय $y^2 = 8(x - 1)$ की अभिलम्ब जीवाओं के सिरों पर खींची गयी स्पर्शीयों के प्रतिच्छेद बिन्दुओं का बिन्दुपथ ज्ञात कीजिए।

Ans. $(x + 3)y^2 + 32 = 0$

Sol. Let parabola $Y^2 = 4aX$

$X = x - 1, Y = y, a = 2$

$$t_1 + t_2 = -\frac{2}{t_1}$$



$$t_1^2 + t_1t_2 = -2$$

$h = at_1t_2, k = a(t_1 + t_2)$

$$\text{or } h = (-2 - t_1^2)a, \ k = -\frac{2a}{t_1}$$

$$\text{or } t_1 = -\frac{2a}{k}$$

$$\text{or } h = -2a - a\left(-\frac{2a}{k}\right)^2$$

$$X = -4 - \frac{32}{Y^2}$$

$$x - 1 = -4\frac{32}{y^2}$$

$(x + 3)y^2 + 32 = 0$

21. Normals are drawn from the point 'P' with slopes m_1, m_2, m_3 to the parabola $y^2 = 4x$. If locus of P with $m_1 m_2 = \alpha$ is a part of the parabola itself then find α.

बिन्दु 'P' से परवलय $y^2 = 4x$ पर m_1, m_2, m_3 प्रवणता के अभिलम्ब खींचे जाते हैं। यदि P का बिन्दुपथ इस प्रकार है, कि $m_1 m_2 = \alpha$ तथा परवलय का ही एक भाग है, तो α ज्ञात कीजिए।

Ans. $\alpha = 2$

Sol. Let P be the point (h, k). Then equation of normal to parabola $y^2 = 4x$ from point (h, k), if m is the slope of normal, is $y = mx - 2m - m^3 = 0$

As it passes through (h, k), therefore

$mh - k - 2m - m^3 = 0$

or $m^3 + (2 - h)m + k = 0$ _____(1)

which is cubic in m, giving three values of m say m_1, m_2 and m_3. Then $m_1m_2m_3 = -k$ (from equation)

βυτγιωεν τηατ $\mu_1 m_2 = \alpha$

$\therefore$ We get $m_3 = -\frac{k}{\alpha}$

But m_3 must satisfy equation (1)

$\therefore$ $\frac{-k^3}{\alpha^3} + (2-h)\left(\frac{-k}{\alpha}\right) + k = 0$

$\Rightarrow$ $k^2 - 2\alpha^2 - h\alpha^2 - \alpha^3 = 0$

$\therefore$ Locus of P(h, k) is $y^2 = \alpha^2 x + (\alpha^3 - 2\alpha^2)$ _____(2)

But ATQ, locus of P is a part of parabola $y^2 = 4x$, therefore comparing the two, we get $\alpha^2 = 4$ and $\alpha^3 - 2\alpha^2 = 0$

$\Rightarrow$ $a = 2$

22. PG is the normal to a standard ellipse at P, G being on the major axis. GP is produced outwards to Q so that PQ = GP. Show that the locus of Q is an ellipse whose eccentricity is $\frac{a^2-b^2}{a^2+b^2}$.

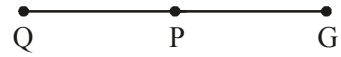
एक मानक दीर्घवृत के बिन्दु P पर अभिलम्ब PG है, जहाँ G दीर्घअक्ष पर स्थित है। GP को बाहर की ओर Q तक इस प्रकार बढ़ाया गया gSid PQ = GP है। दर्शाइये कि Q का बिन्दुपथ एक दीर्घवृत होगा, जिसकी उत्केन्द्रता $\frac{a^2-b^2}{a^2+b^2}$ होगी।

Ans.

Sol. Normal at P $ax\sec\theta - by\,\text{cosec}\,\theta = a^2e^2$

Put $y = 0$

ξ = αε² cos θ



(h,k) (a cosθ, b sinθ) ($ae^2\cos\theta$, 0)

Let

$a\cos\theta = \frac{h + ae^2\cos\theta}{2}$, $b\sin\theta = \frac{K}{2}$

$(2a - ae^2)\cos\theta = h$, $\sin\theta = \frac{K}{2b}$

$\cos^2\theta + \sin^2\theta = \frac{h^2}{a^2\left[2 - 1 + \frac{b^2}{a^2}\right]^2} + \frac{K^2}{4b^2}$

$$\frac{x^2}{a^2\left(2-e^2\right)^2}+\frac{y^2}{4b^2}=1$$

$$E=\sqrt{1-\frac{4b^2}{a^2\left(2-e^2\right)^2}}$$

$$=\sqrt{1-\frac{4b^2}{a^2\left(\frac{a^2+b^2}{a^2}\right)^2}}$$

$$=\sqrt{1-\frac{4a^2b^2}{\left(a^2+b^2\right)^2}}=\sqrt{\frac{\left(a^2+b^2\right)^2-4a^2b^2}{\left(a^2+b^2\right)^2}}=\sqrt{\frac{\left(a^2-b^2\right)^2}{\left(a^2+b^2\right)^2}}$$

$$E=\frac{a^2-b^2}{a^2+b^2}$$

23. A normal inclined at 45° to the axis of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is drawn. It meets the x-axis & the y-axis in P & Q respectively. If C is the centre of the ellipse, show that the area of triangle CPQ is $\frac{(a^2-b^2)^2}{2(a^2+b^2)}$ sq. units.

दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ का एक अभिलम्ब x-अक्ष से 45° के कोण पर खींचा गया है। यह दीर्घ–अक्ष और लघु–अक्ष से क्रमशः P एवं Q में मिलता है। यदि दीर्घवृत्त का केन्द्र C हो, तो दिखाइये कि त्रिभुज CPQ का क्षेत्रफल $\frac{(a^2-b^2)^2}{2(a^2+b^2)}$ वर्ग इकाई है।

Ans.

Sol. Equation of normal at R

$ax\sec\theta - by\,\text{cosec}\,\theta = a^2 - b^2$

now $P\left(\frac{a^2-b^2}{a}\cos\theta, 0\right)$

$$Q\left(0, -\frac{a^2-b^2}{b}\sin\theta\right)$$

$$\text{area} = \frac{1}{2}CP \times CQ$$

$$\Delta = \frac{\left(a^2-b^2\right)^2 |\sin\theta\cdot\cos\theta|}{2ab} \qquad \text{_____(1)}$$

Slope of normal

$$\frac{a}{b}\tan\theta = \tan 45°$$

$$\& \ \sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta} = \frac{2ab}{a^2+b^2}$$

Put in equation (1)

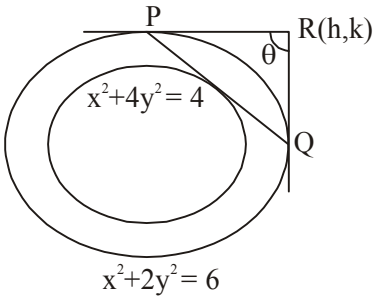
$$\Delta = \frac{\left(a^2-b^2\right)^2 \cdot \frac{ab}{\left(a^2+b^2\right)}}{2ab} = \frac{\left(a^2-b^2\right)^2}{2\left(a^2+b^2\right)}$$

24. A tangent to the ellipse $x^2 + 4y^2 = 4$ meets the ellipse $x^2 + 2y^2 = 6$ at P & Q. Prove that the tangents at P & Q of the ellipse $x^2 + 2y^2 = 6$ are at right angles.

दीर्घवृत $x^2 + 4y^2 = 4$ पर एक स्पर्श रेखा दीर्घवृत $x^2 + 2y^2 = 6$ को P तथा Q पर मिलती है। सिद्ध कीजिए कि दीर्घवृत $x^2 + 2y^2 = 6$ पर P तथा Q पर बनायी गई स्पर्श रेखाऐं समकोणीय प्रतिच्छेद करती है।

Ans.

Sol. Let R (h, k)



PQ is C.O.c. w.r.t. R

equation T = 0

$hx + 2ky - 6 = 0$ _____(1)

It is tangent to ellipse

$x^2 + 4y^2 = 4$

or $2\cos\theta\, x + 4\sin\theta\, y - 4 = 0$ _____(2)

Comparing (1) & (2)

$$\frac{h}{2\cos\theta}=\frac{k}{4\sin\theta}=\frac{-6}{-4}$$

$\Rightarrow h^2+k^2=9$

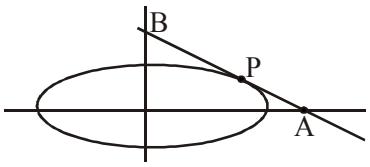
Hence (h, k) lies on director circle.

25. A tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ touches at the point P on it in the first quadrant & meets the coordinate axes in A & B respectively. If P divides AB in the ratio 3 : 1 reckoning from the x-axis find the equation of the tangent.

दीर्घवृत $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ पर एक स्पर्श रेखा P पर स्पर्श करती है एवं प्रथम चतुर्थांश में अक्षों को A तथा B पर क्रमशः काटती है। यदि बिन्दु P, AB को अनुपात 3 : 1 में x-अक्ष की ओर से विभाजित करता है, तो स्पर्श रेखा का समीकरण ज्ञात कीजिए।

Ans. $bx+a\sqrt{3}\,y=2ab$

Sol. Equation of tangent



$$\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$$

$$A\left(\frac{a}{\cos\theta},0\right),\ B\left(0,\frac{b}{\sin\theta}\right)$$

$$P\left(\frac{a}{4\cos\theta},\frac{3b}{4\sin\theta}\right)\equiv P\left(a\cos\theta,b\sin\theta\right)$$

$$\frac{a}{4\cos\theta}=a\cos\theta,\ \frac{3b}{4\sin\theta}=b\sin\theta$$

$$\cos^2\theta=\frac{1}{4},\ \sin^2\theta=\frac{3}{4}$$

Equation of tangent

$$\frac{x}{2a}+\frac{\sqrt{3}y}{2b}=1$$

$$bx+a\sqrt{3}y=2ab$$

26. Let P_i and P_i' be the feet of the perpendiculars drawn from foci S, S' on a tangent T_i to an ellipse whose length of semi-major axis is 20. If $\sum_{i=1}^{10}(SP_i)(S'P_i') = 2560$, then find the value of 100e.

(Where 'e' denotes eccentricity).

माना एक दीर्घवृत्त जिसका दीर्घअक्ष 20 है। बिन्दुओं P_i तथा P_i', नाभियों S, S' से स्पर्श रेखा T_i पर डाले गये लम्बों के पाद है। यदि $\sum_{i=1}^{10}(SP_i)(S'P_i') = 2560$, तो 100e का मान ज्ञात कीजिए। (जहाँ 'e' उत्केन्द्रता है।)

Ans. 60

Sol. $\sum_{i=1}^{10}(SP_i)\cdot(S'P_i) = 2560$

We know that $SP \cdot S'P = b^2$

So $10b^2 = 2560$

$b^2 = 256$

So, $e = \sqrt{1-\frac{256}{400}} = \frac{12}{20} = \frac{3}{5}$

$100e = 60$

27. An ellipse $x^2 + 4y^2 = 4$ is rotated anticlockwise through a right angle in its own plane about its centre. If the locus of the point of intersection of a tangent to ellipse in its origin position with the tangent at the same point of the ellipse in in its new position is given by the curve $(x^2 + y^2)^2 = \lambda(x^2 + y^2) + \mu xy$ where λ and μ are positive integers. Find the value of $(\lambda + \mu)$.

एक दीर्घवृत $x^2 + 4y^2 = 4$ को समतल में केन्द्र के सापेक्ष वामावृत दिशा में समकोण से घुमा दिया जाता है। यदि दीर्घवृत पर स्थित किसी बिन्दु पर दोनों स्थितियों में बनाई गई स्पर्श रेखाओं के प्रतिच्छेदन बिन्दु का बिन्दुपथ यदि वक्र $(x^2 + y^2)^2 = \lambda(x^2 + y^2) + \mu xy$ जहाँ λ तथा μ धनात्मक पूर्णांक है। $(\lambda + \mu)$ मान ज्ञात कीजिए।

Ans. 11

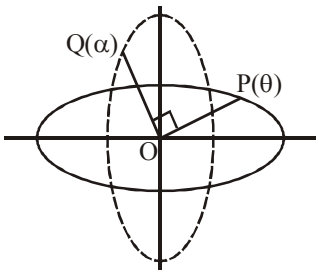
Sol. $E_O : \frac{x^2}{4} + \frac{y^2}{1} = 1$; $E_N : \frac{x^2}{1} + \frac{y^2}{4} = 1$ (where O = original, N = new)

P : $(2\cos\theta, \sin\theta)$

Q : $(\cos\alpha, 2\sin\alpha)$ say

$\vec{V}_1 = 2\cos\theta\,\hat{i} + \sin\theta\,\hat{j}$

$\vec{V}_2 = \cos\alpha\hat{i} + 2\sin\alpha\hat{j}$



———————— as $\vec{V}_1$ and $\vec{V}_2$ are perpendicular

$\vec{V}_1 \cdot \vec{V}_2 = 2\cos(\alpha - \theta) = 0 \Rightarrow \alpha - \theta = \frac{\pi}{2}$

$\alpha = \frac{\pi}{2} + \theta$

Q is $(-\sin\theta, 2\cos\theta)$

tangent at P(2 cosθ, sinθ) on $\frac{x^2}{4} + \frac{y^2}{1} = 1$, is $\frac{xx_1}{4} + \frac{yy_1}{1} = 1$ or $\frac{x\cos\theta}{2} + y\sin\theta = 1$ .....(1)

Tangent at Q(– sinθ, 2 cos θ) on $\frac{x^2}{1} + \frac{y^2}{4} = 1$, is $\frac{xx_1}{1} + \frac{yy_1}{2} = 1$ or $-x\sin\theta + \frac{y\cos\theta}{2} = 1$ ....(2)

From (1) $x\cos\theta + 2y\sin\theta = 2$

from (2) $y\cos\theta - 2x\sin\theta = 2$

solving for cosθ and sinθ, we have (using crammer's rule)

$D = \begin{vmatrix} x & 2y \\ y & -2x \end{vmatrix} = -2(x^2 + y^2)$, $D_1 = \begin{vmatrix} 2 & 2y \\ 2 & -2x \end{vmatrix} = -4(x + y)$, $D_2 = \begin{vmatrix} x & 2 \\ y & 2 \end{vmatrix} = 2(x - y)$

$\cos\theta = \frac{D_1}{D} = \frac{-4(x+y)}{-2(x^2+y^2)} = \frac{2(x+y)}{x^2+y^2}$

$\sin\theta = \frac{D_2}{D} = \frac{2(x-y)}{-2(x^2+y^2)} = \frac{(x-y)}{-(x^2+y^2)}$

Now $\sin^2\theta + \cos^2\theta = 1$

Hence $\frac{(x-y)^2}{(x^2+y^2)^2} + \frac{4(x+y)^2}{(x^2+y^2)^2} = 1 \Rightarrow \frac{x^2+y^2-2xy+4x^2+4y^2+8xy}{(x^2+y^2)^2} = 1$

$\Rightarrow \frac{5(x^2+y^2)+6xy}{(x^2+y^2)^2} = 1 \Rightarrow (x^2+y^2)^2 = 5(x^2+y^2) + 6xy$

$\Rightarrow$ $l = 5$ and $\mu = 6$

So, $(\lambda + \mu) = 5 + 6 = 11$ Ans.

28. Line L_1 is parallel to the line L_2. Slope of L_1 is 9. Also L_3 is parallel to L_4. Slope of L_4 is $\frac{-1}{25}$. All these lines touch the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. Find the area of the parallelogram formed by these lines.

रेखा L_1 रेखा L_2 के समानान्तर है। L_1 की प्रवणता 9 है। साथ ही L_3, L_4 के समानान्तर है जिनकी प्रवणता $\frac{-1}{25}$ है ये सभी रेखाऐं दीर्घवृत $\frac{x^2}{25}+\frac{y^2}{9}=1$ को स्पर्श करती है। रेखाओं द्वारा बनाये गये समानान्तर चतुर्भुज का क्षेत्रफल ज्ञात कीजिए।

Ans. 60

Sol. $\frac{x^2}{25}+\frac{y^2}{9}=1$

$y = mx + c$ touches the ellipse

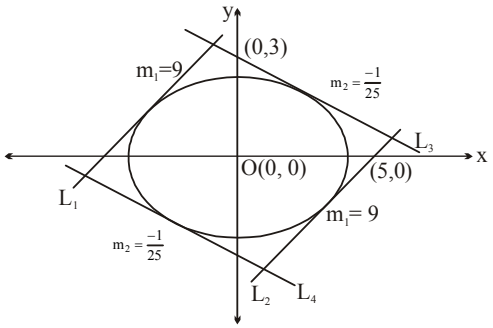
$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ if $c^2 = a^2m^2 + b^2$ (C.O.T.)

Hence $c^2 = (25)(81) + 9 = 9(225 + 1)$

$c = 3\sqrt{226}$ or $-3\sqrt{226}$

$|c_1 - c_2| = 6\sqrt{226}$

|||ly $m_2 = \frac{-1}{25}$



$\therefore \quad d^2 = 25\cdot\frac{1}{25^2}+9 = \frac{226}{25}$

$\Rightarrow \quad d = \frac{\sqrt{226}}{5}$ or $\frac{-\sqrt{226}}{5}$

$\therefore \quad |d_1 - d_2| = \frac{2\sqrt{226}}{5}$

$\therefore$ Area of parallelogram $= \frac{|c_1 - c_2||d_1 - d_2|}{|m_1 - m_2|}$

$$= \frac{6\sqrt{226}\cdot 2\sqrt{226}}{5\left(9+\frac{1}{25}\right)} = \frac{6(226)\cdot 2\cdot 25}{5(226)} = 60 \text{ (sq. units) Ans.}$$

29. Given the equation of the ellipse $\frac{(x-3)^2}{16}+\frac{(y+4)^2}{49}=1$, a parabola is such that its vertex is the lowest point of the ellipse and it passes through the ends of the minor axis of the ellipse. The equation of the parabola is in the form $16y = a(x-h)^2 - k$. Determine the value of $(a + h + k)$.

दिया गया है कि दीर्घवृत का समीकरण $\frac{(x-3)^2}{16}+\frac{(y+4)^2}{49}=1$ है। एक परवलय इस प्रकार है कि इसका शीर्ष दीर्घवृत का न्यूनतम बिन्दु है तथा यह दीर्घवृत के लघुअक्ष के सिरों से होकर गुजरता है। उस परवलय का समीकरण यदि $16y = a(x-h)^2 - k$ के रूप में व्यक्त किया जा सकता है, तो $(a + h + k)$ का मान ज्ञात कीजिए।

Ans. 186

Sol. Parabola is $(x-3)^2 = 4a(y+11)$

Passes through $(7, -4)$

$16 = 4a \times 7$

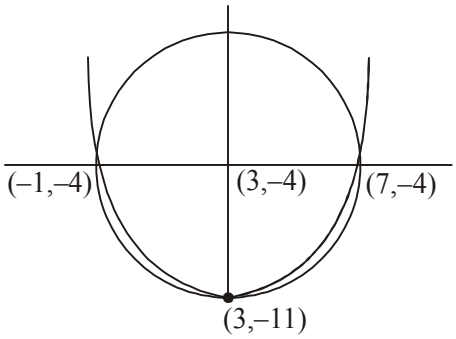
$a = \frac{4}{7}$

$(x-3)^2 = \frac{16}{7}(y+11)$

$7(x-3)^2 - 176 = 16y$

$a = 7, h = 3, k = 176$

$a + h + k = 186$



30. Rectangle ABCD has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. Find the perimeter of the rectangle.

आयत ABCD का क्षेत्रफल 200 है। एक दीर्घवृत जिसका क्षेत्रफल 200π है, जो A तथा C से गुजरता है, एवं नाभियाँ B एवं D है। आयत का परिमाप ज्ञात कीजिए।

Ans. 80

Sol. Let the length of rectangle be x, ty.

Let axis of ellipse be 2a & 2b.

$\because$ Area of ellipse is πab.

We have $xy = ab = 200$

From definition we have $x + y = 2a$

also, length of diagonal $= \sqrt{x^2 + y^2}$

Comparing the length of axes & distance of focii to the centre, we have

$a^2 = \frac{x^2 + y^2}{4} + b^2$

$(x + y)^2 = x^2 + y^2 + 4b^2$

$2xy = 4b^2$

$b = 10,\ a = 20$

Perimeter $= 2(a + b) = 80$

31. Consider the parabola $y^2 = 4x$ and the ellipse $2x^2 + y^2 = 6$, intersecting at P and Q.

(a) Prove that the two curves are orthogonal.

(b) Find the area enclosed by the parabola and the common chord of the ellipse and parabola.

(c) If tangent and normal at the point P on the ellipse intersect the x-axis at T and G respectively then find the area of the triangle PTG.

माना कि परवलय $y^2 = 4x$ है तथा दीर्घवृत $2x^2 + y^2 = 6$ परस्पर बिन्दुओं P एवं Q पर प्रतिच्छेद करते है

(a) दशाईये की दोनों वक्र लाम्बिक है।

(b) परवलय द्वारा दीर्घवृत तथा परवलय की उभयनिष्ठ जीवा के साथ परिबद्ध क्षेत्रफल ज्ञात कीजिए।

(c) यदि दीर्घवृत पर स्थित बिन्दु P से स्पर्श रेखा एवं अभिलम्ब x-अक्ष को क्रमशः T तथा G पर प्रतिच्छेद करते है, तो त्रिभुज PTG का {kskQy Kkr d hft , A

Ans. (b) 8/3, (c) 4

Sol. $y^2 = 4x,\ 2x^2 + y^2 = 6$

Solving $2x^2 + 4x - 6 = 0$

$x^2 + 2x - 3 = 0$

$x = 1, -3$

$P(1,2),\ Q(1,-2)$

(a) tangent to parabola at P $\quad 2y = \dfrac{4(x+1)}{2}$

$y = x + 1$

tangent to ellipse at P $\quad 2x + 2y = 6$

$y = -x + 3$

both are perpendicular. So curves orthogonal.

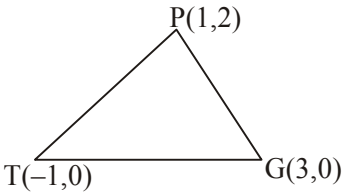
(b) area $= 2\int_0^1 2\sqrt{x}\,dx = 4 \times \dfrac{2}{3}\left(x^{\frac{3}{2}} \right)_0^1 = \dfrac{8}{3}$

(c) tangent $\quad x + y = 3$

normal $y = x + 1$

T(3,0), G(–1,0)

area $= \frac{1}{2} \times 4 \times 2 = 4$



32. Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre 'O' where $a > b > 0$. Tangent at any point P on the ellipse meets the coordinate axes at X and Y and N is the foot of the perpendicular from the origin on the tangent at P. Minimum length of XY is 36 and maximum length of PN is 4.

(a) Find the eccentricity of the ellipse.

(b) Find the maximum area of an isosceles triangle inscribed in the ellipse if one of its vertex coincides with one end of the major axis of the ellipse.

(c) Find the maximum area of the triangle OPN.

माना कि दीर्घवृत $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ जिसका केन्द्र 'O' है, जहाँ $a > b > 0$ है। दीर्घवृत पर स्थिति किसी बिन्दु P से स्पर्श रेखा निर्देशांक अक्षों को X एवं Y पर मिलती है एवं N मूल बिन्दु से स्पर्श रेखा के ऊपर डाला गया लम्बपाद है। XY का न्यूनतम मान 36 तथा PN की लम्बाई 4 है।

(a) दीर्घवृत की उत्केन्द्रता ज्ञात कीजिए।

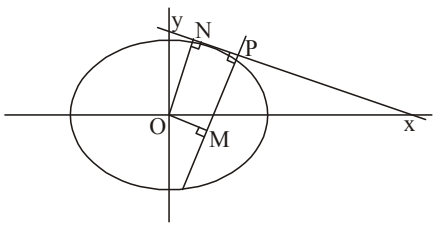
(b) दीर्घवृत्त में अर्न्तनिष्ठ एक समद्विबाहु त्रिभुज जिसका एक शीर्ष दीर्घअक्ष का एक सिरा है, तो त्रिभुज का अधिकतम क्षेत्रफल ज्ञात कीजिए।

(c) त्रिभुज OPN का अधिकतम क्षेत्रफल ज्ञात कीजिए।

Ans. (a) $\frac{3}{5}$; (b) $240\sqrt{3}$; (c) 36

Sol. tangent $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$

X(a sec θ, 0), Y(0, b cosec θ)



$XY = \sqrt{a^2 \sec^2\theta + b^2 \operatorname{cosec}^2\theta}$

$= \sqrt{a^2 + b^2 + a^2\tan^2\theta + b^2\cot^2\theta}$

Minimum value of $a^2\tan^2\theta + b^2\cot^2\theta = 2ab$

$(XY)_{min} = \sqrt{(a+b)^2} = 36$

$a + b = 36$ _____(1)

normal at P

$ax\sec\theta - by\,\text{cosec}\,\theta = a^2 - b^2$

$$NP = OM = \left|\frac{-a^2+b^2}{\sqrt{a^2\sec^2\theta + b^2\,\text{cosec}^2\theta}}\right|$$

$$(NP)_{max} = \frac{a^2-b^2}{a+b} = 4 \text{ (given)}$$

$a - b = 4$ _____(2)

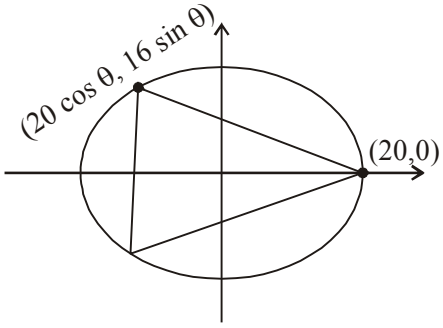
from (1) & (2)

$a = 20,\ b = 16$

(a) $e = \sqrt{1-\frac{16^2}{20^2}} = \frac{3}{5}$

(b) area $= \frac{1}{2}\times(20+20\cos\theta)\cdot 32\sin\theta$

$A = 16 \times 20\,(1+\cos\theta)\sin\theta$ _____(1)



A_{max} at $\theta = \frac{\pi}{3}$

$$A_{max} = 16\times 20\times\left(1+\frac{1}{2}\right)\times\frac{\sqrt{3}}{2}$$

$$= 240\sqrt{3}$$

(c) $\Delta = \frac{1}{2}\times OP\times PN$

$$\Delta = \frac{1}{2}\times\frac{ab\left(a^2-b^2\right)}{a^2\tan\theta + b^2\cot\theta}$$

$$\Delta_{max} = \frac{1}{2}\frac{ab\left(a^2-b^2\right)}{2ab}$$

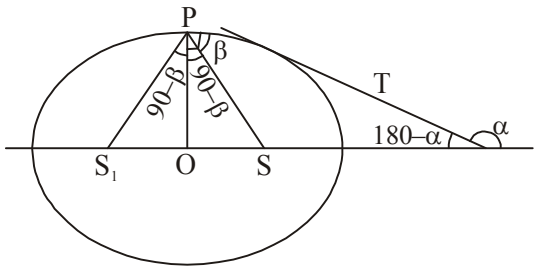
$$= \frac{400-256}{4} = 36$$

33. A straight line AB touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ & the circle $x^2+y^2=r^2$; where $a > r > b$. A focal chord of the ellipse, parallel to AB intersects the circle in P & Q, find the length of the perpendicular drawn from the centre of the ellipse to PQ. Hence show that PQ = 2b.

एक सरल रेखा AB दीर्घवृत $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ को स्पर्श करती है तथा वृत $x^2+y^2=r^2$ जहाँ $a > r > b$ है। दीर्घवृत की एक नाभिय जीवा AB के समानान्तर है, वृत को P तथा Q पर काटती है, दीर्घवृत के केन्द्र से PQ पर डाले गये लम्ब की लम्बाई ज्ञात करें। तदानुसार सिद्ध djsaPQ = 2b है।

Ans. $\sqrt{r^2-b^2}$

Sol. Let equation of tangent



$y=mx+\sqrt{a^2m^2+b^2}$

it touches circle $C^2=r^2(m^2+1)$

$\therefore\ a^2m^2+b^2=r^2m^2+r^2$

$$m^2=\frac{r^2-b^2}{a^2-r^2}$$

Equation of PQ

$y = m(x - ae)$

or $mx - y - mae = 0$

Length of perpendicular from centre to PQ(LP)

$$=\frac{+mae}{\sqrt{m^2+1}}=\sqrt{\frac{r^2-b^2}{a^2-b^2}}\times a\times\frac{1}{a}\sqrt{a^2-b^2}$$

$$=\sqrt{r^2-b^2}$$

$$PQ=2\sqrt{r^2-LP^2}=2\sqrt{b^2}=b$$

34. A ray emanating from the point (– 4, 0) is incident on the ellipse $9x^2+25y^2=225$ at the point P with abscissa 3. Find the equation of the reflected ray after first reflection.

एक किरण जो बिन्दु (– 4, 0) से निर्गमन कर रही है, दीर्घवृत $9x^2+25y^2=225$ के बिन्दु P पर जिसका भुज 3 है, पर आपतित होती gSA i jkofrZr fd j.kd kl ehd j.kKkr d hft , A

Ans.

Sol. For point P, x-coordinate = 3

Given ellipse $9x^2 + 25y^2 = 225$

$9(3)^2 + 25y^2 = 225$

$y = \pm\frac{12}{5}$

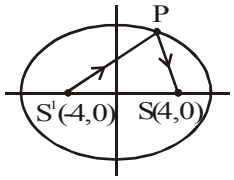
Coordinate of P is $\left(3, \pm\frac{12}{5}\right)$

Now $e = \frac{4}{5}$ & $ae = 4$

so foci is $(\pm 4, 0)$

Now equation of reflected ray (PS) is

$12x + 5y = 48$ or $12x - 5y = 48$



35. Variable pairs of chords at right angles are drawn through any point P (with eccentric angle $\pi/4$) on the ellipse $\frac{x^2}{4} + y^2 = 1$, to meet the ellipse at two points say A and B. If the line joining A and B passes through a fixed point Q(a, b) such that $a^2 + b^2$ has the value equal to $\frac{m}{n}$, where m, n are relatively prime positive integers, find (m + n).

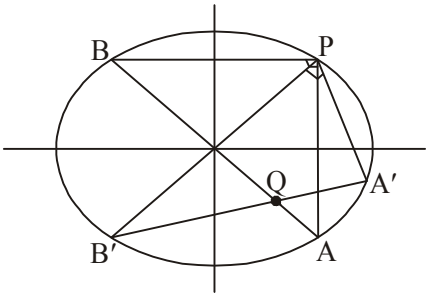
किसी बिन्दु P (जिसका उत्केन्द्रीय कोण $\pi/4$ है) से चर दो परस्पर लम्बवत् चर जीवाऐं दीर्घवृत $\frac{x^2}{4} + y^2 = 1$ को दो बिन्दुओं माना A तथा B पर मिलती है। यदि A तथा B को मिलाने वाली रेखा एक नियत बिन्दु Q(a, b) से होकर जाती है कि $a^2 + b^2$ का मान $\frac{m}{n}$ के : i esa,Q Dr fd ; kt k, st gk,m, n सापेक्षतः अभाज्य संख्याऐं हैं, तो (m + n) का मान ज्ञात कीजिए।

Ans. 19

Sol. $\frac{x^2}{4} + \frac{y^2}{1} = 1$

$P = \left(2\cdot\cos\frac{\pi}{4}, 1\cdot\sin\frac{\pi}{4}\right) = \left(\sqrt{2}, \frac{1}{\sqrt{2}}\right)$

$A\left(\sqrt{2},-\frac{1}{\sqrt{2}}\right), B\left(-\sqrt{2},\frac{1}{\sqrt{2}}\right)$



$m_{AB}=\frac{\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}}{-\sqrt{2}-\sqrt{2}}=-\frac{1}{2}$

$AB: y+\frac{1}{\sqrt{2}}=-\frac{1}{2}\left(x-\sqrt{2}\right)$

$x+2y=0$ _____(!)

passes through Q(a, b)

$b=-\frac{a}{2}$

also $a^2+b^2=a^2+\frac{a^2}{4}=\frac{5a^2}{4}$

now take another point $B'\left(-\sqrt{2},-\frac{1}{\sqrt{2}}\right)$

$m_{(PB')}=\frac{1}{2}$

$\Rightarrow m(PA')=-2$

equation of PA′ : $y-\frac{1}{\sqrt{2}}=-2\left(x-\sqrt{2}\right)$

$2x+y=\frac{5}{\sqrt{2}}$

put it into equation of ellipse to get

$A'\left(\frac{23\sqrt{2}}{17},-\frac{7}{17\sqrt{2}}\right)$

equation of A′B′

$y+\frac{1}{\sqrt{2}}=\frac{1}{8}\left(x+\sqrt{2}\right)$ _____(2)

Solving (1) & (2)

$a = \sqrt{3}$

$\frac{5a^2}{4} = \frac{15}{4} = \frac{m}{n}$

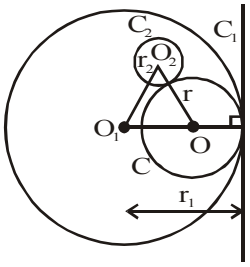
$m + n = 19$

36. Let C_1 and C_2 be two circles with C_2 lying inside C_1. A circle C lying inside C_1 touches C_1 internally and C_2externally. Identify the locus of the centre of C.

माना दो वृत्त C_1 एवं C_2 इस प्रकार है कि C_2 , वृत्त C_1 के अन्दर स्थित है। C_1 के अन्दर एक अन्य वृत्त C, वृत्त C_1 को आन्तरिक रूप से Li ' kZd j r kgS vkSj oÙk C_2 के बाह्य स्पर्श करता है। तो वृत्त C के केन्द्र का बिन्दुपथ ज्ञात कीजिये।

Ans. Locus is an ellipse with foci as the centres of the circles C_1a nd C_2

Sol. Let the given circles C_1 and C_2 have centres O_1 and O_2 and radii r_1 and r_2 respectively.
Let the variable circle C touching C_1 internally, C_2 externally have a radius r and centre at O.
Now , $OO_2 = r + r_2$ and $OO_1 = r_1 - r$.



$\Rightarrow OO_1 + OO_2 = r_1 + r_2$
which is greater than O_1O_2 as $O_1O_2 < r_1 + r_2$
($\because$ C_2 lies inside C_1)
$\Rightarrow$ Locus of O is an ellipse with foci O_1 and O_2.

37. Find the condition so that the line $px + qy = r$ intersects the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in points whose eccentric angles differ by $\frac{\pi}{4}$.

वह शर्त ज्ञात करे जिसके लिए रेखा $px + qy = r$ दीर्घवृत $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ को उन बिन्दुओं पर प्रतिच्छेद करती है जिनके उत्केन्द्रीय कोण esa vUrj $\frac{\pi}{4}$ है।

Ans. $a^2p^2 + b^2q^2 = r^2\sec^2\frac{\pi}{8} = (4 - 2\sqrt{2})r^2$

Sol. equation of chord

$$\frac{x}{a}\cos\left(\frac{\theta_1+\theta_2}{2}\right)+\frac{y}{b}\sin\left(\frac{\theta_1+\theta_2}{2}\right)=\cos\left(\frac{\theta_1-\theta_2}{2}\right)$$

$$\theta_1-\theta_2=\frac{\pi}{4}$$

$$\left.\frac{x}{a}\cos\left(\frac{\theta_1+\theta_2}{2}\right)+\frac{y}{b}\sin\left(\frac{\theta_1+\theta_2}{2}\right)=\cos\frac{\pi}{8}\right] \text{compare}$$
$$px+qy=r$$

$$\frac{pa}{\cos\left(\frac{\theta_1+\theta_2}{2}\right)}=\frac{qb}{\sin\left(\frac{\theta_1+\theta_2}{2}\right)}=\frac{r}{\cos\frac{\pi}{8}}$$

$$\frac{ap}{r}\cos\frac{\pi}{8}=\cos\left(\frac{\theta_1+\theta_2}{2}\right) \qquad \text{_____(1)}$$

$$\frac{bq}{r}\cos\frac{\pi}{8}=\sin\left(\frac{\theta_1+\theta_2}{2}\right) \qquad \text{_____(2)}$$

$(1)^2+(2)^2$

$$a^2p^2+b^2q^2=r^2\sec^2\frac{\pi}{8}=\left(4-2\sqrt{2}\right)r^2$$

38. Prove that, in an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, the perpendicular from a focus upon any tangent and the line joining the centre of the ellipse to the point of contact meet on the corresponding directrix.

सिद्ध कीजिए कि एक दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ में, नाभि से किसी स्पर्श रेखा पर डाला गया लम्ब और दीर्घवृत्त के केन्द्र को स्पर्श बिन्दु से

t kMrsoky hj [kkl axr fu; r ki j feyr sgSA

Ans.

Sol. $SQ: y-0=+\frac{a}{b}\tan\theta(x-ae)$ _____(1)

$OP: y=\left(\frac{b}{a}\tan\theta\right)x$ _____(2)

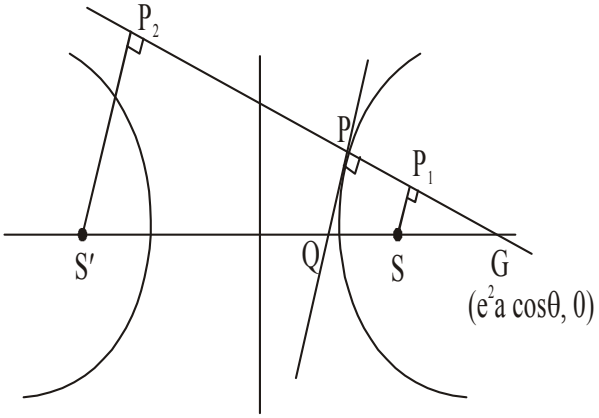
For POI solve (1) & (2)

$$\frac{a}{b}(x-ae)=\frac{b}{a}x$$

$$\frac{a^2-b^2}{ab}x=\frac{a^2e}{b}$$

$a^2e^2x = a^3e$

$$x=\frac{a}{e} \qquad \text{H.P.}$$

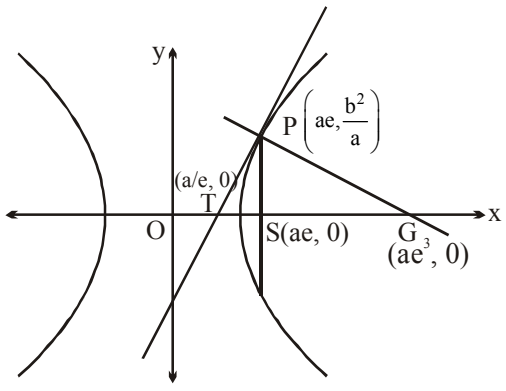


39. Tangent and normal are drawn at the upper end (x_1, y_1) of the latus rectum P with $x_1 > 0$ and $y_1 > 0$, of the hyperbola $\frac{x^2}{4}-\frac{y^2}{12}=1$, intersecting the transverse axis at T and G respectively. Find the area of the triangle PTG.

अतिपरवलय $\frac{x^2}{4}-\frac{y^2}{12}=1$ के नाभिलम्ब के ऊपरी सिरे P (x_1, y_1) जहाँ पर $x_1 > 0$ तथा $y_1 > 0$ स्पर्शी एवं अभिलम्ब डाले जाते है t ksfd vuiqZHkv {kd ksØe'k% T एवं G पर काटती है। त्रिभुज PTG का क्षेत्रफल बताओं।

Ans. 45

Sol. $P\left(ae, \frac{b^2}{a}\right)$



T: $\frac{xx_1}{a^2}-\frac{yy_1}{b^2}=1$

$$\frac{x\,a\,e}{a^2}-\frac{y\,b^2}{b^2a}=1$$

$$\frac{ex}{a}-\frac{y}{a}=1$$

put $y=0$ to get

$$T=\left(\frac{a}{e}, 0\right)$$

N: $\frac{a^2x}{ae}+\frac{b^2y}{y_1}=a^2+b^2$; put $y=0$ to get

$G = \dfrac{e\left(a^2+b^2\right)}{a} = (ae^3, 0)$

G: $(ae^3, 0)$

$\text{Area} = \left(ae^3 - \dfrac{a}{e}\right)\cdot\dfrac{b^2}{a}\cdot\dfrac{1}{2} = \dfrac{\left(e^4-1\right)b^2}{2e}$

Now $e^2 = 1 + \dfrac{12}{4} = 4 \Rightarrow e = 2$ and $b^2 = 12$

$\text{Area} = \dfrac{15\cdot 12}{2\cdot 2} = 45$ Ans.

40. Find the equations of the tangents to the hyperbola $x^2 - 9y^2 = 9$ that are drawn from (3, 2). Find the area of the triangle that these tangents form with their chord of contact.

बिन्दु (3, 2) से अतिपरवल $x^2 - 9y^2 = 9$ पर खीचे गये स्पर्श रेखाओं का समीकरण ज्ञात करो। इन स्पर्श रेखाओं एवं उनके स्पर्श जीवा l scus=Hkq d k{ks=kQy Kkr d hft , A

Ans. $y = \dfrac{5}{12}x + \dfrac{3}{4}$; $x - 3 = 0$; 8 sq. unit

Sol. C.O.C. w.r.t. P(3,2)

$T = 0$

$x - 6y = 3$ _____(1)

H : $x^2 - 9y^2 = 9$ _____(2)

Solving (1) & (2)

$(3 + 6y)^2 - 9y^2 = 9$

$y = 0$ $\qquad y = -\dfrac{4}{3}$

$x = 3$ $\qquad x = -5$

Q(3,0) $\qquad R\left(-5, -\dfrac{4}{3}\right)$

PR : $y = \dfrac{5}{12}x + \dfrac{3}{4}$

PQ : $x - 3 = 0$

$$\Delta = \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 3 & 3 & -5 \\ 2 & 0 & -\frac{4}{5} \end{vmatrix} = 8 \text{ sq. unit}$$

41. The normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ drawn at an extremity of its latus rectum is parallel to an asymptote.

Show that the eccentricity is equal to the square root of $\frac{(1+\sqrt{5})}{2}$.

अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ के नाभिलम्ब के एक सिरे पर अभिलम्ब, इसके एक अनन्तस्पर्शी के समानान्तर है। सिद्ध करो कि इसकी

mRd Uiz k $\frac{(1+\sqrt{5})}{2}$ का वर्ग मूल है।

Ans.

Sol. Normal at $\left(ae, \frac{b^2}{a}\right)$

$$\frac{a^2x}{ae} + \frac{b^2y}{\frac{b^2}{a}} = a^2 + b^2$$

$$\frac{ax}{e} + ay = a^2 + b^2 \qquad ...(1)$$

It is parallel to $\frac{x}{a} + \frac{y}{b} = 0$

$$\Rightarrow \frac{b}{a} = \frac{1}{e} \Rightarrow \frac{b^2}{a^2} = \frac{1}{e^2} \qquad ...(2)$$

We know that

$b^2 = a^2(e^2 - 1)$ ...(3)

from (2) & (3)

$$\frac{a^2\left(e^2-1\right)}{a^2}=\frac{1}{e^2}$$

Let $e^2 = t$

$t^2 - t - 1 = 0$

$$t=\frac{1\pm\sqrt{5}}{2}$$

$$e^2=\frac{1+\sqrt{5}}{2}$$

42. A line through the origin meets the circle $x^2+y^2=a^2$ at P & the hyperbola $x^2-y^2=a^2$ at Q. If the locus of the point of intersection of the tangent at P to the circle and the tangent at Q to the hyperbola is the curve $a^4(x^2-a^2)+\lambda x^2 y^4=0$, then find the value of λ.

मूल बिन्दु से जाने वाली रेखा वृत्त $x^2+y^2=a^2$ को P पर तथा अतिपरवलय $x^2-y^2=a^2$ को Q पर मिलती है। यदि वृत्त के बिन्दु P पर Li ' kZ Sk kr Hkkv fr i j oy ; d sfcUnq Q पर स्पर्श रेखा के प्रतिच्छेद बिन्दु का बिन्दुपथ $a^4(x^2-a^2)+\lambda x^2y^4=0$ तो λ का मान बताओं।

Ans. 4

Sol. Let line $y = mx$ meets the circle $x^2 + y^2 = a^2$ at $P\left(\frac{a}{\sqrt{1+m^2}},\frac{am}{\sqrt{1+m^2}}\right)$

equation of tangent at P

$$x + my = a\sqrt{1+m^2} \qquad \text{_____(1)}$$

$y = mx$ meets hyperbola at $Q\left(\frac{a}{\sqrt{1+m^2}},\frac{am}{\sqrt{1+m^2}}\right)$

tangent at Q

$$x - my = a\sqrt{1+m^2} \qquad \text{_____(2)}$$

$(1)^2-(2)^2$

$$m=\frac{2xy}{a^2}$$

from (1)

$$x + \frac{2xy}{a^2} y = a\sqrt{1 + \frac{4x^2y^2}{a^2}}$$

$$\left(a^2 + 2y^2\right)x = a\sqrt{a^4 + 4x^2y^2}$$

$(a^4 + 4y^4)x^2 = a^6$

43. The graphs of $x^2 + y^2 + 6x - 24y + 72 = 0$ & $x^2 - y^2 + 6x + 16y - 46 = 0$ intersect at four points. Compute the sum of the distances of these four points from the point $(-3, 2)$.

वक्रो $x^2 + y^2 + 6x - 24y + 72 = 0$ तथा $x^2 - y^2 + 6x + 16y - 46 = 0$ के रेखा चित्र एक दूसरे को चार बिन्दुओं पर काटते है। bu pkjLijdVus fcUnqvksa ls fcUnq $(-3, 2)$ से दूरियो का योग बताओं।

Ans. 40

Sol. $x^2 + y^2 + 6x - 24y + 72 = 0$ _____(1)

$x^2 - y^2 + 6x + 16y - 46 = 0$ _____(2)

(1) + (2)

$2x^2 + 12x - 8y + 26 = 0$

$(x + 3)^2 = 4(y - 1)$ _____(3)

(1) – (2)

$$2y^2 - 40y + 118 = 0 \left\langle \begin{matrix} y_1 \\ y_2 \end{matrix} \right.$$

$y_1 + y_2 = 20$

Now according to question

$$D = \sum_{i=1}^{4} \sqrt{\left(x_i + 3\right)^2 + \left(y_i - 2\right)^2}$$

from (3)

$$D = \sum \sqrt{4\left(y_i - 1\right) + y_i^2 - 4y_i + 4}$$

$$= \sum_{i=1}^{4} y_i = 2\left(y_1 + y_2\right) = 2 \times 20 = 40$$

44. An ellipse and a hyperbola have their principal axes along the coordinate axes and have a common foci separated by a distance $2\sqrt{13}$, the difference of their focal semi axes is equal to 4. If the ratio of their eccentricities is 3/7. Find the equation of these curves.

एक दीर्घवृत तथा अतिपरवलय के मुख्य अक्ष निर्देशांक्षो के संलग्न है तथा उनके उभयनिष्ठ नाभियों के बीच की दूरी $2\sqrt{13}$ है तथा उनके ukHkh; v/kZ {ks=kad kv Urj 4 है। यदि उत्केन्द्रताओं का अनुपात 3/7 है तो वक्रो का समीकरण ज्ञात कीजिए।

Ans. $\frac{x^2}{49}+\frac{y^2}{36}=1\ ;\ \frac{x^2}{9}-\frac{y^2}{4}=1$

Sol. Let ellipse : a, b, e

hyperbola : A, B, E

$2ae = 2\sqrt{13}$ $\qquad$ $2AE = 2\sqrt{13}$

$a - A = 4$ $\qquad$ & $\qquad$ $\frac{e}{E}=\frac{3}{7}$

using $b^2 = a^2(1 - e^2)$ $\qquad$ $B^2 = A^2(E^2 - 1)$

and solving, we get

ellipse : $\frac{x^2}{49}+\frac{y^2}{36}=1$

H : $\frac{x^2}{9}-\frac{y^2}{4}=1$

45. From the centre C of the hyperbola $x^2 - y^2 = 9$, CM is drawn perpendicular to the tangent at any point of the curve, meeting the tangent at M and the curve at N. Find the value of the product (CM)(CN).

अतिपरवलय $x^2 - y^2 = 9$ के केन्द्र C से, वक्र के किसी बिन्दु स्पर्श रेखा पर CM लम्ब डाला जाता है। जो कि स्पर्श रेखा को M पर r HkkoØ d ksN पर मिलता है। गुणनफल (CM)(CN) का मान बताइये।

Ans. 9

Sol. Equation of a tangent at P

$x\sec\theta - y\tan\theta = a$ (where a = 3)

ξ □ ψσανθ = a cosθ .....(1)

slope = $\frac{1}{\sin\theta}$

perpendicular slope = $-\sin\theta$

a line perpendicular to (1) and passing through C

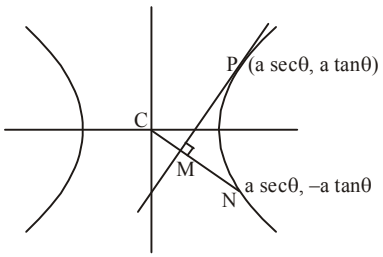
equation is $y = mx + c$ passes through C, then $c = 0$

$x \sin\theta + y = 0$ ....(2)

Solving (2) with

$x^2 - y^2 = a^2$

$x^2 - x^2\sin^2\theta = a^2 \quad \Rightarrow \quad x^2 = a^2\sec^2\theta \quad \Rightarrow \quad x = a\sec\theta$



Hence $M \equiv (a\sec\theta, -a\tan\theta)$

now $CN \cdot CM$ = power of point C w.r.t. the circle on PN as diameter

now equation of circle is $(x - a\sec\theta)^2 + (y - a\tan\theta)(y + a\tan\theta) = 0$

$\therefore \quad CN \cdot CM = a^2\sec^2\theta - a^2\tan^2\theta = a^2 = 9$ Ans.

46. A hyperbola has one focus at the origin and its eccentricity = $\sqrt{2}$ and one of its directrix is $x + y + 1 = 0$. Find the equation to its asymptotes.

एक अतिपरवलय की नाभि मूल बिन्दु पर तथा उत्केन्द्रता = $\sqrt{2}$ है तथा इसकी एक नियता $x + y + 1 = 0$ इसके अन्तस्पर्शीयों का l ehdj.kKkr dhft, A

Ans. $x + 1 = 0$ and $y + 1 = 0$

Sol. Equation to the hyperbola where S = (0, 0) ; directrix is $x + y + 1 = 0$ and $e = \sqrt{2}$ is

$$\sqrt{x^2 + y^2} = \sqrt{2}\left(\frac{x + y + 1}{\sqrt{2}}\right)$$

$\therefore \quad x^2 + y^2 = (x + y + 1)^2$

$2xy + 2x + 2y + 1 = 0$

Let the combined equation of the asymptotes is

$2xy + 2x + 2y + c = 0$

put $D = 0$ to get $c = 2$

hence combined equation of the asymptotes are

$xy + x + y + 1 = 0$

$(x + 1)(y + 1) = 0 \quad \Rightarrow \quad x + 1 = 0$ and $y + 1 = 0$

47. If a rectangular hyperbola have the equation, $xy = c^2$, then the locus of the middle points of its chords of constant length 6 is $(x^2 + y^2)(xy - c^2) = \lambda xy$, find the value of λ.

यदि आयताकार अतिपरवलय $xy = c^2$, के लिये नियत लम्बाई 6 वाले जीवाओं के मध्य बिन्दु का बिन्दुपथ $(x^2 + y^2)(xy - c^2) = \lambda xy$ हो तो λ का मान ज्ञात कीजिए।

Ans. 9

Sol. Let end points of chord

$$P\left(ct_1, \frac{c}{t_1}\right),\ Q\left(ct_2, \frac{c}{t_2}\right)$$

then $2h = c(t_1 + t_2),\ 2k = c\left(\frac{1}{t_1} + \frac{1}{t_2}\right)$

according to question

$$c^2(t_1 - t_2)^2 + c^2\left(\frac{1}{t_1} + \frac{1}{t_2}\right)^2 = 36$$

$$c^2\left[(t_1 + t_2)^2 - 4t_1t_2\right]\left[1 + \frac{1}{(t_1t_2)^2}\right] = 36$$

$$c^2\left[\frac{4h^2}{c^2} - \frac{4h}{k}\right]\left(1 + \frac{k^2}{h^2}\right) = 36$$

$(xy - c^2)(x^2 + y^2) = 9xy$

48. If a chord joining the points P (a secθ, a tanθ) & Q (a secϕ, a tanϕ) on the hyperbola $x^2 - y^2 = a^2$ is a normal to it at P, then show that $\tan\phi = \tan\theta(4\sec^2\theta - 1)$.

यदि अतिपरवलय $x^2 - y^2 = a^2$ के बिन्दुओं P (a secθ, a tanθ) तथा Q (a secϕ, a tanϕ) को जोड़ने वाली जीवा बिन्दु P पर अभिलम्ब gksr ksfl) djsstan $\phi = \tan\theta(4\sec^2\theta - 1)$.

Ans.

Sol. Slope of chord PQ = slope of normal at P

$$\frac{\tan\phi - \tan\theta}{\sec\phi - \sec\theta} = \frac{\tan\theta}{\sec\theta}$$

$$\tan\phi - \tan\theta = -k\tan\theta \Rightarrow (1-k)\tan\theta = \tan\phi \qquad \text{_____(1)}$$

$$\&\ \sec\phi - \sec\theta = k\sec\theta \Rightarrow (1+k)\sec\theta = \sec\phi \qquad \text{_____(2)}$$

now, $(2)^2 - (1)^2$

$$\left[(1+k)\sec\theta\right]^2 - \left[(1-k)\tan\theta\right]^2 = \sec^2\phi - \tan^2\phi$$

$$k = -2\left(\sec^2\theta + \tan^2\theta\right) = -4\sec^2\theta + 2$$

from (1)

$$\tan\phi = \tan\theta\left(1+4\sec^2\theta-2\right) = \tan\theta\left(4\sec^2\theta-1\right)$$

49. Prove that the part of the tangent at any point of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ intercepted between the point of contact and the transverse axis is a harmonic mean between the lengths of the perpendiculars drawn from the foci on the normal at the same point.

सिद्ध करो कि स्पर्श बिन्दु एवं अनुप्रस्थ अक्ष के बीच, अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ के किसी स्पर्श रेखा का अन्तखण्ड, स्पर्श बिन्दु पर vfHkyEc dh nwjh ds fy, nh xbZ nwfj;ksa ds e/; gksus okyh gjkRed ek/; gksrh gSA

Ans.

Sol. $T: \frac{a\sec\theta}{a}-\frac{y\tan\theta}{b}=1$

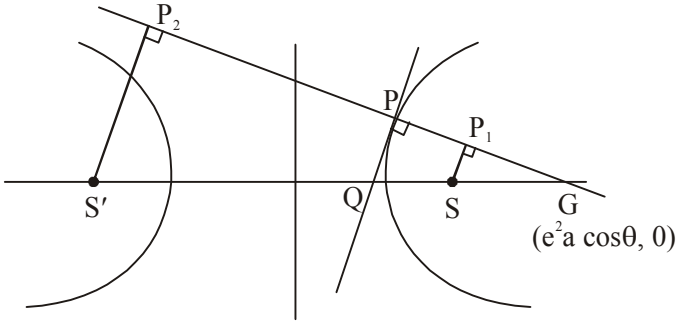
$$N: \frac{ax}{\sec\theta}+\frac{by}{\tan\theta}=a^2e^2$$

$$\frac{P}{P_1}=\frac{QG}{SG}=\frac{e^2a\sec\theta-a\cos\theta}{e^2a\sec\theta-ae}$$

$$=\frac{e^2-\cos^2\theta}{e^2-e\cos\theta}=\frac{e+\cos\theta}{e}$$

$$=1+\frac{\cos\theta}{e}$$

$$\frac{P}{P_2}=\frac{QG}{S'G}=1-\frac{\cos\theta}{e}$$



Now :

$$\frac{P}{P_1}+\frac{P}{P_2}=2$$

$$\frac{1}{P_1}+\frac{1}{P_2}=\frac{2}{P}$$ H.P.

50. Ascertain the co-ordinates of the two points Q and R, where the tangent to the hyperbola $\frac{x^2}{45}-\frac{y^2}{20}=1$ at the point P(9, 4) intersects the two asymptotes. Finally prove that P is the middle point of QR. Also compute the area of the triangle CQR where C is the centre of the hyperbola.

बिन्दुओं Q एवं R के निर्देशांक ज्ञात करो जहाँ पर अतिपरवलय $\frac{x^2}{45}-\frac{y^2}{20}=1$ के बिन्दु P(9, 4) पर स्पर्शी दोनों अनन्तस्पर्शीयों को प्रतिच्छेदित करते है। अन्ततः सिद्ध करो कि P, QR का मध्य बिन्दु है। यदि C अतिपरवलय का केन्द्र हो तो त्रिभुज CQR का क्षेत्रफल cr kvk gA

Ans. (15, 10) and (3, –2) and 30 sq. units

Sol. Tangent at P(9,4)

$$T:\frac{9x}{45}-\frac{4y}{20}=1$$

$$T: x-y=5 \qquad \text{_____(1)}$$

Cuts asymptotes $y=\pm\frac{2}{3}x$ at Q & R

Q (3, –2) R(15,10)

Mid point of QR $\Rightarrow$ (9,4) = P H.P.

$$\Delta CQR=\frac{1}{2}\begin{vmatrix}3 & -2\\ 0 & 0\\ 15 & 10\\ 3 & -2\end{vmatrix}=\frac{1}{2}|-30-30|=30 \text{ sq. units}$$

51. The tangent at P on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ meets one of the asymptote in Q. If the locus of the mid-point of PQ is a hyperbola, $\frac{x^2}{a^2}-\frac{y^2}{b^2}=\lambda$, find the value of 4λ.

अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ के बिन्दु P पर स्पर्शी,, एक अनन्तस्पर्शी को Q पर मिलती है। यदि PQ के मध्य बिन्दु का बिन्दुपथ vfr i j oy ; $\frac{x^2}{a^2}-\frac{y^2}{b^2}=\lambda$ हो तो 4λ का मान बताइये।

Ans. 3

Sol. $T: \frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$ ...(1)

$A: \frac{x}{a} + \frac{y}{b} = 0$ ...(2)

P (a sec θ, b tan θ) Q(a(sec θ – tan θ), –b(sec θ – tan θ))

$h = \frac{a\sec\theta + a\sec\theta - a\tan\theta}{2}$

$\frac{h}{a} = \sec\theta - \frac{\tan\theta}{2}$ ...(3)

Similarly $\frac{k}{b} = \tan\theta - \frac{\sec\theta}{2}$ ...(4)

(3) + (4) $\frac{h}{a} + \frac{k}{b} = \frac{\sec\theta + \tan\theta}{2}$ ...(5)

(3) – (4) $\frac{h}{a} - \frac{k}{b} = \frac{3}{2}(\sec\theta - \tan\theta)$ ...(6)

Now (5) × (6) $\frac{h^2}{a^2} - \frac{k^2}{b^2} = \frac{3}{4}$

$\lambda = \frac{3}{4}$

$4\lambda = 3$

52. If the lines $x + y + 1 = 0$ and $2x - y + 2 = 0$ are the asymptotes of a hyperbola. If the line $x - 2 = 0$ touches the hyperbola then the equation of the hyperbola is $4(x + y + 1)(2x - y + 2) = \lambda$. Find the value of λ.

रेखाएं $x + y + 1 = 0$ तथा $2x - y + 2 = 0$ किसी अतिपरवलय के अनन्तस्पर्शी है। यदि रेखा $x - 2 = 0$ अतिपरवलय को स्पर्श करे r ksv fr i j oy ; d kl ehdj . k $4(x + y + 1)(2x - y + 2) = \lambda$ होगा। λ का मान ज्ञात कीजिए।

Ans. 81

Sol. equation is $(x + y + 1)(2x - y + 2) = c$

solving it with $x = 2$, we get

$y^2 - 3y + (c - 18) = 0$

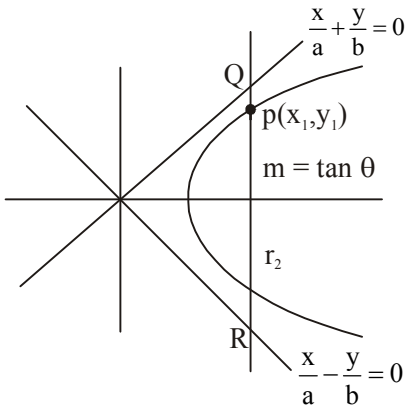
put $D = 0 \Rightarrow c = \frac{81}{4}$

53. Through any point P of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ a line QPR is drawn with a fixed gradient m, meeting the asymptotes in Q & R. Show that the product, (QP) (PR) $=\frac{a^2b^2(1+m^2)}{b^2-a^2m^2}$.

अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ के किसी बिन्दु P से एक रेखा QPR खीचीं जाती है जिसकी प्रवणता नियत है। जो कि अनन्तस्पर्शियों को Q एव R पर मिलती है। सिद्ध करो कि गुणनफलन (QP) (PR) $=\frac{a^2b^2(1+m^2)}{b^2-a^2m^2}$.

Ans.

Sol.



$\frac{x-x_1}{\cos\theta}=\frac{y-y_1}{\sin\theta}=r$

$(x, y)=\left(x_1+r\cos\theta, y_1+r\sin\theta\right)$

$\cos\theta=\frac{1}{\sqrt{1+m^2}}$

$\sin\theta=\frac{m}{\sqrt{1+m^2}}$

$$\frac{\left(x_1+r\cos\theta\right)^2}{a^2}-\frac{\left(y_1+r\cos\theta\right)^2}{b^2}=0$$

$$\left(\frac{\cos^2\theta}{a^2}-\frac{\sin^2\theta}{b^2}\right)r^2+2\left(\frac{x_1\cos\theta}{a^2}-\frac{y_1\sin\theta}{b^2}\right)r+\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}=0 \quad \begin{matrix}\nearrow r_1\\ \searrow r_2\end{matrix}$$

$$QP\cdot QR=\frac{1}{\frac{\cos^2\theta}{a^2}-\frac{\sin^2\theta}{b^2}}$$

putting values of sin θ and cos θ

$$QP \cdot QR = \frac{a^2b^2\left(1+m^2\right)}{b^2 - a^2m^2}$$

54. A point P on a parabola $y^2 = 4x$, the foot of the perpendicular from it upon the directrix, and the focus are the vertices of an equilateral triangle, find the area of the equilateral triangle.

परवलय $y^2 = 4x$ पर स्थित एक बिन्दु P तथा बिन्दु P से नियता पर डाले गये लम्ब का पाद तथा नाभि यदि किसी समबाहु त्रिभुज के शीर्ष gksa rks=kHkqt d k{ks=kQy cr kb; sA

Ans. $4\sqrt{3}$

Sol. $PM = 1 + t^2$

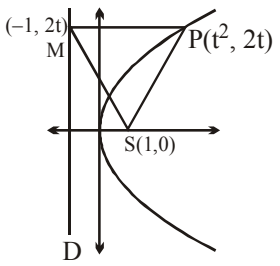
$PS = \sqrt{(t^2-1)^2 + 4t^2} = (t^2+1)$

$MS = \sqrt{4+4t^2} = 2\sqrt{1+t^2}$

$\Rightarrow \quad 2\sqrt{1+t^2} = 1 + t^2$

$\therefore \quad PM = 1 + t^2 = 4 \quad \Rightarrow \quad t^2 = 3$

$\therefore$ Area of $\Delta PMS = \frac{\sqrt{3}}{4}(4^2) = 4\sqrt{3}$ Ans.



55. Consider two concentric circles $S_1 : |z| = 1$ and $S_2 : |z| = 2$ on the Argand plane. A variable parabola is drawn through the points where 'S_1' meets the real axis and having arbitrary tangent drawn to 'S_2' as its directrix. If the locus of the focus of the parabola is a conic C then find the area of the quadrilateral formed by the tangents at the ends of the latus-rectum of conic C.

माना दो सकेन्द्रिय वृत्त $S_1 : |z| = 1$ और $S_2 : |z| = 2$ सममित तल में है तथा 'S_1' के वास्तविक अक्ष के साथ प्रतिच्छेद बिन्दुओं से xtq j usokyk ki j oy ; ft l dh hfu; r koØ 'S_2' की स्पर्श रेखा है। यदि इस प्रकार के परवलय के केन्द्र का बिन्दुपथ C है तो वक्र C के नाभिलम्बों के सिरों पर खींची गयी स्पर्श रेखाओं से बने चर्तुभुज का क्षेत्रफल होगा।

Ans. 16

Sol. Clearly the parabola should pass through (1, 0) and (–1,0). Let directrix of this parabola be $x\cos\theta + y\sin\theta = 2$. If M (h,k) be the focus of this parabola, then distance of (±1, 0) from 'M' and from the directrix should be same.

$\Rightarrow \quad (h-1)^2 + k^2 = (\cos\theta - 2)^2 \quad$ _____(1)

and $\quad (h+1)^2 + k^2 = (\cos\theta + 2)^2 \quad$ _____(2)

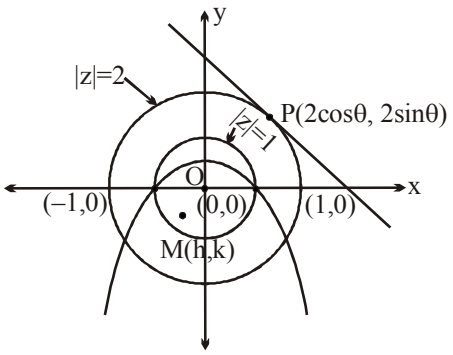
Now (2) – (1) $\Rightarrow \cos\theta = \frac{h}{2}$ _____(3)

Also (2) + (1) $\Rightarrow (h^2 + k^2 + 1) = (\cos^2\theta + 4)$ _____(4)

$\therefore$ From (3) and (4), we get

$h^2 + k^2 + 1 = 4 + \frac{h^2}{4} \Rightarrow \frac{3h^2}{4} + k^2 = 3$

Hence locus of focus M(h, k) is $\frac{x^2}{4} + \frac{y^2}{3} = 1$ (Ellipse)



Also we know that area of the quadrilateral formed by the tangents at the ends of the latus-rectum is $\frac{2a^2}{e}$ (where e is eccentricity of ellipse)

$\therefore$ Requred area = $\frac{2(4)}{1/2} = 16$ (square units) (As $e^2 = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow e = \frac{1}{2}$)

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 56 to 58

From a point 'P' three normals are drawn to the parabola $y^2 = 4x$ such that two of them make angles with the abscissa axis, the product of whose tangents is 2. Suppose the locus of the point 'P' is a part of a conic 'C'. Now a circle S = 0 is described on the chord of the conic 'C' as diameter passing through the point (1, 0) and with gradient unity. Suppose (a, b) are the coordinates of the centre of this circle. If L_1 and L_2 are the two asymptotes of the hyperbola with length of its transverse axis 2a and 2b (principal axes of the hyperbola along the coordinate axes) then answer the following questions.

परवलय $y^2 = 4x$ पर, किसी बिन्दु P से, तीन अभिलम्ब इस प्रकार खींचे जाते है। उनमें से दो की प्रवणताओं का गुणनफल 2 है। माना बिन्दु P का बिन्दुपथ शांकव 'C' का भाग है। अब वृत्त S = 0 शांकव 'C' की जीवा को व्यास मानकर बनाया गया वृत्त है जो बिन्दु (1, 0) से गुजरता है तथा जिसकी प्रवणता इकाई है। माना (a, b) वृत्त के केन्द्र के निर्देशांक हो तो तथा L_1 तथा L_2 अतिपरवलय के दो अनन्त स्पर्शी है। जिसके अनुप्रस्थ अक्ष की लम्बाई 2a है तथा संयुग्मी अक्ष की लम्बाई 2b (अतिपरवलय की मुख्य अक्ष निदेर्शी अक्षों के अनुदिश है) तो fuEu i'uksa ds mRrj nhft , A

56. Locus of P is a :

(A) circle (B) parabola (C) ellipse (D) hyperbola

बिन्दु P का बिन्दुपथ है :

(A) वृत्त (B) परवलय (C) दीर्घवृत्त (D) अतिपरवलय

Ans. B

57. Radius of the circle $S = 0$ is :

वृत्त $S = 0$ की त्रिज्या :

(A) 4 (B) 5 (C) $\sqrt{17}$ (D) $\sqrt{23}$

Ans. A

58. The angle $\alpha \in (0, \pi/2)$ between the two asymptotes of the hyperbola lies in the interval :

अतिपरवलय के अनन्त स्पर्शियों के मध्य कोण $\alpha \in (0, \pi/2)$ निम्न अन्तराल में होगा :

(A) $(0, 15°)$ (B) $(30°, 45°)$ (C) $(45°, 60°)$ (D) $(60°, 75°)$

Ans. D

Sol. Equation of a normal $y = mx - 2m - m^3$

passes through (h, k)]

$$m^3 + (2 - h)m + k = 0$$

$$m_1m_2m_3 = -k$$

but $m_1 m_2 = 2$

$\Rightarrow$ $m_3 = -k/2$

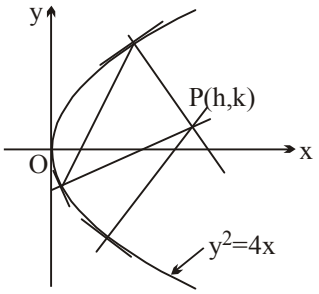
this must satisfy equation (1)

$$\frac{k^3}{8} - (2 - h)\frac{k}{2} + k = 0$$

$$k^3 - 4k(2 - h) + 8k = 0 \quad (k \neq 0)$$

$$k^2 - 8 - 4h + 8 = 0 \Rightarrow y^2 - 4x = 0 \Rightarrow y^2 = 4x$$

locus of 'P' is $y^2 = 4x$ which is a parabola Ans.



now chord passing through $(1, 0)$ is the focal chord.

Given that gradient of focal chord is 1

$$\therefore \quad \frac{2}{t_1 + t_2} = 1 \quad \Rightarrow \quad t_1 + t_2 = 2, \text{ Also } t_1t_2 = -1$$

equation of circle described on t_1t_2 as diameter is

$$(x - t_1^2)(x - t_2^2) + (y - 2t_1)(y - 2t_2) = 0$$

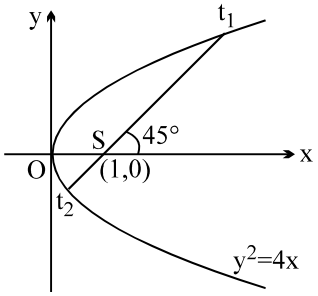
$$x^2 + y^2 - x(t_1^2 + t_2^2) + t_1^2 t_2^2 - 2y(t_1 + t_2) + 4t_1t_2 = 0$$

$$x^2 + y^2 - x[4 + 2] + 1 - 2y(2) - 4 = 0$$

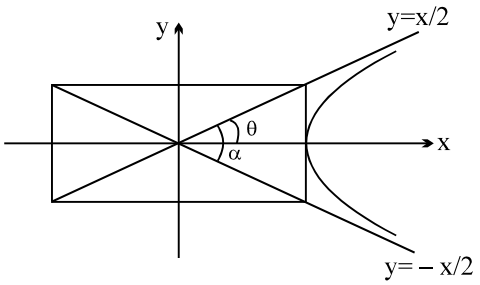
$$x^2 + y^2 - 6x - 4y - 3 = 0$$

centre $a = 3$ and $b = 2$; $r = 4$ Ans.

now the hyperbola is $\frac{x^2}{9} - \frac{y^2}{4} = 1$



asymptotes are $y = \frac{2x}{3}$ and $y = -\frac{2x}{3}$



now $\tan\theta = 2/3$

$\therefore \quad \alpha = 2\theta$

$\tan\alpha = \frac{2\cdot(2/3)}{1-(4/9)}; \qquad \tan\alpha = \frac{12}{5}; \qquad \alpha = \tan^{-1}\left(\frac{12}{5}\right)$

hence $\alpha \in (60°, 75°)$ Ans.

Paragraph for question nos. 59 to 61

A conic C passes through the point (2, 4) and is such that the segment of any of its tangents at any point contained between the co-ordinate axes is bisected at the point of tangency. Let S denotes circle described on the foci F_1 and F_2 of the conic C as diameter.

एक शांकव C बिन्दु (2, 4) से गुजरता तथा उस पर स्थित किसी बिन्दु पर खींची स्पर्श रेखा का अक्षों के साथ काटा गया अन्तःखण्ड़ स्पर्श fcUnq j l ef} Hkkx r gks kgS Sekuk' kadk o C की नाभियों को व्यास मानकर बनाया गया वृत्त S है। तो।

59. Vertex of the conic C is :

शांकव C का शीर्ष होगा :

(A) (2, 2), (–2, – 2) (B) $\left(2\sqrt{2}, 2\sqrt{2}\right), \left(-2\sqrt{2}, -2\sqrt{2}\right)$

(C) (4, 4), (–4, – 4) (D) $\left(\sqrt{2}, \sqrt{2}\right), \left(-\sqrt{2}, -\sqrt{2}\right)$

Ans. B

60. Director circle of the conic is :

(A) $x^2 + y^2 = 4$ (B) $x^2 + y^2 = 8$ (C) $x^2 + y^2 = 2$ (D) None

शांकव का नियामक वृत्त होगा :

(A) $x^2 + y^2 = 4$ (B) $x^2 + y^2 = 8$ (C) $x^2 + y^2 = 2$ (D) कोई नहीं

Ans. D

61. Equation of the circle S is :

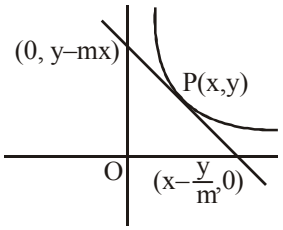
वृत्त S का समीकरण होगा :

(A) $x^2 + y^2 = 16$ (B) $x^2 + y^2 = 8$ (C) $x^2 + y^2 = 32$ (D) $x^2 + y^2 = 4$

Ans. C

Sol. $Y - y = m(X - x)$; if $Y = 0$ then

$X = x - \frac{y}{m}$ and if $X = 0$ then $Y = y - mx$.



Hence $x - \frac{y}{m} = 2x \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$

$\int \frac{dy}{y} + \int \frac{dx}{x} = c \Rightarrow xy = c$

passes through (2, 4)

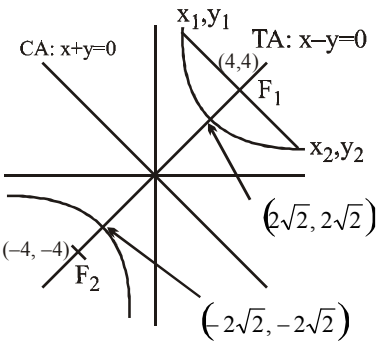
$\Rightarrow$ equation of conic is $xy = 8$

which is a rectangular hyperbola with $e = \sqrt{2}$.

Hence the two vertices are $(2\sqrt{2}, 2\sqrt{2})$, $(-2\sqrt{2}, -2\sqrt{2})$

focii are (4, 4) & (–4, 4)

$\therefore$ Equation of S is $x^2 + y^2 = 32$ Ans.



MATCH THE COLUMN TYPE QUESTION(S)

62. Consider the parabola $y^2 = 12x$:

Column-I	**Column-II**
(A) Tangent and normal at the extremities of the latus rectum intersect the x axis at T and G respectively. The coordinates of the middle point of T and G are	(P) (0, 0)
(B) Variable chords of the parabola passing through a fixed point K on the axis, such that sum of the squares of the reciprocals of the two parts of the chords through K, is a constant. The coordinate of the point K are	(Q) (3, 0) (R) (6, 0)
(C) All variable chords of the parabola subtending a right angle at the origin are concurrent at the point	
(D) AB and CD are the chords of the parabola which intersect at a point E on the axis. The radical axis of the two circles described on AB and CD as diameter always passes through	(S) (12, 0)

परवलय $y^2 = 12x$ के लिए :

Column-I	**Column-II**
(A) नाभिलम्ब के अन्तः बिन्दु पर स्पर्श रेखा और अभिलम्ब x-अक्ष को क्रमशः T और G पर प्रतिच्छेद करते हैं तब T और G के मध्य बिन्दु के निर्देशांक है।	(P) (0, 0)
(B) परवलय की चर जीवा, अक्ष पर स्थित स्थिर बिन्दु K से गुजरती है। इस प्रकार की जीवा के K द्वारा दो भागों के व्युत्क्रमों के वर्गो का योग अचर है। तब k के निर्देशांक होंगे।	(Q) (3, 0)

(C) परवलय की सभी चर जीवाऐं केन्द्र पर समकोण निर्मित करती है तथा एक ही बिन्दु से गुजरती है। उस बिन्दु के निर्देशांक होंगे। (R) (6, 0)

(D) यदि AB और CD परवलय की दो जीवाऐं है। जो अक्ष पर स्थित बिन्दु E पर प्रतिच्छेद करती है। तब AB और CD को व्यास मानकर खींचे गये वृत्तों की मूलाक्ष (radical axis) सदैव किस बिन्दु से जायेगी। (S) (12, 0)

Ans. (A) Q; (B) R; (C) S; (D) P

Sol. (A) $L(a, 2a) \equiv (3,6)$

$T : 6y = 6(x + 3)$ $\qquad$ $N : y - 6 = -1(x - 3)$

$x - y + 3 = 0$ $\qquad$ $x + y - 9 = 0$

$T(-3,0)$ $\qquad$ $G(9,0)$

mid point $= (3,0)$

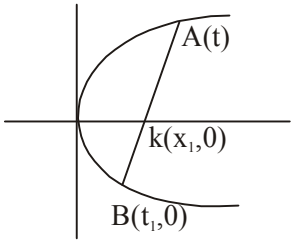
(B) $\dfrac{1}{Ak^2} + \dfrac{1}{Bk^2} = \text{constant}$

$$AB : y - 2at = \frac{2}{t_1 + t}\left(x - at^2\right)$$

Put $y = 0$

$\Rightarrow x = -att_1$

$K(-att_1, 0)$



$$AK = \sqrt{\left(at_1^2 + att_1\right)^2 + (2at)^2}$$

$$= at\sqrt{(t + t_1)^2 + 4}$$

$$BK = at_1\sqrt{(t + t_1)^2 + 4}$$

$$\frac{1}{AK^2} + \frac{1}{BK^2} = \text{constant}$$

$$\frac{1}{a\left[(t_1 + t)^2 + 4\right]}\left(\frac{1}{t_1^2} + \frac{1}{t^2}\right) = \text{constant}$$

$$\frac{t_1^2 + t^2}{(tt_1)^2\left(t + t_1^2 + 4 + 2tt_1\right)} = \text{constant}$$

if $4 + 2tt_1 = 0 \Rightarrow t_1t = -2$, $a = 3$

$K(-att_1, 0) = K(6,0)$

(C) Let equation of chord $y = mx + c$

homogenting $\quad y^2 - 12x\left(\frac{y - mx}{c}\right) = 0$

$cy^2 - 12xy + 12mx^2 = 0$

$12m + c = 0 \Rightarrow c = -12m$

Chord $\quad y = m(x - 12)$

$y = 0$, $x = 12$

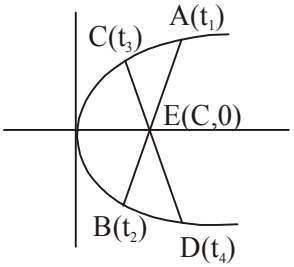
(12,0)

(D) AB : $y - 2at_1 = \frac{2}{t_1 + t_2}\left(x - at_1^2\right)$

$2x - (t_1 + t_2)y + 2at_1t_2 = 0$

(c, 0)

$t_1t_2 = -\frac{c}{a}$



Similarly $\quad t_3t_4 = -\frac{c}{a}$

Circle with AB diameter

$S_1 : x^2 + y^2 - a(t_1^2 + t_2^2)x - 2a(t_1 + t_2)y + c^2 - 4ac = 0$

Circle with CD diameter

$S_2 : x^2 + y^2 - a(t_3^2 + t_4^2)x - 2a(t_3 + t_4)y + c^2 - 4ac = 0$

Radical axis

$S_1 - S_2 = 0$

$a\left(t_1^2 + t_2^2 - t_3^2 - t_4^2\right)x - 2a\left(t_1 + t_2 - t_3 - t_4\right)y = 0$

which passes through (0,0).

63. Match the properties given in column-I with the corresponding curves given in the column-II.

Column-I	**Column-II**
(A) The curve such that product of the distances of any of its tangent from two given points is constant, can be	(P) Circle
(B) A curve for which the length of the subnormal at any of its point is equal to 2 and the curve passes through (1, 2), can be	(Q) Parabola
(C) A curve passes through (1, 4) and is such that the segment joining any point P on the curve and the point of intersection of the normal at P with the x-axis is bisected by the y-axis. The curve can be	(R) Ellipse
(D) A curve passes through (1, 2) is such that the length of the normal at any of its point is equal to 2. The curve can be	(S) Hyperbola

Match the properties given in column-I with the corresponding curves given in the column-II.

Column-I	**Column-II**
(A) एक वक्र जिस पर खींची गयी स्पर्श रेखाओं पर किन्ही दिय बिन्दुओं से लम्बाईयों का गुणनफल अचर है, हो सकता है	(P) वृत्त
(B) एक वक्र पर स्थित किसी बिन्दु से खींचे गये अधोलम्ब का मान 2 है तथा वक्र बिन्दु (1, 2) से गुजरता है, हो सकता है	(Q) परवलय
(C) एक वक्र बिन्दु (1, 4) से गुजरता है इस प्रकार है कि P बिन्दु तथा P पर खींचे अभिलम्ब का x-अक्ष के साथ प्रतिच्छेद बिन्दु से बना रेखा खण्ड y-अक्ष पर समद्विभाजित होता है। तो वक्र हो सकता है	(R) दीर्घवृत्त
(D) एक वक्र जो बिन्दु (1, 2) से गुजरता है तथा इस बिन्दु पर खींचे गये v fHkyEc d hy Ecbl2 है तो वक्र हो सकता है	(S) अतिपरवलय

Ans. (A) R, S; (B) Q; (C) R; (D) P

Sol. (A) Very important property of ellipse and hyperbola $(p_1p_2 = b^2) \Rightarrow$ (R), (S)

(B) $y\frac{dy}{dx} = 2 \quad \Rightarrow \quad \frac{y^2}{2} = 2x + C$

$x = 1, y = 2 \quad \Rightarrow \quad C = 0$

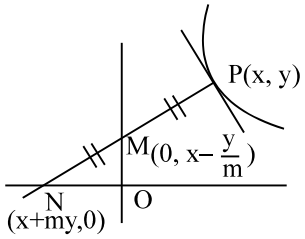
$\Rightarrow \quad y^2 = 4x \Rightarrow$ parabola $\Rightarrow$ (Q)

(C) Equation of normal at P

$$Y - y = -\frac{1}{m}(X - x)$$

$Y = 0, \; X = x + my$

$X = 0, \; Y = y - \frac{x}{m}$



hence $x + my + x = 0 \Rightarrow \quad 2x + y\frac{dy}{dx} = 0$

$2x\,dx + y\,dy = 0$

$$x^2 + \frac{y^2}{2} = C \quad \text{passes through } (1, 4)$$

$$1 + 8 = C$$

hence $x^2 + \frac{y^2}{2} = 9 \Rightarrow \frac{x^2}{9} + \frac{y^2}{18} = 1 \Rightarrow$ ellipse $\Rightarrow$ (R)

(D) length of normal

$$(x + my - x)^2 + y^2 = 4$$

$$m^2y^2 + y^2 = 4$$

$$m^2 = \frac{4 - y^2}{y^2}; \quad \frac{dy}{dx} = \frac{\sqrt{4 - y^2}}{y}; \quad \int \frac{y\,dy}{\sqrt{4 - y^2}} = \int dx$$

$$-\sqrt{4 - y^2} = x + C$$

$x = 1, y = 4 \Rightarrow C = -1$

$\therefore \quad (x - 1)^2 = 4 - y^2$

$(x - 1)^2 + y^2 = 4 \Rightarrow$ circle $\Rightarrow$ (P)

64. Consider an ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ with centre C and a point P on it with eccentric angle $\frac{\pi}{4}$. Normal drawn at P intersects the major and minor axes in A and B respectively. N_1 and N_2 are the feet of the perpendiculars from the foci S_1 and S_2 respectively on the tangent at P and N is the foot of the perpendicular from the centre of the ellipse on the normal at P. Tangent at P intersects the axis of x at T.

Match the entries of Column-I with the entries of Column-II.

Column-I	**Column-II**
(A) (CA)(CT) is equal to	(P) 9
(B) (PN)(PB) is equal to	(Q) 16
(C) $(S_1N_1)(S_2N_2)$ is equal to	(R) 17
(D) $(S_1P)(S_2P)$ is equal to	(S) 25

माना एक दीर्घवृत $\frac{x^2}{25} + \frac{y^2}{9} = 1$ जिसका केन्द्र C तथा इस पर स्थित एक बिन्दु P का उत्केन्द्रीय कोण $\frac{\pi}{4}$ है। P पर बनाया गया अभिलम्ब दीर्घ–अक्ष एवं लघु–अक्ष को क्रमशः A तथा B पर मिलता है। यदि P पर बनायी गई स्पर्श रेखा पर नाभियों S_1 एवं S_2 से डाले गये लम्बों के पाद क्रमशः N_1 तथा N_2 है तथा N दीर्घवृत के केन्द्र से P पर बनाए गए अभिलम्ब पर डाला गया लम्ब पाद है। P पर स्पर्श रेखा x-अक्ष को T पर प्रतिच्छेद करती है

Ir Hkk I के अवयवों को स्तम्भ-II से मिलान कीजिये

Column-I	**Column-II**
(A) ;s । ds ctkcj gS	(P) 9
(B) ;s PS ds cjkcj gS	(Q) 16
(C) $(S_1N_1)(S_2N_2)$ cjkcj gS	(R) 17
(D) $(S_1P)(S_2P)$ cjkcj gS	(S) 25

Ans. (A) Q; (B) S; (C) P; (D) R

Sol. (A) $(CA)(CT) = \left(e^2 a\cos\theta\right)\cdot\left(\frac{a^2}{a\cos\theta}\right)$

$= a^2 - b^2 = 16$

(B) (PN) (PB $= a^2 = 25$

(C) (S_1N_1) $(S_2N_2) = b^2 = 9$

(D) (S_1P) $(S_2P) = [a + e(a\cos\theta)]\,[a - e(a\cos\theta)]$

$= a^2\sin^2\theta + b^2\cos^2\theta = 17$

65. Match the conics in Column I with the statements/expressions in Column II : **[JEE-2009]**

Column I	**Column II**
(A) Circle	(P) The locus of the point (h, k) for which the line $hx + ky = 1$ touches the circle $x^2 + y^2 = 4$
(B) Parabola	(Q) Points z in the complex plane satisfying $\|z + 2\| - \|z - 2\| = \pm 3$
(C) Ellipse	(R) Points of the conic have parametric representation $x = \sqrt{3}\left(\frac{1-t^2}{1+t^2}\right),\ y = \frac{2t}{1+t^2}$
(D) Hyperbola	(S) Τηε εχχεντριχιτψοφτηε χονιχ λεσιν τηε ιντερβαλ $1 \le x < \infty$
	(T) Points z in the complex plane satisfying $\mathrm{Re}(z + 1)^2 = \|z\|^2 + 1$

स्तम्भ–I में दिये गये शाँकवों (conics) को स्तम्भ–II में दिए गये वक्तव्यों/व्यंजको में सुमेल कीजिए : **[JEE-2009]**

स्तम्भ–I	**स्तम्भ–II**
(A) वृत्त	(P) ऐसे बिन्दु (h, k) का बिन्दु पथ जिसके लिये रेखा $hx + ky = 1$ वृत्त $x^2 + y^2 = 4$ को स्पर्श करती है
(B) परवलय	(Q) सम्मिश्र तल में $\|z + 2\| - \|z - 2\| = \pm 3$ को संतुष्ट करने वाले बिन्दु z
(C) दीर्घवृत्त	(R) शाँकव के बिन्दुओं का प्राचलिक–निरूपण $x = \sqrt{3}\left(\frac{1-t^2}{1+t^2}\right),\ y = \frac{2t}{1+t^2}$
(D) अतिपरवलय	(S) शाँकव की उत्केन्द्रता (eccentricity) अन्तराल $1 \le x < \infty$ में है
	(T) सम्मिश्र–तल में $\mathrm{Re}(z + 1)^2 = \|z\|^2 + 1$ को संतुष्ट करने वाले बिन्दु z

Ans. (A) P, (B) S, T; (C) R ; (D) Q, S

Sol. (p) $hx + ky = 1$ touches $x^2 + y^2 = 4$

$$\Rightarrow \left|\frac{-1}{\sqrt{h^2+k^2}}\right| = 2 \qquad \Rightarrow h^2 + k^2 = \frac{1}{4}(\text{circle})$$

(q) $||z + 2| - |z - 2|| = 3$

$e = \frac{4}{3} > 1$ i.e. (hyperbola)

(r) $x = \sqrt{3}\left(\frac{1-t^2}{1+t^2}\right), \qquad y = \frac{2t}{1+t^2}$ put $t = \tan\theta$

$\therefore \quad x = \sqrt{3}\cos 2\theta$

$y = \sin 2\theta$

$\Rightarrow \left(\frac{x}{\sqrt{3}}\right)^2 + y^2 = 1$ (ellipse)

(s) Clearly parabola or hyperbola

(t) Let $z = x + iy$

$\text{Re}(z+1)^2 = (x+1)^2 - y^2 = x^2 + y^2 + 1$

$y^2 = x$ (parabola)